

Fedokawaii — learn

Every key, every program, and two languages for the age of AI: from your first command to a small language model of your own

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How to read this book

This book is the practical one. It teaches by doing: a short example, what you should see, then an exercise of your own. The full handbook — how every part of the desktop is built and why — stays in the project's repository on GitHub (`docs/fedokawaii-guide.md`), for the day you want to know how something works inside. This book is about *using* it.

It has five parts. Read Part I first if you have never typed a command. After that, go where you need to: each chapter stands on its own and says what it expects you to know.

Part	You learn	Chapters
I — From zero	what a terminal, a file, a path, a program and a variable are; how to read an error	1–3
II — The desktop	the keys, windows, the bar and menus, the clipboard, screenshots, updates, snapshots	4–8
III — The programs	each program, its best moves, real uses and exercises	9–22
IV — Python	a daily development routine, then a full course in six levels: from your first number to classes, tests, async and packaging, then data, NumPy, a neural network, a transformer, fine-tuning and an agent	23–47
V — Rust	a full course in six levels: from cargo to ownership, traits, lifetimes, tests, iterators and threads, then a CLI, your own autograd, models in Rust, and Rust inside Python	48–66
Appendices	a glossary; the worked solution to every exercise of Parts IV and V, with its output	A–C

How a lesson looks

Every new idea comes as an example first: you type it, then you see what it printed.

```
$ echo hello
hello
```

(the `$` is the prompt — do not type it; the line under it is what you should see)

A line starting with `$` is what you type, without the `$`. The lines under it are what the computer answers. Your answer may differ in details (a name, a date, a number): what matters is its shape. A block without `$` is a file's contents, to type into an editor.

The boxes

`Word — prompt.` The text the terminal shows when it waits for your command. Each new word gets a box like this the first time it appears.

Checkpoint. A short test at the end of a section. If you can do it without looking back, go on. If not, read the section again: it costs five minutes now and saves an hour later.

Careful. Something that can lose work or break a setting. Read it before you type.

The exercises

Each chapter ends with exercises, graded by stars:

Stars	Means	Takes
★	do exactly what the chapter showed, on your own files	a minute
★★	combine two things from the chapter	five minutes
★★★	a small project: you decide how	half an hour
★★★★	a challenge: it stretches what the chapter taught	an evening

Every exercise says what you should see when it works. In Parts IV and V, every exercise has a worked solution at the back — appendix B for Python, C for Rust, under the exercise's number — with the output it printed. Try

for real before you look: the struggle is where the learning happens; the solution is there to check, or to unstick you after twenty honest minutes. Keep a line for each one you finish in your notes (`Super+N`): `- [x] 10.3 every PDF I own`. Watching that list grow is half the fun.

Three rules that make learning fast

1. **Type, never paste.** Your fingers learn the shapes: brackets, quotes, the order of words.
2. **One small step, then check.** Run it after every change. A mistake found at once is a mistake you understand.
3. **Read the error.** The last line of an error message almost always says what is wrong (chapter 3 shows how).



Part I — From zero

Never typed a command? Start here. Three short chapters: what the pieces are called, your first ten minutes in the terminal, and how to read the computer's complaints.

始 1. What the pieces are

In this chapter

- what a file, a folder, a path, a program, a command and a variable are
- one example of each, on this computer

You need: nothing.

Everything on a computer is one of a few kinds of thing. Once you can name them, every manual becomes readable.

始 1.1 Files and folders

A **file** is a named piece of stored data: a photo, a song, a letter, a program. A **folder** (also called a *directory*) holds files and other folders. Your own files live in your **home folder**, `/home/you` (with your user name instead of `you`), written `~` for short.

Thing	Example on this desktop	What it is
a file	`~/Notes/notes.md`	your notes: plain text
a folder	`~/Pictures/Screenshots`	where screenshots land
your home	`~` = `/home/you`	everything that is yours
the top of everything	`/`	every file on the computer sits somewhere below it

始 1.2 Paths

A **path** is a file's address: the folders you walk through to reach it, separated by `/`.

```
/home/you/Notes/notes.md
 |  |     |    |      └──── the file
 |  |     |    └─────── a folder inside your home
 |  |     └──────── your home folder
 |  └─────── the folder that holds every home
 └──────── the top: the root
```

`Word — absolute and relative path.` An *absolute* path starts at `/` (or `~`) and works from anywhere. A *relative* path starts where you are: standing in your home, `Notes/notes.md` means `/home/you/Notes/notes.md`. `..` means "one folder up", `.` means "here".

始 1.3 Programs and commands

A **program** is a file the computer can run. `nvim` (the editor), `mpv` (the video player) and `ls` (lists files) are programs. A **command** is a line you type to run one, often with **arguments** — words that tell it what to work on — and **options**, which start with `-`.

```
ls  -l  ~/Pictures
 |    |    └──── an argument: which folder
 |    └──── an option: the long listing
 └──── the program
```

始 1.4 Variables

A **variable** is a name that holds a value. The shell has some already: `$HOME` holds your home's path, `$USER` your name. In programming (Part IV), variables are the first thing you make.

```
$ echo $HOME
/home/you
$ name=Kyoto
$ echo "I would like to visit $name"
I would like to visit Kyoto
```

(no spaces around `=`: `name = Kyoto` would run a program called `name`)

始 1.5 Text, and why it matters here

Almost everything on this desktop is plain text: your notes, every setting, every script. Plain text is readable by any program, searchable (`rg`, chapter 10), and its history can be kept (git, chapter 14). That is why this book keeps returning to the terminal and the editor: they are the two tools that work on everything.

Exercises

1.1 ★ Name the parts. In the command `cp -r ~/Pictures /run/media/stick`, which word is the program, which the option, which the arguments?

Expected: the program is `cp` (copy), `-r` is an option (recursive: folders too), the two paths are arguments: from where, to where.

1.2 ★ Absolute or relative? Which of these work from any folder: `~/Notes`, `Notes`, `/etc/hosts`, `../Music`?

Expected: `~/Notes` and `/etc/hosts` (absolute); the other two depend on where you stand.

端 2. Your first ten minutes in the terminal

In this chapter

- open a terminal and run your first commands
- move between folders, make and remove a folder
- get help on any command

You need: chapter 1.

端 2.1 Open it

Press `Super+Return` (Super is the key with the Windows logo). A terminal window opens: foot, the desktop's terminal. It shows a **prompt**, then a blinking cursor.

```
⛩  ~ ------------------------------------------------ ☾  08:52:07 PM
›
```

(the top line shows the folder you are in — `~`, your home — and the time; you type after the `›`. After a command that failed, the top line also shows a vermilion `×` and a number: chapter 3)

`Word — shell.` The program inside the terminal that reads your commands and runs them. Here it is **bash**. The terminal is the window; the shell is what talks to you inside it.

端 2.2 Five commands, one at a time

Type each line (without the `$`) and press `Enter`. After each one, compare with the answer.

```
$ pwd
/home/you
```

(print working directory: where you are)

```
$ ls
Books  Documents  Downloads  Music  Notes  Pictures  repos  Videos
```

(list: what is here; your list may differ)

```
$ mkdir practice
$ cd practice
$ pwd
/home/you/practice
```

(make a folder, change into it, check)

```
$ echo "my first file" > first.txt
$ ls
first.txt
$ cat first.txt
my first file
```

(`>` sends what echo prints into a file; `cat` shows a file)

```
$ cd ..
$ pwd
/home/you
```

(`..` is one folder up)

`Checkpoint.` Without looking: make a folder `test2` in your home, go into it, make a file `hello.txt` holding `hi`, show it, go back home. Then delete it all: `rm -r ~/test2`.

端 2.3 The keys that save the most time

Keys	Does
`Tab`	completes a name: type `cd Doc` then Tab, it becomes `cd Documents/`
Up · **Down**	earlier commands; type the start first (`git`) and Up finds only those
`Ctrl+R`	search every command you ever ran (fzf): type a few letters, Enter
`→` or **End**	take the grey suggestion the shell offers as you type
`Ctrl+C`	stop the running command (it does *not* copy here)
`Ctrl+Shift+C` · `Ctrl+Shift+V`	copy · paste in the terminal
`Ctrl+L`	clear the screen
`Ctrl+Shift+N`	a new terminal, in this same folder

端 2.4 Help is always one command away

You want	Type
the five most useful examples of a command	`tldr tar`
its whole manual	`man tar` (`/word` searches, `n` the next match, `q` quits)
a short list of its options	`tar --help`
the desktop's keys	`Super+/`

```
$ tldr mkdir
  Create directories and set their permissions.
  - Create specific directories:
    mkdir path/to/directory1 path/to/directory2 ...
  - Create specific directories and their parents if needed:
    mkdir -p path/to/directory1 path/to/directory2 ...
```

(for example; shortened)

Exercises

2.1 ★ A small tree. Make `~/practice/a/b/c` in one command (`mkdir -p`), then go into `c` using Tab to complete each name.

Expected: the prompt ends in `c`; `pwd` prints `/home/you/practice/a/b/c`.

2.2 ★ History. Press `Ctrl+R`, type `mkdir`, and run the command you typed in 2.1 again.

Expected: no error (`-p` does not mind that the folders exist).

2.3 ★★ Count. `ls ~/Pictures/Screenshots | wc -l` counts your screenshots. Take one with `Print` (drag a box), then count again.

Expected: one more than before. `|` sends what `ls` prints into `wc -l`, which counts lines.

3. Reading errors

In this chapter

- read an error message from its last line up
- recognise the ten errors you will meet most

You need: chapter 2.

An error is not a failure: it is the computer telling you, quite precisely, what it could not do. Programmers spend much of their day reading them. The trick is to read in the right order.

3.1 The method: last line first

1. Read the **last line**. It names the problem.
2. Find the **file and line number** it mentions. Go there.
3. Only then read the rest, from the bottom up: it is the path that led there.

```
$ python3 hello.py
Traceback (most recent call last):
  File "/home/you/hello.py", line 3, in <module>
    print(nmae)
          ^^^^
NameError: name 'nmae' is not defined
```

(the last line says what: a name nobody defined; the line above says where: line 3)

3.2 The ten you will meet most

You see	It means	Do
`command not found`	no program by that name: a typo, or not installed	check the spelling; `tldr` the name
`No such file or directory`	the path is wrong	`ls` the folder; use Tab to complete paths
`Permission denied`	not yours to change	system files: `sudoedit FILE`; a script: `chmod +x FILE`
`Is a directory`	you gave a folder where a file was wanted	add the file's name
`SyntaxError` (Python)	the code is not well formed: a missing `:`, `)` or quote	look at the line named, *and the one before*
`IndentationError` (Python)	the spaces at a line's start do not match	use the editor's indent (4 spaces) everywhere
`NameError` (Python)	a name used before it was given a value, or misspelt	compare the spelling with where you made it
`ModuleNotFoundError` (Python)	a library is not in this project	`uv add NAME` (chapter 24)
`error[E0382]: borrow of moved value` (Rust)	a value was given away, then used again	chapters 51–52: borrow it with `&`
`mismatched types` (Rust)	a number where text was expected, or similar	read the `expected` ··· `found` ··· lines

`Careful.` A long red error is not worse than a short one. Python and Rust print the whole path of calls; only the top and the bottom usually matter.

3.3 Ask for an explanation

When an error makes no sense, copy it (select it in the terminal: selecting copies) and ask Claude Code in the same folder (chapter 21): *"explain this error and what I should look at — do not fix it"*. You learn more from the explanation than from a fix.

Exercises

3.1 ★ Break it on purpose. Run `cd Nowhere`, `lss`, and `cat ~/Pictures`. Read each message.

Expected: `No such file or directory`, `command not found`, `Is a directory`.

3.2 ★★ Find the line. Save this as `~/practice/oops.py` (in neovim: `nvim ~/practice/oops.py`, `i`, type, `Esc`, `:wq`), run `python3 ~/practice/oops.py`, and fix it from the message alone:

```
prices = [3, 5, 8]
total = sum(prices)
print("total:" total)
```

Expected: `SyntaxError: invalid syntax. Perhaps you forgot a comma?` on line 3. With the comma (`print("total:", total)`) it prints `total: 16`.



Part II — The desktop

Five chapters on the desktop itself: the keys and windows, the bar and its menus, copying and capturing, starting programs, and looking after the machine. Learn the keys of chapter 4 first: everything else is faster once your hands know them.

窓 4. Keys, windows and layouts

In this chapter

- open, move, resize and close windows from the keyboard
- use nine workspaces as nine desks
- choose a layout for the work in front of you

You need: chapter 2.

Word — compositor. The program that draws every window and decides where it goes. Here it is **dwl**: it *tiles* windows — places them side by side, never overlapping — so you never drag a window to make room.

窓 4.1 The ten keys to learn first

Keys	Does
`Super+Return`	a terminal
`Super+Space`	the launcher: type a program's name, Enter
`Super+B` · `Super+E`	the browser (Brave) · the files (yazi)
`Super+Q`	close the window
`Super+J` · `Super+K`	focus the next · previous window
`Super+Shift+Return`	move this window to the first, largest place
`Super+F`	fullscreen, and back
`Super+1…9`	go to workspace 1 to 9
`Super+Shift+1…9`	send this window to that workspace
`Super+/`	every key, searchable

窓 4.2 How the windows arrange themselves

The default layout is a **golden spiral**: the first window takes the largest place on the left, each next one a smaller place, winding inward. Three layouts, one key each:

Keys	Layout	Good for
`Super+S`	spiral (the default)	one main window and helpers: an editor, then a terminal, then a browser
`Super+T`	columns, left to right	comparing side by side: two files, two terminals
`Super+Y`	floating	windows you move and size yourself, as on Windows
`Super+Shift+T`	the tiling layout used before	

`Super+H` and `Super+L` make the first window narrower or wider. `Super+Shift+Space` floats one window on its own; **Super+drag** moves it, **Super+right-drag** resizes it.

窓 4.3 Workspaces: nine desks

`Word — workspace.` A separate screen of windows. Only one shows at a time; the others keep their windows exactly as you left them. The bar shows them as kanji numerals: 一 二 三 …

A habit that works: `1` for the browser, `2` for code (editor and terminal), `3` for files, `4` for chat and music, `9` for games. `Super+Tab` jumps back to the workspace you were on: with two desks, it flips between them. `Super+0` shows all at once.

With two monitors, `Super+,` and `Super+.` move the focus left and right, and `Super+Shift+,` · `Super+Shift+.` send the window there.

4.4 The session

Keys	Does
`Super+Escape`	lock the screen (asleep after 90 s)
`Super+Shift+E`	the power menu: lock, sleep, log out, restart, shut down
`Super+Shift+N`	dismiss every notice
`Alt+Shift`	switch the keyboard layout (us / ar)
`Ctrl+Alt+F2…F6`	a text console, should the desktop ever freeze (`Ctrl+Alt+F1` back)

Exercises

4.1 ★ Three windows. Open a terminal, then yazi (`Super+E`), then another terminal. Press `Super+J` five times and watch the vermilion frame move.

Expected: the frame visits each window in turn: the focus follows the order of the spiral.

4.2 ★ Promote. Focus the smallest window and press `Super+Shift+Return`.

Expected: it swaps into the largest place on the left.

4.3 ★★ A coding desk. On workspace 2, open neovim in a terminal (`nvim`) and a second terminal. Switch to columns (`Super+T`), back to the spiral (`Super+S`). Send the second terminal to workspace 3 (`Super+Shift+3`) and follow it (`Super+3`), then go back (`Super+Tab`).

Expected: the bar's 三 lights while it holds a window; Super+Tab returns you to 二.

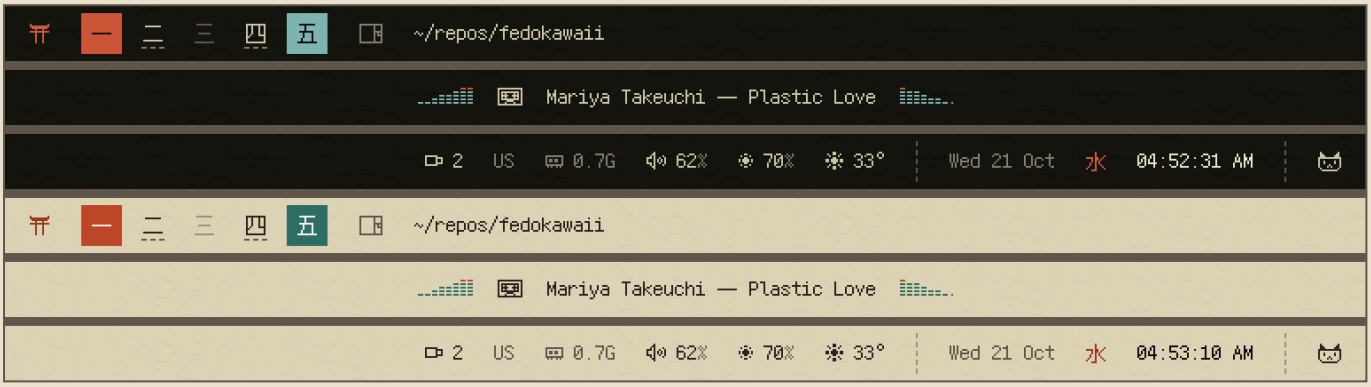
4.4 ★★★ A week without the mouse. For one week, open, move and close every window by key. Keep the help sheet (`Super+/`) a key away; write down the three keys you looked up most.

5. The bar, menus and notices

In this chapter

- read every part of the bar
- open quick settings, sound, calendar and power by key or click
- hold notices back while you work

You need: chapter 4.



The bar, its left, middle and right ends: dark theme above, light below.

5.1 The bar, left to right

Part	Shows	Click
the torii ⛩		the launcher
workspaces 一 二 三 …	the desks; 四–九 only while they hold a window	go there
the layout icon	spiral, columns or floating	switch spiral ⇄ columns
the title	the focused window	
now playing	the song, with a turning cassette and a level meter	play/pause; right: next
a seal 静 or 共有	do-not-disturb on · your screen is being shared	
the drive mark	a USB stick, disk or SD card is plugged in	open or eject it
the download mark	something is downloading: the percent of all of it	the downloads centre (chapter 7)
memory	how much the programs use (5.2G); vermilion past 85 %	btop
volume · brightness	speaker and sun, with the level	the sound panel · the brightness slider
weather	the sky and the temperature where you are	
date, day, time	the day as a kanji seal (月 Monday … 日 Sunday)	the calendar; the time: a timer
the cat	quick settings	quick settings



Quick settings, power, the calendar and the sound panel, on both themes.

5.2 The menus

Menu	Key	Inside
quick settings	`Super+X`	brightness and sound, theme, night light, network, Bluetooth, drives, backup, do-not-disturb
sound panel	`Super+A`	the volume, every output and input, the mixer
calendar	`Super+C`	the month; *next*, *today*
power	`Super+Shift+E`	lock · sleep · log out · restart · shut down
help	`Super+/`	every key; type to filter
wallpaper · theme	`Super+W` · `Super+Shift+W`	a picture for this theme · dark ⇄ light

Every menu works the same: arrows or `j` `k` move, `Enter` chooses, `Esc` closes, and typing filters the list.

The wallpaper picker (`Super+W`) lists the theme's pictures, with the one under the cursor shown underneath, as the screen will show it. Hold ↓ to run through them; `Enter` sets one, `Esc` leaves everything as it was. Its first row is a switch, *a new one every 4 hours* (☐ off, ▣ on): `Enter` turns it over, and from then on the pack's next picture goes up every four hours of a session, by itself. It stays as you left it across logins.

The one transition. A theme switch (`Super+Shift+W`) and a new wallpaper share the desktop's only animation: the old screen is *washed away* in 8 dithered steps — about a quarter of a second, stepped, never a fade. It never holds a click: press `Super+Shift+W` again during it and it jumps to its end, then switches back.

5.3 Notices

A notice appears at the top right and fades by itself; a click does what it offers (opens the download, the drive). `Super+Shift+N` dismisses them all. While you need quiet — a call, an exam — turn on *dnd* in quick settings: notices wait, and **静** stands in the bar.

Exercises

5.1 ★ Every menu once. Open each menu of 5.2 by its key, then close it with `Esc`.

Expected: six menus, each in the same pixel style, each closed by Esc.

5.2 ★ Quiet. Turn on *dnd* (`Super+X`, *dnd*), then run `notify-send hello` in a terminal. Turn *dnd* off.

Expected: no notice while 静 is in the bar; the notice shows once dnd is off.

5.3 ★★ A 25-minute focus timer. Click the time in the bar and choose *25 min*.

Expected: the time left shows in the timer's heading; a notice when it ends.

5.4 ★ A new wallpaper every four hours. `Super+W`, `Enter` on the first row so it shows ▣, `Esc`. Then `Super+Shift+W` twice, and watch the wash each time.

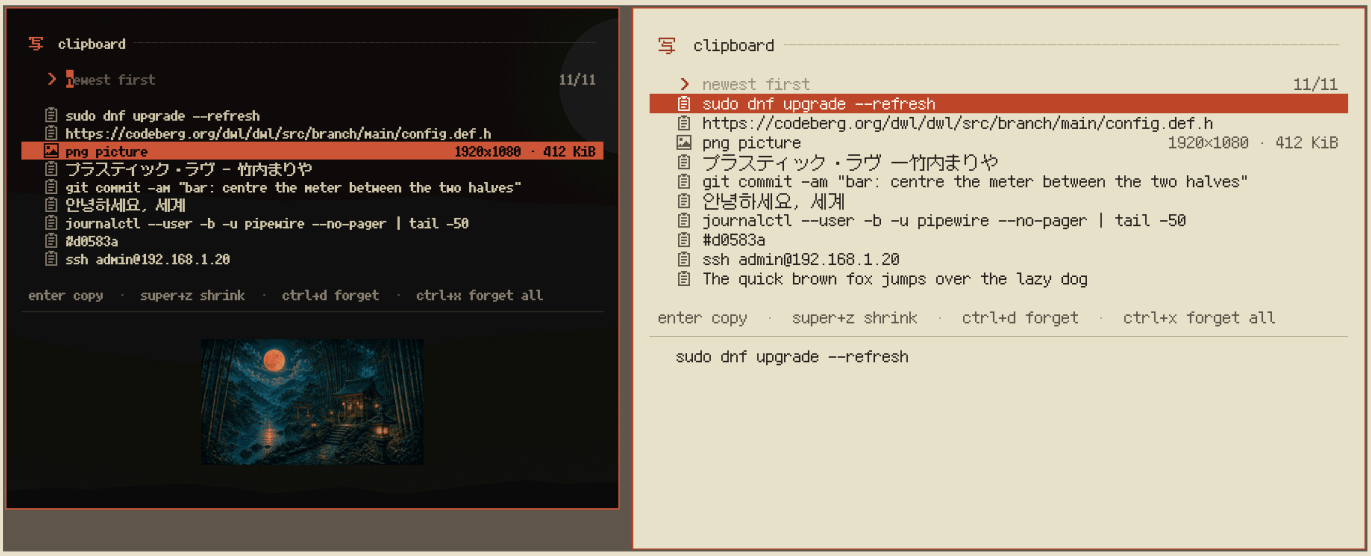
Expected: ▣ stays on after you close the picker; each theme switch washes the screen in a quarter of a second, in steps.

6. Clipboard, screenshots and recordings

In this chapter

- paste anything you copied earlier
- take, mark up and share a screenshot
- record the screen
- shrink a picture or PDF for email

You need: chapter 4.



The clipboard history (Super+V): text and pictures, filtered as you type.

6.1 The clipboard and its history

Where	Copy	Paste
every program	`Ctrl+C`	`Ctrl+V`
the terminal	`Ctrl+Shift+C`, or just select the text	`Ctrl+Shift+V`
anything copied earlier	—	`Super+V`, choose, `Enter`, then paste as usual

The history (`Super+V`) keeps text *and* pictures. Type to filter, `Ctrl+D` forgets one entry, `Ctrl+X` forgets them all.

6.2 Screenshots

Press	Takes	Then
`Print`	a region: drag a box (`Esc` gives up)	saved to `~/Pictures/Screenshots` and copied
`Shift+Print`	the whole screen	saved and copied
`Super+Print`	a region, opened in satty to mark it up	`Enter` copies, `Ctrl+S` saves
`Super+Shift+Print`	a region's **text**, read by tesseract (English and Arabic)	the text copied, nothing saved

No Print key? A small (61-key) keyboard has the same four on letters:

Keys	Same as
`Super+Shift+S`	Print: a region, saved and copied
`Super+Shift+F`	Shift+Print: the whole screen
`Super+Shift+A`	Super+Print: a region to annotate
`Super+Shift+X`	Super+Shift+Print: a region's text

The text key reads what you cannot select — a picture of a document, a video's subtitle, an error in a window that will not let you copy it. *no text found* means the region held none.

In satty: `z` an arrow, `r` a rectangle, `t` text, `m` numbered markers, `b` blur (for a password or a name), **1…5** the colour, `Ctrl+Z` undo.

写 6.3 Recording the screen

`Super+Shift+R` opens a small menu: the whole screen or a region, with the desktop's sound or without. Choose, and it records: **録** and the time gone by stand in the bar, in vermilion. `Super+Shift+R` again — or a click on 録 — stops it. A notice says where the file is (a click plays it), and its path is on the clipboard.

`Super+Shift+P` pauses it: the bar shows **録 停 00:12**, and nothing is recorded — not the screen, not the sound. Press it again and the bar counts `録 3`, `録 2`, `録 1`, a second each, then records on. The pieces become one file when you stop.

What	Where
the recordings	`~/Videos/Recordings/2026-10-20_14-05-31.mp4` (the date and time it started)
the format	MP4, the picture encoded by the graphics card (H.264), the sound AAC

Use it for a bug report, a short tutorial for a friend, or to watch your own coding session back.

写 6.4 Smaller pictures and PDFs: Super+Z

Copied a photo that is too big for an email? `Super+Z` offers: *email · under 1 MB*, *chat · under 500 KB*, a size you type, another format (JPEG, WebP, PNG), or a PDF at three sizes. The result is on the clipboard, ready to paste. In yazi, `Alt+S` does the same to selected files.

写 6.5 Archives and PDFs: Super+D

One menu does what you would otherwise look for a website for. Copy the files (`Ctrl+C` in yazi) and press `Super+D` — or select them in yazi and press `Alt+F`, or use the entry under the cursor in the clipboard history (`Super+V`). *File tools* in the launcher works on what you copied, or opens a file dialog if nothing is. Only the parts that fit your files are shown:

Part	Does
解 unpack	extract here · into a folder · list its contents · test it (zip, 7z, tar.*, rar, iso …)
紙 pages	merge into one PDF (PDFs and pictures) · split · take some pages (`3-7,10`) · rotate · reorder · delete pages
抜 take out	its pictures · its text · pages as pictures
変 change	make a scan searchable · make it smaller (Super+Z's sizes) · lock with a password · unlock
署 sign	write or draw on it · sign by hand · draw your signature once, then stamp it · check digital signatures
包 pack	zip (opens anywhere) · 7z (smaller) · tar.zst, with a password if you want; then the compression: fast, balanced, smallest

The result goes next to the original, named for what was done (`report-merged.pdf`, `report.pages-3-7.pdf`, `photos.zip`) — nothing is ever overwritten — and its path is copied to the clipboard. The notice opens it when you click it.

Exercises

6.1 ★ Annotate. `Super+Print`, drag over this window, put an arrow (`z`) and a number marker (`m`), `Enter`. Paste it into a note (`Super+N` cannot hold pictures — paste it into a Brave chat or email instead).

Expected: the marked-up picture appears where you paste.

6.2 ★ The history. Copy three different words, one after another. Open `Super+V` and paste the first.

Expected: the first word, though you copied two others since.

6.3 ★★ Under 500 KB. Take a full screenshot (`Shift+Print`), then `Super+Z` → *chat*. Check both sizes: `ls -lh ~/Pictures/Screenshots | tail -2`.

Expected: the shrunk copy is under 500 KB.

6.4 ★★ A 20-second demo. Record yourself (`Super+Shift+R`, whole screen, no sound) opening the launcher, starting btop and closing it; stop, and click the notice.

Expected: the recording plays in mpv; it is in `~/Videos/Recordings`.

6.5 ★ Read the screen. Open any picture with text in it (a screenshot of this page will do), press `Super+Shift+Print` (or `Super+Shift+X`), drag over a line of text, and paste it into a note.

Expected: the line arrives as text you can edit; nothing was saved in `~/Pictures`.

6.6 ★★ A pause in the middle. Record a region, pause it with `Super+Shift+P` for ten seconds while you open something private, resume, and stop.

Expected: the bar shows 録 停 while paused, then 録 3 · 2 · 1; the file plays straight through, without the private part.

6.7 ★★ One PDF from three. Copy three PDFs (or pictures) in yazi, `Super+D` → 紙 *pages* → *merge into one PDF*, in the order you chose them. Then take pages `1-2` out of the result.

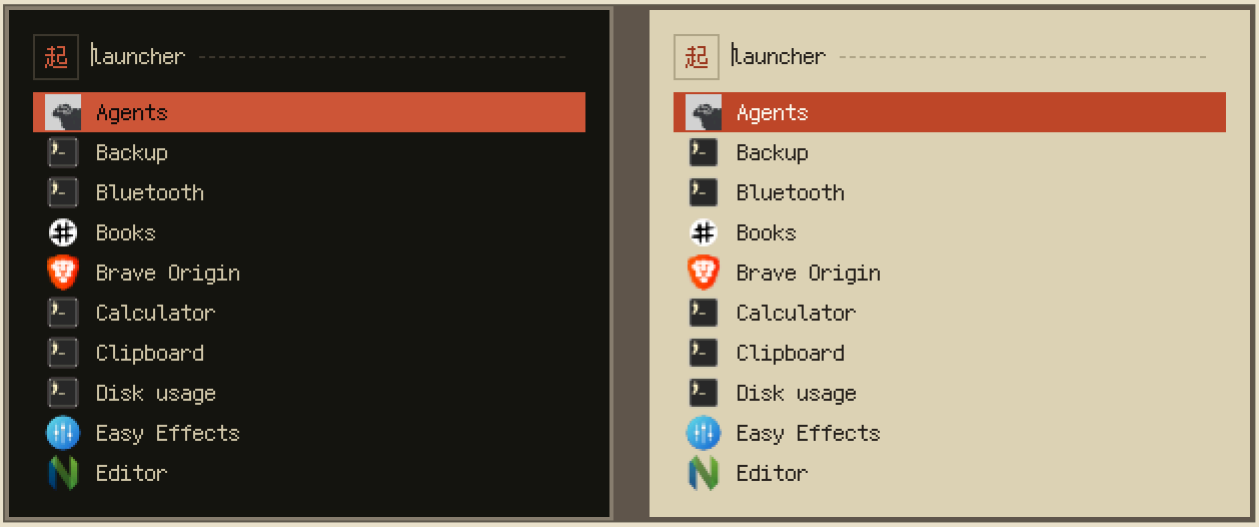
Expected: `····-merged.pdf` next to the first file, then `····-merged.pages-1-2.pdf`; each path was on the clipboard in turn.

起 7. Starting things: launcher, calculator, downloads

In this chapter

- start any program by a few letters of its name
- calculate without opening a program
- follow every download in one place

You need: chapter 4.



The launcher (Super+Space).

起 7.1 The launcher

`Super+Space` (or the torii). Type a few letters — `bra` for Brave, `ste` for Steam, `syn` for Syncplay — and `Enter`. The programs you start most rise to the top. `Shift+Delete` makes one forget its rank.

Each entry has its icon: a window program shows its own (Brave, Steam, LibreOffice); every other entry — the terminal programs, the pages in Brave — a **seal** of its own, drawn in the desktop's pixels: one kanji in a thin frame, in the theme's colours. 端 Terminal, 編 Editor, 箱 Files, 枝 Git, 記 Notes, 写 Clipboard, 算 Calculator, 具 File tools, 壁 Wallpaper, 視 System monitor — after a week you find them by their seal before you have read a word.

Sums, without the calculator. Type a sum in the launcher itself — `15% of 2400`, `120 usd to aed` — and press `Enter`. Nothing matches a program, so the line goes to the calculator, and a notice says *15% of 2400 = 360 · copied*: the answer is on the clipboard. (The answer is not drawn in the launcher's list as you type; for that, open the calculator.)

起 7.2 The calculator

Calculator in the launcher (`Super+Space`, `calc`): type a sum, a percentage or a conversion and the answer shows as you type; `Enter` copies it. `↑` brings back the sum before, `Ctrl+U` clears, `Esc` closes.

You type	You see
`15% of 2400`	`360`
`120 usd to aed`	the amount in dirhams (rates refreshed weekly)
`5 ft to cm · 3.5 h to min`	`152.4 cm · 210 min`
`sqrt(2) · 255 to hex`	`1.414213562 · 0xFF`

起 7.3 The downloads centre

Anything that downloads — Brave, Steam, a torrent — shows in the bar's download mark as one percent. Click it for the list: each download with its source, size and progress; a click opens where it lands. A notice says when each one finishes.

Exercises

7.1 ★ No mouse. Start Brave, yazi and btop from the launcher alone.

7.2 ★ Quick sums. In the calculator: how many seconds in a week (`1 week to s`)?

Expected: `604800 s`.

7.4 ★ A sum in the launcher. `Super+Space`, type `15% of 2400`, `Enter`; paste into a terminal.

Expected: the notice *15% of 2400 = 360 · copied*, and `360` pasted.

7.3 ★★ Watch it arrive. Download a Fedora image's checksum file in Brave (`https://fedoraproject.org`) and watch the bar's mark, then click the finished notice.

Expected: the mark appears while it downloads and goes when it is done; the notice opens `~/Downloads`.

8. Looking after it: Super+U

In this chapter

- update the whole system in one step
- check its health, and repair what is wrong
- go back in time with snapshots

You need: chapter 4.

`Super+U` opens the system's own control: one window for everything that keeps the machine healthy. Each entry shows its state beside it — when the update is due, whether verify passed.



Super+U: the status on the dark theme, the move to the next Fedora on the light.

8.1 The entries you will use

Entry	Does	How often
status	how things stand, on one page	when curious
update	everything up to date: Fedora, Brave, Claude Code, the desktop's tools, neovim's plugins — with a snapshot first	each season, when the menu says *due*
update fedokawaii	the desktop itself, newest from GitHub	when you hear of a change
upgrade to Fedora N	the move to the next release	once it says *ready*
verify	every check; changes nothing	after something odd
repair	verify, then put right what it found	when verify fails
snapshots	the list of snapshots of the system	before and after a risky change
backup	your home to the backup drive	weekly
learn	this book	

From a terminal: `update`, `verify`, `repair`, `snaps`, and `fedokawaii report` (below).

8.2 Snapshots: an undo for the whole system

`Word — snapshot`. A frozen picture of every file at one moment. It costs almost no space (it shares unchanged data with the live files) and lets you copy any file back as it was.

Snapshots are taken by themselves: every hour, around every package install, before every update. Two everyday uses:

```
$ snaps
 #  | Type   | Pre # | Date                     | Description
────┼────────┼───────┼──────────────────────────┼──────────────
57  | single |       | Mon 20 Oct 2026 10:00:01 | timeline
58  | pre    |       | Mon 20 Oct 2026 10:12:40 | dnf install htop
59  | post   |    58 | Mon 20 Oct 2026 10:12:52 | dnf install htop
```

(for example; shortened)

You want	Do
undo one package install	`sudo snapper -c root undochange 58..59`, then reboot and `verify`
a file of yours as it was an hour ago	`sudo ls /home/.snapshots/` to find a number, then copy it back: `sudo cp -a /home/.snapshots/57/snapshot/$USER/Documents/file.ods ~/Documents/`
a mark before a risky change	`sudo snapper -c root create -p -d "before my experiment"` (prints its number)
everything back to that mark	`sudo snapper -c root undochange NUMBER..0`, then reboot

> Careful. `~/Downloads` is not in the snapshots: move what you keep out of it.

守 8.3 The health report

When something is wrong and you want help, `fedokawaii report` writes `~/fedokawaii-report.txt`: hundreds of checks, each with an ID and a verdict. Search it for `FAIL` (in neovim: `/FAIL`). Give that file to whoever helps you — or to Claude Code.

Exercises

8.1 ★ Status. Open `Super+U`, read *status*.

Expected: your Fedora release, until when it is supported, and when the next update is due.

8.2 ★ Verify. Run `verify` in a terminal.

Expected: a list of checks ending in *passed*, or the few to fix with what to do.

8.3 ★★ A safe experiment. Make a mark (`sudo snapper -c root create -p -d "before cowsay"`), install a toy (`sudo dnf install cowsay`), run `cowsay hi`, then put everything back to your mark (`sudo snapper -c root undochange NUMBER..0`, with the number the first command printed) and reboot.

Expected: after the reboot, `cowsay` answers `command not found`, and `verify` passes.

8.4 ★★ A lost file. Make `~/Documents/test.txt`, wait for the next hourly snapshot (or `sudo snapper -c home create -p`), delete the file, then copy it back from the snapshot.

Expected: `cat ~/Documents/test.txt` shows it again.



Part III — The programs

Each chapter here is one program, in the same order: what it is best at, the moves that unlock most of its power, real uses, then exercises. Learn a chapter's moves table by heart and you have most of the program; the rest is in its manual.

9. bash and ble.sh: the shell

In this chapter

- chain commands with pipes, and save output to files
- write your own commands: aliases and functions
- let ble.sh's colours catch mistakes before Enter

You need: chapter 2.

Best at: gluing programs together. Every program in this book can be run, combined and repeated from bash; a task you do twice by hand is a one-line command the third time.

9.1 The moves

Move	Example	Does
pipe `|`	`ls ~/Pictures | wc -l`	one program's output into the next
redirect `>` · `>>`	`date >> ~/log.txt`	output into a file: replace · append
`&&` · `||`	`mkdir x && cd x`	the second only if the first worked · only if it failed
`$( … )`	`echo "today is $(date +%A)"`	a command's output, in the middle of another
a loop	`for f in *.png; do echo "$f"; done`	the same command for each file
quotes	`"$f"`	keeps a name with spaces whole — quote every variable
`!!` · `Alt+.`	`sudo !!`	the last command · its last argument
an alias	`alias gs='git status'`	a short name for a long command
a function	`mkcd() { mkdir -p "$1" && cd "$1"; }`	a command of your own, with arguments
exit status	`echo $?`	0 means it worked; anything else, it failed

As you type, **ble.sh** colours the line: a known command in teal, an unknown one in vermilion, an existing file underlined. Vermilion before you press Enter is a typo caught for free. The grey suggestion is an earlier command: `→` takes it, `Ctrl+→` one word.

Your own aliases and functions go in `~/.bashrc.d/local.sh` — yours; the desktop never touches it. Every new terminal reads it.

9.2 Use cases

- rename a hundred photos by their date in one loop
- a `backup-notes` function that copies `~/Notes` to a USB stick
- `history | rg ffmpeg` to find that clever command from last month

Exercises

9.1 ★ mkcd. Add the `mkcd` function above to `~/.bashrc.d/local.sh`, open a new terminal, and run `mkcd ~/practice/new`.

Expected: the prompt shows `~/practice/new` at once.

9.2 ★★ Count by kind. How many files of each kind are in your pictures?

```
fd -t f . ~/Pictures | sed 's/.*\.//' | sort | uniq -c | sort -rn
```

Expected: lines such as `42 png`, the most common first. Read it as a sentence: list the files, keep what follows the last dot, sort, count each run, sort by the count.

9.3 ★★★ A daily line. Write a function `today` that prints the date, the weather line (`kawa-weather`), and how many lines your notes file has (`wc -l < ~/Notes/notes.md`).

探 10. Finding anything: fzf, fd, ripgrep

In this chapter

- find any file by a few letters of its name
- find any word inside any file, in a second
- pick anything from a list by typing

You need: chapter 9.

Best at: the question you ask a computer most — *where is it?* `fd` finds files by name, `rg` (ripgrep) finds text inside files, `fzf` turns any list into a searchable menu.

探 10.1 The moves

Command	Does
`fd notes`	files whose name contains *notes*, below here
`fd -e pdf . ~`	every PDF in your home
`fd -t d conf`	folders only
`fd -S +500m . ~`	files over 500 MB
`rg "rice" ~/Notes`	every line holding *rice*, with file and line number
`rg -i -t py "todo"`	ignore case, only Python files
`rg -l word` · `rg -C 2 word`	file names only · two lines of context around each match
`nvim "$(fzf)"`	pick a file by typing, edit it
`Ctrl+T` · `Alt+C` · `Ctrl+R`	insert a path · go into a folder · find an old command — all fzf

In every fzf list: words in any order narrow it; `'word` exact; `^start` and `end$`; `!not`.

探 10.2 Use cases

- `rg -i "invoice" ~/Documents -l` — which files mention invoices
- `fd -e mkv . ~/Videos -x du -h` — every film and its size
- in a code project, `rg "def train"` finds where a function is defined in a second

Exercises

10.1 ★ Every PDF you own. `fd -e pdf . ~`

Expected: a list of paths; nothing if you have no PDFs yet (try `-e png`).

10.2 ★ Grep your notes. `rg -i` a word you wrote in your notes.

Expected: `~/Notes/notes.md`, then the line number and the line.

10.3 ★★ Pick and edit. In your home, `nvim "$(fd -e md | fzf)"`.

Expected: a list of Markdown files only; Enter opens the one you chose.

10.4 ★★ The big ones. List the five largest files in your home: `fd -t f -S +100m . ~ -x du -h | sort -h | tail -5`.

10.5 ★★★ A finder of your own. Write a function `nf` (*note find*) that runs `rg -i "$1" ~/Notes` and, if nothing is found, says *nothing about $1*.

11. Moving and reading: zoxide, eza, bat, tldr

In this chapter

- jump to any folder by one word
- list folders with sizes, dates and git state
- read a file with colours

You need: chapter 2.

Best at: the small daily moves. `cd` here is **zoxide**: it remembers the folders you visit and jumps by a word. `ls`, `ll` and `lt` are **eza**. `cat` is **bat**.

11.1 The moves

Command	Does
`cd proj`	jump to the folder you use most whose name matches *proj*, from anywhere
`cdi`	the folders zoxide knows, in fzf
`ll`	everything, with sizes, dates and git state (`N` new, `M` modified)
`lt · eza -T -L 3`	a tree, two levels · three
`ll -s size -r · ll -s modified`	biggest first · newest last
`bat file.py`	the file with line numbers and syntax colours
`tldr ffmpeg`	the five commands you actually wanted
`gdu ~`	what fills the disk, folder by folder, with a delete key

Exercises

11.1 ★ Teach zoxide. `cd ~/Documents`, `cd /`, then `cd doc`.

Expected: you are back in `~/Documents`.

11.2 ★ A tree. `lt ~/repos` once you have a project (chapter 24).

11.3 ★★ Clean-up hunt. `gdu ~`: find the biggest folder in your home and decide whether it should stay. `q` quits.

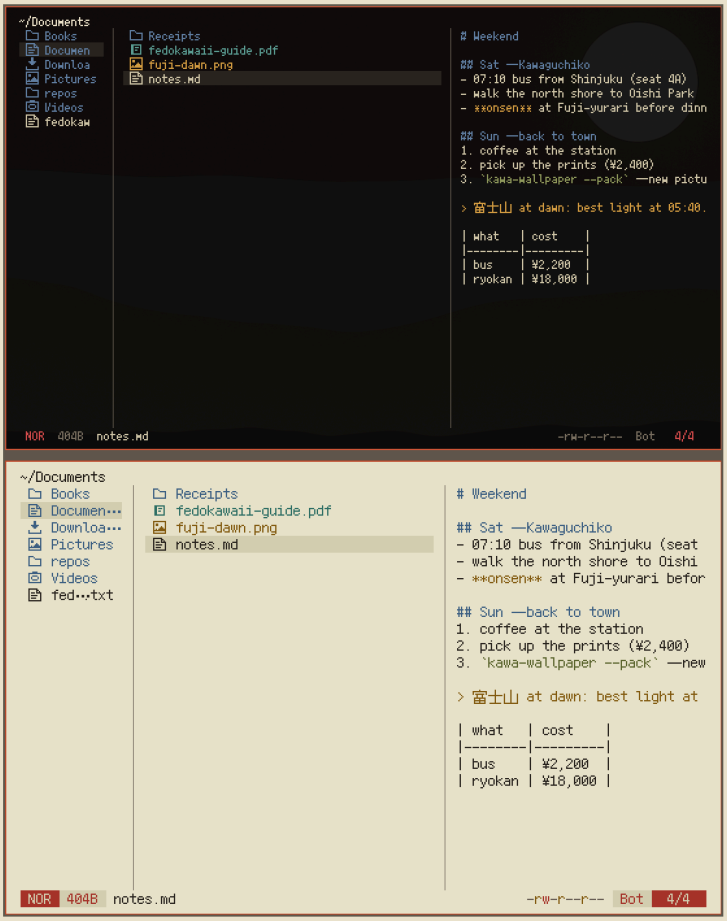
12. yazi: the files

In this chapter

- move through folders with h j k l, preview anything
- copy, move, rename and delete many files at once
- jump anywhere by typing

You need: chapter 4.

Best at: working on many files fast, with a preview of each — pictures, PDFs, videos and code — without leaving the keyboard. `Super+E` opens it. It is also every program's file dialog here.



yazi: the parent folder, this folder, and a preview of the file under the cursor.

12.1 The moves

Keys	Does
`h` `j` `k` `l`	up a folder · down · up · open
`Space` · `v` · `Ctrl+A`	select one · a range · all
`y` · `x` · `p`	copy · cut · paste
`d` · **`D`**	to the trash · delete for good
`a`	new file (end with `/` for a folder)
`r`	rename; several selected: rename them all in neovim
`z` · **`Z`**	jump to anything below here (fzf) · to a folder you use (zoxide)

Keys	Does
`s` · **S**	search names (fd) · contents (ripgrep)
`.`	hidden files on / off
`g` `h` · `g` `d` · **M**	home · your drives · drives: open or eject
`Enter` on a film	it plays in mpv; **W**: watch it together (Syncplay)
`Alt+S`	shrink or convert pictures and PDFs
`Alt+F`	archives and PDFs: unpack, pack, merge, sign (Super+D's menu)
`;`	run a command on the selection: `%s` the selected files



12.2 Use cases

- sort a holiday's photos: select by range, cut, paste into a new folder
- rename 30 episode files at once with `r` and neovim's `:%s` (chapter 13)
- clean `~/Downloads`: sort by size (`,` then `s`), trash the big ones you watched

Exercises

12.1 ★ Three moves. In `~/practice`, make a folder `pics/` (`a`, `pics/`), copy two screenshots into it (`y`, then `p` inside it).

12.2 ★★ Rename many. Select three files, press `r`: in neovim, change all three names at once with `:%s/^/2026-/`, then `:wq`.

Expected: each name now starts with `2026-`.

12.3 ★★ Pack them. Select the files of 12.1, press `;`, type `7z a pics.7z %s`.

Expected: `pics.7z` appears in the folder.

筆 13. neovim: the editor

In this chapter

- edit without getting stuck: modes, save, quit, undo
- the grammar that makes editing fast: verb + motion
- LazyVim's Space menu: files, search, git, code

You need: chapter 2.

Best at: editing text — notes, settings, code — faster than any mouse, with the same keys everywhere: your notes window (`Super+N`), man pages and git messages all open in it. Here it runs **LazyVim**, a ready-made setup: a file finder, language servers, lazygit inside.



neovim with LazyVim: the file tree, a file, and the status line with the mode.

筆 13.1 Never stuck

Situation	Do
keys do odd things	`Esc`, twice: you are in Normal mode
save · save and quit · quit without saving	`:w` · `:wq` · `:q!` (or `Ctrl+S` to save)
undo · redo	`u` · `Ctrl+R`
recording @q at the bottom	press `q` once

> **Word — mode.** In *Normal* mode, keys are commands; in *Insert* mode (`i`), keys type text. `Esc` always goes back to Normal.

筆 13.2 The moves

Keys	Does
`i` · `a` · `o`	type before the cursor · after it · on a new line below
`w` · `b` · `0` · `$` · `gg` · `G`	next word · back a word · line start · end · first · last line
`s` and two letters	**flash**: jump anywhere visible by a label
`dd` · `yy` · `p`	cut · copy a line · paste
`ciw` · `ci"` · `dap`	change this word · inside the quotes · delete a paragraph
`.`	repeat the last change — the most useful key
`/word` · `n`	search · next match
`:%s/old/new/g`	replace everywhere in the file
`Space` `Space` · `Space` `/`	find a file · search text in the whole project
`Space` `e` · `Space` `g` `g`	the file tree · lazygit
`gd` · `K` · `Space` `c` `a` · `Space` `c` `r`	go to the definition · its documentation · a fix · rename everywhere

The grammar: a **verb** (`d` delete, `c` change, `y` copy) + a **motion** or **object** (`w` a word, `ip` a paragraph, `i(` inside brackets). `d3w` deletes three words; `yi"` copies what is inside the quotes. Learn five verbs and five objects, and you have twenty-five commands.

筆 13.3 Use cases

- edit any setting: `nvim ~/.bashrc.d/local.sh`
- write code with completion, errors underlined as you type, and a test runner (chapter 24)
- change a word everywhere in a project: `Space` `s` `r`

Exercises

13.1 ★ The tutor. Run `nvim +Tutor` and do lesson 1.

Expected: you can move, delete, and quit with `:q!` without thinking.

13.2 ★★ Fix a list. Make a file with five lines `tea`, `rice`, `tea`, `milk`, `tea`. Turn every *tea* into *coffee* with one command, then undo it, then do it with `*`, `ciw`, `n` and `.` instead.

Expected: the same result both ways; the second teaches you the dot.

13.3 ★★ Project search. In `~/repos/fedokawaii`, `Space` `/`, type `kawa-notes`.

Expected: every file that mentions it, with a preview; Enter jumps there.

13.4 ★★★ Thirty days. Use neovim for everything for a month. Each week, add one new move to your notes and use it on purpose all week.

史 14. git, lazygit and GitHub

In this chapter

- commit, branch and undo with lazygit
- put a project on GitHub with `gh`
- see who changed what, and when

You need: chapter 13.

Word — commit. A saved snapshot of a project, with a message saying why. **git** keeps every commit, so any version can come back. A **repository** is a project folder git looks after.

Best at: never losing work, and trying things without fear: commit, experiment, and if it goes wrong, go back. **lazygit** is git with a screen: `Space` `g` `g` in neovim, or `lazygit` in a terminal.



lazygit: the five panels on the left, the selected change on the right.

史 14.1 Once

```
$ git config --global user.name "Your Name"
$ git config --global user.email you@example.com
$ gh auth login
```

(`gh` asks a few questions: GitHub.com, HTTPS, log in with a web browser)

史 14.2 The moves in lazygit

Keys	Does
`1` … `5`	the panels: status, files, branches, commits, stash
`Space` · `a`	stage a file · all files
`Enter` on a file	its lines: stage only some of them
`c`	commit (type the message, Enter)
`P` · `p`	push · pull
`n` (branches) · `Space`	a new branch · switch to one
`z`	undo the last git action
`s`	stash the changes away for a moment
`?`	every key of this panel

史 14.3 GitHub: gh

lazygit does the work on your own machine; **gh** talks to GitHub.

Command	Does
`gh repo create NAME --private --source . --push`	put this project on GitHub
`gh repo clone USER/NAME`	a copy of a project
`gh pr create` · `gh pr list`	a pull request · the open ones

史 14.4 Use cases

- every project in `~/repos` is a repository: the prompt shows its branch and changes
- an experiment on a branch: if it works, merge; if not, delete the branch
- `git log -p FILE`: why a line is the way it is

Exercises

14.1 ★ First commit. `mkcd ~/repos/hello`, `git init`, make `README.md` with one line, commit it in lazygit (`a`, `c`).

Expected: lazygit's commits panel (`4`) shows one commit.

14.2 ★★ Undo. Change the README, commit; then undo that commit with `z` in lazygit.

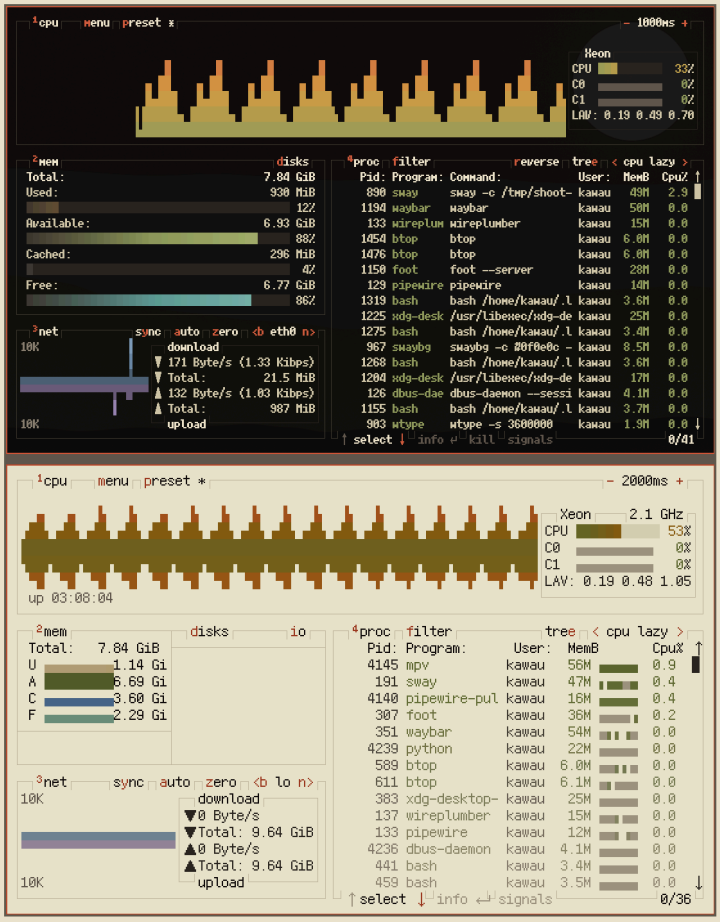
14.3 ★★ A branch. Make a branch `idea`, commit a new file there, switch back to `main`.

Expected: the file disappears on `main` and returns on `idea`.

14.4 ★★★ On GitHub. Create the repository on GitHub with `gh repo create`, push, and open it in Brave (`gh repo view --web`).

動 15. btop: the machine

Best at: seeing at once what the processor, memory, disks, network and graphics card are doing, and which program is to blame. Click the bar's memory, or run `btop`.



btop: the processor, memory, network and the process list.

Keys	Does
1 · 2 · 3 · 4	the processor, memory, network, process boxes on / off
f	filter the processes: type `brave`
e	show processes as a tree
↑ ↓ then t · k	end a process politely · kill it at once
p	cycle layout presets
m · o	menu · options

On AMD it shows the graphics card's load, memory, heat and watts: watch it while a game or a training run (chapter 28) works.

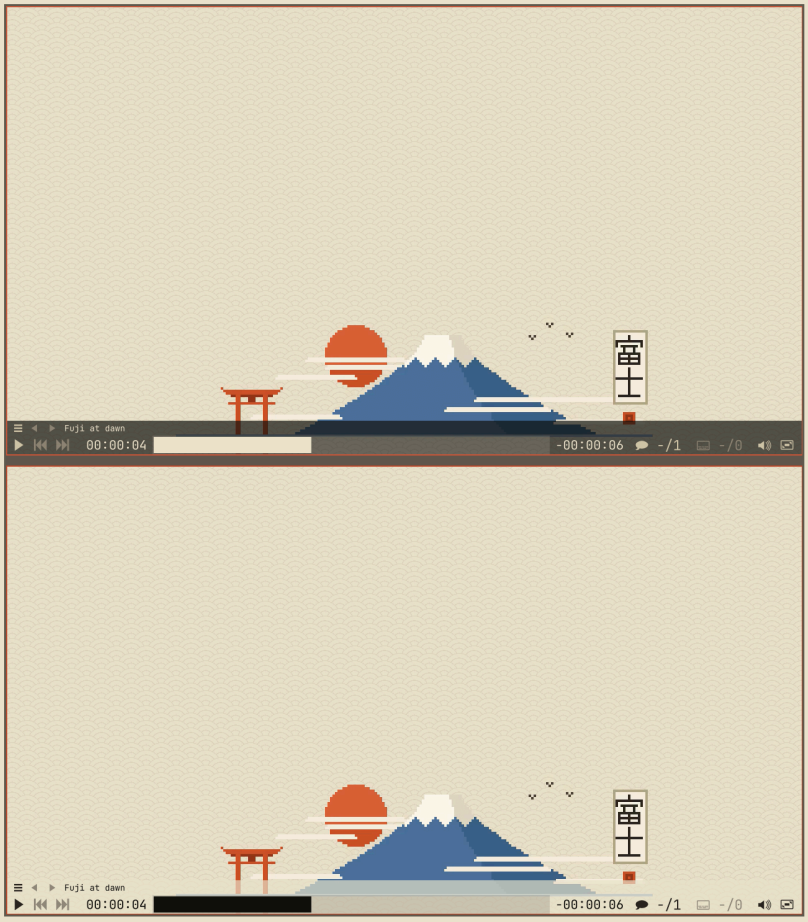
Exercises

15.1 ★ Who uses memory? Sort by memory (← → change the sorted column until it is *mem*, or click its heading) and name the top three programs.

15.2 ★★ Watch the card. Start a game or a video, then watch the GPU line rise.

音 16. mpv, yt-dlp and Syncplay: video and sound

Best at: playing anything — files, links, playlists — with the keyboard, and keeping your place. `mpv URL` plays a YouTube link directly (yt-dlp fetches it).



mpv, its controls shown on the picture.

Keys	Does
`Space`	pause
`←` `→` · `↑` `↓`	5 s · 1 min
`[` `]` · `Backspace`	slower · faster · normal speed
`v` · `j` · `#`	subtitles on/off · next subtitle track · next audio track
`z` · `Z`	subtitles earlier · later
`s`	a screenshot of the frame
`Ctrl+P`	a menu of everything
`q`	quit, keeping the position

Command	Does
`mpv "https://youtu.be/…"`	play a web video
`yt-dlp "URL"`	keep it, in the best quality
`yt-dlp -x --audio-format opus "URL"`	the sound only
`mpv --speed=1.5 lecture.mp4`	a lecture at 1.5×

Syncplay: a film with a friend elsewhere, each from your own copy. In yazi, **W** on the film; agree on a room name; one person's pause pauses both. `Enter` in mpv chats.

Exercises

16.1 ★ A lecture faster. Play a long talk at 1.5× and slow it back with `Backspace`.

16.2 ★★ Keep the audio. Download a talk's sound as Opus and play it in mpv.

16.3 ★★ Together. Watch a short film with a friend through Syncplay.

網 17. Brave: the web

Best at: the web, privately, with ads and trackers blocked. `Super+B`.

Keys	Does
`Ctrl+L`	the address bar: type a search or an address
`Ctrl+T` · `Ctrl+W` · `Ctrl+Shift+T`	new tab · close · reopen the last closed
`Ctrl+Shift+A`	search your open tabs
`Ctrl+F` · `Ctrl+J` · `Ctrl+H`	find in page · downloads · history
`Ctrl+Shift+N`	private window
`F12`	the developer tools: a page's code, network, console

For the developer: the developer tools' *Console* runs JavaScript on any page; *Network* shows every request a page makes — the fastest way to learn how a website talks to its server.

Exercises

17.1 ★ Tab search. Open ten tabs, then find one with `Ctrl+Shift+A`.

17.2 ★★ Look inside. On any page, `F12` → *Network*, reload: which request is the largest?

遊 18. Steam: games

Best at: Windows games on Linux through Proton. The gamepad in the bar shows Steam is on; right-click it for its menu.

Launch option (right-click a game → Properties)	Does
`gamemoderun %command%`	the processor tuned for the game
`mangohud %command%`	the overlay: frames per second, temperatures (`Right Shift+F12`)
`gamescope -W 2560 -H 1440 -r 144 -f -- %command%`	the game in its own compositor, fixed size and rate

If a game will not start: Properties → Compatibility → force a Proton version (GE-Proton is installed). ProtonDB (`protondb.com`) lists what others did for each game.

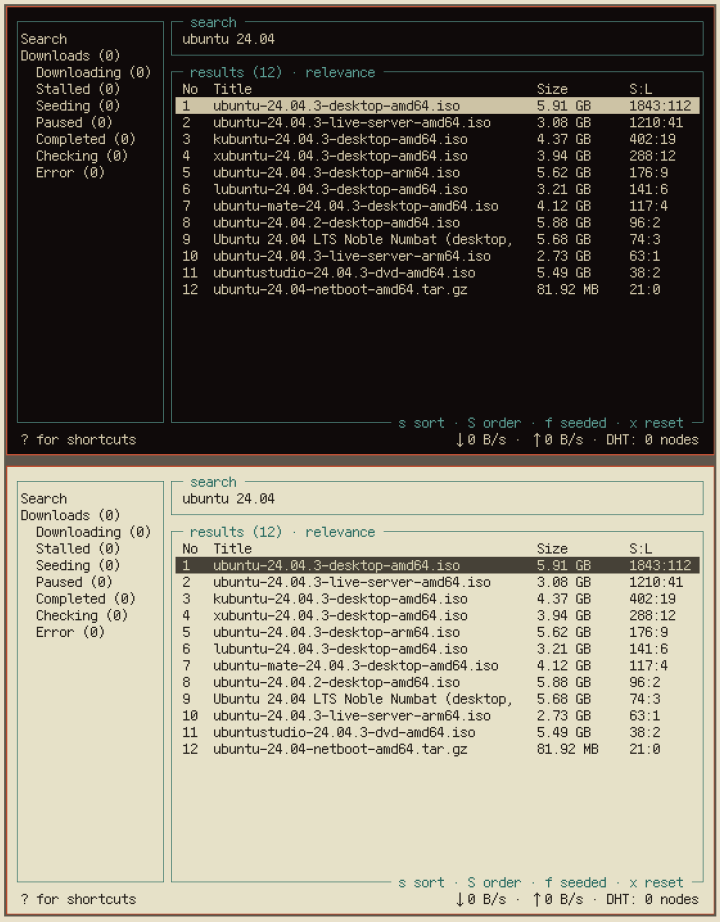
Exercises

18.1 ★ The overlay. Add `mangohud %command%` to a game and find its frame rate.

18.2 ★★ Compare. Play a minute with and without `gamemoderun` and compare the frame rate.

流 19. torrra: torrents

Best at: searching many torrent sites at once and downloading, in a terminal. *Torrents* in the launcher, or `torrra`. The first time, Prowlarr (its search service) asks for indexers.



torrra: one search across every indexer, seeders and size per result.

Keys	Does
type, Enter	search every indexer
j k	move
Enter	the torrent's details, then its files; Enter again downloads
s · f	sort menu · hide results nobody seeds

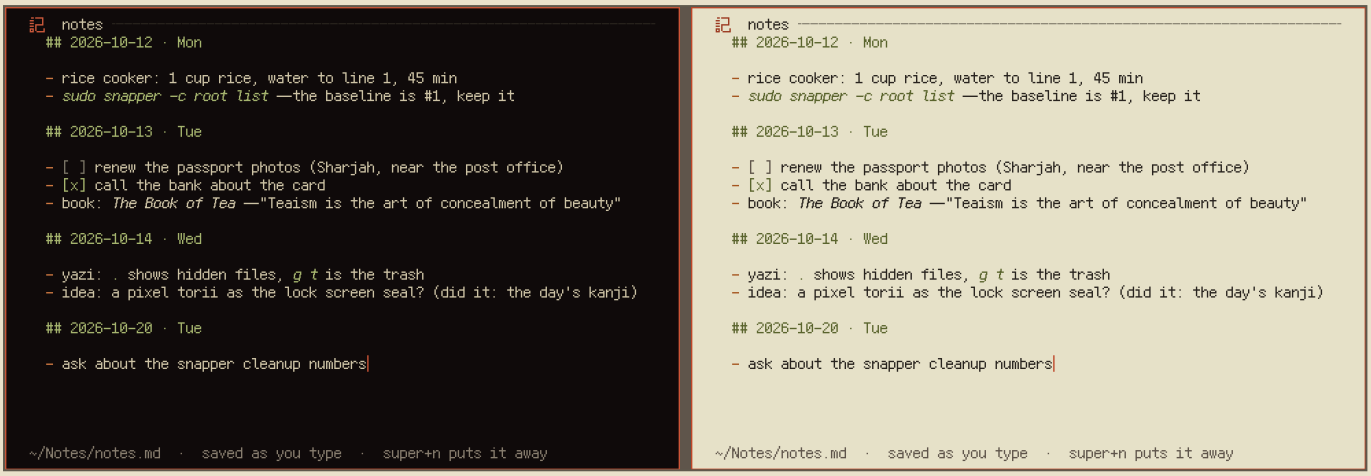
The `S:L` column is seeders:leechers: more seeders, faster. The bar's download mark follows it.

Exercises

19.1 ★ A legal one. Search `debian netinst` and download the image.

Expected: it appears in the downloads centre, then a notice when it is done.

20. Notes, books and the small tools



The notes window (Super+N): one file, a heading per day, ready to type.

Tool	Opens with	Best at
notes	`Super+N`	one Markdown file, a heading per day: type at once, saved as you type
books	`Super+R`	your PDFs, EPUBs and comics; `Ctrl+A` brings books in from Downloads
zathura	opening a PDF	reading; `Ctrl+R` recolours the page to the theme
VisiData	`vd file.csv`	tables: sort (`[ ]`), select rows by a pattern (`|`), sum, group (`Shift+F`)
jq	`jq . file.json`	reading and reshaping JSON
gdu	`gdu ~`	what fills the disk
LibreOffice	opening a .docx, .xlsx, .pptx	office files

20.1 PDFs: one tool for each job

Job	Tool	Command
merge	qpdf	`qpdf --empty --pages a.pdf b.pdf -- merged.pdf`
some pages only	qpdf	`qpdf in.pdf --pages . 3-5 -- part.pdf`
split into pages	qpdf	`qpdf --split-pages in.pdf page-%d.pdf`
rotate	qpdf	`qpdf in.pdf --rotate=+90:2 out.pdf` (page 2; leave out `:2` for all)
lock with a password	qpdf	`qpdf --encrypt --user-password=OPEN --owner-password=OWNER --bits=256 -- in.pdf locked.pdf`
the text out	pdftotext (poppler)	`pdftotext -layout in.pdf out.txt`
the pictures out	pdfimages (poppler)	`pdfimages -png in.pdf img` → `img-000.png`, `img-001.png` …
smaller	ghostscript	`gs -q -dNOPAUSE -dBATCH -dSAFER -sDEVICE=pdfwrite -dPDFSETTINGS=/ebook -sOutputFile=small.pdf in.pdf`

(`/ebook` is 150 dpi pictures; `/screen` smaller, `/printer` sharper. Super+Z runs the same ghostscript on a PDF you copied, and Super+D (§6.5) runs the qpdf and poppler commands above)

Exercises

20.1 ★ A to-do. In your notes, write `- [ ] finish chapter 20`. Later, make it `- [x]`.

20.2 ★★ A table. Save a few expenses as `~/practice/money.csv` (`date,item,amount`), open it in VisiData, sort by amount, and find the total (on the amount column: **z+** then *sum*).

20.3 ★★ JSON. `curl -s "https://api.github.com/repos/neovim/neovim" | jq .stargazers_count`

Expected: one number: neovim's stars on GitHub.

20.4 ★★ A booklet. Take pages 1–3 of this guide (`docs/Fedokawaii-Learn.pdf` in your working copy) and pages 1–2 of another PDF, merge them into one file, rotate its second page, lock it with a password, and shrink it.

Expected: zathura asks for the password; `ls -l` shows the shrunk file smaller.

問 21. Claude Code and herdr: AI agents

`Word — agent.` A program that uses an AI model to work on a task in steps: it reads files, runs commands and edits, asking you first. **Claude Code** is one, in the terminal.

Best at: explaining code and errors, planning a change, writing tests, doing a well-described chore. You stay the author: you read every change before you keep it.

In a terminal	Does
`cd ~/repos/hello && claude`	start it in a project
type a request, `Enter`	it plans, asks, then works
`Esc`	stop it at once
`Shift+Tab`	switch modes: ask before edits · accept edits · plan only
`/clear` · `/init`	a fresh conversation · write a `CLAUDE.md` describing the project
`/exit`	leave

herdr runs several agents side by side, with a sidebar marking each *working*, *blocked* (it needs you) or *idle*. *Agents* in the launcher. Every herdr key is `Ctrl+B`, then one key: `Ctrl+B` `a` a new Claude, `Ctrl+B` `v` a split, **Ctrl+B h/j/k/l** move, `Ctrl+B` `o` the agent that just called, `Ctrl+B` `q` close the window (the agents keep working).

Habits that keep you learning:

1. Commit before you ask for a change: `git diff` then shows exactly what it did.
2. Ask it to *explain* first, *write* second: "explain what this function does, line by line".
3. Ask for tests with every change, and run them yourself.
4. Never paste a secret into a prompt.

Exercises

21.1 ★ Explain. In `~/repos/fedokawaii`, ask Claude Code: *"explain bin/kawa-weather to a beginner, step by step — do not change anything"*.

21.2 ★★ Two at once. In herdr, run one agent writing tests for your `hello` project while another explains git's reflog to you.

箱 22. podman: containers

> `Word — container.` A program packed with everything it needs, run apart from the rest of the system. Removing it leaves nothing behind.

Best at: trying software without installing it, and running services (Prowlarr runs in one here). podman runs containers as you, never as root.

Command	Does
`podman run --rm -it fedora:44 bash`	a throw-away Fedora, gone when you `exit`
`podman run --rm -it python:3.14 python`	a Python of another version, for a minute
`podman ps -a` · `podman images`	running and stopped containers · the images kept
`podman run --rm -v ~/practice:/w:Z -w /w python:3.14 python script.py`	run a script of yours inside
`podman image prune`	delete images no container uses

Exercises

22.1 ★ A throw-away system. Start the Fedora container, `dnf install -y cowsay`, run it, `exit`.

Expected: `cowsay` exists only inside; on your desktop, `command not found`.

22.2 ★★ Your script inside. Run `~/practice/oops.py` (fixed) in the `python:3.14` container with `-v`.



Part IV — Python, for the age of AI

From your first line of Python to a language model you trained yourself, an AI fine-tuned on your own notes, and an agent that uses tools — in six levels. Levels 1–5 (chapters 25–39) are the language itself, one idea at a time, each with tiny examples, the mistakes everyone makes, graded exercises and a checkpoint project. Level 6 (chapters 40–47) turns it to data and machine learning. Every chapter builds on the one before; every step has something to run and something to see.

23. Why Python and Rust

In this chapter

- why these two languages, and not others
- the plan of Parts IV and V

You need: nothing.

The AI world runs on two layers. On top, people describe models, data and experiments — in **Python**: PyTorch, NumPy, Hugging Face, every research paper's code, the SDKs of Claude and the other models. Underneath, the parts that must be fast and must never crash — tokenizers, inference servers, data loaders, the tools that run AI on your own machine — are increasingly written in **Rust**: Hugging Face's `tokenizers` and `safetensors`, the `candle` and `burn` frameworks, `uv` (the Python tool you will use) and `ruff` are Rust.

	Python	Rust
you write	what to do, briefly	what to do, and exactly who owns each piece of memory
speed	slow itself; fast through libraries written in C, C++ and Rust	as fast as C
mistakes found	when the line runs	before it runs: the compiler refuses unsafe code
for AI	the language of training, research, data, agents	the language of fast, safe tools and inference
learn it	in weeks	in months — and it teaches you how computers really work

Together they cover the whole range: experiment in Python, make the slow part fast in Rust, call it from Python (chapter 65). Other good choices exist — TypeScript for the web, C++ for CUDA kernels, Mojo — but no other pair covers both research and production as well, today.

The plan. Chapter 24 sets up the daily routine once. Then each language is a course in six levels, and each level ends with a checkpoint project that uses everything in it:

Level	Python (Part IV)	Rust (Part V)
1	numbers, text, decisions, loops (25–27)	variables, types, functions, control flow (49–50)
2	functions, lists, dictionaries, sets (28–30)	ownership, borrowing, structs, enums (51–54)
3	files, errors, modules and uv (31–33)	collections, errors (55–56)
4	classes, pytest, type hints (34–36)	traits, generics, lifetimes (57–58)
5	generators, async, CLIs and packaging (37–39)	modules, tests, iterators, threads (59–61)
6	data, NumPy, a network from scratch, PyTorch, a transformer, fine-tuning, an agent (40–47)	a CLI with clap, your own autograd, candle and burn, PyO3 (62–65)

Learn Python to level 3 first; then the two can go side by side — a Rust level after each Python one is a good rhythm, because each language explains the other. Chapter 66 is a week-by-week plan.

How each lesson works. A concept in plain words; three to six tiny programs, each with what it prints under *You should see*; the mistakes beginners make, with the real error message; then exercises from ★ to ★★★★, with worked solutions in appendices B and C. Type every example. Then change one thing and predict what it will print before you run it — that habit, more than any other, is what makes a programmer.

24. The daily development loop

In this chapter

- make a project with its own Python and libraries (uv)
- edit with completion and errors shown as you type (neovim)
- run, test on every save, debug with breakpoints
- commit and push (lazygit, GitHub)

You need: chapters 13 and 14.

Professional programmers repeat one loop all day: *edit → run → test → commit*. Set it up once here; every later chapter uses it.

24.1 The tools, and what each does

Tool	Job
uv	makes a project, picks its Python, installs its libraries into the project's own folder, runs it
neovim (LazyVim)	the editor: completion, errors as you type, formatting, tests, the debugger
pytest	runs your tests — from neovim, once or on every save
ruff	finds mistakes and formats the code (neovim runs it on save)
git, **lazygit**, **gh**	history, and GitHub

`Word — virtual environment (venv).` A folder (`.venv`) holding one project's Python and its libraries, apart from every other project's. uv makes and fills it; you never activate it by hand: `uv run` uses it.

24.2 A new project

```
$ cd ~/repos
$ uv init hello
Initialized project `hello` at `/home/you/repos/hello`
$ cd hello
$ ls -a
.  ..  .git  .gitignore  main.py  pyproject.toml  .python-version  README.md
$ uv run main.py
Hello from hello!
```

(uv made a git repository too, with a `.gitignore` that keeps `.venv` out of it)

File	Holds
`main.py`	your program
`pyproject.toml`	the project's name, its Python version and its libraries
`.python-version`	which Python this project uses
`uv.lock`	(after the first `uv add`) the exact version of every library: the same everywhere

Libraries come in with `uv add`; tools you only need while developing, with `--dev`:

```
$ uv add --dev pytest
Resolved 6 packages in 0.4s
Installed 5 packages in 12ms
```

環 24.3 Edit in neovim

```
$ nvim .
```

Python support is set up already: the language servers (basedpyright and ruff), formatting on save, the test runner and the debugger load with the first `.py` file. In one:

Keys	Does
typing	completion appears; `Enter` takes it, `Ctrl+Space` asks for it
`K` · `gd` · `gr`	the documentation of a name · its definition · everywhere it is used
`]d` · `Space` `c` `d`	the next problem · the full text of this line's problem
`Space` `c` `a` · `Space` `c` `r`	a fix the language server offers · rename everywhere
`Space` `c` `f`	format the file (also on every save)
`Space` `r` `f`	run this file (`uv run`: the project's `.venv`)
`Space` `r` `r` · `Space` `r` `c` · `Space` `r` `l`	a Python prompt (REPL) under the code · send it the cell (`# %%` lines part a file into cells) · the line
`Space` `t` `t` · `Space` `t` `r` · `Space` `t` `w`	run this file's tests · the test under the cursor · run them on every save
`Space` `t` `s` · `Space` `t` `o`	the test summary · the last test's output
`Space` `d` `b` · `Space` `d` `c`	a breakpoint here · run with the debugger
`Space` `t` `d`	debug the test under the cursor
`Space` `d` `i` · `Space` `d` `O` · `Space` `d` `o`	step into · over · out of
`Space` `d` `u` · `Space` `d` `t`	the debugger's panels · stop it
`Space` `c` `v`	choose the project's `.venv` (when a project has more than one)
`Space` `g` `g`	lazygit
`Ctrl+/`	a terminal at the bottom, and back

環 24.4 Test on every save

Open a test file and press `Space` `t` `w`: neovim now runs its tests every time you save (`Ctrl+S`), in a fraction of a second, and marks each test in the margin — ✓ green, ✗ red, the failure's message beside the line. `Space` `t` `s` keeps the summary open on the right. Green means your last change kept everything working. This loop — change, save, glance at the margin — is the fastest way to learn a language there is. `Space` `t` `w` again stops watching.

環 24.5 Debug: look inside a running program

When a result is wrong and you cannot see why, stop the program in the middle and look:

1. Cursor on a line inside the function, `Space` `d` `b`: a red dot in the margin.
2. `Space` `d` `c`: the program runs and stops there; the panels show every variable's value.
3. `Space` `d` `O` steps one line; watch the values change. `Space` `d` `t` stops.

環 24.6 Commit and push

`Space` `g` `g`, `a` (stage all), `c`, a message that says *why*, `Enter`, **P**. The first time, put the project on GitHub:

```
$ gh repo create hello --private --source . --push
✓ Created repository you/hello on GitHub
✓ Pushed commits to https://github.com/you/hello.git
```

環 24.7 The whole loop, on one card

Step	Command or keys
new project	`uv init NAME && cd NAME && uv add --dev pytest`
a library	`uv add numpy`
edit	`nvim .` · Space Space a file
run	Space r f
test on save	Space t w in a test file
debug	Space d b, Space d c
commit · push	Space g g, a, c, **P**

Exercises

24.1 ★ The loop once. Make the `hello` project, add pytest, write `tests/test_one.py` with `def test_one(): assert 1 + 1 == 2`, run `uv run pytest`.

Expected: `1 passed`.

24.2 ★★ Red, then green. Watch the test file (Space t w), change the test to `1 + 1 == 3`, save: see it fail, read the message, put it back.

Expected: the failure shows `assert 2 == 3` — pytest prints both sides.

24.3 ★★ A breakpoint. In `main.py`, make a variable, put a breakpoint on the next line, run the debugger, and read the variable's value in the panel.

25. Python level 1: numbers, variables, print

In this chapter

- Python's prompt, and how to run a file
- numbers and arithmetic, with the four mistakes everyone makes once
- variables: names for values
- `print` and `input`: talk to the person running your program

You need: chapter 24 (a project made with `uv init`).

This is level 1 of six. Each level has three chapters and ends with a small project that uses everything in it — the *checkpoint*. Do not skip ahead: every later chapter assumes the earlier ones. If you can do a level's checkpoint without looking back, you are ready for the next level.

Level	You learn	Chapters	Checkpoint
1	numbers, text, decisions, loops	25–27	a guessing game
2	functions, lists, dictionaries, sets	28–30	a contacts book
3	files, errors, modules, uv	31–33	an expense tracker
4	classes, tests, type hints	34–36	a tested library
5	comprehensions, generators, async, packaging, CLIs	37–39	a command of your own
6	data, NumPy, then machine learning	40–47	a network that learns

Every example in this part was run with Python 3.14 as printed, and the output under *You should see* is what it printed.

25.1 Two ways to run Python

Python reads your code and does it, one line at a time. There are two ways to give it code.

1. The prompt (the REPL). Type a line, press Enter, see the answer at once. Use it to try things.

```
$ uv run python
Python 3.14.0 (main, ···) [GCC 15.2.1] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

(`>>>` is Python waiting for you. `exit()` or Ctrl+D leaves.)

> **Word — `REPL`.** *Read, Evaluate, Print, Loop*: Python reads a line, works it out, prints the result, and waits for the next. At the prompt, the value of an expression is printed by itself; in a file it is not — there you must call `print`.

```
>>> 2 + 3
5
>>> 10 - 4
6
>>> "hello"
'hello'
```

2. A file. Write the lines into a file ending in `.py`, then run the whole file. Use it for anything longer than a few lines, and for anything you want to keep.

`hello.py`:

```
print("hello")
print(2 + 3)
```

You should see:

```
hello
5
```

```
$ uv run hello.py
hello
5
```

In the examples below, a block that starts with `>>>` is for the prompt; any other block is a file: save it (as `example.py`, say) and run it.

数 25.2 Numbers and arithmetic

Python is a calculator first. It knows two kinds of numbers:

- **int** — a whole number: `7`, `-3`, `2026`. As big as you like.
- **float** — a number with a point: `3.5`, `0.1`, `-2.0`.

```
>>> 7 + 3
10
>>> 7 - 3
4
>>> 7 * 3
21
>>> 7 / 2
3.5
>>> 7 // 2
3
>>> 7 % 2
1
>>> 2 ** 10
1024
```

Sign	Does	Example	Gives
`+ - *`	add, subtract, multiply	`7 * 3`	`21`
`/`	divide — always gives a float	`6 / 2`	`3.0`
`//`	divide and drop the remainder	`7 // 2`	`3`
`%`	the remainder	`7 % 2`	`1`
`**`	power	`2 ** 10`	`1024`

Python follows the order you learnt at school: powers, then `* / // %`, then `+ -`. Brackets go first, so use them whenever you are unsure.

```
>>> 2 + 3 * 4
14
>>> (2 + 3) * 4
20
>>> 10 - 2 - 3
5
>>> 2 ** 3 ** 2
512
```

(the last one is `2 ** 9`: powers are worked out from the right)

Ints can be huge — Python never overflows them:

```
>>> 2 ** 100
1267650600228229401496703205376
>>> 1_000_000 * 3
3000000
```

(underscores in a number are only for your eyes: `1_000_000` is one million)

Floats are close, not exact. A computer stores `0.1` in binary, where it cannot be written exactly, just as 1/3 cannot be written exactly in decimal:

```
>>> 0.1 + 0.2
0.30000000000000004
>>> round(0.1 + 0.2, 2)
0.3
>>> round(3.14159, 3)
3.142
```

This is not a bug in Python: every language does it. For money, count in whole fils or cents (ints), or round when you show the result.



25.3 Variables: names for values

A **variable** is a name that points at a value. `=` means *let this name point at this value* — not "equals" as in maths.

```
>>> price = 12.5
>>> count = 4
>>> price * count
50.0
>>> count = count + 1
>>> count
5
```

Read `count = count + 1` from the right: *work out* `count + 1` *(that is 5), then let* `count` *point at 5*. It is so common that it has a short form:

```
>>> score = 10
>>> score += 5
>>> score
15
>>> score -= 3
>>> score
12
>>> score *= 2
>>> score
24
```

Good names say what the value *is*: `price`, `total_hours`, `user_name`. The rules:

- letters, digits and _, not starting with a digit: `day2` yes, `2day` no;
- lowercase words joined by _ (*snake case*): `total_hours`, not `TotalHours`;
- case matters: `Price` and `price` are two different names.

> **Word — expression.** Anything that has a value: `3`, `price * count`, `"hi"`. A **statement** does something: `count = 4` makes a variable; it has no value of its own.

25.4 Types: what kind of value

Every value has a **type**, and the type decides what you can do with it. `type()` tells you.

```
>>> type(7)
<class 'int'>
>>> type(7.0)
<class 'float'>
>>> type("7")
<class 'str'>
>>> type(True)
<class 'bool'>
```

Type	Holds	Examples
`int`	whole numbers	`0`, `42`, `-7`
`float`	numbers with a point	`3.5`, `0.1`
`str`	text (a *string* of characters)	`"hello"`, `'7'`
`bool`	true or false	`True`, `False`

The same sign can mean different things for different types:

```
>>> 3 + 4
7
>>> "3" + "4"
'34'
>>> "ab" * 3
'ababab'
```

You change a value's type with the type's name, used like a function:

```
>>> int("42") + 1
43
>>> float("2.5") * 2
5.0
>>> str(42) + "!"
'42!'
>>> int(3.99)
3
```

(`int(3.99)` cuts the decimals off; it does not round. `round(3.99)` gives 4.)

数 25.5 print: show something

In a file, nothing appears unless you `print` it. `print` takes any number of values, separated by commas, and puts a space between them:

```
name = "Mona"
age = 31
print("Hello")
print(name, age)
print("Next year", name, "will be", age + 1)
```

You should see:

```
Hello
Mona 31
Next year Mona will be 32
```

The best way to put values into text is an **f-string**: an `f` before the quote, and names or expressions inside `{ }`:

```
item = "tea"
price = 8.5
count = 3
print(f"{count} × {item} cost {price * count} dirhams")
print(f"{price:.2f}")
print(f"{1234567:,}")
print(f"[{item:>6}]")
```

You should see:

```
3 × tea cost 25.5 dirhams
8.50
1,234,567
[   tea]
```

Inside { }	Means	x = 3.14159 gives
{x}	the value	3.14159
{x:.2f}	2 digits after the point	3.14
{x:8.2f}	the same, 8 characters wide	3.14
{n:,}	thousands with commas (n = 1234567)	1,234,567
{s:>6}	right-aligned in 6 characters	tea for s = "tea"

`print` has two useful settings: `sep` (what goes between the values) and `end` (what goes at the end, normally a new line):

```
print(2026, 10, 21, sep="-")
print("no new line here", end="")
print(" — continued")
```

You should see:

```
2026-10-21
no new line here — continued
```

数 25.6 input: ask a question

`input("question ")` shows the question, waits for the person to type a line and press Enter, and gives back what they typed — **always as text**:

```
name = input("Your name? ")
print(f"Hello, {name}!")
```

You should see:

```
Your name? Sara
Hello, Sara!
```

(in these examples the typed answer is shown after the question, as you would see it)

To use a typed number, change its type first:

```
year = int(input("Born in? "))
print("This year you turn", 2026 - year)
```

You should see:

```
Born in? 1990
This year you turn 36
```

数 25.7 Common mistakes

1. Adding text and a number. `input` gives text; text and a number cannot be added.

```
age = input("Age? ")
print(age + 1)
```

Python says:

```
Age? 30
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    print(age + 1)
          ~~~~^~~
TypeError: can only concatenate str (not "int") to str
```

Python names the problem in the last line: `can only concatenate str (not "int") to str`. Fix: `int(age) + 1`. (This version was fed `30` as the answer.)

2. A name used before it exists.

```
print(total)
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 1, in <module>
    print(total)
          ^^^^^
NameError: name 'total' is not defined
```

3. A typo in a name. Python does not guess: `Price` and `price` are different names.

```
price = 5
print(Price * 2)
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    print(Price * 2)
          ^^^^^
NameError: name 'Price' is not defined. Did you mean: 'price'?
```

(Python 3.14 even suggests the right name: `Did you mean: 'price'?`*)*

4. A missing bracket or quote. The error is a `SyntaxError`, and it points near — often just after — the real mistake:

```
print("hello"
```

Python says:

```
  File "example.py", line 1
    print("hello"
          ^
SyntaxError: '(' was never closed
```

`Checkpoint.` Without running it, say what this prints; then run it and check: `x = 7`, `x = x // 2 + x % 2`, `print(x * 2)`. *(It prints 8.)*

`Exercises`

25.1 ★ Seconds in a day. Use the prompt to work out how many seconds there are in a day, then in a week. *Expected:* `86400` and `604800`.

25.2 ★ Your first file. Write `me.py` that prints your name on one line and your city on the next.

25.3 ★ Even or odd. Use `%` to find the remainder of 17, 18 and 2026 divided by 2. Which are even? *Expected:* `1 0 0` — 18 and 2026 are even.

25.4 ★★ Shopping total. Make variables for 3 kg of rice at 7.25 a kilo and 2 packs of tea at 12.5 each. Print the total with two decimals. *Expected:* `Total: 46.75`.

25.5 ★★ Minutes and seconds. Given `seconds = 754`, print it as minutes and seconds using `//` and `%`. *Expected:* `12 min 34 s`.

25.6 ★★ Celsius to Fahrenheit. Ask for a temperature in Celsius and print it in Fahrenheit (F = C × 9 / 5 + 32), with one decimal. *Expected:* for `31`: `31.0 °C is 87.8 °F`.

25.7 ★★★ Swap without a third name. Given `a = 5` and `b = 9`, make `a` 9 and `b` 5. First with a third variable; then look up Python's `a, b = b, a`. *Expected:* `9 5`.

25.8 ★★★ A receipt. Ask for an item's name, its price and how many; print a receipt line with the name left in 12 characters and the total right in 8, with two decimals (`{name:<12}` and `{total:>8.2f}`). *Expected:* for `tea`, `8.5`, `3`: `tea 25.50`.

25.9 ★★★★ Change in coins. Ask for an amount in fils (an int, say `385`) and print the fewest coins for it using 100, 50, 25, 10, 5 and 1 fils coins — only `//`, `%` and variables, no loops yet. *Expected:* for 385: `100×3 50×1 25×1 10×1 5×0 1×0`.

字 26. Python level 1: text

In this chapter

- strings: making them, joining them, cutting pieces out of them
- the string methods you will use every day
- why text cannot be changed in place, and what to do instead

You need: chapter 25.

Most data you will ever handle is text: names, files, web pages, messages to a model. Python calls a piece of text a **string** (`str`): a row of characters, in order.

字 26.1 Making strings

Single or double quotes — the same thing. Choose the one that is not inside the text:

```
>>> 'hello'
'hello'
>>> "it's here"
"it's here"
>>> 'she said "yes"'
'she said "yes"'
>>> len("hello")
5
>>> len("")
0
```

Three quotes make a string that spans lines. `\n` inside a string is a new line, `\t` a tab:

```
poem = """old pond
a frog jumps in
the sound of water"""
print(poem)
print("one\ntwo\tthree")
```

You should see:

```
old pond
a frog jumps in
the sound of water
one
two three
```

`Word — escape.` A `\` with a letter after it, standing for a character you cannot type in a string directly: `\n` new line, `\t` tab, `\\` a backslash, `\"` a double quote.

字 26.2 Indexing and slicing

Each character has a position, its **index**, counted **from 0**. Negative indexes count from the end:

```
 K   y   o   t   o
 0   1   2   3   4
-5  -4  -3  -2  -1
```

```
>>> city = "Kyoto"
>>> city[0]
'K'
>>> city[4]
'o'
>>> city[-1]
'o'
>>> city[-2]
't'
```

A **slice** `[start:stop]` takes from `start` up to *but not including* `stop`. Leave one out for "from the beginning" or "to the end":

```
>>> word = "fedokawaii"
>>> word[0:4]
'fedo'
>>> word[4:]
'kawaii'
>>> word[:4]
'fedo'
>>> word[-3:]
'aii'
>>> word[::2]
'fdkwi'
>>> word[::-1]
'iiawakodef'
```

(`[::-1]` steps backwards: the string reversed)

The rule *up to but not including* has a nice result: `word[:4] + word[4:]` is always the whole word.



26.3 Methods: what a string can do

A **method** is a function that belongs to a value, called with a dot: `value.method()`.

```
>>> s = "  Hello, World  "
>>> s.strip()
'Hello, World'
>>> s.lower()
'  hello, world  '
>>> s.upper()
'  HELLO, WORLD  '
>>> s.strip().replace("World", "Sharjah")
'Hello, Sharjah'
>>> "hello world".title()
'Hello World'
```

Asking questions about text:

```
>>> name = "fedokawaii"
>>> name.startswith("fedo")
True
>>> name.endswith(".py")
False
>>> "kawa" in name
True
>>> name.count("a")
2
>>> name.find("kawa")
4
>>> "2026".isdigit()
True
```

Cutting text into pieces and gluing them back:

```
>>> "rice,tea,milk".split(",")
['rice', 'tea', 'milk']
>>> "one two  three".split()
['one', 'two', 'three']
>>> "-".join(["2026", "10", "21"])
'2026-10-21'
>>> ", ".join(["rice", "tea"])
'rice, tea'
```

Method	Does	`"a,b".…`
`.strip()`	removes spaces and new lines at both ends	
`.lower()` / `.upper()`	all small / all capital	
`.replace(a, b)`	every `a` replaced by `b`	`.replace(",", ";")` → `'a;b'`
`.split(sep)`	a list of the pieces between `sep`	`.split(",")` → `['a', 'b']`
`sep.join(list)`	the pieces glued with `sep` between	
`.startswith(x)` / `.endswith(x)`	`True` or `False`	
`.find(x)`	the index of `x`, or `-1`	`.find("b")` → `2`
`.count(x)`	how many times `x` appears	



26.4 Strings never change

A string is **immutable**: once made, it cannot be changed. Every method gives back a *new* string and leaves the old one alone:

```
>>> s = "tea"
>>> s.upper()
'TEA'
>>> s
'tea'
>>> s = s.upper()
>>> s
'TEA'
```

(to keep the change, point the name at the new string: `s = s.upper()`*)*

```
word = "cat"
word[0] = "b"
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    word[0] = "b"
    ~~~~^^^
TypeError: 'str' object does not support item assignment
```

To change a letter, build a new string from pieces:

```
>>> word = "cat"
>>> "b" + word[1:]
'bat'
```

字 26.5 Text and numbers together

An f-string is the easy way. `str()` and `int()` change one into the other:

```
count = 3
print("I have " + str(count) + " cats")
print(f"I have {count} cats")
year = "2026"
print(int(year) + 1)
```

You should see:

```
I have 3 cats
I have 3 cats
2027
```

`ord` and `chr` go between a character and its number (its Unicode *code point*) — every character has one, Arabic and kanji too:

```
>>> ord("A")
65
>>> chr(66)
'B'
>>> ord("é")
233
>>> chr(0x732B)
'猫'
```

字 26.6 Common mistakes

1. Forgetting that methods return a new string.

```
name = "  sara "
name.strip()
print(f"[{name}]")
```

You should see:

```
[  sara ]
```

The spaces are still there: the stripped string was made and thrown away. Write `name = name.strip()`.

2. An index past the end.

```
s = "abc"
print(s[3])
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    print(s[3])
          ~^^^
IndexError: string index out of range
```

(the last index of a 3-letter string is 2; slices never complain: `s[1:99]` *is* `'bc'`*)*

3. Comparing a typed number with a number. `"10" == 10` is `False`: text is never equal to a number. Change the type first.

```
>>> "10" == 10
False
>>> int("10") == 10
True
```

Exercises

26.1 ★ Initials. From `first = "mohamed"` and `last = "eldoush"`, print the initials in capitals. *Expected:* `M.E`.

26.2 ★ Shout. Ask for a sentence and print it in capitals with three `!` after it. *Expected:* for `hello`: `HELLO!!!`

26.3 ★ The extension. From `filename = "report.final.pdf"`, print the extension (`pdf`) using `split` and `[-1]`.

26.4 ★★ Palindrome check. Ask for a word and print `True` if it reads the same backwards (ignore capitals). *Expected:* `Level` → `True`, `kyoto` → `False`.

26.5 ★★ Count the vowels. Count a, e, i, o, u in `"Fedokawaii is a desktop"` using `.count` five times. *Expected:* `10`.

26.6 ★★ Date flip. Turn `"2026-10-21"` into `"21/10/2026"` with `split` and `join`.

26.7 ★★★ Title case for a sentence. Ask for a sentence, and print it with only the first letter capital and the rest small, and the number of words. *Expected:* `tHE quick FOX` → `The quick fox (3 words)`.

26.8 ★★★ Mask a card number. Given `"4111222233334444"`, print it with all but the last 4 digits as `*`. *Expected:* `**********4444`.

26.9 ★★★★ Caesar shift, one letter. Using `ord` and `chr`, shift the small letter `c = "y"` forward by `k = 3` places, wrapping from `z` back to `a` (use `% 26`). *Expected:* `b`.

27. Python level 1: decisions and loops

In this chapter

- `True` and `False`, and the comparisons that make them
- `if`, `elif`, `else`: do something only sometimes
- `while` and `for`: do something many times
- `break` and `continue`; `range`
- the checkpoint for level 1: a guessing game

You need: chapters 25 and 26.

27.1 True and False

A **comparison** asks a question and gets a `bool`: `True` or `False`.

```
>>> 5 > 3
True
>>> 5 == 3
False
>>> 5 != 3
True
>>> 4 <= 4
True
>>> "apple" < "banana"
True
```

Sign	Asks
`==`	equal? (two signs — one `=` makes a variable)
`!=`	not equal?
`< > <= >=`	smaller, larger, or equal too

Combine questions with `and`, `or`, `not`:

```
>>> age = 20
>>> age >= 18 and age < 65
True
>>> age < 13 or age > 60
False
>>> not age > 18
False
>>> 13 <= age <= 19
False
```

(the last line is Python's short way to ask "between 13 and 19?")

27.2 if, elif, else

`if` runs its block only when the question is `True`. The block is the **indented** lines under it — four spaces:

```
temperature = 38
if temperature > 35:
    print("too hot")
    print("stay inside")
print("this line always runs")
```

You should see:

```
too hot
stay inside
this line always runs
```

`else` is what to do otherwise; `elif` (*else if*) asks another question. Python checks them from the top and runs **only the first** that is true:

```
score = 72
if score >= 90:
    grade = "A"
elif score >= 75:
    grade = "B"
elif score >= 60:
    grade = "C"
else:
    grade = "F"
print("grade:", grade)
```

You should see:

```
grade: C
```

Truthy and falsy. In an `if`, Python treats `0`, `0.0`, `""` (empty text), and empty collections as false; everything else as true:

```
name = ""
if name:
    print("hello", name)
else:
    print("no name given")
```

You should see:

```
no name given
```



27.3 while: repeat while true

`while` repeats its block as long as its question stays true:

```
n = 1
while n <= 5:
    print(n)
    n += 1
print("done")
```

You should see:

```
1
2
3
4
5
done
```

Careful. If the question never becomes false, the loop runs forever. `Ctrl+C` stops a running program in the terminal.

A loop that asks until the answer is good:

```
answer = input("A number? ")
while not answer.isdigit():
    print("that is not a number")
    answer = input("A number? ")
print("thanks:", int(answer))
```

You should see:

```
A number? abc
that is not a number
A number? 42
thanks: 42
```

 27.4 for: once for each thing

`for` takes each item of a collection in turn — the characters of a string, the items of a list:

```
for ch in "tea":
    print(ch)
for city in ["Sharjah", "Dubai", "Kyoto"]:
    print("I like", city)
```

You should see:

```
t
e
a
I like Sharjah
I like Dubai
I like Kyoto
```

`range` gives numbers to loop over:

```
>>> list(range(5))
[0, 1, 2, 3, 4]
>>> list(range(2, 6))
[2, 3, 4, 5]
>>> list(range(0, 20, 5))
[0, 5, 10, 15]
>>> list(range(5, 0, -1))
[5, 4, 3, 2, 1]
```

(like slices: `range(2, 6)` stops before 6)

```
total = 0
for n in range(1, 101):
    total += n
print(total)
```

You should see:

```
5050
```

(the sum of 1 to 100. Gauss did it in his head as a boy: 50 pairs of 101)

A loop inside a loop — a times table:

```
for row in range(1, 4):
    for col in range(1, 4):
        print(f"{row * col:3}", end="")
    print()
```

You should see:

```
  1  2  3
  2  4  6
  3  6  9
```



27.5 break and continue

`break` leaves the loop at once. `continue` skips the rest of this turn and goes to the next:

```
for n in range(1, 10):
    if n == 5:
        break
    print(n, end=" ")
print()
for n in range(1, 10):
    if n % 3 == 0:
        continue
    print(n, end=" ")
print()
```

You should see:

```
1 2 3 4
1 2 4 5 7 8
```

A common shape: loop forever, and `break` when done:

```
while True:
    item = input("item (q to stop)? ")
    if item == "q":
        break
    print("added", item)
```

You should see:

```
item (q to stop)? tea
added tea
item (q to stop)? rice
added rice
item (q to stop)? q
```

1. `=` where you meant `==`.

```
x = 5
if x = 5:
    print("five")
```

Python says:

```
  File "example.py", line 2
    if x = 5:
       ^^^^^
SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?
```

2. A missing colon, or a missing indent.

```
for i in range(3)
    print(i)
```

Python says:

```
  File "example.py", line 1
    for i in range(3)
                    ^
SyntaxError: expected ':'
```

```
for i in range(3):
print(i)
```

Python says:

```
  File "example.py", line 2
    print(i)
    ^^^^^
IndentationError: expected an indented block after 'for' statement on line 1
```

3. Changing the wrong variable in a while loop. This loop would never end, because `i` never grows — so here it is stopped after three turns with `break`:

```
i = 0
turns = 0
while i < 3:
    turns += 1
    print("turn", turns, "i is still", i)
    if turns == 3:
        break
```

You should see:

```
turn 1 i is still 0
turn 2 i is still 0
turn 3 i is still 0
```

The fix is `i += 1` inside the loop.

4. Off by one. `range(10)` is 0 to 9 — ten numbers, but not 10. If you want 1 to 10, write `range(1, 11)`.

27.7 Checkpoint project — a guessing game

Everything from level 1 in one program: the computer picks a number from 1 to 100; you guess; it says *higher* or *lower* until you have it, and counts your tries.

`random.randint(1, 100)` picks the number. `random.seed(7)` makes the "random" choice the same every run, so you can check your output against this one — remove it to play for real.

```
import random

random.seed(7)
secret = random.randint(1, 100)
tries = 0
while True:
    guess = int(input("Your guess? "))
    tries += 1
    if guess < secret:
        print("higher")
    elif guess > secret:
        print("lower")
    else:
        print(f"yes! {secret} in {tries} tries")
        break
```

You should see:

```
Your guess? 50
lower
Your guess? 25
higher
Your guess? 37
higher
Your guess? 43
lower
Your guess? 40
higher
Your guess? 42
yes! 42 in 6 tries
```

`Checkpoint`. Add three things, one at a time, running after each: (1) refuse a guess that is not a number (`isdigit`), without counting it; (2) give up after 7 tries and show the number; (3) at the end, ask *play again? (y/n)* and loop the whole game. If you can do all three without looking back, level 1 is done.

Exercises

27.1 ★ Positive, negative, zero. Ask for a number and print `positive`, `negative` or `zero`.

27.2 ★ Count down. Print 10 down to 1, then `lift off!`, with `range`.

27.3 ★ Sum of evens. Add up the even numbers from 2 to 100. *Expected:* `2550`.

27.4 ★★ FizzBuzz. Print 1 to 15, but `Fizz` for multiples of 3, `Buzz` for multiples of 5, and `FizzBuzz` for both.

27.5 ★★ Leap year. A year is a leap year if it divides by 4, except centuries, which must divide by 400. Ask for a year and say whether it is one. *Expected:* 2024 yes, 1900 no, 2000 yes.

27.6 ★★ Vowels by loop. Count the vowels of a typed word with a `for` loop and `in "aeiou"`.

27.7 ★★★ Password check. Ask for a password until it has at least 8 characters and at least one digit; then print `saved`.

27.8 ★★★ Is it prime?. Ask for a number and say whether it is prime (no divisor from 2 up to its square root — `d * d <= n`). *Expected:* 97 is prime, 91 is not (7 × 13).

27.9 ★★★ A triangle of stars. Ask for a height and print a right triangle of `*` that high.

27.10 ★★★★ The Collatz walk. Start from a number `n`. If it is even, halve it; if odd, make it `3n + 1`. Repeat until it is 1, printing each step, then the number of steps. Nobody has ever proved this always reaches 1. *Expected:* for 6: `6 3 10 5 16 8 4 2 1` and `8 steps`.

関 28. Python level 2: functions

In this chapter

- define a function, give it values, get a result back
- default and keyword arguments
- where a variable lives (scope)
- small functions that each do one thing — and how to check them with `assert`

You need: level 1 (chapters 25–27).

関 28.1 What a function is

A **function** is a piece of work with a name. You write it once with `def`, then **call** it — run it — as often as you like, each time with different values.

```
def greet():
    print("hello!")
    print("welcome to fedokawaii")

greet()
greet()
```

You should see:

```
hello!
welcome to fedokawaii
hello!
welcome to fedokawaii
```

The lines under `def` (indented) are the function's **body**. Defining it does nothing yet; calling it — `greet()`, with brackets — runs the body.

関 28.2 Parameters: values going in

Names in the brackets of `def` are **parameters**: they receive the values you pass when you call it (the **arguments**).

```
def greet(name):
    print(f"hello, {name}!")

greet("Sara")
greet("Omar")
```

You should see:

```
hello, Sara!
hello, Omar!
```

```
def rectangle(width, height):
    print(f"{width} × {height} = {width * height}")

rectangle(3, 4)
rectangle(10, 2)
```

You should see:

```
3 × 4 = 12
10 × 2 = 20
```

Word — parameter, argument. In `def rectangle(width, height)`, `width` and `height` are parameters: the names. In `rectangle(3, 4)`, `3` and `4` are arguments: the values.

28.3 return: a value coming back

`print` shows a value to the person; `return` hands it back to the code that called, so it can be used again. Most functions should `return`, not print:

```
def area(width, height):
    return width * height

a = area(3, 4)
print(a)
print(area(2, 5) + area(1, 1))
```

You should see:

```
12
11
```

`return` ends the function at once. A function with no `return` gives back `None` — Python's value for *nothing*:

```
def sign(n):
    if n > 0:
        return "positive"
    if n < 0:
        return "negative"
    return "zero"

def shout(text):
    print(text.upper())

print(sign(5), sign(-2), sign(0))
result = shout("hi")
print(result)
```

You should see:

```
positive negative zero
HI
None
```

A function can return several values at once, as a tuple (chapter 29):

```
def min_max(a, b, c):
    return min(a, b, c), max(a, b, c)

low, high = min_max(7, 2, 9)
print(low, high)
```

You should see:

```
2 9
```

関 28.4 Defaults and keyword arguments

A parameter can have a **default**: the value it takes when the call leaves it out.

```
def price_with_vat(price, rate=0.05):
    return round(price * (1 + rate), 2)

print(price_with_vat(100))
print(price_with_vat(100, 0.15))
print(price_with_vat(rate=0.2, price=50))
```

You should see:

```
105.0
115.0
60.0
```

Naming the argument in the call (`rate=0.2`) is a **keyword argument**: clearer to read, and the order no longer matters. Python's own functions use them all the time: `print(..., sep="-")`, `sorted(..., reverse=True)`.

関 28.5 Scope: where a name lives

A variable made inside a function exists only inside it, and only while it runs. This is good: two functions can each have their own `total` without touching each other.

```
def make_total():
    total = 42
    return total

make_total()
print(total)
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 6, in <module>
    print(total)
          ^^^^^
NameError: name 'total' is not defined
```

A function can **read** a name from outside, but should not change it. Pass values in and return results out — then each function can be understood on its own:

```
VAT = 0.05

def with_vat(price):
    return price * (1 + VAT)

print(with_vat(200))
```

You should see:

```
210.0
```

(names in CAPITALS are, by habit, constants: values the program never changes)

関 28.6 Docstrings and assert: say it, then check it

A **docstring** — a string as the first line of the body — says what the function does. `help()` shows it, and so does neovim's `K`:

```
def bmi(weight_kg, height_m):
    """Body mass index: weight divided by height squared."""
    return weight_kg / height_m ** 2

print(round(bmi(70, 1.75), 1))
print(bmi.__doc__)
```

You should see:

```
22.9
Body mass index: weight divided by height squared.
```

`assert` checks something you believe is true. If it is, nothing happens; if not, the program stops and tells you. It is the smallest possible test (chapter 35 grows them into pytest):

```
def double(n):
    return n * 2

assert double(4) == 8
assert double(0) == 0
assert double(-3) == -6
print("all good")
```

You should see:

```
all good
```

```
def half(n):
    return n // 2

assert half(7) == 3.5, "half of 7 should be 3.5"
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 4, in <module>
    assert half(7) == 3.5, "half of 7 should be 3.5"
           ^^^^^^^^^^^^^^^
AssertionError: half of 7 should be 3.5
```

(the message after the comma says what you expected; `//` dropped the .5 — the fix is `/`)

関 28.7 Functions calling functions

Big programs are small functions that call each other. Each one does one thing, has a name that says what, and can be checked on its own:

```
def is_vowel(ch):
    return ch.lower() in "aeiou"

def count_vowels(text):
    n = 0
    for ch in text:
        if is_vowel(ch):
            n += 1
    return n

def describe(word):
    return f"{word}: {count_vowels(word)} vowels, {len(word)} letters"

print(describe("Fedokawaii"))
print(describe("rhythm"))
```

You should see:

```
Fedokawaii: 6 vowels, 10 letters
rhythm: 0 vowels, 6 letters
```

A function may even call itself: that is **recursion**. Each call works on a smaller problem until one is so small it is answered directly:

```
def factorial(n):
    if n <= 1:
        return 1
    return n * factorial(n - 1)

print(factorial(5))
print(factorial(20))
```

You should see:

```
120
2432902008176640000
```

関 28.8 Common mistakes

1. Forgetting the brackets. Without `()`, the function is not called — you get the function itself:

```
def now():
    return "12:00"

print(now)
print(now())
```

You should see:

```
<function now at 0x7fb08aad3480>
12:00
```

2. Printing instead of returning. The value is shown, but the caller gets `None`:

```
def add(a, b):
    print(a + b)

total = add(2, 3) * 10
```

Python says:

```
5
Traceback (most recent call last):
  File "example.py", line 4, in <module>
    total = add(2, 3) * 10
            ~~~~~~~~~~^~~~
TypeError: unsupported operand type(s) for *: 'NoneType' and 'int'
```

3. Wrong number of arguments.

```
def area(width, height):
    return width * height

area(5)
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 4, in <module>
    area(5)
    ~~~~^^^
TypeError: area() missing 1 required positional argument: 'height'
```

4. Code after `return` never runs. Python stops at `return`; lines below it in the same block are dead. neovim greys them out.

Exercises

28.1 ★ Square. Write `square(n)` returning `n * n`; assert it for 0, 3 and -4.

28.2 ★ Greeting. Write `greeting(name, hour)` returning `Good morning, NAME` before 12 and `Good evening, NAME` from 12 on.

28.3 ★ Is even. Write `is_even(n)` returning `True` or `False`, in one line of body.

28.4 ★★ Pay with overtime. `pay(hours, rate)`: hours over 40 are paid 1.5 × the rate. Assert three cases.
Expected: rate 10: 30 h → `300`, 40 h → `400`, 45 h → `475.0`.

28.5 ★★ Clamp. `clamp(x, low=0, high=100)` returns `x`, but never below `low` or above `high`.

28.6 ★★ Reverse words. `reverse_words("one two three")` returns `"three two one"`.

28.7 ★★★ Power without .** Write `power(base, exp)` with a loop (exp a whole number ≥ 0), then again with recursion. Assert both against `**`.

28.8 ★★★ Grade. `grade(score)` returns A (90+), B (75+), C (60+) or F, and refuses scores outside 0–100 with `assert 0 <= score <= 100`.

28.9 ★★★★ Fibonacci, two ways. `fib(n)`: 0, 1, 1, 2, 3, 5, 8 … (each the sum of the two before). Write it with recursion, then with a loop, and print `fib(30)` with each. Which is slower, and why? (The recursive one works out the same values again and again: over a million calls for 30.)

29. Python level 2: lists and tuples

In this chapter

- lists: many values, in order, that you can change
- add, remove, sort, search
- loop over a list with its index (`enumerate`) or with another list (`zip`)
- tuples: lists that never change

You need: chapter 28.

29.1 A list

A **list** holds many values in order, inside `[ ]`, separated by commas. It indexes and slices exactly like a string (chapter 26):

```
>>> fruits = ["apple", "banana", "cherry"]
>>> len(fruits)
3
>>> fruits[0]
'apple'
>>> fruits[-1]
'cherry'
>>> fruits[1:]
['banana', 'cherry']
>>> "banana" in fruits
True
```

A list can hold any values, even a mix — but one kind per list is easier to work with:

```
>>> scores = [72, 95, 61]
>>> empty = []
>>> sum(scores), max(scores), min(scores)
(228, 95, 61)
>>> len(empty)
0
```

列 29.2 Changing a list

Unlike a string, a list can be changed in place — it is **mutable**:

```
>>> todo = ["shop", "cook"]
>>> todo.append("clean")
>>> todo
['shop', 'cook', 'clean']
>>> todo.insert(0, "wake up")
>>> todo
['wake up', 'shop', 'cook', 'clean']
>>> todo[1] = "shop for rice"
>>> todo
['wake up', 'shop for rice', 'cook', 'clean']
>>> todo.remove("cook")
>>> todo
['wake up', 'shop for rice', 'clean']
>>> last = todo.pop()
>>> last
'clean'
>>> todo
['wake up', 'shop for rice']
```

Method	Does
`.append(x)`	adds `x` at the end
`.insert(i, x)`	puts `x` at index `i`, pushing the rest along
`.remove(x)`	removes the first `x` (an error if there is none)
`.pop()` · `.pop(i)`	removes and gives back the last item · the one at `i`
`.extend(other)`	adds every item of another list
`.index(x)`	the index of the first `x`
`.count(x)`	how many times `x` is in it
`.sort()` · `.reverse()`	sorts · reverses, in place

列 29.3 Sorting

`.sort()` sorts the list itself; `sorted(list)` makes a new sorted list and leaves the old one alone:

```
>>> nums = [5, 2, 9, 1]
>>> sorted(nums)
[1, 2, 5, 9]
>>> nums
[5, 2, 9, 1]
>>> nums.sort(reverse=True)
>>> nums
[9, 5, 2, 1]
>>> words = ["banana", "Apple", "cherry"]
>>> sorted(words)
['Apple', 'banana', 'cherry']
>>> sorted(words, key=str.lower)
['Apple', 'banana', 'cherry']
>>> sorted(words, key=len)
['Apple', 'banana', 'cherry']
```

(capitals sort before small letters; `key=` *says what to sort by: here, the word in small letters, or its length)*

列 29.4 Looping over lists

```
prices = [3.5, 12.0, 7.25]
total = 0
for p in prices:
    total += p
print("total", total)
```

You should see:

```
total 22.75
```

`enumerate` gives the index and the item together; `zip` walks two lists side by side:

```
names = ["Sara", "Omar", "Lina"]
scores = [88, 92, 79]
for i, name in enumerate(names, start=1):
    print(i, name)
for name, score in zip(names, scores):
    print(f"{name:<5} {score}")
```

You should see:

```
1 Sara
2 Omar
3 Lina
Sara  88
Omar  92
Lina  79
```

Building a new list from an old one, the long way (chapter 37 shows the short way):

```
celsius = [20, 25, 31, 40]
fahrenheit = []
for c in celsius:
    fahrenheit.append(c * 9 / 5 + 32)
print(fahrenheit)
```

You should see:

```
[68.0, 77.0, 87.8, 104.0]
```

列 29.5 Lists of lists

A list can hold lists: a table, a board, a grid.

```
grid = [
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9],
]
print(grid[1])
print(grid[1][2])
for row in grid:
    print(sum(row))
```

You should see:

```
[4, 5, 6]
6
6
15
24
```

列 29.6 Tuples: fixed lists

A **tuple** is like a list in `( )`, but it can never change. Use one for a few values that belong together: a point, a date, a name and an age.

```
>>> point = (3, 4)
>>> point[0]
3
>>> x, y = point
>>> x + y
7
>>> date = (2026, 10, 21)
>>> year, month, day = date
>>> month
10
```

(taking a tuple apart into names is called unpacking)

```
point = (3, 4)
point[0] = 10
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    point[0] = 10
    ~~~~~^^^
TypeError: 'tuple' object does not support item assignment
```

`Word — mutable, immutable.` Mutable values can be changed in place: lists, dictionaries, sets. Immutable ones cannot: numbers, strings, tuples. "Changing" an immutable value always means making a new one.

列 29.7 Two names, one list

This surprises everyone once. `=` does not copy a list; it gives the same list a second name:

```
a = [1, 2, 3]
b = a
b.append(4)
print(a)
c = a.copy()
c.append(5)
print(a, c)
```

You should see:

```
[1, 2, 3, 4]
[1, 2, 3, 4] [1, 2, 3, 4, 5]
```

(`b = a` — one list, two names: change it through `b` and `a` sees it too. `.copy()` (or `list(a)`, or `a[:]`) makes a second list)

列 29.8 Common mistakes

1. Index out of range.

```
items = ["a", "b", "c"]
print(items[3])
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    print(items[3])
          ~~~~~^^^
IndexError: list index out of range
```

2. Saving the result of `.sort()`. It sorts in place and returns `None`:

```
nums = [3, 1, 2]
result = nums.sort()
print(result, nums)
```

You should see:

```
None [1, 2, 3]
```

3. Changing a list while looping over it. Items get skipped. Loop over a copy, or build a new list:

```
nums = [1, 2, 2, 3]
for n in nums:
    if n == 2:
        nums.remove(n)
print("wrong:", nums)
nums = [1, 2, 2, 3]
kept = []
for n in nums:
    if n != 2:
        kept.append(n)
print("right:", kept)
```

You should see:

```
wrong: [1, 2, 3]
right: [1, 3]
```

Exercises

29.1 ★ The middle. Print the middle item of `[4, 8, 15, 16, 23]` using `len` and `//`. *Expected:* `15`.

29.2 ★ Average. Write `average(nums)` with `sum` and `len`, and print the average of `[72, 95, 61]`. *Expected:* `76.0`.

29.3 ★ Shopping list. Start with an empty list, append three items, remove one, print the list sorted.

29.4 ★★ Only the long words. Given `words = ["a", "kawaii", "dwl", "fedora", "is"]`, build a list of words longer than 3 letters, with a loop. *Expected:* `['kawaii', 'fedora']`.

29.5 ★★ Second largest. Write `second_largest(nums)` without sorting: one loop, two variables. *Expected:* `[3, 9, 4, 9, 7]` → `7`: the largest value *different* from the largest.

29.6 ★★ Report card. With `names` and `scores` lists, print each name with its score and `pass` (60+) or `fail`, using `zip`.

29.7 ★★★ Remove duplicates, keep order. `unique([3, 1, 3, 2, 1])` returns `[3, 1, 2]`.

29.8 ★★★ Rotate. `rotate(items, k)` moves the last `k` items to the front, with slices. *Expected:* `rotate([1, 2, 3, 4, 5], 2)` → `[4, 5, 1, 2, 3]`.

29.9 ★★★★ Tic-tac-toe winner. A board is a list of three lists of `"X"`, `"O"` or `" "`. Write `winner(board)` returning `"X"`, `"O"` or `None`: check rows, columns and both diagonals.

鍵 30. Python level 2: dictionaries and sets

In this chapter

- dictionaries: look a value up by a key
- counting and grouping, the two jobs dictionaries do best
- sets: collections without repeats, and fast `in`
- the checkpoint for level 2: a contacts book

You need: chapters 28 and 29.

鍵 30.1 A dictionary

A **dictionary** (`dict`) maps **keys** to **values**, like a word to its meaning, or a name to a phone number. Write it in `{}` with `key: value` pairs:

```
>>> phone = {"Sara": "050 111", "Omar": "055 222"}
>>> phone["Sara"]
'050 111'
>>> phone["Lina"] = "052 333"
>>> phone
{'Sara': '050 111', 'Omar': '055 222', 'Lina': '052 333'}
>>> len(phone)
3
>>> "Omar" in phone
True
```

Looking up by key is instant, however big the dictionary — no searching through it.

鍵 30.2 Getting values safely

A missing key is an error; `.get` gives a default instead:

```
stock = {"rice": 2, "tea": 0}
print(stock["milk"])
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    print(stock["milk"])
          ~~~~~^^^^^^^^
KeyError: 'milk'
```

```
>>> stock = {"rice": 2, "tea": 0}
>>> stock.get("milk")
>>> stock.get("milk", 0)
0
>>> stock.get("rice", 0)
2
```

鍵 30.3 Changing and removing

```
>>> stock = {"rice": 2, "tea": 0}
>>> stock["tea"] += 5
>>> stock["milk"] = 1
>>> stock
{'rice': 2, 'tea': 5, 'milk': 1}
>>> del stock["rice"]
>>> stock
{'tea': 5, 'milk': 1}
>>> stock.pop("milk")
1
>>> stock
{'tea': 5}
```

鍵 30.4 Looping over a dictionary

A loop over a dictionary gives its keys. `.items()` gives key and value together:

```
ages = {"Sara": 31, "Omar": 28, "Lina": 35}
for name in ages:
    print(name)
for name, age in ages.items():
    print(f"{name} is {age}")
print(list(ages.values()), sum(ages.values()))
```

You should see:

```
Sara
Omar
Lina
Sara is 31
Omar is 28
Lina is 35
[31, 28, 35] 94
```

(a dictionary remembers the order keys were added in)

Sorted by value — `key=` again, here a small `lambda` function that gives the age:

```
ages = {"Sara": 31, "Omar": 28, "Lina": 35}
for name, age in sorted(ages.items(), key=lambda pair: pair[1]):
    print(name, age)
```

You should see:

```
Omar 28
Sara 31
Lina 35
```

Word — `lambda`. A tiny function without a name, written in one line: `lambda pair: pair[1]` is the same as `def f(pair): return pair[1]`.

鍵 30.5 Counting and grouping

The job dictionaries do most: count how often each thing appears.

```
text = "the cat and the dog and the bird"
counts = {}
for word in text.split():
    counts[word] = counts.get(word, 0) + 1
print(counts)
```

You should see:

```
{'the': 3, 'cat': 1, 'and': 2, 'dog': 1, 'bird': 1}
```

The standard library's `Counter` does the same, and finds the most common:

```
from collections import Counter

counts = Counter("the cat and the dog and the bird".split())
print(counts.most_common(2))
print(Counter("mississippi"))
```

You should see:

```
[('the', 3), ('and', 2)]
Counter({'i': 4, 's': 4, 'p': 2, 'm': 1})
```

Grouping: a dictionary of lists.

```
words = ["apple", "avocado", "banana", "blueberry", "cherry"]
by_letter = {}
for w in words:
    by_letter.setdefault(w[0], []).append(w)
print(by_letter)
```

You should see:

```
{'a': ['apple', 'avocado'], 'b': ['banana', 'blueberry'], 'c': ['cherry']}
```

(`setdefault(key, []`) gives the list for that key, making an empty one first if needed)

鍵 30.6 Nested data

Real data — a JSON file, a web API's answer — is dictionaries and lists inside each other:

```
user = {
    "name": "Sara",
    "langs": ["Python", "Rust"],
    "address": {"city": "Sharjah", "country": "UAE"},
}
print(user["langs"][1])
print(user["address"]["city"])
user["langs"].append("SQL")
print(len(user["langs"]))
```

You should see:

```
Rust
Sharjah
3
```

鍵 30.7 Sets

A **set** holds values without repeats and without order. It answers *is this in it?* fast, and does the maths of groups:

```
>>> colours = {"red", "green", "red", "blue"}
>>> len(colours)
3
>>> "red" in colours
True
>>> set([3, 1, 3, 2, 1]) == {1, 2, 3}
True
>>> a = {"python", "rust", "sql"}
>>> b = {"rust", "go"}
>>> sorted(a | b)
['go', 'python', 'rust', 'sql']
>>> a & b
{'rust'}
>>> sorted(a - b)
['python', 'sql']
```

Sign	Gives
`a | b`	everything in either (union)
`a & b`	what is in both (intersection)
`a - b`	what is in `a` but not `b` (difference)

(an empty set is `set()`; `{}` is an empty dictionary)

鍵 30.8 Common mistakes

1. A list as a key. Keys must be immutable; use a tuple:

```
places = {}
places[[25, 55]] = "Dubai"
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    places[[25, 55]] = "Dubai"
    ~~~~~~^^^^^^^^^^
TypeError: cannot use 'list' as a dict key (unhashable type: 'list')
```

```
places = {}
places[(25.2, 55.3)] = "Dubai"
print(places)
```

You should see:

```
{(25.2, 55.3): 'Dubai'}
```

2. Expecting a set to keep order. It does not; sort it when you show it: `sorted(my_set)`.

3. Changing a dictionary's size while looping over it. Loop over `list(d)` instead:

```
stock = {"rice": 2, "tea": 0, "milk": 0}
for item in list(stock):
    if stock[item] == 0:
        del stock[item]
print(stock)
```

You should see:

```
{'rice': 2}
```

鍵 30.9 Checkpoint project — a contacts book

Level 2 in one program: functions, a list of commands, a dictionary of contacts, a loop reading commands. Each command is one small function.

```
def add(book, name, number):
    book[name] = number
    return f"added {name}"

def find(book, part):
    hits = []
    for name in book:
        if part.lower() in name.lower():
            hits.append(f"{name}: {book[name]}")
    return ", ".join(hits) or "nobody"

def show(book):
    lines = []
    for name in sorted(book):
        lines.append(f"{name:<8}{book[name]}")
    return "\n".join(lines) or "(empty)"

def delete(book, name):
    if name in book:
        del book[name]
        return f"deleted {name}"
    return f"no {name}"

book = {}
while True:
    words = input("> ").split()
    if not words:
        continue
    cmd = words[0]
    if cmd == "quit":
        break
    elif cmd == "add" and len(words) == 3:
        print(add(book, words[1], words[2]))
    elif cmd == "find" and len(words) == 2:
        print(find(book, words[1]))
    elif cmd == "list":
        print(show(book))
    elif cmd == "del" and len(words) == 2:
        print(delete(book, words[1]))
    else:
        print("commands: add NAME NUMBER · find TEXT · list · del NAME · quit")
```

You should see:

```
> add Sara 050111
added Sara
> add Omar 055222
added Omar
> find sa
Sara: 050111
> list
Omar    055222
Sara    050111
> del Omar
deleted Omar
> list
Sara    050111
> quit
```

(`", ".join(hits) or "nobody"` — `or` gives its right side when the left is empty or false: a short way to say "this, or else that")

`Checkpoint.` Extend it: (1) each contact holds a dictionary `{"phone": …, "email": …}` instead of one number; (2) `add` refuses a name already there; (3) `count` prints how many contacts start with each letter. Write an `assert` for each function before you change it.

`Exercises`

30.1 ★ Capitals. Make a dictionary of four countries and their capitals and print each as `UAE → Abu Dhabi`.

30.2 ★ Letter count. Count each letter of `"banana"` with a dictionary and `.get`. *Expected:* `{'b': 1, 'a': 3, 'n': 2}`.

30.3 ★ Unique visitors. Given a list of names with repeats, print how many different people came, with a set.

30.4 ★★ Price list total. `prices = {"rice": 7.25, "tea": 12.5}`, `basket = {"rice": 3, "tea": 2}`: print the total. *Expected:* `46.75`.

30.5 ★★ Invert a dictionary. Turn `{"a": 1, "b": 2}` into `{1: "a", 2: "b"}`.

30.6 ★★ Common friends. Two people's friends as sets: print the friends they share, and those only the first has, both sorted.

30.7 ★★★ Word lengths. Group the words of a sentence by their length: `{3: ['the', 'cat'], 5: ['house']}`.

30.8 ★★★ Anagrams. Group `["listen", "silent", "enlist", "google", "gooegl", "cat"]` into anagram families: words with the same letters. Hint: `"".join(sorted(word))` is the same for anagrams — use it as the key.

30.9 ★★★★ The most common word. Read a paragraph (a string), make everything small, remove `.` and `,`, ignore the words `the`, `a`, `and`, `of`, and print the three most common words with their counts — first with a plain dictionary and `sorted`, then with `Counter`.

31. Python level 3: files

In this chapter

- read and write text files, the whole file or line by line
- paths with `pathlib`: join, look, list
- CSV and JSON: the two formats you will meet everywhere

You need: level 2 (chapters 28–30).

A variable lives only while the program runs. A **file** stays on the disk: notes, settings, data, results. Every program that remembers anything between runs uses files.

31.1 Paths with pathlib

`Path` (from the `pathlib` module) is Python's way to name a file. `/` joins pieces of a path — on any system:

```
from pathlib import Path

p = Path("notes") / "2026" / "october.md"
print(p)
print(p.name, p.stem, p.suffix)
print(p.parent)
print(Path.home().is_absolute())
```

You should see:

```
notes/2026/october.md
october.md october .md
notes/2026
True
```

Part	Path("notes/2026/october.md")
.name	october.md
.stem · .suffix	october · .md
.parent	notes/2026
.exists() · .is_file() · .is_dir()	True/False

31.2 Write, then read, a whole file

`write_text` makes (or replaces) a file; `read_text` gives the whole file as one string. Always say `encoding="utf-8"`, so Arabic, kanji and emoji come back as they went in:

```
from pathlib import Path

f = Path("hello.txt")
f.write_text("first line\nsecond line\n日本語\n", encoding="utf-8")
text = f.read_text(encoding="utf-8")
print(text)
print(len(text.splitlines()), "lines")
```

You should see:

```
first line
second line
日本語

3 lines
```

> Careful. `write_text` replaces the whole file without asking. Write to a new name when you are not sure.

31.3 Line by line, with open

For big files, or to add to the end, use `open` inside a `with` block. `with` closes the file for you when the block ends — even if an error happens inside:

```
with open("log.txt", "w", encoding="utf-8") as f:
    f.write("started\n")
with open("log.txt", "a", encoding="utf-8") as f:
    f.write("step 1 done\n")
    f.write("step 2 done\n")
with open("log.txt", encoding="utf-8") as f:
    for number, line in enumerate(f, start=1):
        print(number, line.rstrip())
```

You should see:

```
1 started
2 step 1 done
3 step 2 done
```

Mode	Means
`"r"` (the default)	read
`"w"`	write: a new, empty file (replacing one there)
`"a"`	append: add to the end
`"rb"` · `"wb"`	read · write bytes (pictures, zips — not text)

(each line read keeps its `\n`; `.rstrip()` takes it off)

31.4 Folders

```
from pathlib import Path

d = Path("project")
(d / "data").mkdir(parents=True, exist_ok=True)
(d / "data" / "a.csv").write_text("x\n1\n")
(d / "data" / "b.csv").write_text("x\n2\n")
(d / "readme.md").write_text("# project\n")
for p in sorted(d.rglob("*")):
    print(p, "(folder)" if p.is_dir() else f"{p.stat().st_size} bytes")
print([p.name for p in sorted(d.glob("data/*.csv"))])
```

You should see:

```
project/data (folder)
project/data/a.csv 4 bytes
project/data/b.csv 4 bytes
project/readme.md 10 bytes
['a.csv', 'b.csv']
```

(`glob("*.csv")` finds matching names in one folder; `rglob` searches every folder below)

31.5 CSV: tables in text

A **CSV** file is a table: one row per line, values separated by commas. Spreadsheets save and open it. The `csv` module reads it properly — quotes, commas inside values and all:

```
import csv
from pathlib import Path

Path("people.csv").write_text("name,city,age\nSara,Sharjah,31\nOmar,Dubai,28\n", encoding="utf-8")
with open("people.csv", encoding="utf-8", newline="") as f:
    for row in csv.DictReader(f):
        print(row["name"], "lives in", row["city"], "and is", int(row["age"]))
```

You should see:

```
Sara lives in Sharjah and is 31
Omar lives in Dubai and is 28
```

(`DictReader` gives each row as a dictionary keyed by the first line's names. Every value comes as text: change numbers with `int` or `float`)

Writing one:

```
import csv

rows = [{"item": "tea", "price": 8.5}, {"item": "rice", "price": 7.25}]
with open("prices.csv", "w", encoding="utf-8", newline="") as f:
    w = csv.DictWriter(f, fieldnames=["item", "price"])
    w.writeheader()
    w.writerows(rows)
print(open("prices.csv", encoding="utf-8").read())
```

You should see:

```
item,price
tea,8.5
rice,7.25
```

31.6 JSON: nested data in text

JSON is how programs exchange data — web APIs, settings, Claude's tool calls. It looks almost exactly like Python's dictionaries and lists, and `json` turns one into the other:

```
import json
from pathlib import Path

settings = {"theme": "dark", "font_size": 12, "langs": ["python", "rust"], "beta": False}
Path("settings.json").write_text(json.dumps(settings, indent=2), encoding="utf-8")
print(Path("settings.json").read_text(encoding="utf-8"))
loaded = json.loads(Path("settings.json").read_text(encoding="utf-8"))
print(loaded["langs"][0], loaded == settings)
```

You should see:

```
{
  "theme": "dark",
  "font_size": 12,
  "langs": [
    "python",
    "rust"
  ],
  "beta": false
}
python True
```

(`dumps` — dump to string — makes JSON text; `loads` reads it back. Note `False` in Python is `false` in JSON)

31.7 Common mistakes

1. A file that is not there.

```
from pathlib import Path
print(Path("nope.txt").read_text())
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    print(Path("nope.txt").read_text())
          ~~~~~~~~~~~~~~~~~~~~~~~~~~^^
  File "/root/.local/share/uv/python/cpython-3.14.6-linux-x86_64-gnu/lib/python3.14/pathlib/__init__.py",
line 787, in read_text
    with self.open(mode='r', encoding=encoding, errors=errors, newline=newline) as f:
         ~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/root/.local/share/uv/python/cpython-3.14.6-linux-x86_64-gnu/lib/python3.14/pathlib/__init__.py",
line 771, in open
    return io.open(self, mode, buffering, encoding, errors, newline)
           ~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: 'nope.txt'
```

Check first with `.exists()`, or catch the error (chapter 32).

2. A path relative to the wrong folder. `Path("data.csv")` means *in the folder the program was started from*, not the folder the script is in. For a file next to the script, use `Path(__file__).parent / "data.csv"`.

3. Forgetting that CSV values are text. `row["age"] + 1` fails; `int(row["age"]) + 1` works.

4. Forgetting `encoding="utf-8"`. On Linux it is usually the default, but not everywhere; writing it costs nothing and makes your program work on every system.

Exercises

31.1 ★ Write and read. Write three lines of your choice into `me.txt`, then read it back and print the number of characters.

31.2 ★ Number the lines. Print every line of a file with its line number in front, like `cat -n`.

31.3 ★ A diary. Append today's line to `diary.txt` each run (mode `"a"`); run it twice and print the file.

31.4 ★★ Word count. Write `wc(path)` returning lines, words and characters of a file, as a tuple.

31.5 ★★ Average from a CSV. Make `marks.csv` with `name,mark` rows and print the average mark.

31.6 ★★ Settings with defaults. Load `settings.json` if it exists, otherwise use `{"theme": "dark"}`; set `"font_size"` to 14 and save it back.

31.7 ★★★ Biggest files. List the three biggest files under a folder (`rglob("*")`, `.stat().st_size`), biggest first.

31.8 ★★★ CSV to JSON. Turn `people.csv` (name, city, age) into a JSON list of dictionaries, with `age` as a number.

31.9 ★★★★ Find in files. Write `grep(word, folder)`: every `.txt` file under the folder, every line containing the word (any case), printed as `file:line: text` — your own small ripgrep.

例 32. Python level 3: errors and exceptions

In this chapter

- read a traceback from the bottom up
- catch an error you expect with `try` / `except`
- raise your own errors, with a message that helps
- `else`, `finally`, and why not to catch everything

You need: chapter 31.

例 32.1 What an exception is

When Python cannot do what a line asks, it **raises an exception**: it stops and prints a **traceback** — where it was, and what went wrong. Read it from the bottom: the last line names the kind of error and the reason; the lines above say where.

```
def average(nums):
    return sum(nums) / len(nums)

print(average([3, 4]))
print(average([]))
```

Python says:

```
3.5
Traceback (most recent call last):
  File "example.py", line 5, in <module>
    print(average([]))
          ~~~~~~~^^^^
  File "example.py", line 2, in average
    return sum(nums) / len(nums)
           ~~~~~~~~~~^~~~~~~~~~
ZeroDivisionError: division by zero
```

(read up from the bottom: a `ZeroDivisionError`, in `average`, line 2, called from line 5 with an empty list)

Exception	Means	Example
`NameError`	no such name	a typo
`TypeError`	the wrong kind of value	`"3" + 4`
`ValueError`	the right kind, a wrong value	`int("abc")`
`IndexError` · `KeyError`	no such index · key	`[1][5]`, `{}["a"]`
`ZeroDivisionError`	divided by zero	`1 / 0`
`FileNotFoundError`	no such file	`open("nope")`
`AttributeError`	no such method or field	`"a".push()`

例 32.2 try and except

When you *expect* a problem — a person typing letters where you wanted a number, a file that may be missing — catch it with `try` / `except` and do something sensible:

```
while True:
    text = input("How many? ")
    try:
        n = int(text)
        break
    except ValueError:
        print(f"{text!r} is not a whole number, try again")
print("ok:", n)
```

You should see:

```
How many? twelve
'twelve' is not a whole number, try again
How many? 12
ok: 12
```

(`{text!r}` shows the value with quotes, so an empty answer is visible as `''`)

Several kinds, each its own answer; `as e` names the exception so you can show its message:

```
def safe_divide(a, b):
    try:
        return a / b
    except ZeroDivisionError:
        return None
    except TypeError as e:
        print("type problem:", e)
        return None

print(safe_divide(10, 4))
print(safe_divide(1, 0))
print(safe_divide("1", 2))
```

You should see:

```
2.5
None
type problem: unsupported operand type(s) for /: 'str' and 'int'
None
```

例 32.3 else and finally

`else` runs only if nothing went wrong; `finally` runs **always** — for clean-up:

```
def read_number(text):
    try:
        n = float(text)
    except ValueError:
        print("  not a number:", text)
    else:
        print("  got", n)
    finally:
        print("  (checked", repr(text) + ")")

read_number("3.5")
read_number("abc")
```

You should see:

```
  got 3.5
  (checked '3.5')
  not a number: abc
  (checked 'abc')
```

例 32.4 Raising your own

When your function receives something it cannot handle, `raise` an exception with a message that says what was wrong — the person who sees it is often you, next month:

```
def set_volume(level):
    if not 0 <= level <= 100:
        raise ValueError(f"volume must be 0-100, got {level}")
    print("volume", level)

set_volume(40)
set_volume(140)
```

Python says:

```
volume 40
Traceback (most recent call last):
  File "example.py", line 7, in <module>
    set_volume(140)
    ~~~~~~~~~~^^^^^
  File "example.py", line 3, in set_volume
    raise ValueError(f"volume must be 0-100, got {level}")
ValueError: volume must be 0-100, got 140
```

Your own kind of error is a small class (classes come in chapter 34; this is all you need now):

```
class OutOfStock(Exception):
    pass

stock = {"tea": 0}

def buy(item):
    if stock.get(item, 0) == 0:
        raise OutOfStock(item)
    stock[item] -= 1

try:
    buy("tea")
except OutOfStock as e:
    print("sorry, no", e)
```

You should see:

```
sorry, no tea
```

例 32.5 When to catch, and when not

The rule: **catch only what you expect, and only where you can do something useful about it**. Everything else should stop the program with its traceback — that traceback is how you find the bug.

```
data = {"a": 1}
try:
    print(data["b"])
except Exception:
    pass
print("the bug is hidden: nothing told you key 'b' was missing")
```

You should see:

```
the bug is hidden: nothing told you key 'b' was missing
```

(never write `except: pass` *— it swallows every error, typos included, and the program goes on with wrong values)*

例 32.6 Common mistakes

1. Too much inside `try`**.** Put only the line that can fail there; the rest after it, or in `else`. Otherwise a bug in another line is caught by mistake.

2. Catching the wrong kind. `int("x")` raises `ValueError`, not `TypeError`. Run it once at the prompt to see which.

```
try:
    int("x")
except TypeError:
    print("caught")
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    int("x")
    ~~~^^^^^
ValueError: invalid literal for int() with base 10: 'x'
```

3. Losing the message. `except ValueError as e: print("error")` throws away what went wrong; print `e` too.

Exercises

32.1 ★ Safe int. `to_int(text, default=0)` returns the number, or `default` if the text is not one.

32.2 ★ Read or say so. Read `notes.txt`; if it is missing, print `no notes yet` instead of a traceback.

32.3 ★ Which error?. At the prompt, find the exception each raises: `[1, 2][9]`, `{"a": 1}["b"]`, `"a" * "b"`, `int("1.5")`, `None.upper()`.

32.4 ★★ Age check. `check_age(age)` raises `ValueError` with a clear message for a negative age or one over 150; call it in a `try` with three values.

32.5 ★★ Average of good lines. A list of strings, some numbers and some not: average only the numbers, and count the bad ones. *Expected:* `["3", "x", "5", "", "10"]` → `average 6.0, 2 bad`.

32.6 ★★★ Retry. `ask_int(prompt, low, high)` keeps asking until the answer is a whole number from `low` to `high`, saying what was wrong each time.

32.7 ★★★ Your own exception. Make `InsufficientFunds(Exception)` and `withdraw(balance, amount)` that raises it with the missing amount in the message.

32.8 ★★★★ A safe calculator. Read lines like `12 / 4`, `3 * x`, `5 / 0`; print the answer or a clear message for each problem (a bad number, an unknown sign, dividing by zero, a wrong number of parts) — each problem caught by its own `except`.

권 33. Python level 3: modules, venv and uv

In this chapter

- `import`: use code from another file, the standard library, or the internet
- your own module, and `if __name__ == "__main__"`
- projects with uv: their own Python, their own libraries, a lock file
- the checkpoint for level 3: an expense tracker

You need: chapters 31 and 32 (and chapter 24's loop).

권 33.1 import: code from elsewhere

A **module** is a file of Python code you can use from another. Python comes with hundreds — the **standard library** — and `import` brings one in:

```
import math
import random
from datetime import date

print(math.sqrt(16), math.pi)
random.seed(1)
print(random.choice(["tea", "coffee", "water"]))
print(date(2026, 12, 2) - date(2026, 10, 21))
```

You should see:

```
4.0 3.141592653589793
tea
42 days, 0:00:00
```

Form	Then you write
`import math`	`math.sqrt(2)`
`from math import sqrt`	`sqrt(2)`
`import numpy as np`	`np.array(…)` (a short name, by habit)

Some standard modules worth knowing by name:

Module	For
`math`, `statistics`, `random`	numbers, averages, chance
`datetime`, `time`	dates, times, timing
`pathlib`, `shutil`, `os`	files and folders
`json`, `csv`	data formats
`collections`	`Counter`, `defaultdict`, `deque`
`itertools`, `functools`	tools for loops and functions (chapter 37)
`argparse`	command-line options (chapter 39)
`urllib.request`	the web
`sqlite3`	a database in one file

권 33.2 Your own module

Any `.py` file is a module. Put functions in one file and import them from another:

shapes.py:

```
"""Areas of simple shapes."""
import math

def circle(r):
    return math.pi * r ** 2

def square(side):
    return side * side
```

main.py:

```
import shapes
from shapes import square

print(round(shapes.circle(2), 2))
print(square(3))
print(shapes.__doc__)
```

```
$ uv run main.py
12.57
9
Areas of simple shapes.
```

33.3 if **name** == "**main**"

A module's top-level code runs when it is imported. To have code that runs only when the file is run *itself* — a demo, or the program's start — put it under this line:

temperature.py:

```
def c_to_f(c):
    return c * 9 / 5 + 32

if __name__ == "__main__":
    print("demo:", c_to_f(100))
```

use_it.py:

```
from temperature import c_to_f
print(c_to_f(37))
```

```
$ uv run temperature.py
demo: 212.0
$ uv run use_it.py
98.6
```

(run directly, `__name__` is `"__main__"`; imported, it is the module's name, so the demo is skipped)

33.4 Packages from the internet: PyPI

Beyond the standard library are over half a million packages on **PyPI** (the Python Package Index) — NumPy, requests, PyTorch. You install them into a project with uv.

Word — package, library, dependency. A *package* is code you install; a *library* is a package meant to be used by other code; a *dependency* is a package your project needs.

Word — virtual environment (venv). A project's own folder (`.venv`) holding its Python and its packages, apart from every other project and from the system. Two projects can need two versions of the same library: each venv has its own.

권 33.5 A project with uv

`uv` (chapter 24) does the whole job: makes the project, picks its Python, makes its venv, installs packages, writes the lock file, runs your code inside the venv:

```
$ uv init --app expenses
Initialized project `expenses` at `/home/you/repos/expenses`
$ ls expenses
README.md  main.py  pyproject.toml
$ cat expenses/pyproject.toml
[project]
name = "expenses"
version = "0.1.0"
description = "Add your description here"
readme = "README.md"
requires-python = ">=3.14"
dependencies = []
```

(`dependencies` *fills as you* `uv add`*;* `ls -a` *would show* `.python-version`*,* `.gitignore` *and* `.git` *too)*

Command	Does
`uv init NAME`	a new project: `pyproject.toml`, `main.py`, `.python-version`, git
`uv add NAME`	install a package, and write it into `pyproject.toml` and `uv.lock`
`uv add --dev NAME`	a package needed only while developing (pytest, ruff)
`uv remove NAME`	the opposite
`uv run FILE` · `uv run CMD`	run inside the project's venv (made on the first run)
`uv sync`	make `.venv` match the lock file exactly (after `git pull`, say)
`uv tree`	the dependencies, and theirs
`uv python list`	the Pythons uv knows and has
`uvx TOOL`	run a tool once without installing it: `uvx ruff check .`

Careful. Never `sudo pip install`. It mixes packages into the system's Python, which Fedora's own tools use. Always a project and `uv add`; for a tool you want everywhere, `uv tool install NAME`.

The lock file. `uv.lock` records the exact version of every package, and of every package those need. Commit it to git: on another machine, `uv sync` rebuilds exactly the same `.venv`, so the code runs the same.

권 33.6 Common mistakes

1. Naming your file after a module. A file called `random.py` or `json.py` in your folder is found *before* the real one, and `import random` imports yours:

`random.py`:

```
print("this is my random.py, not Python's")
```

```
import random
print(random.randint(1, 6))
```

Python says:

```
this is my random.py, not Python's
Traceback (most recent call last):
  File "example.py", line 2, in <module>
    print(random.randint(1, 6))
          ^^^^^^^^^^^^^^
AttributeError: module 'random' has no attribute 'randint' (consider renaming 'random.py' since it has the
same name as the standard library module named 'random' and prevents importing that standard library module)
```

Rename yours (`dice.py`), and delete any `__pycache__` folder next to it.

```
$ rm random.py && rm -rf __pycache__
```

2. A missing package. `ModuleNotFoundError: No module named 'numpy'` means it is not in *this* project's venv: `uv add numpy`, and run with `uv run`.

3. Running with the system's `python3` instead of `uv run`: the project's packages are not there. In a uv project, always `uv run`.



33.7 Checkpoint project — an expense tracker

Level 3 in one program: files (a CSV that survives between runs), errors (bad input, a missing file), and a module of functions kept apart from the part that talks to the person.

ledger.py:

```
"""Expenses kept in a CSV file: date, category, amount."""
import csv
from pathlib import Path

FIELDS = ["date", "category", "amount"]

def load(path):
    p = Path(path)
    if not p.exists():
        return []
    with open(p, encoding="utf-8", newline="") as f:
        rows = list(csv.DictReader(f))
    for r in rows:
        r["amount"] = float(r["amount"])
    return rows

def save(path, rows):
    with open(path, "w", encoding="utf-8", newline="") as f:
        w = csv.DictWriter(f, fieldnames=FIELDS)
        w.writeheader()
        w.writerows(rows)

def parse(line):
    """'2026-10-21 food 32.5' -> a row; ValueError if it is not one."""
    parts = line.split()
    if len(parts) != 3:
        raise ValueError("write: DATE CATEGORY AMOUNT")
    date, category, amount = parts
    return {"date": date, "category": category, "amount": float(amount)}

def totals(rows):
    out = {}
    for r in rows:
        out[r["category"]] = out.get(r["category"], 0) + r["amount"]
    return out
```

track.py:

```
from ledger import load, parse, save, totals

PATH = "expenses.csv"
rows = load(PATH)
while True:
    line = input("expense (or 'report', 'quit')? ").strip()
    if line == "quit":
        break
    if line == "report":
        for cat, amount in sorted(totals(rows).items()):
            print(f"  {cat:<10}{amount:>8.2f}")
        continue
    try:
        rows.append(parse(line))
    except ValueError as e:
        print("  ", e)
        continue
    save(PATH, rows)
    print(f"  saved ({len(rows)} expenses)")
```

```
$ uv run track.py
expense (or 'report', 'quit')? 2026-10-21 food 32.5
  saved (1 expenses)
expense (or 'report', 'quit')? 2026-10-21 taxi 18
  saved (2 expenses)
expense (or 'report', 'quit')? lunch
   write: DATE CATEGORY AMOUNT
expense (or 'report', 'quit')? 2026-10-22 food 12.25
  saved (3 expenses)
expense (or 'report', 'quit')? report
  food         44.75
  taxi         18.00
expense (or 'report', 'quit')? quit
```

```
$ cat expenses.csv
date,category,amount
2026-10-21,food,32.5
2026-10-21,taxi,18.0
2026-10-22,food,12.25
```

(run it again: the three expenses are loaded from the file, and the next one is added to them)

`Checkpoint.` Add (1) `report MONTH` (`report 2026-10`) to total one month only; (2) a refusal of negative amounts and of dates not shaped `YYYY-MM-DD` (`datetime.date.fromisoformat` raises `ValueError` for you); (3) an `assert` for `parse` and `totals` in a `__main__` block of `ledger.py`. Level 3 done.

Exercises

33.1 ★ Dice. With `random.randint`, roll two dice ten times and print each pair and its sum (seed it with `random.seed(3)` to compare).

33.2 ★ Days to go. With `datetime.date`, print how many days from today to 31 December this year.

33.3 ★ Statistics. Use `statistics.mean`, `median` and `stdev` on `[2, 4, 4, 4, 5, 5, 7, 9]`.

33.4 ★★ Your module. Make `textutil.py` with `vowels(s)` and `words(s)`, a `__main__` demo, and use both from `main.py`.

33.5 ★★ Counter and defaultdict. Count the letters of `"hello world"` with `Counter`, then group words by first letter with `collections.defaultdict(list)`.

33.6 ★★ A project of your own. `uv init quotes`, `uv add --dev pytest`, write `main.py` that prints a random line from `quotes.txt`. Check `pyproject.toml` lists pytest under the dev group, and commit with `uv.lock`.

33.7 ★★★ A web request. With `urllib.request` and `json`, fetch `https://api.github.com/repos/python/cpython` and print its stars and its description. (Needs the network.)

33.8 ★★★ SQLite. With `sqlite3`, make a table `expenses(date, category, amount)` in `money.db`, insert three rows, and print the total per category with `GROUP BY`.

33.9 ★★★★ Back up a folder. Write `backup.py SRC DEST`: copy every file of SRC (keeping sub-folders) into a new folder `DEST/SRC-YYYY-MM-DD`, skip files over 10 MB, and print how many were copied and skipped. Use `pathlib`, `shutil.copy2` and `datetime.date.today()`.

類 34. Python level 4: classes and objects

In this chapter

- a class: your own kind of value, with data and behaviour together
- `__init__`, `self`, methods, and the special methods that make `print` and `==` work
- dataclasses: the short way
- inheritance, briefly — and when not to use it

You need: level 3 (chapters 31–33).

類 34.1 Why classes

Until now, data and the functions working on it lived apart: a dictionary for an account, and functions that take the dictionary. A **class** keeps them together: it describes a new kind of value — an *account* — with its data (**attributes**) and what it can do (**methods**). Each value made from it is an **object** (or *instance*).

```
class Dog:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def bark(self):
        return f"{self.name} says woof"

rex = Dog("Rex", 3)
luna = Dog("Luna", 5)
print(rex.name, rex.age)
print(luna.bark())
```

You should see:

```
Rex 3
Luna says woof
```

Piece	Means
`class Dog:`	a new kind of value, named with a capital by habit
`__init__`	runs when an object is made: `Dog("Rex", 3)` calls it
`self`	the object itself — Python passes it in for you
`self.name = name`	an attribute: data stored on this object
`def bark(self)`	a method: a function every Dog can do

類 34.2 self, step by step

`rex.bark()` is Python's short way to write `Dog.bark(rex)`: the object before the dot becomes `self`. That is how one method serves every dog:

```
class Dog:
    def __init__(self, name):
        self.name = name

    def bark(self):
        return f"{self.name}: woof"

rex = Dog("Rex")
print(rex.bark())
print(Dog.bark(rex))
```

You should see:

```
Rex: woof
Rex: woof
```

Attributes can change; each object has its own:

```
class Counter:
    def __init__(self):
        self.count = 0

    def click(self):
        self.count += 1

a = Counter()
b = Counter()
a.click(); a.click(); b.click()
print(a.count, b.count)
```

You should see:

```
2 1
```

Methods keep the rules in one place — here, *no withdrawing more than the balance*:

```
class Account:
    def __init__(self, owner, balance=0):
        self.owner = owner
        self.balance = balance
        self.history = []

    def deposit(self, amount):
        if amount <= 0:
            raise ValueError("deposit must be positive")
        self.balance += amount
        self.history.append(("in", amount))

    def withdraw(self, amount):
        if amount > self.balance:
            raise ValueError(f"only {self.balance} available")
        self.balance -= amount
        self.history.append(("out", amount))

acc = Account("Sara", 100)
acc.deposit(50)
acc.withdraw(30)
print(acc.balance, acc.history)
try:
    acc.withdraw(500)
except ValueError as e:
    print("refused:", e)
```

You should see:

```
120 [('in', 50), ('out', 30)]
refused: only 120 available
```

類 34.4 Special methods: print, ==, +

Methods with double underscores (*dunder* methods) let your objects work with Python's own signs and functions:

```
class Money:
    def __init__(self, amount, currency="AED"):
        self.amount = amount
        self.currency = currency

    def __repr__(self):
        return f"Money({self.amount}, {self.currency!r})"

    def __str__(self):
        return f"{self.amount:.2f} {self.currency}"

    def __eq__(self, other):
        return self.amount == other.amount and self.currency == other.currency

    def __add__(self, other):
        return Money(self.amount + other.amount, self.currency)

a = Money(10)
b = Money(5.5)
print(a + b)
print(repr(a + b))
print(a == Money(10), a == b)
```

You should see:

```
15.50 AED
Money(15.5, 'AED')
True False
```

Method	Used by
`__init__`	making the object
`__repr__`	the prompt, `repr()`, lists: an exact description, for programmers
`__str__`	`print`, f-strings: a friendly one
`__eq__` · `__lt__`	`==` · `<` (and so `sorted`)
`__add__`	`+`
`__len__`	`len()`

類 34.5 Dataclasses: the short way

Most classes start as "a few fields, with a nice `repr` and `==`". `@dataclass` writes `__init__`, `__repr__` and `__eq__` for you from the fields you list:

```
from dataclasses import dataclass, field

@dataclass
class Item:
    name: str
    price: float
    count: int = 1

    def cost(self):
        return self.price * self.count

@dataclass
class Basket:
    items: list[Item] = field(default_factory=list)

    def total(self):
        return sum(i.cost() for i in self.items)

b = Basket()
b.items.append(Item("rice", 7.25, 3))
b.items.append(Item("tea", 12.5))
print(b.items[0])
print(b.total())
print(Item("tea", 12.5) == Item("tea", 12.5))
```

You should see:

```
Item(name='rice', price=7.25, count=3)
34.25
True
```

(`field(default_factory=list)` gives each basket its own new list; a plain `= []` would be one list shared by every basket — Python refuses it)

`@dataclass(frozen=True)` makes objects that cannot change — safe as dictionary keys:

```
from dataclasses import dataclass

@dataclass(frozen=True)
class Point:
    x: float
    y: float

p = Point(1, 2)
print(p, {p: "home"})
p.x = 5
```

Python says:

```
Point(x=1, y=2) {Point(x=1, y=2): 'home'}
Traceback (most recent call last):
  File "example.py", line 10, in <module>
    p.x = 5
    ^^^
  File "<string>", line 16, in __setattr__
dataclasses.FrozenInstanceError: cannot assign to field 'x'
```

34.6 Inheritance, briefly

A class can **inherit** from another: it gets all its methods, and can add or replace some. `super()` calls the parent's version:

```
class Animal:
    def __init__(self, name):
        self.name = name

    def speak(self):
        return "..."

    def intro(self):
        return f"{self.name} says {self.speak()}"

class Cat(Animal):
    def speak(self):
        return "meow"

class Kitten(Cat):
    def speak(self):
        return super().speak() + " (tiny)"

for a in [Animal("thing"), Cat("Tama"), Kitten("Mochi")]:
    print(a.intro())
print(isinstance(Kitten("x"), Animal))
```

You should see:

```
thing says ...
Tama says meow
Mochi says meow (tiny)
True
```

Use it sparingly. Most of the time a class that *holds* another (an Account holding a list of transactions) is clearer than one that *is* another. Inheritance is right when the new thing truly is a kind of the old one — as your own exceptions are kinds of `Exception` (chapter 32).

34.7 Common mistakes

1. Forgetting `self`. Every method's first parameter is `self`, and every attribute is reached through it:

```
class Light:
    def __init__(self):
        self.on = False

    def toggle():
        self.on = not self.on

Light().toggle()
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 8, in <module>
    Light().toggle()
    ~~~~~~~~~~~~~~~^^
TypeError: Light.toggle() takes 0 positional arguments but 1 was given
```

2. `name` instead of `self.name` inside a method — a `NameError`, or worse, a different variable of the same name.

3. Calling the class without its arguments.

```
class Dog:
    def __init__(self, name):
        self.name = name

d = Dog()
```

Python says:

```
Traceback (most recent call last):
  File "example.py", line 5, in <module>
    d = Dog()
TypeError: Dog.__init__() missing 1 required positional argument: 'name'
```

4. A list as a plain default. In a normal class, `def __init__(self, items=[])` shares one list between all objects. Use `items=None` and `self.items = items or []`; in a dataclass, `field(default_factory=list)`.

Exercises

34.1 ★ A rectangle. `Rectangle(width, height)` with `area()` and `perimeter()`.

34.2 ★ A nice print. Give the rectangle a `__str__` that prints `3×4 rectangle`.

34.3 ★ As a dataclass. Write the rectangle again as a `@dataclass`, and compare two equal ones with `==`.

34.4 ★★ A to-do list. `TodoList` with `add(text)`, `done(n)` and `__str__` that prints `[ ]` / `[x]` lines, numbered.

34.5 ★★ Temperature. `Temperature(celsius)` with a `fahrenheit()` method and `__lt__`, so a list of them can be `sorted`.

34.6 ★★ Shapes. `Shape` with `area()` returning 0, and `Circle` and `Square` inheriting from it. Print the total area of a list of mixed shapes.

34.7 ★★★ A stack. `Stack` with `push`, `pop` (raising `IndexError("empty stack")` when empty), `peek`, and `__len__`. Use it to check that brackets in `"(a[b]{c})"` are balanced.

34.8 ★★★ Money that refuses to mix. Make `Money.__add__` raise `ValueError` when the currencies differ, and `__mul__` for `Money(10) * 3`.

34.9 ★★★★ An inventory. `Product` (a frozen dataclass: name, price) and `Inventory` holding a dictionary product → count, with `add(product, n)`, `remove(product, n)` (an error if too few), `value()` (the total worth) and `low(limit)` (products below `limit`, sorted by name).

35. Python level 4: testing with pytest

In this chapter

- why tests: change code without fear
- pytest: test functions, `assert`, running, reading a failure
- testing errors, many cases at once, and temporary files
- test first: red, green, tidy

You need: chapter 34, and chapter 24's loop (Space t w).

35.1 Why tests

A **test** is a small program that runs your code and checks the answer. You write it once; then, every time you change anything, all the tests run in a second and tell you whether everything still works. Without tests, every change is a risk; with them, you can rebuild a program boldly.

pytest finds files named `test_*.py`, runs every function in them named `test_*`, and reports. Inside, a plain `assert` is all you need.

35.2 The first test

shop.py:

```
def total(prices, discount=0.0):
    """The sum of the prices, less a discount (0.1 is 10 %)."""
    return round(sum(prices) * (1 - discount), 2)
```

test_shop.py:

```
from shop import total

def test_total():
    assert total([3, 5, 8]) == 16

def test_discount():
    assert total([10, 10], discount=0.1) == 18

def test_empty():
    assert total([]) == 0
```

```
$ uv run pytest -q
...                                                                  [100%]
3 passed in 0.01s
```

(each `.` is a test that passed. In a uv project, `uv add --dev pytest` *once; in neovim, Space t t runs them, Space t w on every save)*

35.3 Reading a failure

Break `total` on purpose — forget the discount — and run again:

shop.py:

```
def total(prices, discount=0.0):
    return round(sum(prices), 2)
```

```
$ uv run pytest -q
.F                                                                   [100%]
================================== FAILURES ===================================
_______________________________ test_discount ________________________________

    def test_discount():
>       assert total([10, 10], discount=0.1) == 18
E       assert 20 == 18
E        +  where 20 = total([10, 10], discount=0.1)

test_shop.py:7: AssertionError
=========================== short test summary info ===========================
FAILED test_shop.py::test_discount - assert 20 == 18
1 failed, 2 passed in 0.01s
```

Read it like a traceback, from the bottom: which test failed, and pytest's `assert 20 == 18` shows *both* sides: the code gave 20, the test wanted 18. Put the discount back and it is green again.

shop.py:

```
def total(prices, discount=0.0):
    """The sum of the prices, less a discount (0.1 is 10 %)."""
    return round(sum(prices) * (1 - discount), 2)
```

35.4 Testing errors

`pytest.raises` checks that something *fails* the way it should:

prices.py:

```
def parse_price(text):
    try:
        value = float(text)
    except ValueError:
        raise ValueError(f"not a price: {text!r}") from None
    if value < 0:
        raise ValueError("a price cannot be negative")
    return value
```

test_prices.py:

```
import pytest
from prices import parse_price

def test_number():
    assert parse_price("12.5") == 12.5

def test_words_are_not_prices():
    with pytest.raises(ValueError, match="not a price"):
        parse_price("twelve")

def test_negative():
    with pytest.raises(ValueError, match="negative"):
        parse_price("-3")
```

```
$ uv run pytest -q test_prices.py
...                                                                  [100%]
3 passed in 0.01s
```

試 35.5 Many cases, one test

`@pytest.mark.parametrize` runs one test with many inputs — each counts as a test of its own, so a failure names the case:

`test_grade.py`:

```
import pytest

def grade(score):
    if score >= 90: return "A"
    if score >= 75: return "B"
    if score >= 60: return "C"
    return "F"

@pytest.mark.parametrize("score, expected", [
    (100, "A"), (90, "A"), (89, "B"), (75, "B"), (60, "C"), (59, "F"), (0, "F"),
])
def test_grade(score, expected):
    assert grade(score) == expected
```

```
$ uv run pytest -v test_grade.py
test_grade.py::test_grade[100-A] PASSED                                  [ 14%]
test_grade.py::test_grade[90-A] PASSED                                   [ 28%]
test_grade.py::test_grade[89-B] PASSED                                   [ 42%]
test_grade.py::test_grade[75-B] PASSED                                   [ 57%]
test_grade.py::test_grade[60-C] PASSED                                   [ 71%]
test_grade.py::test_grade[59-F] PASSED                                   [ 85%]
test_grade.py::test_grade[0-F] PASSED                                    [100%]
============================== 7 passed in 0.01s ===============================
```

(the header lines pytest prints first — its version, the folder — are left out here. The edges — 90, 89, 75, 60, 59 — are where mistakes hide; test them)

試 35.6 Files in tests: tmp_path

A test must not touch your real files. Name a parameter `tmp_path` and pytest hands it a new, empty folder for that test alone — a **fixture**:

`notes.py`:

```
from pathlib import Path

def add_note(path, text):
    with open(path, "a", encoding="utf-8") as f:
        f.write(f"- {text}\n")

def count_notes(path):
    p = Path(path)
    if not p.exists():
        return 0
    return sum(1 for line in p.read_text(encoding="utf-8").splitlines() if line.startswith("- "))
```

test_notes.py:

```
from notes import add_note, count_notes

def test_new_file_has_none(tmp_path):
    assert count_notes(tmp_path / "n.md") == 0

def test_add_two(tmp_path):
    f = tmp_path / "n.md"
    add_note(f, "buy rice")
    add_note(f, "call Omar")
    assert count_notes(f) == 2
    assert f.read_text(encoding="utf-8").splitlines()[1] == "- call Omar"
```

```
$ uv run pytest -q test_notes.py
..                                                                       [100%]
2 passed in 0.01s
```



35.7 Test first: red, green, tidy

The surest way to write a function: write its test **first**, watch it fail (*red*), write the simplest code that passes (*green*), then improve the code while the test keeps it honest (*tidy*). The test says what you want before the code says how.

test_slug.py:

```
from slug import slugify

def test_spaces_become_dashes():
    assert slugify("Hello World") == "hello-world"

def test_punctuation_goes():
    assert slugify("Rust, Python & AI!") == "rust-python-ai"

def test_no_double_dashes():
    assert slugify("a  --  b") == "a-b"
```

```
$ uv run pytest -q test_slug.py
ERROR test_slug.py
!!!!!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!
1 error in 0.08s
```

Red: there is no `slug` module yet. Now the code:

slug.py:

```
def slugify(title):
    words = "".join(ch if ch.isalnum() else " " for ch in title.lower()).split()
    return "-".join(words)
```

```
$ uv run pytest -q test_slug.py
...                                                                      [100%]
3 passed in 0.01s
```

試 35.8 What to test

Test	Example
the normal case	`total([3, 5])`
the edges	an empty list, zero, one item, the largest allowed
wrong input	text where a number belongs: it should raise
every bug you fix	write the test that shows the bug first; it stays, so the bug never comes back

Keep each test small, named for what it checks (`test_empty_basket_costs_nothing`), and independent of the others.

試 35.9 Common mistakes

1. A test file or function not named `test_…` — pytest silently skips it. Check the count of tests it reports.

2. Comparing floats with `==`. `0.1 + 0.2 == 0.3` is `False` (chapter 25). Use `pytest.approx`:

`test_float.py`:

```
import pytest

def test_float():
    assert 0.1 + 0.2 == pytest.approx(0.3)
```

```
$ uv run pytest -q test_float.py
.                                                                        [100%]
1 passed in 0.01s
```

3. Tests that depend on each other, or on the order they run in, or on today's date. Each test makes what it needs (`tmp_path` helps), and passes alone.

Exercises

35.1 ★ Your first test file. Test `is_even` (chapter 28) with three asserts in one test, and run pytest.

35.2 ★ Parametrize it. Rewrite the test with `@pytest.mark.parametrize` and five cases.

35.3 ★ Expect an error. Test that `int("x")` raises `ValueError`, with `pytest.raises`.

35.4 ★★ Test the Account. Tests for chapter 34's `Account`: a deposit, a withdrawal, and that overdrawing raises with `match="available"`.

35.5 ★★ Files with tmp_path. Test the `wc(path)` function of exercise 31.4 with `tmp_path`: an empty file, and a file of two lines.

35.6 ★★ Find the bug with a test. This function should return the largest number, but has a bug. Write tests until one fails, then fix it: `def biggest(nums): m = 0; for n in nums: m = max(m, n); return m`.

35.7 ★★★ Test first: Roman numerals. Write tests first for `roman(n)`: 1 → I, 4 → IV, 9 → IX, 14 → XIV, 40 → XL, 1994 → MCMXCIV, 2026 → MMXXVI. Then make them pass.

35.8 ★★★ A fixture of your own. Write a pytest fixture `basket` (with `@pytest.fixture`) that returns a `Basket` holding two items, and two tests that use it.

35.9 ★★★★ Tests for the expense tracker. Write a `test_ledger.py` for level 3's checkpoint: `parse` (good line, three bad shapes, parametrized), `totals`, and a save-then-load round trip through `tmp_path`. Aim for every function covered; run `uv add --dev pytest-cov` and `uv run pytest --cov` to see how much of `ledger.py` the tests run.

型 36. Python level 4: type hints

In this chapter

- write down what kind of value each name holds
- let the editor (and a checker) find mistakes before you run anything
- lists, dictionaries, optional values and your own types
- the checkpoint for level 4: a small, typed, tested library

You need: chapters 34 and 35.

型 36.1 What a type hint is

A **type hint** says what kind of value a name should hold: `price: float`, `def total(prices: list[float]) -> float`. Python itself ignores them when it runs. But your editor reads them — neovim's basedpyright underlines a wrong call as you type — and a checker can test a whole project at once. They are also the best documentation: they say exactly what goes in and what comes out.

```
def greet(name: str, times: int = 1) -> str:
    return ", ".join([f"hello {name}"] * times)

age: int = 31
print(greet("Sara", 2))
print(greet.__annotations__)
```

You should see:

```
hello Sara, hello Sara
{'name': <class 'str'>, 'times': <class 'int'>, 'return': <class 'str'>}
```

型 36.2 Checking: the mistake found before it runs

Python runs this without complaint until the bad line, and the result is nonsense:

`typed.py`:

```
def area(width: float, height: float) -> float:
    return width * height

print(area(3, 4))
print(area("3", 4))
```

```
$ uv run typed.py
12
3333
```

(`"3" * 4` is `"3333"` — no error, just a wrong answer)

A type checker finds it without running anything. neovim shows it as you type; on the command line, `mypy` (or `basedpyright`) checks a file or a whole project:

```
$ uvx mypy typed.py
typed.py:5: error: Argument 1 to "area" has incompatible type "str"; expected "float"  [arg-type]
Found 1 error in 1 file (checked 1 source file)
```

型 36.3 The common types

Hint	Means	Example
`int` `float` `str` `bool`	the basic types	`n: int = 3`
`list[int]`	a list of ints	`[1, 2, 3]`
`dict[str, float]`	keys str, values float	`{"tea": 8.5}`
`tuple[int, int]`	exactly two ints	`(3, 4)`
`set[str]`	a set of strings	`{"a"}`
`int | None`	an int, or `None`	`.get()` results
`float | str`	either	
`Callable[[int], int]`	a function taking an int, giving one	`lambda n: n * 2`
`Any`	anything — the checker gives up here	

(`int` is accepted where `float` is asked: every int is a fine float.)

```
from collections.abc import Callable

def apply_twice(f: Callable[[int], int], x: int) -> int:
    return f(f(x))

def find(names: list[str], prefix: str) -> str | None:
    for n in names:
        if n.startswith(prefix):
            return n
    return None

print(apply_twice(lambda n: n * 3, 2))
print(find(["sara", "omar"], "om"), find(["sara"], "x"))
```

You should see:

```
18
omar None
```

型 36.4 Optional values: None handled

`str | None` says *maybe nothing*. The checker then insists you handle the `None` before using the value — the commonest bug in any language, found for free:

`maybe.py`:

```
def find(names: list[str], prefix: str) -> str | None:
    for n in names:
        if n.startswith(prefix):
            return n
    return None

def shout_first(names: list[str]) -> str:
    hit = find(names, "s")
    return hit.upper()
```

```
$ uvx mypy maybe.py
maybe.py:9: error: Item "None" of "str | None" has no attribute "upper"  [union-attr]
Found 1 error in 1 file (checked 1 source file)
```

The fix says what happens when nothing was found:

`maybe.py`:

```
def find(names: list[str], prefix: str) -> str | None:
    for n in names:
        if n.startswith(prefix):
            return n
    return None

def shout_first(names: list[str]) -> str:
    hit = find(names, "s")
    if hit is None:
        return "NOBODY"
    return hit.upper()
```

```
$ uvx mypy maybe.py
Success: no issues found in 1 source file
```

型 36.5 Your own types

Your classes are types too. A dataclass's fields are already typed; a `type` statement names a longer type once:

```
from dataclasses import dataclass
from typing import Literal

type Grid = list[list[int]]
Colour = Literal["red", "green", "blue"]

@dataclass
class Pixel:
    x: int
    y: int
    colour: Colour

def size(g: Grid) -> tuple[int, int]:
    return len(g), len(g[0])

print(size([[0, 1, 2], [3, 4, 5]]))
print(Pixel(1, 2, "red"))
```

You should see:

```
(2, 3)
Pixel(x=1, y=2, colour='red')
```

(`Literal["red", "green", "blue"]` *allows only those three strings: a typo like* `"rde"` *is flagged)*

型 36.6 Generic functions

A function that works for any type, keeping it: `T` stands for "whatever type comes in":

```
def first[T](items: list[T]) -> T:
    return items[0]

def pairs[K, V](d: dict[K, V]) -> list[tuple[K, V]]:
    return sorted(d.items())

n = first([3, 1, 2])
s = first(["b", "a"])
print(n + 1, s.upper())
print(pairs({"b": 2, "a": 1}))
```

You should see:

```
4 B
[('a', 1), ('b', 2)]
```

(the checker knows `n` is an `int` and `s` a `str`. The `[T]` after the name is Python 3.12's short way to declare it)

型 36.7 How much to type

Where	Advice
function parameters and returns	always: it is where mistakes cross from one piece of code to another
class fields	always (dataclasses need them anyway)
local variables	rarely: the checker works them out (`x = 3` is an `int`)
quick scripts at the prompt	not needed

型 36.8 Common mistakes

1. Expecting Python to enforce them. It does not: `area("3", 4)` runs. Hints help only if something checks them — your editor, `mypy`, or `basedpyright`.

2. `list` instead of `list[int]`. A bare `list` says "a list of anything": the checker learns nothing about the items.

3. Forgetting `| None`. If a function can return `None`, say so in its return type; otherwise every caller believes it never does.

型 36.9 Checkpoint project — a tested, typed library

Level 4 in one: a small library with a class, type hints everywhere, tests for every function, and a clean type check. The subject is a **grade book**.

gradebook.py:

```
"""A grade book: students, their marks, and averages."""
from dataclasses import dataclass, field
from statistics import mean

@dataclass
class Student:
    name: str
    marks: dict[str, float] = field(default_factory=dict)

    def average(self) -> float | None:
        return mean(self.marks.values()) if self.marks else None

class GradeBook:
    def __init__(self) -> None:
        self.students: dict[str, Student] = {}

    def add(self, name: str) -> Student:
        if name in self.students:
            raise ValueError(f"{name} is already in the book")
        s = self.students[name] = Student(name)
        return s

    def mark(self, name: str, subject: str, value: float) -> None:
        if not 0 <= value <= 100:
            raise ValueError(f"a mark is 0-100, got {value}")
        self.students[name].marks[subject] = value

    def ranking(self) -> list[tuple[str, float]]:
        scored = [(s.name, a) for s in self.students.values() if (a := s.average()) is not None]
        return sorted(scored, key=lambda pair: pair[1], reverse=True)
```

test_gradebook.py:

```
import pytest
from gradebook import GradeBook

@pytest.fixture
def book() -> GradeBook:
    b = GradeBook()
    for name in ("Sara", "Omar", "Lina"):
        b.add(name)
    b.mark("Sara", "maths", 90); b.mark("Sara", "art", 80)
    b.mark("Omar", "maths", 95)
    return b

def test_average(book: GradeBook) -> None:
    assert book.students["Sara"].average() == 85

def test_no_marks_no_average(book: GradeBook) -> None:
    assert book.students["Lina"].average() is None

def test_ranking_skips_unmarked(book: GradeBook) -> None:
    assert book.ranking() == [("Omar", 95), ("Sara", 85)]

def test_duplicate_name(book: GradeBook) -> None:
    with pytest.raises(ValueError, match="already"):
        book.add("Sara")

@pytest.mark.parametrize("bad", [-1, 101])
def test_mark_range(book: GradeBook, bad: float) -> None:
    with pytest.raises(ValueError, match="0-100"):
        book.mark("Sara", "maths", bad)
```

```
$ uv run pytest -q test_gradebook.py
......                                                                   [100%]
6 passed in 0.01s
$ uvx mypy gradebook.py test_gradebook.py
Success: no issues found in 2 source files
```

(`(a := s.average())` is the walrus *operator: it names a value inside an expression, so the average is worked out once)*

`Checkpoint.` Add `subject_average(subject) -> float | None` and `report() -> str` (a table of every student and their average, — for none). Tests first; then make pytest and mypy both clean. Level 4 done: you can now write code other people can trust.

Exercises

36.1 ★ Hint it. Add type hints to `def pay(hours, rate)` and `def greeting(name, hour)` from chapter 28.

36.2 ★ Read the hints. Without running: what does `def f(xs: list[str]) -> dict[str, int]` take and give? Write a body that fits: each string's length.

36.3 ★ Find the bug by checking. Run `mypy` on a file where `def half(n: int) -> int: return n / 2`. What does it say, and why?

36.4 ★★ Optional. `def price_of(item: str, prices: dict[str, float]) -> float | None` returns the price or `None`; then `def total(...)` that skips unknown items — with a check `if p is not None`.

36.5 ★★ Typed dataclass. A `Book` dataclass (title: str, year: int, tags: list[str]) and a function `by_tag(books: list[Book], tag: str) -> list[str]` returning titles.

36.6 ★★★ A generic helper. `def chunks[T](items: list[T], n: int) -> list[list[T]]` splits a list into pieces of `n`. Test it with ints and with strings.

36.7 ★★★ Literal states. A traffic light: `State = Literal["red", "green", "amber"]` and `def next_state(s: State) -> State`. Show that mypy flags `next_state("blue")`.

36.8 ★★★★ Type the expense tracker. Add full hints to level 3's `ledger.py` — a `Row` as a `TypedDict` (`from typing import TypedDict`: date: str, category: str, amount: float) — and make `mypy --strict ledger.py` clean.

生 37. Python level 5: comprehensions, iterators and generators

In this chapter

- comprehensions: build a list, dictionary or set in one line
- what `for` really does: iterables and iterators
- generators: values made one at a time, as they are asked for
- `itertools` and `functools`: ready-made tools for loops and functions

You need: level 4 (chapters 34–36).

生 37.1 List comprehensions

You have written this shape many times: an empty list, a loop, an `append`. A **list comprehension** says the same in one line:

```
squares = []
for n in range(6):
    squares.append(n * n)
print(squares)

squares = [n * n for n in range(6)]
print(squares)
```

You should see:

```
[0, 1, 4, 9, 16, 25]
[0, 1, 4, 9, 16, 25]
```

Read it as English: *n × n, for each n in range(6)*. Add `if` to keep only some:

```
>>> [n for n in range(20) if n % 3 == 0]
[0, 3, 6, 9, 12, 15, 18]
>>> [w.upper() for w in ["tea", "rice", "milk"] if len(w) > 3]
['RICE', 'MILK']
>>> [c * 9 / 5 + 32 for c in [20, 25, 31]]
[68.0, 77.0, 87.8]
```

An `if … else` *before* the `for` chooses a value for every item (it does not filter):

```
>>> ["even" if n % 2 == 0 else "odd" for n in range(4)]
['even', 'odd', 'even', 'odd']
```

Shape	Does
`[f(x) for x in xs]`	change every item
`[x for x in xs if test(x)]`	keep some items
`[a if test(x) else b for x in xs]`	choose for every item
`[(x, y) for x in xs for y in ys]`	every pair (loops in the same order as nested `for`s)

生 37.2 Dictionary and set comprehensions

The same with `{ }`: `key: value` makes a dictionary, a single value makes a set:

```
>>> words = ["tea", "rice", "milk", "tea"]
>>> {w: len(w) for w in words}
{'tea': 3, 'rice': 4, 'milk': 4}
>>> {len(w) for w in words}
{3, 4}
>>> prices = {"tea": 12.5, "rice": 7.25, "cake": 30}
>>> {k: v for k, v in prices.items() if v < 20}
{'tea': 12.5, 'rice': 7.25}
>>> {v: k for k, v in prices.items()}
{12.5: 'tea', 7.25: 'rice', 30: 'cake'}
```

`Careful.` A comprehension is for *building* a collection. If the line grows past one screen-width, or has two `if`s and two `for`s, write the plain loop: it is easier to read.

生 37.3 What for really does

A `for` loop works on anything **iterable**: lists, strings, dictionaries, files, `range`. It asks the value for an **iterator** with `iter()`, then asks the iterator for one item at a time with `next()`, until the iterator says *no more* (by raising `StopIteration`):

```
>>> it = iter(["a", "b"])
>>> next(it)
'a'
>>> next(it)
'b'
>>> next(it)
Traceback (most recent call last):
  File "<python-input-3>", line 1, in <module>
StopIteration
```

That is all a `for` loop is: `iter` once, then `next` until `StopIteration`, which it catches for you. An iterator is used up once it has given everything:

```
>>> it = iter([1, 2, 3])
>>> list(it)
[1, 2, 3]
>>> list(it)
[]
```

生 37.4 Generators: values on demand

A function with `yield` instead of `return` is a **generator**. Calling it runs nothing yet; it gives an iterator. Each `next` runs the body up to the next `yield`, hands out that value, and *pauses* there until asked again:

```
def count_up(limit):
    n = 1
    while n <= limit:
        print(f"  (making {n})")
        yield n
        n += 1

gen = count_up(3)
print("made the generator; nothing has run")
for value in gen:
    print("got", value)
```

You should see:

```
made the generator; nothing has run
  (making 1)
got 1
  (making 2)
got 2
  (making 3)
got 3
```

Why it matters: a generator holds **one** value at a time, not all of them. It can walk a file of a billion lines, or even go on forever, in almost no memory:

```
import itertools

def fibonacci():
    a, b = 0, 1
    while True:
        yield a
        a, b = b, a + b

print(list(itertools.islice(fibonacci(), 12)))
```

You should see:

```
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89]
```

(`islice(it, 12)` takes the first 12 values and stops asking: the infinite generator is never run to its end)

A pipeline of generators, each passing values to the next — reading a log, keeping the errors, taking their messages — with only one line in memory at a time:

```
from pathlib import Path

Path("app.log").write_text("INFO start\nERROR disk full\nINFO retry\nERROR no network\n")

def lines(path):
    with open(path, encoding="utf-8") as f:
        for line in f:
            yield line.rstrip("\n")

def errors(rows):
    for row in rows:
        if row.startswith("ERROR"):
            yield row

def messages(rows):
    for row in rows:
        yield row.split(" ", 1)[1]

for m in messages(errors(lines("app.log"))):
    print(m)
```

You should see:

```
disk full
no network
```

生 37.5 Generator expressions

A comprehension in `( )` is a generator: nothing is built; values are made as `sum`, `max` or a loop asks for them. Use it when you only need to pass over the values once:

```
import sys

big_list = [n * n for n in range(1_000_000)]
lazy = (n * n for n in range(1_000_000))
print(sum(lazy))
print(sys.getsizeof(big_list) > 1_000_000, sys.getsizeof(lazy) < 1_000)
print(any(n > 5 for n in [1, 7, 3]), all(n > 0 for n in [1, 7, 3]))
```

You should see:

```
333332833333500000
True True
True True
```

(inside a call, the extra brackets can go: `sum(n * n for n in ⋯)`*)*

生 37.6 Tools: zip, enumerate, sorted, itertools

```
import itertools

names = ["Sara", "Omar", "Lina"]
scores = [88, 92, 79]
print(dict(zip(names, scores)))
print(max(zip(scores, names)))
print(list(itertools.pairwise([1, 4, 9, 16])))
print(list(itertools.batched("abcdefg", 3)))
print(list(itertools.combinations("ABC", 2)))
print(list(itertools.accumulate([1, 2, 3, 4])))
groups = itertools.groupby(sorted(["apple", "avocado", "banana"]), key=lambda w: w[0])
print({k: list(v) for k, v in groups})
```

You should see:

```
{'Sara': 88, 'Omar': 92, 'Lina': 79}
(92, 'Omar')
[(1, 4), (4, 9), (9, 16)]
[('a', 'b', 'c'), ('d', 'e', 'f'), ('g',)]
[('A', 'B'), ('A', 'C'), ('B', 'C')]
[1, 3, 6, 10]
{'a': ['apple', 'avocado'], 'b': ['banana']}
```

Tool	Gives
`zip(a, b)`	pairs, side by side
`enumerate(xs)`	(index, item)
`itertools.pairwise(xs)`	each item with the next
`itertools.batched(xs, n)`	pieces of `n`
`itertools.combinations(xs, k)` · `permutations` · `product`	choices, orders, every pairing
`itertools.accumulate(xs)`	running totals
`itertools.groupby(sorted_xs, key)`	runs of items with the same key
`itertools.chain(a, b)`	one after the other

生 37.7 functools: tools for functions

`functools.cache` remembers a function's answers: the second call with the same arguments is instant. It turns chapter 28's slow recursive Fibonacci fast:

```
from functools import cache, reduce, partial
import time

@cache
def fib(n):
    return n if n < 2 else fib(n - 1) + fib(n - 2)

t = time.perf_counter()
print(fib(200))
print("fast:", time.perf_counter() - t < 0.1)

print(reduce(lambda a, b: a * b, [1, 2, 3, 4, 5]))
base2 = partial(int, base=2)
print(base2("1010"))
```

You should see:

```
280571172992510140037611932413038677189525
fast: True
120
10
```

Word — decorator. `@cache` above a `def` is a **decorator**: a function that takes your function and gives back an improved one. `@dataclass` (chapter 34) and `@pytest.fixture` (chapter 35) are decorators too.

A decorator of your own — one that times any function:

```
import time
from functools import wraps

def timed(f):
    @wraps(f)
    def wrapper(*args, **kwargs):
        start = time.perf_counter()
        result = f(*args, **kwargs)
        print(f"{f.__name__} took {time.perf_counter() - start:.1f} s")
        return result
    return wrapper

@timed
def slow_sum(n):
    return sum(range(n))

print(slow_sum(1_000_000))
```

You should see:

```
slow_sum took 0.0 s
499999500000
```

(`*args` and `**kwargs` catch any positional and keyword arguments and pass them on, so the wrapper fits any function)

生 37.8 Common mistakes

1. Using an iterator twice. The second pass finds it empty (see `list(it)` above). If you need the values twice, make a list: `values = list(gen)`.

2. A comprehension for its side effects. `[print(x) for x in xs]` builds a list of `None`s for nothing. Use a plain `for` loop for doing; comprehensions for building.

3. `return` in a generator. It ends the generator; the value is not handed out. Use `yield`.

Exercises

37.1 ★ Squares of evens. With one comprehension, the squares of the even numbers from 1 to 20.

37.2 ★ Lengths. `{word: len(word)}` for the words of a sentence, as a dictionary comprehension.

37.3 ★ Flatten. Turn `[[1, 2], [3], [4, 5, 6]]` into `[1, 2, 3, 4, 5, 6]` with one comprehension (two `for`s).

37.4 ★★ Countdown generator. `countdown(n)` yields n, n−1, … 1. Use it in a `for`, and with `list()`.

37.5 ★★ Matrix transpose. Transpose `[[1, 2, 3], [4, 5, 6]]` with a comprehension — then with `zip(*m)`.

37.6 ★★ Running average. A generator `running_avg(nums)` yielding the average so far after each number. *Expected:* `[2, 4, 6]` → `2.0 3.0 4.0`.

37.7 ★★★ Primes forever. A generator of all prime numbers; print the first 15 with `islice`.

37.8 ★★★ A decorator that counts. `@counted` adds a `.calls` counter to any function.

37.9 ★★★ Words in a big file. Without reading the whole file into memory, count how many lines of a file contain a word — with a generator expression inside `sum`.

37.10 ★★★★ Memoise by hand. Write your own `memo` decorator (a dictionary from arguments to results) and show it makes `fib(35)` fast. Compare with `functools.cache`.

待 38. Python level 5: async basics

In this chapter

- why waiting is the slow part of most programs
- `async def`, `await`, and `asyncio.run`
- many waits at once: `gather`, `TaskGroup`, timeouts
- when async helps, and when it does not

You need: chapter 37.

待 38.1 Waiting, not working

Most programs that feel slow are not computing: they are **waiting** — for a web page, a database, a model's answer. While one request waits, the computer could send ten more. **async** code lets one program have many waits going at once, and switch to whichever is ready.

Here is the slow way: three downloads that each take one second, one after the other. `time.sleep` stands in for the network:

```
import time

def download(name):
    time.sleep(1)
    return f"{name} done"

start = time.perf_counter()
for name in ["a", "b", "c"]:
    print(download(name))
print(f"took {time.perf_counter() - start:.0f} s")
```

You should see:

```
a done
b done
c done
took 3 s
```

待 38.2 async and await

The same with `asyncio`. `async def` makes a **coroutine** — a function that can pause. `await` marks the place where it waits, and lets others run meanwhile. `asyncio.gather` starts several at once and waits for all:

```
import asyncio
import time

async def download(name):
    await asyncio.sleep(1)
    return f"{name} done"

async def main():
    results = await asyncio.gather(download("a"), download("b"), download("c"))
    print(results)

start = time.perf_counter()
asyncio.run(main())
print(f"took {time.perf_counter() - start:.0f} s")
```

You should see:

```
['a done', 'b done', 'c done']
took 1 s
```

Three seconds became one: the three waits overlapped.

Piece	Means
`async def f():`	`f()` makes a coroutine; nothing runs yet
`await x`	run `x`, and let other work go on while it waits
`asyncio.run(main())`	start the event loop and run `main` to its end — once, at the top
`asyncio.gather(a, b, c)`	run them together; their results, in order
`asyncio.sleep(s)`	wait without blocking the others (never `time.sleep` inside async code)

> `Word — event loop.` The part of asyncio that keeps a list of waiting coroutines, and runs whichever is ready. `asyncio.run` makes one, runs your `main`, and closes it.

待 38.3 Seeing the switching

Each coroutine runs until its `await`, then the loop runs another:

```
import asyncio

async def worker(name, delay):
    for i in range(3):
        print(f"{name} step {i}")
        await asyncio.sleep(delay)

async def main():
    await asyncio.gather(worker("fast", 0.1), worker("slow", 0.25))

asyncio.run(main())
```

You should see:

```
fast step 0
slow step 0
fast step 1
fast step 2
slow step 1
slow step 2
```

待 38.4 Tasks and TaskGroup

`asyncio.TaskGroup` starts tasks that run in the background, and waits for them all when its block ends. If one fails, the others are cancelled and the error comes out — no task is left running unnoticed:

```
import asyncio

async def fetch(n):
    await asyncio.sleep(0.1 * n)
    return n * n

async def main():
    async with asyncio.TaskGroup() as tg:
        tasks = [tg.create_task(fetch(n)) for n in range(1, 6)]
    print([t.result() for t in tasks])

asyncio.run(main())
```

You should see:

```
[1, 4, 9, 16, 25]
```

待 38.5 Timeouts

A network call can hang. `asyncio.timeout` gives up after a time you choose:

```
import asyncio

async def slow_answer():
    await asyncio.sleep(5)
    return 42

async def main():
    try:
        async with asyncio.timeout(0.5):
            print(await slow_answer())
    except TimeoutError:
        print("gave up after 0.5 s")

asyncio.run(main())
```

You should see:

```
gave up after 0.5 s
```

待 38.6 Limiting: a semaphore

Starting a thousand requests at once can overload a server (or get you blocked). A **semaphore** lets only N run at a time:

```
import asyncio, time

async def job(i, limit):
    async with limit:
        await asyncio.sleep(0.2)
        return i

async def main():
    limit = asyncio.Semaphore(3)
    start = time.perf_counter()
    results = await asyncio.gather(*(job(i, limit) for i in range(9)))
    print(results, f"{time.perf_counter() - start:.1f} s")

asyncio.run(main())
```

You should see:

```
[0, 1, 2, 3, 4, 5, 6, 7, 8] 0.6 s
```

(9 jobs of 0.2 s, 3 at a time: three rounds, about 0.6 s)

待 38.7 Real use: many web requests

The real network needs an async library; the standard one is `httpx` (`uv add httpx`). The shape is always the same — one client, many `await`ed requests gathered together:

```
import asyncio
import httpx

async def main():
    urls = [f"https://httpbin.org/delay/1?n={i}" for i in range(5)]
    async with httpx.AsyncClient(timeout=10) as client:
        replies = await asyncio.gather(*(client.get(u) for u in urls))
    print([r.status_code for r in replies])

asyncio.run(main())
```

(not run here: it needs the network. Five one-second requests finish in about one second. The Claude SDK has the same: `anthropic.AsyncAnthropic`*, chapter 47)*

待 38.8 When async helps

Your program is slow because it	Use
waits for the network, disk, a database, an API	**async** (or threads)
computes: maths, images, training	not async: NumPy, multiprocessing, or Rust (Part V)

async does not make computing faster: there is still only one thing running at a time. It makes *waiting* overlap.

侍 38.9 Common mistakes

1. Forgetting `await`. You get a coroutine object, not its result, and a warning:

```
import asyncio

async def answer():
    return 42

async def main():
    x = answer()
    print(x)
    print(await answer())

asyncio.run(main())
```

You should see:

```
<coroutine object answer at 0x7f4288b99d80>
42
/root/.local/share/uv/python/cpython-3.14.6-linux-x86_64-gnu/lib/python3.14/asyncio/events.py:94:
RuntimeWarning: coroutine 'answer' was never awaited
  self._context.run(self._callback, *self._args)
RuntimeWarning: Enable tracemalloc to get the object allocation traceback
```

(Python also warns, on the error stream, that the first coroutine "was never awaited")

2. `time.sleep` (or any slow normal call) inside async code. It stops the whole event loop: nothing else runs. Use `await asyncio.sleep`, and async libraries for I/O.

3. Calling `asyncio.run` inside async code. There is one `asyncio.run`, at the top; inside, always `await`.

Exercises

38.1 ★ Hello, later. An async `greet(name, delay)` that waits then prints; run three with `gather`, delays 0.3, 0.1, 0.2. In which order do they print?

38.2 ★ Results in order. Show that `gather` returns results in the order you passed them, not the order they finish.

38.3 ★★ Time it. Ten coroutines that each sleep 0.5 s: time them one after another (`await` in a loop), then with `gather`.

38.4 ★★ Timeout per job. Five jobs with delays 0.1 to 0.5 s; give each a 0.35 s timeout and print which finished.

38.5 ★★★ A producer and a consumer. With `asyncio.Queue`: a producer puts 5 numbers (one every 0.1 s), a consumer takes and prints their squares; stop with a `None` sentinel.

38.6 ★★★ A failure in a group. In a `TaskGroup`, one task raises `ValueError`. Catch it with `except* ValueError` and show the other tasks were cancelled.

38.7 ★★★★ A polite crawler, simulated. Simulate fetching 20 pages (`await asyncio.sleep(random.uniform(0.05, 0.2))`), at most 4 at a time (a semaphore), each with a 0.15 s timeout, and report how many succeeded, how many timed out, and the total time.

令 39. Python level 5: command-line tools and packaging

In this chapter

- a real command-line tool with `argparse`: options, help, exit codes
- reading from a pipe, writing to another
- a package layout with `src/`, an entry point, and `uv tool install`
- the checkpoint for level 5: a command of your own, installed

You need: chapters 33, 35 and 37.

令 39.1 A program that takes arguments

A command-line program gets its words from the command line: `sys.argv` is the list of them, the program's own name first:

`args.py`:

```
import sys
print(sys.argv)
```

```
$ uv run args.py hello 42 --loud
['args.py', 'hello', '42', '--loud']
```

Parsing that list by hand gets messy fast. `argparse` does it, and writes `--help` for you.

令 39.2 argparse

`greet.py`:

```
"""greet — say hello, a number of times."""
import argparse

def main() -> None:
    parser = argparse.ArgumentParser(description="Say hello.")
    parser.add_argument("name", help="who to greet")
    parser.add_argument("-n", "--times", type=int, default=1, help="how many times (default 1)")
    parser.add_argument("--loud", action="store_true", help="in capitals")
    args = parser.parse_args()
    text = f"hello, {args.name}"
    for _ in range(args.times):
        print(text.upper() if args.loud else text)

if __name__ == "__main__":
    main()
```

```
$ uv run greet.py Sara
hello, Sara
$ uv run greet.py Sara -n 2 --loud
HELLO, SARA
HELLO, SARA
$ uv run greet.py --help
usage: greet.py [-h] [-n TIMES] [--loud] name

Say hello.

positional arguments:
  name                who to greet

options:
  -h, --help          show this help message and exit
  -n, --times TIMES   how many times (default 1)
  --loud              in capitals
$ uv run greet.py
usage: greet.py [-h] [-n TIMES] [--loud] name
greet.py: error: the following arguments are required: name
```

add_argument(…)	Makes
`"name"`	a positional argument: required, in order
`"-n", "--times", type=int`	an option with a value, changed to an int
`default=1`	its value when left out
`action="store_true"`	a flag: `True` when given
`choices=["a", "b"]`	only these values
`nargs="+"` · `nargs="*"`	one or more · any number of values, as a list



39.3 Exit codes and the error stream

A program tells the shell how it went with its **exit code**: `0` for success, anything else for failure. Scripts and `&&` rely on it. Errors go to the **error stream** (`stderr`), so they are not mixed into output piped to the next program:

`check_age.py`:

```
import sys

def main() -> int:
    if len(sys.argv) != 2 or not sys.argv[1].isdigit():
        print("usage: check_age.py AGE", file=sys.stderr)
        return 2
    age = int(sys.argv[1])
    print("adult" if age >= 18 else "minor")
    return 0

if __name__ == "__main__":
    sys.exit(main())
```

```
$ uv run check_age.py 20; echo "exit code $?"
adult
exit code 0
$ uv run check_age.py old; echo "exit code $?"
usage: check_age.py AGE
exit code 2
```

令 39.4 Pipes: stdin and stdout

A good tool reads from a pipe when given no file, so it fits between others (`cat file | yourtool | sort`). `sys.stdin` is the pipe coming in, line by line:

`upper.py`:

```
import sys

for line in sys.stdin:
    sys.stdout.write(line.upper())
```

```
$ printf 'tea\nrice\n' | uv run upper.py | sort
RICE
TEA
```

`argparse.FileType`, or the convention `-` for "the pipe", lets one tool do both:

`count.py`:

```
import argparse, sys

p = argparse.ArgumentParser()
p.add_argument("files", nargs="*", default=["-"])
args = p.parse_args()
for name in args.files:
    f = sys.stdin if name == "-" else open(name, encoding="utf-8")
    print(f"{sum(1 for _ in f):>6} {name}")
```

```
$ printf 'a\nb\nc\n' > three.txt
$ uv run count.py three.txt
     3 three.txt
$ cat three.txt three.txt | uv run count.py
     6 -
```

令 39.5 A package: src layout and an entry point

A script runs with `uv run file.py`. A **package** can be installed as a command you type anywhere, like `rg` or `bat`. uv makes the layout:

```
$ uv init --package wordfreq
Initialized project `wordfreq` at `/home/you/repos/wordfreq`
$ find wordfreq -type f | sort
wordfreq/.python-version
wordfreq/README.md
wordfreq/pyproject.toml
wordfreq/src/wordfreq/__init__.py
```

File	Holds
`src/wordfreq/__init__.py`	the package; `main()` is the command's start
`pyproject.toml`	name, version, dependencies — and `[project.scripts]`: the command's name

`wordfreq/src/wordfreq/__init__.py`:

```
"""wordfreq — the most common words of a text."""
import argparse
import re
import sys
from collections import Counter

def top_words(text: str, n: int) -> list[tuple[str, int]]:
    words = re.findall(r"[a-z']+", text.lower())
    return Counter(words).most_common(n)

def main() -> None:
    p = argparse.ArgumentParser(prog="wordfreq", description=__doc__)
    p.add_argument("file", nargs="?", help="a text file (default: the pipe)")
    p.add_argument("-n", type=int, default=5, help="how many words")
    args = p.parse_args()
    text = open(args.file, encoding="utf-8").read() if args.file else sys.stdin.read()
    for word, count in top_words(text, args.n):
        print(f"{count:>5}  {word}")
```

```
$ grep -A1 'project.scripts' wordfreq/pyproject.toml
[project.scripts]
wordfreq = "wordfreq:main"
$ cd wordfreq && echo "the cat and the hat and the bat" | uv run -q wordfreq -n 2
    3  the
    2  and
```

(`uv run wordfreq` runs the command inside the project. `[project.scripts]` says the command `wordfreq` calls `wordfreq:main`)

令 39.6 Install it as your own command

```
$ uv tool install --editable .
Installed 1 executable: wordfreq
$ cd ~ && wordfreq -n 3 ~/Notes/notes.md
```

`uv tool install` puts the command in `~/.local/bin`, which is on your `PATH` — it now works from any folder, in its own venv. `--editable` means a change to the code works at once, without installing again. `uv tool list` shows your tools; `uv tool uninstall wordfreq` removes it.

Sharing it. `uv build` makes a *wheel* (`dist/wordfreq-0.1.0-py3-none-any.whl`): one file anyone can install with `uv tool install wordfreq-0.1.0-py3-none-any.whl`. Pushed to GitHub, anyone can install it straight from there: `uv tool install git+https://github.com/you/wordfreq`.

```
$ cd wordfreq && uv build -q && ls dist
wordfreq-0.1.0-py3-none-any.whl
wordfreq-0.1.0.tar.gz
```

令 39.7 Common mistakes

1. Printing errors to stdout. They end up in the next program's input. `print(…, file=sys.stderr)`.

2. Exiting with 0 after a failure. `sys.exit(1)` (or any non-zero) after an error, so scripts and `&&` know.

3. Code at the top level of the package (outside functions): it runs on every import, even for `--help`. Keep everything in functions, started from `main()`.

令 39.8 Checkpoint project — notes, from the command line

Level 5 in one tool: `note add TEXT`, `note list [--grep WORD]`, `note today`; notes in a JSON lines file (one JSON object per line), read with a generator; tested with pytest; installed with `uv tool install`. The core, without the packaging:

`note.py`:

```
"""note — tiny notes from the command line."""
import argparse
import json
import sys
from collections.abc import Iterator
from datetime import date
from pathlib import Path

STORE = Path("notes.jsonl")

def read(path: Path) -> Iterator[dict]:
    if path.exists():
        with open(path, encoding="utf-8") as f:
            for line in f:
                yield json.loads(line)

def add(path: Path, text: str, day: str) -> None:
    with open(path, "a", encoding="utf-8") as f:
        f.write(json.dumps({"day": day, "text": text}) + "\n")

def main(argv: list[str] | None = None) -> int:
    p = argparse.ArgumentParser(prog="note")
    sub = p.add_subparsers(dest="cmd", required=True)
    a = sub.add_parser("add"); a.add_argument("text", nargs="+")
    ls = sub.add_parser("list"); ls.add_argument("--grep")
    sub.add_parser("today")
    args = p.parse_args(argv)
    today = "2026-10-21"  # date.today().isoformat() in your version
    if args.cmd == "add":
        add(STORE, " ".join(args.text), today)
        return 0
    notes = read(STORE)
    if args.cmd == "today":
        notes = (n for n in notes if n["day"] == today)
    elif args.grep:
        notes = (n for n in notes if args.grep.lower() in n["text"].lower())
    found = 0
    for found, n in enumerate(notes, 1):
        print(f"{n['day']}  {n['text']}")
    if not found:
        print("no notes", file=sys.stderr)
        return 1
    return 0

if __name__ == "__main__":
    sys.exit(main())
```

```
$ uv run note.py add buy rice
$ uv run note.py add call Omar about Rust
$ uv run note.py list
2026-10-21  buy rice
2026-10-21  call Omar about Rust
$ uv run note.py list --grep rust
2026-10-21  call Omar about Rust
$ uv run note.py list --grep tea; echo "exit $?"
no notes
exit 1
```

(`main(argv)` takes the arguments as a list, so a test can call `main(["add", "x"])` directly — with `STORE` pointed into `tmp_path` by `monkeypatch.setattr(note, "STORE", tmp_path / "n.jsonl")`)

`Checkpoint.` Turn it into a package (`uv init --package note`), add `done N` and `--since DATE`, write pytest tests through `main([...])`, and `uv tool install --editable .`. When `note list` works from your home folder, level 5 is done.

Exercises

39.1 ★ Add an option. Give `greet.py` a `--lang` option with `choices=["en", "ar"]` that greets with `hello` or `marhaba`.

39.2 ★ Exit codes. A script `isnum.py X` that exits 0 if X is a number and 1 if not; try it with `&& echo yes`.

39.3 ★★ Sum from a pipe. `sumcol.py` reads numbers from stdin, one per line, and prints their sum, skipping blank lines. `seq 1 100 | uv run sumcol.py` should print `5050`.

39.4 ★★ Many files. Give `count.py` a `--words` flag that counts words instead of lines, and a total line when given two or more files.

39.5 ★★★ Subcommands. A `calc` tool with subcommands `add A B`, `mul A B`, `sqrt X` (`add_subparsers`).

39.6 ★★★ Test the CLI. Test `note.main` with pytest: `add` then `list --grep`, using `tmp_path`, `monkeypatch` and `capsys` (pytest's fixture that captures printed output: `capsys.readouterr().out`).

39.7 ★★★★ wordfreq, finished. Finish `wordfreq`: a `--ignore FILE` of words to skip, `--min-length`, output as JSON with `--json`, tests for `top_words`, `uv tool install --editable .`, and run it on this book's source (`docs/fedokawaii-learn.md` in the toolkit).

統 40. Python level 6: data with the standard library

In this chapter

- load, clean, group and summarise a table with nothing but Python itself
- `statistics`, `collections`, `datetime`: the analyst's standard tools
- a text chart in the terminal

You need: level 5 (chapters 37–39).

Level 6 turns everything you know toward data and, from chapter 42, machine learning. Before any library, do the whole job once by hand: it is the only way to understand what pandas or NumPy then do for you, and many small jobs need nothing more.

統 40.1 The data

A small CSV of sales — the kind of file a spreadsheet exports. Note the problems real data has: a blank amount, a typo'd city, an extra space.

`sales.csv`:

```
date,city,product,amount
2026-10-01,Sharjah,tea,12.5
2026-10-01,Dubai,rice,7.25
2026-10-02,Sharjah,rice,14.5
2026-10-02,dubai,tea,25
2026-10-03,Sharjah,tea,
2026-10-03,Abu Dhabi,milk,6
2026-10-04, Dubai,tea,12.5
2026-10-04,Sharjah,milk,9
2026-10-05,Abu Dhabi,tea,37.5
2026-10-05,Dubai,rice,21.75
```

統 40.2 Load and clean

Clean as you load: trim spaces, fix the case, change types, and set aside rows that cannot be used — counting them, never silently dropping them:

`load.py`:

```
import csv
from datetime import date

def load(path):
    good, bad = [], []
    with open(path, encoding="utf-8", newline="") as f:
        for row in csv.DictReader(f):
            try:
                good.append({
                    "date": date.fromisoformat(row["date"].strip()),
                    "city": row["city"].strip().title(),
                    "product": row["product"].strip().lower(),
                    "amount": float(row["amount"]),
                })
            except ValueError:
                bad.append(row)
    return good, bad

if __name__ == "__main__":
    rows, bad = load("sales.csv")
    print(len(rows), "rows,", len(bad), "set aside:", bad)
    print(rows[0])
```

```
$ uv run load.py
9 rows, 1 set aside: [{'date': '2026-10-03', 'city': 'Sharjah', 'product': 'tea', 'amount': ''}]
{'date': datetime.date(2026, 10, 1), 'city': 'Sharjah', 'product': 'tea', 'amount': 12.5}
```



40.3 Summaries: statistics

```
import statistics as st
from load import load

rows, _ = load("sales.csv")
amounts = [r["amount"] for r in rows]
print("count ", len(amounts))
print("total ", sum(amounts))
print("mean  ", round(st.mean(amounts), 2))
print("median", st.median(amounts))
print("stdev ", round(st.stdev(amounts), 2))
print("min/max", min(amounts), max(amounts))
print("quartiles", st.quantiles(amounts, n=4))
```

You should see:

```
count  9
total  146.0
mean   16.22
median 12.5
stdev  10.17
min/max 6.0 37.5
quartiles [8.125, 12.5, 23.375]
```

`Word — mean, median.` The *mean* is the sum divided by the count; the *median* the middle value once sorted. One huge sale pulls the mean up but barely moves the median — compare both.

統 40.4 Group by

The analyst's most common question — *the total per city* — is chapter 30's grouping, with a `defaultdict`:

```
from collections import defaultdict
from load import load

rows, _ = load("sales.csv")
by_city = defaultdict(float)
by_product = defaultdict(list)
for r in rows:
    by_city[r["city"]] += r["amount"]
    by_product[r["product"]].append(r["amount"])

for city, total in sorted(by_city.items(), key=lambda kv: -kv[1]):
    print(f"{city:<10}{total:>8.2f}")
print({p: len(v) for p, v in by_product.items()})
```

You should see:

```
Dubai       66.50
Abu Dhabi   43.50
Sharjah     36.00
{'tea': 4, 'rice': 3, 'milk': 2}
```

Two keys at once — a tuple as the key gives a pivot table:

```
from collections import defaultdict
from load import load

rows, _ = load("sales.csv")
pivot = defaultdict(float)
for r in rows:
    pivot[(r["city"], r["product"])] += r["amount"]
cities = sorted({c for c, _ in pivot})
products = sorted({p for _, p in pivot})
print(f"{'':<10}" + "".join(f"{p:>8}" for p in products))
for c in cities:
    print(f"{c:<10}" + "".join(f"{pivot.get((c, p), 0):>8.2f}" for p in products))
```

You should see:

```
            milk    rice     tea
Abu Dhabi   6.00    0.00   37.50
Dubai       0.00   29.00   37.50
Sharjah     9.00   14.50   12.50
```

統 40.5 Dates

`datetime` reads, compares and steps dates; group by week or weekday with its methods:

```
from collections import Counter
from datetime import date, timedelta
from load import load

rows, _ = load("sales.csv")
print(rows[0]["date"].strftime("%A %d %B %Y"))
print(Counter(r["date"].strftime("%a") for r in rows))
first = min(r["date"] for r in rows)
last = max(r["date"] for r in rows)
print((last - first).days + 1, "days;", "a week later:", first + timedelta(weeks=1))
```

You should see:

```
Thursday 01 October 2026
Counter({'Thu': 2, 'Fri': 2, 'Sun': 2, 'Mon': 2, 'Sat': 1})
5 days; a week later: 2026-10-08
```

統 40.6 A chart in the terminal

No library needed for a quick look: a bar of ■ per value, scaled to the largest:

```
from collections import defaultdict
from load import load

rows, _ = load("sales.csv")
per_day = defaultdict(float)
for r in rows:
    per_day[r["date"]] += r["amount"]
top = max(per_day.values())
for day, total in sorted(per_day.items()):
    bar = "■" * round(total / top * 30)
    print(f"{day:%m-%d} {bar} {total:.2f}")
```

You should see:

```
10-01 ■■■■■■■■■■■■■■■ 19.75
10-02 ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 39.50
10-03 ■■■■■ 6.00
10-04 ■■■■■■■■■■■■■■■■ 21.50
10-05 ■■■■■■■■■■■■■■■■■■■
```

1. Forgetting to clean keys. `"Dubai"`, `"dubai"` and `" Dubai"` are three different keys until you `strip().title()` them — three cities in your report instead of one.

2. Dropping bad rows silently. Count and show them: a report that quietly ignored half the file is worse than one that crashed.

3. Summing floats for money and printing all the digits. Round when you *show* (`:.2f`); or count in fils as ints.

Exercises

40.1 ★ The biggest sale. Print the row with the largest amount (`max` with `key=`).

40.2 ★ Sharjah only. Total the Sharjah sales with a generator expression in `sum`.

40.3 ★★ Average per product. The mean amount per product, rounded to 2 places.

40.4 ★★ Share of total. Each city's share of all sales, as a percentage with one decimal; they should add to 100.

40.5 ★★★ Outliers. Flag amounts more than 1.5 × the interquartile range above the third quartile (`statistics.quantiles`).

40.6 ★★★ Day-on-day change. For each day, the total and its change from the day before (`itertools.pairwise`).

40.7 ★★★★ Your own bank statement. Export a statement from your bank as CSV (or make one), and write `spend.py` that loads and cleans it, sets aside bad rows, totals spending per category and per month, prints a terminal bar chart of the months, and writes `summary.json`. Test `load` and the grouping with pytest.

41. Python level 6: NumPy

In this chapter

- arrays: many numbers, one operation, no loop
- shapes, indexing, slicing, and masks
- broadcasting: arrays of different shapes working together
- the maths of machine learning — sums, means, dot products, matrix products

You need: chapter 40. In a project: `uv add numpy`.

Python lists are flexible, but slow for numbers: each item is a full Python object, and every operation is a Python loop. A **NumPy array** stores numbers packed side by side, all of one type, and does each operation in compiled C — often a hundred times faster. Every machine learning library (PyTorch included) is built on this idea, with the same vocabulary.

41.1 Arrays

```
>>> import numpy as np
>>> a = np.array([1, 2, 3, 4])
>>> a
array([1, 2, 3, 4])
>>> a * 10
array([10, 20, 30, 40])
>>> a + a
array([2, 4, 6, 8])
>>> a ** 2
array([ 1,  4,  9, 16])
>>> a.sum(), a.mean(), a.max()
(np.int64(10), np.float64(2.5), np.int64(4))
```

No loop: `a * 10` multiplies every element. Compare a list, where `*` repeats it:

```
>>> import numpy as np
>>> [1, 2, 3] * 2
[1, 2, 3, 1, 2, 3]
>>> np.array([1, 2, 3]) * 2
array([2, 4, 6])
```

Making arrays:

```
>>> import numpy as np
>>> np.zeros(3)
array([0., 0., 0.])
>>> np.ones((2, 3))
array([[1., 1., 1.],
       [1., 1., 1.]])
>>> np.arange(0, 10, 2)
array([0, 2, 4, 6, 8])
>>> np.linspace(0, 1, 5)
array([0.  , 0.25, 0.5 , 0.75, 1.  ])
>>> np.eye(3)
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

猿 41.2 Shape and dtype

Every array has a **shape** (its size along each axis) and a **dtype** (the type of every element):

```
>>> import numpy as np
>>> m = np.array([[1, 2, 3], [4, 5, 6]])
>>> m.shape
(2, 3)
>>> m.ndim, m.size, m.dtype
(2, 6, dtype('int64'))
>>> m.reshape(3, 2)
array([[1, 2],
       [3, 4],
       [5, 6]])
>>> m.T
array([[1, 4],
       [2, 5],
       [3, 6]])
>>> np.arange(12).reshape(3, 4)
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11]])
```

Term	m above
shape	(2, 3): 2 rows, 3 columns
ndim	2 axes
size	6 elements
dtype	int64 (whole numbers); float32/float64 for decimals
T	the transpose: rows become columns

Word — axis. One direction of an array. In a table, axis 0 runs down the rows, axis 1 across the columns. m.sum(axis=0) sums *down* (one total per column).

```
>>> import numpy as np
>>> m = np.array([[1, 2, 3], [4, 5, 6]])
>>> m.sum()
np.int64(21)
>>> m.sum(axis=0)
array([5, 7, 9])
>>> m.sum(axis=1)
array([ 6, 15])
>>> m.mean(axis=0)
array([2.5, 3.5, 4.5])
```

猿 41.3 Indexing and slicing

Like lists, but with one index per axis, separated by commas:

```
>>> import numpy as np
>>> m = np.arange(12).reshape(3, 4)
>>> m
array([[ 0,  1,  2,  3],
       [ 4,  5,  6,  7],
       [ 8,  9, 10, 11]])
>>> m[1, 2]
np.int64(6)
>>> m[0]
array([0, 1, 2, 3])
>>> m[:, 1]
array([1, 5, 9])
>>> m[1:, 2:]
array([[ 6,  7],
       [10, 11]])
>>> m[::2, ::-1]
array([[ 3,  2,  1,  0],
       [11, 10,  9,  8]])
```

`Careful.` A slice of an array is a **view**: the same numbers, seen through another window. Changing it changes the original. `.copy()` when you need your own.

```
>>> import numpy as np
>>> a = np.arange(5)
>>> v = a[1:3]
>>> v[0] = 99
>>> a
array([ 0, 99,  2,  3,  4])
```

猿 41.4 Masks: choosing by condition

A comparison gives an array of `True`/`False` — a **mask** — and a mask chooses elements:

```
>>> import numpy as np
>>> t = np.array([28, 35, 41, 22, 38])
>>> t > 30
array([False,  True,  True, False,  True])
>>> t[t > 30]
array([35, 41, 38])
>>> (t > 30).sum()
np.int64(3)
>>> np.where(t > 30, "hot", "ok")
array(['ok', 'hot', 'hot', 'ok', 'hot'], dtype='<U3')
>>> t[(t > 25) & (t < 40)]
array([28, 35, 38])
```

(combine masks with `&` *and* `|`*, in brackets — not* `and`/`or`*)*

猜 41.5 Broadcasting

When two arrays of different shapes meet, NumPy stretches the smaller one along the missing axis — without copying it. This is **broadcasting**:

```
>>> import numpy as np
>>> m = np.array([[1, 2, 3], [4, 5, 6]])
>>> m + 10
array([[11, 12, 13],
       [14, 15, 16]])
>>> m + np.array([100, 200, 300])
array([[101, 202, 303],
       [104, 205, 306]])
>>> m - m.mean(axis=0)
array([[-1.5, -1.5, -1.5],
       [ 1.5,  1.5,  1.5]])
>>> m / m.sum(axis=1, keepdims=True)
array([[0.16666667, 0.33333333, 0.5       ],
       [0.26666667, 0.33333333, 0.4       ]])
```

The rule: compare shapes from the right; each pair must be equal, or one of them 1. `(2, 3)` and `(3,)` fit (the row is repeated down); `(2, 3)` and `(2, 1)` fit (the column across); `(2, 3)` and `(2,)` do not:

```
>>> import numpy as np
>>> m = np.ones((2, 3))
>>> m + np.array([1, 2])
Traceback (most recent call last):
  File "<python-input-2>", line 1, in <module>
ValueError: operands could not be broadcast together with shapes (2,3) (2,)
```

猜 41.6 Dot products and matrices

Machine learning is mostly one operation: multiply and add — the **dot product**, and its big brother the **matrix product** `@`:

```
>>> import numpy as np
>>> w = np.array([0.5, -1.0, 2.0])
>>> x = np.array([4.0, 1.0, 0.5])
>>> w @ x
np.float64(2.0)
>>> (w * x).sum()
np.float64(2.0)
>>> X = np.array([[1, 2], [3, 4], [5, 6]])
>>> W = np.array([[1, 0, 1], [0, 1, 1]])
>>> (X @ W).shape
(3, 3)
>>> X @ W
array([[ 1,  2,  3],
       [ 3,  4,  7],
       [ 5,  6, 11]])
```

A matrix product `(3, 2) @ (2, 3)` gives `(3, 3)`: the inner sizes must match, and each element is a row of the left times a column of the right. A neural network layer is exactly `inputs @ weights + bias` (chapter 43).

猿 41.7 Random numbers, and speed

```
import time
import numpy as np

rng = np.random.default_rng(0)
print(rng.integers(1, 7, size=10))
x = rng.normal(loc=170, scale=10, size=100_000)
print(round(x.mean(), 1), round(x.std(), 1))

nums = list(range(1_000_000))
arr = np.arange(1_000_000)
t = time.perf_counter(); s1 = sum(n * n for n in nums); py = time.perf_counter() - t
t = time.perf_counter(); s2 = int((arr * arr).sum()); npt = time.perf_counter() - t
print(s1 == s2, f"NumPy faster: {py / npt > 10}")
```

You should see:

```
[6 4 4 2 2 1 1 1 2 5]
170.0 10.0
True NumPy faster: False
```

(`default_rng(0)` — a seed — gives the same "random" numbers each run, so results can be repeated; ML code seeds everything)

猿 41.8 Common mistakes

1. Python loops over arrays. `for i in range(len(a)): a[i] *= 2` works, but throws away the speed. Write `a *= 2`.

2. Shape mismatches. Print `.shape` of everything when an error mentions shapes. Most bugs in ML code are a wrong shape.

3. Integer arrays where you wanted floats. `np.array([1, 2]) / 2` is fine (it gives floats), but assigning `0.5` *into* an int array cuts it to 0:

```
>>> import numpy as np
>>> a = np.array([1, 2, 3])
>>> a[0] = 0.5
>>> a
array([0, 2, 3])
>>> np.array([1, 2, 3], dtype=float)
array([1., 2., 3.])
```

Exercises

41.1 ★ First array. Make `np.arange(1, 11)` and print its sum, mean, and the squares of its even numbers.

41.2 ★ Celsius, all at once. Convert `[20, 25, 31, 40]` to Fahrenheit with one array expression.

41.3 ★ Shapes. Make a 4 × 5 array of 0…19; print its last column, its second row, and its shape after `.T`.

41.4 ★★ Normalise. Scale each column of a 5 × 3 random array (seed 0) to mean 0 and standard deviation 1; check with `.mean(axis=0)` and `.std(axis=0)`.

41.5 ★★ Distances. Given 5 points as a (5, 2) array, the distance of each from (0, 0) without a loop (`np.sqrt((p ** 2).sum(axis=1))`), and the nearest.

41.6 ★★ Dice statistics. Roll two dice 100 000 times (`rng.integers(1, 7, size=(100_000, 2))`), sum each pair, and count how often each total 2–12 comes (`np.bincount`). Which is most common?

41.7 ★★★ Matrix product by hand. Write `matmul(A, B)` with three Python loops, and check it equals `A @ B` for random (3, 4) and (4, 2) matrices (`np.allclose`).

41.8 ★★★ Least squares. Fit a line `y = a·x + b` to noisy points (`x = np.linspace(0, 10, 50)`, `y = 3x + 2 + noise`) with `np.polyfit(x, y, 1)`; then with the normal equations `np.linalg.solve(X.T @ X, X.T @ y)`.

41.9 ★★★★ Game of Life, vectorised. Write one step of Conway's Game of Life on a (n, n) array of 0/1 without Python loops: count neighbours by adding the eight shifted copies (`np.roll`), then apply the rules with masks. Check that a *blinker* (three in a row) flips between horizontal and vertical.

42. Small fun projects

In this chapter

- four programs you will actually use, each with tests that run in a blink

You need: levels 1–5 (chapters 25–39).

Each project is one file and one test file, in a project of its own (`uv init`, `uv add --dev pytest`). Watch the test file (`Space` `t` `w`) before you write the first line.

42.1 Your writing, as a chart

How much did you write in your notes each day? The notes file has a heading per day (`## 2026-10-16 · Fri`): count the lines under each.

```
"""notestats — how much you wrote in your notes, day by day"""
import re
import sys
from collections import Counter
from pathlib import Path

HEADING = re.compile(r"^## (\d{4}-\d{2}-\d{2})")

def lines_per_day(text: str) -> Counter:
    """count the non-empty lines under each day's heading"""
    counts = Counter()
    day = None
    for line in text.splitlines():
        if m := HEADING.match(line):
            day = m.group(1)
        elif day and line.strip():
            counts[day] += 1
    return counts

def bar(n: int, most: int, width: int = 30) -> str:
    return "█" * max(1, round(n / most * width))

def main() -> None:
    path = Path(sys.argv[1]) if len(sys.argv) > 1 else Path.home() / "Notes" / "notes.md"
    counts = lines_per_day(path.read_text(encoding="utf-8"))
    if not counts:
        print("no dated notes yet")
        return
    most = max(counts.values())
    for day, n in sorted(counts.items())[-14:]:
        print(f"{day}  {n:3}  {bar(n, most)}")

if __name__ == "__main__":
    main()
```

```
from notestats import bar, lines_per_day

NOTES = """## 2026-10-15 · Thu

buy rice
call the bank

## 2026-10-16 · Fri

- [ ] chapter 37
"""

def test_counts_lines_under_each_day():
    assert lines_per_day(NOTES) == {"2026-10-15": 2, "2026-10-16": 1}

def test_lines_before_any_heading_are_ignored():
    assert lines_per_day("hello\n## 2026-10-15 · Thu\nx\n") == {"2026-10-15": 1}

def test_bar_is_never_empty():
    assert bar(1, 100) == "█"
```

```
$ uv run notestats.py
2026-10-15    2  ████████████████████████████████████████
2026-10-16    1  ████████████████████
```

(for example) New here: a **regular expression** (`re`) finds the headings; `Counter` is a dictionary that counts; `:=` gives a name to a value inside a condition.

Extend it: the weekday you write most (`datetime.date.fromisoformat(day).strftime("%A")`).

楽 42.2 Flashcards

Learning Japanese or Arabic words? A file of `front = back` lines becomes a quiz.

```
"""flash — flashcards from a file of "front = back" lines"""
import random
import sys
from pathlib import Path

def load(text: str) -> list[tuple[str, str]]:
    cards = []
    for line in text.splitlines():
        if "=" in line and not line.startswith("#"):
            front, back = line.split("=", 1)
            cards.append((front.strip(), back.strip()))
    return cards

def check(answer: str, back: str) -> bool:
    return answer.strip().lower() == back.lower()

def main() -> None:
    cards = load(Path(sys.argv[1]).read_text(encoding="utf-8"))
    random.shuffle(cards)
    right = 0
    for front, back in cards:
        if check(input(f"{front} → "), back):
            right += 1
            print("  ✓")
        else:
            print(f"  × {back}")
    print(f"{right} of {len(cards)}")

if __name__ == "__main__":
    main()
```

```
$ uv run flash.py ~/Notes/japanese.txt
water → mizu
  ✓
book → hono
  × hon
1 of 2
```

Test it: `load` and `check` take strings, so their tests need no file and no keyboard. **Extend it:** keep the cards you missed in a second file and quiz those first next time — the idea behind spaced repetition.

楽 42.3 The weather, from a web API

```
"""sky — the next hours' temperature, from Open-Meteo (no key needed)"""
import json
import urllib.request

URL = ("https://api.open-meteo.com/v1/forecast?latitude={lat}&longitude={lon}"
       "&hourly=temperature_2m&forecast_hours=12&timezone=auto")

def hours(data: dict) -> list[tuple[str, float]]:
    h = data["hourly"]
    return [(t[-5:], c) for t, c in zip(h["time"], h["temperature_2m"])]

def main() -> None:
    with urllib.request.urlopen(URL.format(lat=25.35, lon=55.39), timeout=10) as r:
        data = json.load(r)
    for time, celsius in hours(data):
        print(f"{time}  {celsius:5.1f}°  {'▇' * round(celsius / 2)}")

if __name__ == "__main__":
    main()
```

(25.35, 55.39 is Sharjah: put your own place. The test feeds `hours` *a small dictionary shaped like the answer — no network in tests)*

楽 42.4 The Game of Life

A grid of cells; each step, a live cell with two or three live neighbours lives on, a dead cell with exactly three comes alive. Simple rules, endless patterns.

```
"""life — Conway's Game of Life in the terminal"""
import time

def step(live: set[tuple[int, int]]) -> set[tuple[int, int]]:
    near: dict[tuple[int, int], int] = {}
    for x, y in live:
        for dx in (-1, 0, 1):
            for dy in (-1, 0, 1):
                if dx or dy:
                    near[(x + dx, y + dy)] = near.get((x + dx, y + dy), 0) + 1
    return {c for c, n in near.items() if n == 3 or (n == 2 and c in live)}

def show(live: set[tuple[int, int]], w: int = 40, h: int = 20) -> str:
    return "\n".join("".join("█" if (x, y) in live else "·" for x in range(w)) for y in range(h))

if __name__ == "__main__":
    cells = {(1, 0), (2, 1), (0, 2), (1, 2), (2, 2)}           # a glider
    for _ in range(60):
        print("\033[H\033[J" + show(cells))
        cells = step(cells)
        time.sleep(0.1)
```

Test it: a *blinker* — three cells in a row — turns into three in a column and back: `assert step({(0, 1), (1, 1), (2, 1)}) == {(1, 0), (1, 1), (1, 2)}`.

Exercises

42.1 ★ Run all four. Each with its tests green in the margin.

42.2 ★★ Weekday. Extend `notestats` with `--weekday`: lines per weekday.

42.3 ★★ Missed first. Extend `flash` as 42.2 suggests.

42.4 ★★★ Rain alert. Extend `sky` with `precipitation_probability` and print a warning if any of the next 12 hours is above 50 %. Then add it to your `today` function from 9.3.

43. A neural network from scratch, in NumPy

In this chapter

- arrays: many numbers worked on at once
- what a neural network computes: forward, loss, backward, update
- train one on a spiral and on handwritten digits, with no AI library at all

You need: chapter 41 (NumPy). This chapter is level 6's checkpoint: when your network reads digits, the Python course is done.

Before PyTorch does it for you, do it once by hand. Every model in the news — the largest language models included — learns by the four steps in this chapter, repeated billions of times.

```
$ uv init nn && cd nn
$ uv add numpy scikit-learn
$ uv add --dev pytest
```

43.1 Arrays

```
>>> import numpy as np
>>> a = np.array([1.0, 2.0, 3.0])
>>> a * 10
array([10., 20., 30.])
>>> M = np.array([[1, 2], [3, 4]])
>>> M.shape, M.T
((2, 2), array([[1, 3],
       [2, 4]]))
>>> M @ np.array([1, 1])
array([3, 7])
>>> M.sum(axis=0), M.max(axis=1)
(array([4, 6]), array([2, 4]))
```

(`@` multiplies matrices: each row of `M` times the vector, summed. `axis=0` works down the columns, `axis=1` along the rows)

`Word — tensor.` An array of any number of dimensions: a number (0), a list (1), a table (2), a stack of tables (3) … A batch of 64 colour pictures is a 4-dimensional tensor: 64 × 3 × height × width. Neural networks are made of tensor arithmetic.

Broadcasting stretches a smaller array to fit: a table (100 × 3) plus a row (3) adds the row to every line. It is how one line of NumPy does what a loop would do a hundred times — and fast, because the loop runs in C.

43.2 What a network computes

A two-layer network turns an input `x` (here 2 numbers: a point's position) into a score for each class (3: which arm of a spiral it is on):

```
h      = max(0, x @ W1 + b1)     a hidden layer: 64 numbers, negatives cut to 0 (ReLU)
scores = h @ W2 + b2             3 scores, one per class
p      = softmax(scores)         the scores as probabilities that add up to 1
loss   = -log(p[right class])    small when the right class got a high probability
```

Training is four steps, again and again:

Step	Does
forward	compute the scores and the loss for the data
backward	compute how the loss would change if each weight moved a little: its **gradient**
update	move every weight a small step against its gradient: downhill
repeat	until the loss stops falling

`Word — gradient.` For each weight, the slope of the loss: how much the loss rises when the weight rises a little. Step the other way and the loss falls. **Backpropagation** computes all the gradients at once with the chain rule, from the loss backwards.



43.3 The whole network, in 40 lines

```
# mlp.py — a two-layer neural network in NumPy alone: forward, loss, backward, update
import numpy as np

rng = np.random.default_rng(0)

def spiral(n=100, k=3):
    """n points in each of k arms of a spiral: X is (n*k, 2), y the arm of each point"""
    X, y = [], []
    for c in range(k):
        r = np.linspace(0.0, 1.0, n)
        t = np.linspace(c * 4, (c + 1) * 4, n) + rng.normal(0, 0.2, n)
        X.append(np.c_[r * np.sin(t), r * np.cos(t)])
        y.append(np.full(n, c))
    return np.vstack(X), np.concatenate(y)

X, y = spiral()
N, D, H, K = X.shape[0], 2, 64, 3

W1 = rng.normal(0, 1, (D, H)) * np.sqrt(2 / D); b1 = np.zeros(H)
W2 = rng.normal(0, 1, (H, K)) * np.sqrt(2 / H); b2 = np.zeros(K)
lr = 0.5

for step in range(3001):
    # forward
    h = np.maximum(0, X @ W1 + b1)                  # ReLU
    logits = h @ W2 + b2
    logits -= logits.max(axis=1, keepdims=True)     # (for numerical safety)
    p = np.exp(logits); p /= p.sum(axis=1, keepdims=True)
    loss = -np.log(p[np.arange(N), y]).mean()
    # backward: the gradient of the loss with respect to every number above
    dlogits = p.copy(); dlogits[np.arange(N), y] -= 1; dlogits /= N
    dW2 = h.T @ dlogits; db2 = dlogits.sum(axis=0)
    dh = dlogits @ W2.T; dh[h <= 0] = 0
    dW1 = X.T @ dh; db1 = dh.sum(axis=0)
    # update: a small step downhill
    for P, G in ((W1, dW1), (b1, db1), (W2, dW2), (b2, db2)):
        P -= lr * G
    if step % 500 == 0:
        acc = (p.argmax(axis=1) == y).mean()
        print(f"step {step:4d}  loss {loss:.3f}  accuracy {acc:.2f}")
```

```
$ uv run mlp.py
step    0  loss 1.033  accuracy 0.29
step  500  loss 0.116  accuracy 0.98
step 1000  loss 0.075  accuracy 0.99
step 1500  loss 0.059  accuracy 0.99
step 2000  loss 0.050  accuracy 0.99
step 2500  loss 0.044  accuracy 0.99
step 3000  loss 0.040  accuracy 0.99
```

At step 0 it guesses (29 %: about one in three); by step 500 it has learnt the spiral.

43.4 Check the calculus

Backward is where mistakes hide. Check any gradient by *nudging*: move one weight by a tiny `eps` up and down, and see how the loss changes. The two numbers must agree.

```
import numpy as np
rng = np.random.default_rng(1)
X = rng.normal(size=(5, 3)); y = np.array([0, 2, 1, 2, 0]); W = rng.normal(size=(3, 3))

def loss_and_grad(W):
    z = X @ W; z -= z.max(axis=1, keepdims=True)
    p = np.exp(z); p /= p.sum(axis=1, keepdims=True)
    loss = -np.log(p[np.arange(5), y]).mean()
    d = p.copy(); d[np.arange(5), y] -= 1
    return loss, X.T @ d / 5

loss, G = loss_and_grad(W)
eps = 1e-5
W[1, 2] += eps; up, _ = loss_and_grad(W)
W[1, 2] -= 2 * eps; down, _ = loss_and_grad(W)
print("by calculus", G[1, 2], " by nudging", (up - down) / (2 * eps))
```

```
by calculus 0.036057134325191066  by nudging 0.03605713432275337
```

(equal to nine digits: the backward pass is right)

43.5 Handwritten digits

scikit-learn ships 1 797 small pictures of handwritten digits, 8 × 8 pixels each:

```
from sklearn.datasets import load_digits
d = load_digits()
X, y = d.data / 16.0, d.target          # 64 numbers from 0 to 1 per picture; the digit
print(X.shape, y[:10])
```

```
(1797, 64) [0 1 2 3 4 5 6 7 8 9]
```

Exercises

43.1 ★ Too simple. Remove the hidden layer (scores straight from `X @ W + b`). What accuracy does the spiral reach now?

Expected: about 0.5: a straight line cannot separate a spiral. The hidden layer is what lets a network bend.

43.2 ★ Learning rate. Try `lr = 5.0` and `lr = 0.01`.

Expected: 5.0 jumps about or blows up (`nan`); 0.01 learns, but slowly. Choosing it is half the craft.

43.3 ★★ Check another. Gradient-check `b1` of `mlp.py` the same way as 43.4.

43.4 ★★ Digits. Train the same network on the digits: `D = 64`, `K = 10`. Keep the last 297 pictures apart, never trained on, and measure the accuracy on them.

Expected: above 0.9 on pictures it has never seen.

43.5 ★★★ Mini-batches. Instead of all pictures every step, use a random 64 each step. It learns from noisier steps, but each is much cheaper — the way every large model is trained.

44. PyTorch on your graphics card

In this chapter

- install PyTorch for the RX 9060 XT
- tensors on the card, and how much faster they are
- the digits network again, in PyTorch, trained on the card

You need: chapter 43.

44.1 Where things stand (October 2026)

Piece	State on Fedora 44
the card's driver	`amdgpu`, in the kernel: nothing to install
ROCm, AMD's computing stack	Fedora packages it (`rocm-*`, version 7.1); AMD's own list of supported systems does not include Fedora
PyTorch for ROCm	PyTorch's official wheels, `download.pytorch.org/whl/rocm7.2` (PyTorch 2.14), carry their own copy of ROCm 7.2 and include the RX 9060 XT (`gfx1200`, RDNA 4) — so Fedora's ROCm packages are not needed for PyTorch
btop	shows the card's load, memory, heat and watts while you train (through `rocm-smi`, installed)

So: the PyTorch wheel brings everything; Fedora brings the driver. One check of access to the card's compute device:

```
$ ls -l /dev/kfd
crw-rw-rw-. 1 root render 235, 0 Oct 20 09:12 /dev/kfd
```

(every user may open it. If yours shows `crw-rw----`*, you need the* `render` *group:* `sudo usermod -aG render $USER`*, then log out and in)*

The card comes in an 8 GB and a 16 GB model; `rocm-smi --showmeminfo vram` says which is yours. Every project in this book fits in 8 GB.

44.2 Install

```
$ uv init dl && cd dl
```

Add these lines to `pyproject.toml`, so that uv fetches PyTorch from PyTorch's ROCm index and everything else from the usual one:

```
[tool.uv.sources]
torch = [{ index = "pytorch-rocm" }]
pytorch-triton-rocm = [{ index = "pytorch-rocm" }]

[[tool.uv.index]]
name = "pytorch-rocm"
url = "https://download.pytorch.org/whl/rocm7.2"
explicit = true
```

```
$ uv add torch scikit-learn safetensors
$ uv add --dev pytest
```

(the ROCm wheel is several gigabytes: the first `uv add` *takes a while; later projects reuse uv's cache)*

火 44.3 Is the card there?

```
# gpu.py — PyTorch sees the card, and how much faster it is
import time
import torch

print(torch.__version__, "· HIP", torch.version.hip)
print(torch.cuda.is_available(), torch.cuda.get_device_name(0))

def bench(device: str) -> float:
    a = torch.randn(4096, 4096, device=device)
    b = torch.randn(4096, 4096, device=device)
    a @ b                                          # (a first run warms it up)
    torch.cuda.synchronize()
    t = time.perf_counter()
    for _ in range(10):
        a @ b
    torch.cuda.synchronize()                       # (the card works on its own: wait for it)
    return (time.perf_counter() - t) / 10

cpu, gpu = bench("cpu"), bench("cuda")
print(f"4096×4096 matrix product: processor {cpu * 1000:.0f} ms, card {gpu * 1000:.1f} ms")
```

```
$ uv run gpu.py
2.14.1+rocm7.2 · HIP 7.2.53211
True AMD Radeon RX 9060 XT
4096×4096 matrix product: processor 310 ms, card 9.8 ms
```

(for example: your numbers will differ)

> `Word — "cuda" on an AMD card.` PyTorch calls every graphics card `cuda`, NVIDIA's name; on ROCm the same calls go to AMD's HIP. So `device="cuda"` and `torch.cuda` are right here, and every tutorial written for NVIDIA works unchanged.

火 44.4 The digits network, in PyTorch

What took 20 lines of calculus in chapter 43 is three here: PyTorch's **autograd** computes every gradient by itself.

```
# digits.py — the digits network in PyTorch, trained on the card, saved for Rust (chapter 64)
import torch
import torch.nn as nn
import torch.nn.functional as F
from safetensors.torch import save_file
from sklearn.datasets import load_digits

torch.manual_seed(0)
DEVICE = "cuda"

d = load_digits()
X = torch.tensor(d.data / 16.0, dtype=torch.float32)
y = torch.tensor(d.target, dtype=torch.int64)
Xtr, ytr, Xte, yte = X[:1500].to(DEVICE), y[:1500].to(DEVICE), X[1500:], y[1500:]

class Net(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(64, 64)
        self.fc2 = nn.Linear(64, 10)

    def forward(self, x):
        return self.fc2(F.relu(self.fc1(x)))

model = Net().to(DEVICE)
opt = torch.optim.Adam(model.parameters(), lr=1e-2)

for epoch in range(201):
    loss = F.cross_entropy(model(Xtr), ytr)          # forward
    opt.zero_grad()
    loss.backward()                                  # backward: every gradient
    opt.step()                                       # update
    if epoch % 50 == 0:
        with torch.no_grad():
            acc = (model(Xte.to(DEVICE)).argmax(1).cpu() == yte).float().mean()
        print(f"epoch {epoch:3d}  loss {loss.item():.3f}  test accuracy {acc:.3f}")

save_file({k: v.cpu() for k, v in model.state_dict().items()}, "digits.safetensors")
save_file({"x": Xte, "y": yte}, "test.safetensors")
```

```
$ uv run digits.py
epoch   0  loss 2.312  test accuracy 0.391
epoch  50  loss 0.089  test accuracy 0.912
epoch 100  loss 0.014  test accuracy 0.922
epoch 150  loss 0.005  test accuracy 0.919
epoch 200  loss 0.003  test accuracy 0.919
```

(for example) The training loss keeps falling while the test accuracy stops rising: the network has started to memorise rather than learn — **overfitting**. Exercise 44.3 fights it.

Exercises

44.1 ★ Your card. Run `gpu.py` and write your two times in your notes.

44.2 ★ Watch it. Raise the benchmark to 1 000 runs and watch btop's GPU line while it works.

44.3 ★★ Less memorising. Add `weight_decay=1e-3` to the optimiser and `nn.Dropout(0.2)` after the ReLU (call `model.train()` before training and `model.eval()` before testing). Does the test accuracy hold up better?

44.4 ★★ A test for the model. In `tests/test_digits.py`, check that `Net()(torch.zeros(5, 64)).shape == (5, 10)`. Shape tests catch most model bugs, in milliseconds, without the card.

44.5 ★★★ Pictures. Train a small convolutional network (`nn.Conv2d`) on Fashion-MNIST (`uv add torchvision` with the same ROCm index lines). Aim for 90 % on the test set.

語 45. A tiny transformer that writes

In this chapter

- text as numbers: a character tokenizer
- attention, the idea behind every large language model
- train a 3-million-parameter model on the handbook, and let it write

You need: chapter 44.

A large language model does one thing: given some text, predict the next piece. Do that well enough, one piece at a time, and it writes. This chapter builds the same machine — a **transformer** — at a size your card trains in minutes, on a text you know: the full handbook in your working copy, almost a million characters about this desktop.

語 45.1 The ideas, in plain words

Idea	Means
token	a piece of text the model reads as one number. Here, one character; large models use pieces of words
embedding	each token's number turned into a vector of 256 numbers the model learns
attention	each position looks back at every earlier position and takes what it needs from them: in *the cat sat on its mat*, *its* looks at *cat*
causal	a position may look back, never forward: it must not see the answer it is predicting
block	attention, then a small network on each position, each added to what was there (a *residual*)
logits	a score for every possible next token; softmax makes them probabilities
temperature	how adventurous sampling is: low repeats the likeliest, high takes risks

語 45.2 The model

```
# chargpt.py — a character-level transformer: python chargpt.py TEXTFILE
import sys
from pathlib import Path

import torch
import torch.nn as nn
import torch.nn.functional as F

torch.manual_seed(0)
DEVICE = "cuda"
BLOCK, BATCH, D, HEADS, LAYERS = 128, 64, 256, 4, 4

text = Path(sys.argv[1]).read_text(encoding="utf-8")
chars = sorted(set(text))
stoi = {c: i for i, c in enumerate(chars)}
data = torch.tensor([stoi[c] for c in text], dtype=torch.long)
cut = int(0.9 * len(data))
train, val = data[:cut], data[cut:]

def batch(part: torch.Tensor):
    """BATCH random stretches of BLOCK characters, and the same shifted by one: the answers"""
    i = torch.randint(len(part) - BLOCK - 1, (BATCH,))
    x = torch.stack([part[j:j + BLOCK] for j in i])
    y = torch.stack([part[j + 1:j + BLOCK + 1] for j in i])
    return x.to(DEVICE), y.to(DEVICE)
```

The model: a stack of blocks, each attention then a small network:

```
class Block(nn.Module):
    def __init__(self):
        super().__init__()
        self.ln1, self.ln2 = nn.LayerNorm(D), nn.LayerNorm(D)
        self.qkv = nn.Linear(D, 3 * D)
        self.proj = nn.Linear(D, D)
        self.mlp = nn.Sequential(nn.Linear(D, 4 * D), nn.GELU(), nn.Linear(4 * D, D))

    def forward(self, x):
        B, T, _ = x.shape
        q, k, v = self.qkv(self.ln1(x)).split(D, dim=2)
        q, k, v = (t.view(B, T, HEADS, D // HEADS).transpose(1, 2) for t in (q, k, v))
        a = F.scaled_dot_product_attention(q, k, v, is_causal=True)   # attention
        x = x + self.proj(a.transpose(1, 2).reshape(B, T, D))
        return x + self.mlp(self.ln2(x))

class CharGPT(nn.Module):
    def __init__(self, vocab: int):
        super().__init__()
        self.tok = nn.Embedding(vocab, D)
        self.pos = nn.Embedding(BLOCK, D)
        self.blocks = nn.Sequential(*[Block() for _ in range(LAYERS)])
        self.ln = nn.LayerNorm(D)
        self.head = nn.Linear(D, vocab)

    def forward(self, idx):
        x = self.tok(idx) + self.pos(torch.arange(idx.shape[1], device=idx.device))
        return self.head(self.ln(self.blocks(x)))

    @torch.no_grad()
    def generate(self, idx, n: int, temperature: float = 0.8):
        for _ in range(n):
            logits = self(idx[:, -BLOCK:])[:, -1] / temperature
            idx = torch.cat([idx, torch.multinomial(F.softmax(logits, dim=-1), 1)], dim=1)
        return idx
```

The training, and a sample of what it learnt:

```
def loss_on(x, y):
    return F.cross_entropy(model(x).view(-1, len(chars)), y.view(-1))

model = CharGPT(len(chars)).to(DEVICE)
print(f"{len(text):,} characters, {len(chars)} different; "
      f"{sum(p.numel() for p in model.parameters()):,} parameters")
opt = torch.optim.AdamW(model.parameters(), lr=3e-4)

for step in range(5001):
    loss = loss_on(*batch(train))
    opt.zero_grad(set_to_none=True)
    loss.backward()
    opt.step()
    if step % 500 == 0:
        with torch.no_grad():
            print(f"step {step:5d}  train {loss.item():.3f}  val {loss_on(*batch(val)).item():.3f}")

torch.save(model.state_dict(), "chargpt.pt")
start = torch.tensor([[stoi["#"]]], device=DEVICE)
print("".join(chars[i] for i in model.generate(start, 600)[0].tolist()))
```

```
$ uv run chargpt.py ~/repos/fedokawaii/docs/fedokawaii-guide.md
962,051 characters, 246 different; 3,318,518 parameters
step     0  train 5.663  val 5.661
step   500  train 2.012  val 2.087
step  1000  train 1.563  val 1.684
…
step  5000  train 1.012  val 1.198
# 18.4 Practice: the terminal

| Keys | Does |
|---|---|
| **Super+Shift+R** | the mixer opens the folder in yazi, …
```

(for example: the counts depend on your copy of the handbook, and the text it writes is new every time — plausible-looking, often wrong, never copied. That is a language model)

Read the loss as *surprise*: at step 0 it is about `ln(246) ≈ 5.5`, a blind guess among 246 characters; at 1.0 the model is nearly as sure of the next character as a reader who knows the book's style.

Exercises

45.1 ★ Temperature. Generate with `temperature=0.3` and `1.5`.

Expected: 0.3 repeats safe phrases (`the the`, tables of `| **`); 1.5 invents words.

45.2 ★★ Your voice. Train it on your own notes, or on a book from `~/Books` converted to text (`pdftotext book.pdf book.txt`).

45.3 ★★ Bigger. `D = 384`, `LAYERS = 6`. Watch the card's memory in btop, and the validation loss: lower, or overfitting?

45.4 ★★ A prompt. Start generation from a string of your own instead of `#`: encode it with `stoi`.

45.5 ★★★ Words, not letters. Replace the character tokenizer with a byte-pair one (`uv add tokenizers`; train it on the same text). The model now predicts pieces of words.

46. Fine-tuning a small model on your own data

In this chapter

- take a real pretrained language model from Hugging Face
- teach it facts of your own with LoRA, in minutes, on your card
- compare its answers before and after

You need: chapter 45.

Training a useful language model from nothing takes months of computing. **Fine-tuning** takes one that already knows language and teaches it a little more. **LoRA** (low-rank adaptation) makes that cheap: the model's own weights stay frozen, and small extra matrices — under 1 % of the size — learn the change.

46.1 The data: questions and answers about your desktop

Write `keys.jsonl`, one question and answer per line (forty or more is good; the tables of this book are a ready source):

```
{"q": "How do I open a terminal on Fedokawaii?", "a": "Super+Return opens foot, the terminal."}
{"q": "How do I take a screenshot of a region?", "a": "Press Print and drag a box; it is saved and copied."}
{"q": "Where do my notes live?", "a": "In ~/Notes/notes.md; Super+N opens them."}
```

46.2 The code

```
$ uv add transformers peft
```

```
# tune.py — LoRA fine-tuning of a small chat model on keys.jsonl
import json

import torch
from peft import LoraConfig, get_peft_model
from transformers import AutoModelForCausalLM, AutoTokenizer

NAME = "HuggingFaceTB/SmolLM2-360M-Instruct"
DEVICE = "cuda"
torch.manual_seed(0)

tok = AutoTokenizer.from_pretrained(NAME)
model = AutoModelForCausalLM.from_pretrained(NAME, dtype=torch.bfloat16).to(DEVICE)

def prompt(q: str) -> str:
    return tok.apply_chat_template([{"role": "user", "content": q}],
                                   tokenize=False, add_generation_prompt=True)

def ask(q: str) -> str:
    enc = tok(prompt(q), return_tensors="pt", add_special_tokens=False).to(DEVICE)
    out = model.generate(**enc, max_new_tokens=60, do_sample=False)
    return tok.decode(out[0, enc.input_ids.shape[1]:], skip_special_tokens=True)

def example(q: str, a: str):
    """the whole exchange as tokens; the labels hide the question (-100): only the answer is learnt"""
    p = tok(prompt(q), add_special_tokens=False).input_ids
    full = tok(prompt(q) + a + tok.eos_token, add_special_tokens=False).input_ids
    ids = torch.tensor([full], device=DEVICE)
    labels = ids.clone()
    labels[0, :len(p)] = -100
    return ids, labels
```

Then the training itself, in the same file: ask once before, attach LoRA, four passes, ask again:

```
pairs = [json.loads(line) for line in open("keys.jsonl", encoding="utf-8")]
test_q = "How do I open a terminal on Fedokawaii?"
print("before:", ask(test_q))

model = get_peft_model(model, LoraConfig(
    r=16, lora_alpha=32, lora_dropout=0.05,
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj"], task_type="CAUSAL_LM"))
model.print_trainable_parameters()
opt = torch.optim.AdamW([p for p in model.parameters() if p.requires_grad], lr=2e-4)

model.train()
for epoch in range(4):
    total = 0.0
    for pair in pairs:
        ids, labels = example(pair["q"], pair["a"])
        loss = model(input_ids=ids, labels=labels).loss
        opt.zero_grad()
        loss.backward()
        opt.step()
        total += loss.item()
    print(f"epoch {epoch}  loss {total / len(pairs):.3f}")

model.eval()
print("after:", ask(test_q))
model.save_pretrained("keys-lora")
```

```
$ uv run tune.py
before: To open a terminal, press Ctrl+Alt+T or search for "Terminal" in the applications menu.
trainable params: 3,276,800 || all params: 365,097,920 || trainable%: 0.8975
epoch 0  loss 2.104
epoch 1  loss 0.712
epoch 2  loss 0.253
epoch 3  loss 0.118
after: Super+Return opens foot, the terminal.
```

(for example) Before, it answered what most Linux desktops do; after, yours. The adapter in `keys-lora/` is a few megabytes: the base model stays as it was downloaded.

> Careful. With forty examples, it memorises them; ask the same thing in other words and it may still answer the old way. More, more varied examples — each fact asked three ways — teach it rather than make it recite. And a small model invents with confidence: never trust its answer about something it was not taught.

Exercises

46.1 ★ Before and after. Run it with your own forty pairs; ask three questions it was taught and one it was not.

46.2 ★★ Held out. Keep five pairs out of training and ask those after. Which does it get right?

46.3 ★★★ Load it later. Write `chat.py` that loads the base model, adds the adapter with `PeftModel.from_pretrained(model, "keys-lora")`, and answers questions in a loop.

使 47. An agent with the Claude API

In this chapter

- call Claude from Python
- give it tools — functions of yours it may ask to run
- the agent loop: ask, run the tool, give back the result, repeat

You need: chapters 37–39. An API key from `console.anthropic.com` (paid by use).

An **agent** is a model in a loop with tools. You describe the tools; the model decides when to call one; your code runs it and hands back the result; the model carries on until it can answer. Claude Code (chapter 21) is that loop with tools for files and the shell. Here you build a small one, for your notes.

使 47.1 Set up

Put the key in your secrets file — never in code — and open a new terminal:

```
# ~/.config/fedokawaii/secrets.sh (chmod 600)
export ANTHROPIC_API_KEY=sk-ant-…
```

```
$ uv init notes-agent && cd notes-agent
$ uv add anthropic
```

使 47.2 One question

```
import anthropic

client = anthropic.Anthropic()            # reads ANTHROPIC_API_KEY
reply = client.messages.create(
    model="claude-haiku-5-5",
    max_tokens=300,
    messages=[{"role": "user", "content": "Explain a gradient in two sentences."}],
)
print(reply.content[0].text)
```

(Haiku is the quick, inexpensive model — right for learning. `claude-opus-5-5` *is the most capable)*

```
# agent.py — a notes assistant: Claude searches and adds to ~/Notes/notes.md through two tools
import datetime
from pathlib import Path

import anthropic

NOTES = Path.home() / "Notes" / "notes.md"
client = anthropic.Anthropic()

TOOLS = [
    {
        "name": "search_notes",
        "description": "Find lines in the user's notes that contain a word (case ignored). "
                       "Returns matching lines with their line numbers.",
        "input_schema": {
            "type": "object",
            "properties": {"word": {"type": "string", "description": "the word to look for"}},
            "required": ["word"],
        },
    },
    {
        "name": "add_note",
        "description": "Append one line to the user's notes. The user confirms it first.",
        "input_schema": {
            "type": "object",
            "properties": {"line": {"type": "string", "description": "the line to add"}},
            "required": ["line"],
        },
    },
]
```

The tools themselves are ordinary functions, and the loop calls them when Claude asks:

```
def search_notes(word: str) -> str:
    hits = [f"{n}: {line}" for n, line in enumerate(NOTES.read_text(encoding="utf-8").splitlines(), 1)
            if word.lower() in line.lower()]
    return "\n".join(hits[:50]) or "no line contains that word"

def add_note(line: str) -> str:
    if input(f"  add to your notes: {line!r} — y/n? ").strip().lower() != "y":
        return "the user declined"
    with NOTES.open("a", encoding="utf-8") as f:
        f.write(line + "\n")
    return "added"

RUN = {"search_notes": search_notes, "add_note": add_note}

def agent(question: str) -> str:
    messages = [{"role": "user", "content": question}]
    system = f"You help the user with their notes. Today is {datetime.date.today():%A %d %B %Y}."
    while True:
        reply = client.messages.create(model="claude-haiku-5-5", max_tokens=1024,
                                       system=system, tools=TOOLS, messages=messages)
        if reply.stop_reason != "tool_use":
            return "".join(b.text for b in reply.content if b.type == "text")
        messages.append({"role": "assistant", "content": reply.content})
        results = []
        for block in reply.content:
            if block.type == "tool_use":
                print(f"  [{block.name} {block.input}]")
                results.append({"type": "tool_result", "tool_use_id": block.id,
                                "content": RUN[block.name](**block.input)})
        messages.append({"role": "user", "content": results})

if __name__ == "__main__":
    while (q := input("you > ")).strip():
        print(agent(q))
```

```
$ uv run agent.py
you > what did I write about rice?
  [search_notes {'word': 'rice'}]
You noted "rice cooker: compare prices" (line 3) and "Rice, two kilos" (line 8).
you > remind me to buy it on Friday
  [add_note {'line': '- [ ] buy rice (Friday)'}]
  add to your notes: '- [ ] buy rice (Friday)' — y/n? y
Added "- [ ] buy rice (Friday)" to your notes.
you >
```

(for example; an empty line ends it) Notice what you did *not* write: no rule for when to search or which word to use. The model chose the tool and its argument from your question; your code only did what it asked — and asked you before writing.

Exercises

47.1 ★ Watch the loop. Print `reply.stop_reason` on every turn. How many turns does one question take?

47.2 ★★ A third tool. `today_notes`: the lines under today's heading. Ask "what's on my plate today?"

47.3 ★★ Test without the network. Test `search_notes` and `add_note` with a temporary file (pytest's `tmp_path`, and `monkeypatch` for `NOTES` and `input`). Tools are ordinary functions: test them as such.

47.4 ★★★ Your model as a tool. Add a tool that asks your fine-tuned model of chapter 46 about the desktop's keys. One agent, two models: the big one plans, the small one knows your desktop.



Part V — Rust, for speed and safety

Rust asks more of you up front: it wants to know who owns every value. In return, a program that compiles almost never crashes, and runs as fast as C. This part teaches it the same way as Part IV — a course in six levels of tiny programs, mistakes, exercises and checkpoints — and ends with Rust doing AI work: your own autograd, a model run in Rust, and Rust code called from Python.

錆 48. Rust from zero: cargo and the loop

In this chapter

- make, build, run and test a Rust project with cargo
- a loop that checks your code on every save
- read the compiler's messages: Rust's best teacher

You need: chapter 24.

Rust comes with **cargo**, one tool for the whole job: it makes projects, fetches libraries (*crates*), builds, runs, tests, checks style. Setup has already installed Rust for you — the stable toolchain with rust-analyzer, clippy and rustfmt, through rustup, in `~/.local/share/cargo` — and `fedokawaii update` keeps it current. Check it is there:

```
$ cargo --version
cargo 1.98.0 (…)
$ rustc --version
rustc 1.98.0 (…)
```

(for example; a new Rust comes every six weeks: `fedokawaii update` *runs* `rustup update` *for you, or run it yourself)*

錆 48.1 A project

```
$ cd ~/repos && cargo new hello-rs && cd hello-rs
$ lt
.
├──── Cargo.toml
└──── src
   └──── main.rs
$ cargo run
   Compiling hello-rs v0.1.0 (/home/you/repos/hello-rs)
    Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.4s
     Running `target/debug/hello-rs`
Hello, world!
```

Command	Does
`cargo run`	build and run
`cargo run --release`	the same, optimised: often ten times faster — use it for anything heavy
`cargo test`	run every test
`cargo check`	only check the code compiles: the quickest feedback
`cargo clippy`	advice: simpler, more idiomatic ways to write what you wrote
`cargo fmt`	format the code (neovim does it on save)
`cargo add NAME`	add a crate from crates.io
`cargo doc --open`	the documentation of your project and every crate it uses, in Brave

錆 48.2 In neovim

Rust support is set up already: **rust-analyzer** runs with the first `.rs` file. Types appear beside each `let` in grey, errors are underlined as you type, `K` shows any function's documentation and `gd` jumps into the standard library's own source. `Space` `c` `a` offers the fixes the compiler suggests.

Keys	Does
Space r f · Space r c	cargo run · cargo check, in a terminal under the code
Space r t · Space r l	cargo test · cargo clippy
Space r r	rust-analyzer's runnables: pick a test, a binary or an example to run
Space t t · Space t r · Space t w	this file's tests · the one under the cursor · on every save
Space d b · Space d c	a breakpoint · run with the debugger (codelldb), as in chapter 24

錆 48.3 Check on every save

Every save (Ctrl+S) runs clippy over the project: its findings appear in the code as you work, no command needed. For the tests, Space t w in a file with tests: they run on every save too, ✓ or ✗ in the margin.

錆 48.4 Reading the compiler

Rust's compiler explains itself. Make this mistake on purpose:

```
fn main() {
    let name = String::from("Kyoto");
    let other = name;
    println!("{name}");
}
```

```
error[E0382]: borrow of moved value: `name`
 --> src/main.rs:4:16
  |
2 |     let name = String::from("Kyoto");
  |         ---- move occurs because `name` has type `String`, which does not implement the `Copy` trait
3 |     let other = name;
  |                 ---- value moved here
4 |     println!("{name}");
  |                ^^^^ value borrowed here after move
  |
help: consider cloning the value if the performance cost is acceptable
  |
3 |     let other = name.clone();
  |                     ++++++++
```

Read it as a story: line 2 made the value, line 3 *moved* it into `other`, line 4 tried to use it after. And a fix. `rustc --explain E0382` tells the whole story with examples. Chapter 51 explains moving.

Exercises

48.1 ★ Hello. Make `hello-rs`, change the message, `cargo run`.

48.2 ★ Break it. Remove the `!` from `println!` and read the error; put it back.

Expected: the compiler says `println` is a macro and suggests the `!`.

48.3 ★★ Release. Write a loop summing the numbers 1 to 1 000 000 000 into a `u64`, and time `cargo run` against `cargo run --release` (`time cargo run --release`).

Expected: the release build is many times faster.

变 49. Rust level 1: variables, types and functions

In this chapter

- `let`, and why variables cannot change unless you say `mut`
- Rust's number types, `bool`, `char`, and text
- functions with typed parameters and a returned value
- expressions and statements: why the last line has no `;`

You need: chapter 48 (cargo, and reading the compiler).

Rust is learnt in six levels, like Python in Part IV:

Level	You learn	Chapters	Checkpoint
1	variables, types, functions, control flow	49–50	a number-guessing machine
2	ownership, borrowing, structs, enums	51–54	a bank with commands
3	collections, errors	55–56	a word counter that never panics
4	traits, generics, lifetimes	57–58	a generic statistics library
5	modules, tests, iterators, closures, threads	59–61	a parallel file scanner
6	a CLI, autograd, models in Rust, Rust in Python	62–65	your own crate on GitHub

Every program here is complete: put it in `src/main.rs` of a `cargo new` project and `cargo run` — or, for a quick try, save it as `main.rs` and run `rustc main.rs && ./main`. Each was compiled with Rust 1.97 (edition 2024) and printed what is shown.

变 49.1 let: a name for a value

```
fn main() {
    let apples = 5;
    let price = 1.25;
    println!("{apples} apples cost {}", apples as f64 * price);
}
```

You should see:

```
5 apples cost 6.25
```

`let` makes a variable. `println!` prints, filling each `{}` with the next value, and `{apples}` with the variable of that name. (The `!` means `println` is a **macro**: code that writes code. For now, just remember the `!`.)

Variables cannot change — unless you ask. This is Rust's first surprise:

```
fn main() {
    let count = 1;
    count = 2;
    println!("{count}");
}
```

The compiler says:

```
error[E0384]: cannot assign twice to immutable variable `count`
 --> src/main.rs:3:5
  |
2 |     let count = 1;
  |         ----- first assignment to `count`
3 |     count = 2;
  |     ^^^^^^^^^ cannot assign twice to immutable variable
  |
help: consider making this binding mutable
  |
2 |     let mut count = 1;
  |         +++
```

The compiler says exactly what to do: add `mut` (*mutable*, changeable):

```
fn main() {
    let mut count = 1;
    println!("{count}");
    count = 2;
    count += 10;
    println!("{count}");
}
```

You should see:

```
1
12
```

Why the default? Most values never need to change, and a value that cannot change cannot be changed by mistake. When you see `mut`, you know to watch that name.



49.2 Shadowing

A new `let` with the same name makes a **new** variable that hides the old one — even of a different type. It is handy for converting a value step by step:

```
fn main() {
    let input = "42";
    let input: i32 = input.parse().unwrap();
    let input = input * 2;
    println!("{input}");
}
```

You should see:

```
84
```

(`.parse()` turns text into a number; `.unwrap()` means "I am sure it worked" — chapter 56 does this properly)

变 49.3 Number types

Rust wants to know each number's type and size. It usually works it out; you write it when it cannot, or to choose:

Type	Holds	Range
`i32`	whole numbers (the default)	about ±2.1 billion
`i64`	larger whole numbers	about ±9.2 × 10^{18}
`u8` · `u32` · `u64`	whole numbers ≥ 0 (*u*nsigned)	0–255 · 0–4.2 billion · 0–1.8 × 10^{19}
`usize`	sizes and indexes: as wide as the machine	0–1.8 × 10^{19} on 64-bit
`f64`	decimals (the default)	15–16 digits
`f32`	smaller decimals: what GPUs and ML models use	6–7 digits

```
fn main() {
    let a: i32 = 7;
    let b: i32 = 2;
    println!("{} {} {} {}", a + b, a * b, a / b, a % b);
    let x: f64 = 7.0;
    println!("{}", x / 2.0);
    let big: u64 = 2u64.pow(40);
    println!("{big}");
    println!("{}", i32::MAX);
    println!("{:.3}", std::f64::consts::PI);
}
```

You should see:

```
9 14 3 1
3.5
1099511627776
2147483647
3.142
```

(`7 / 2` on whole numbers is `3`: the remainder is dropped, as `//` does in Python)

Rust never mixes types silently. An int and a float cannot be added until you convert one with `as`:

```
fn main() {
    let n: i32 = 3;
    let x: f64 = 0.5;
    println!("{}", n + x);
}
```

The compiler says:

```
error[E0277]: cannot add `f64` to `i32`
 --> src/main.rs:4:22
  |
4 |     println!("{}", n + x);
  |                      ^ no implementation for `i32 + f64`
  |
  = help: the trait `Add<f64>` is not implemented for `i32`
```

```
fn main() {
    let n: i32 = 3;
    let x: f64 = 0.5;
    println!("{}", n as f64 + x);
    println!("{}", 3.99_f64 as i32);
    println!("{}", 300_i32 as u8);
}
```

You should see:

```
3.5
3
44
```

(`as` to a smaller type cuts: 3.99 → 3; 300 does not fit a `u8`, and wraps to 44. Use it knowingly)

Overflow — a number too big for its type — is caught in a debug build: the program stops rather than giving a wrong answer.

```
fn main() {
    let mut level: u8 = 250;
    for _ in 0..10 {
        level += 1;
    }
    println!("{level}");
}
```

You should see:

```
thread 'main' (1) panicked at src/main.rs:4:9:
attempt to add with overflow
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace
```

(a panic is Rust stopping the program on purpose: something happened that must not be ignored)



49.4 bool, char and text

```
fn main() {
    let ready: bool = 3 > 2;
    let letter: char = 'k';
    let kanji: char = '猫';
    let name: &str = "Sharjah";
    let mut owned: String = String::from("hello");
    owned.push_str(", world");
    println!("{ready} {letter} {kanji} {name} {owned}");
    println!("{} {}", name.len(), owned.to_uppercase());
}
```

You should see:

```
true k 猫 Sharjah hello, world
7 HELLO, WORLD
```

Type	Is
`bool`	`true` or `false`
`char`	one character, in single quotes: `'a'`, `'猫'`, `'é'`
`&str`	a piece of text you look at, in double quotes: `"hello"`
`String`	text you own and can grow: `String::from("hello")`

Why two kinds of text? It is about *ownership*, Rust's central idea (chapter 51). For now: literals in quotes are `&str`; when you need to build or change text, make a `String`.

变 49.5 Tuples and arrays

```
fn main() {
    let point: (i32, i32) = (3, 4);
    let (x, y) = point;
    println!("{} {} {x} {y}", point.0, point.1);

    let days: [&str; 3] = ["mon", "tue", "wed"];
    let zeros = [0; 5];
    println!("{} {} {:?} {:?}", days[0], days.len(), days, zeros);
}
```

You should see:

```
3 4 3 4
mon 3 ["mon", "tue", "wed"] [0, 0, 0, 0, 0]
```

(`{:?}` prints a value in its debug form — how Rust shows a whole tuple, array or list. An array's size is part of its type: `[&str; 3]`. For a list that grows, chapter 55's `Vec`)

变 49.6 Functions

```
fn area(width: f64, height: f64) -> f64 {
    width * height
}

fn greet(name: &str) {
    println!("hello, {name}!");
}

fn main() {
    greet("Sara");
    let a = area(3.0, 4.5);
    println!("{a}");
}
```

You should see:

```
hello, Sara!
13.5
```

Piece	Means
`fn area(…)`	a function named `area`
`width: f64`	every parameter has its type — always
`-> f64`	the type it gives back; none means it gives back nothing
`width * height` (no `;`)	the last expression is the value returned

变 49.7 Expressions and the missing semicolon

In Rust, almost everything is an **expression** — it has a value. A block `{ }` has the value of its last line, *if that line has no semicolon*. A `;` turns an expression into a **statement**, whose value is "nothing" — written `()`:

```
fn double(n: i32) -> i32 {
    n * 2
}

fn main() {
    let y = {
        let x = 3;
        x + 1
    };
    println!("{y} {}", double(y));
}
```

You should see:

```
4 8
```

Add a semicolon to `n * 2` and the function returns nothing — and the compiler tells you why:

```
fn double(n: i32) -> i32 {
    n * 2;
}

fn main() {
    println!("{}", double(4));
}
```

The compiler says:

```
error[E0308]: mismatched types
 --> src/main.rs:1:22
  |
1 | fn double(n: i32) -> i32 {
  |    ------            ^^^ expected `i32`, found `()`
  |    |
  |    implicitly returns `()` as its body has no tail or `return` expression
2 |     n * 2;
  |          - help: remove this semicolon to return this value
```

`return` exists too, for leaving early: `return 0;`. At the end of a function, the habit is the plain last expression.

变 49.8 Common mistakes

1. Forgetting `mut`. `cannot assign twice to immutable variable` — add `mut` where the compiler points.

2. Mixing number types. `cannot add a float to an integer` — convert with `as`, or write the literal with a point (`2.0`, not `2`).

3. A `;` after the last expression. `mismatched types ⋯ expected i32, found ()` — remove it.

4. Single quotes for text. `'hello'` is an error: single quotes hold one `char`. Text is in double quotes.

```
fn main() {
    let s = 'hello';
}
```

The compiler says:

```
error: character literal may only contain one codepoint
 --> src/main.rs:2:13
  |
2 |     let s = 'hello';
  |             ^^^^^^^
  |
help: if you meant to write a string literal, use double quotes
  |
2 -     let s = 'hello';
2 +     let s = "hello";
  |
```

Exercises

49.1 ★ Print it. Make variables for your name, your city and your age, and print one sentence with all three.

49.2 ★ Seconds. How many seconds in a week, as a `u32`? And in a century — does it still fit in `u32`? (Try; then use `u64`.)

49.3 ★ Make it change. Fix this so it compiles and prints 3: `let n = 1; n += 2; println!("{n}");`

49.4 ★★ Celsius. `fn to_fahrenheit(c: f64) -> f64`, and print 0, 37 and 100 °C converted, with one decimal (`{:.1}`).

49.5 ★★ Average of three. `fn average(a: i32, b: i32, c: i32) -> f64`, returning the exact average (`7, 8, 8` → `7.666…`).

49.6 ★★ Swap. `fn swap(p: (i32, i32)) -> (i32, i32)` returning the pair the other way round.

49.7 ★★★ Seconds to time. `fn hms(total: u32) -> (u32, u32, u32)`: hours, minutes and seconds; print 3725 as `01:02:05` (`{:02}` pads with zeros).

49.8 ★★★ Find the overflow. What is the largest `n` for which `n * n` fits in an `i32`? Find it with `i32::MAX` and `f64::sqrt`, then check that `(n + 1).checked_mul(n + 1)` is `None`.

49.9 ★★★★ Body mass index, with a verdict. `fn bmi(kg: f64, m: f64) -> f64` and `fn verdict(bmi: f64) -> &'static str` (under 18.5 *under*, under 25 *normal*, under 30 *over*, else *obese* — use `if`/`else if` as an expression, chapter 50 shows them; try it). Print three people in an aligned table (`{:<8}`, `{:>6.1}`).

50. Rust level 1: control flow

In this chapter

- `if` and `else` — as statements and as values
- `loop`, `while` and `for`, with ranges
- `break` with a value, labelled loops
- the checkpoint for level 1: a number-guessing machine

You need: chapter 49.

50.1 if and else

Like Python's, without the colon; the condition needs no brackets, and must be a `bool` — Rust never treats a number as true or false:

```
fn main() {
    let temp = 38;
    if temp > 35 {
        println!("too hot");
    } else if temp > 25 {
        println!("warm");
    } else {
        println!("cool");
    }
}
```

You should see:

```
too hot
```

```
fn main() {
    let n = 3;
    if n {
        println!("non-zero");
    }
}
```

The compiler says:

```
error[E0308]: mismatched types
 --> src/main.rs:3:8
  |
3 |     if n {
  |        ^ expected `bool`, found integer
```

`if` is an expression: it has a value, so it can sit on the right of `let`. Both branches must give the same type:

```
fn main() {
    let age = 20;
    let group = if age >= 18 { "adult" } else { "minor" };
    let fee = if age < 12 { 0 } else { 25 };
    println!("{group} pays {fee}");
}
```

You should see:

```
adult pays 25
```

50.2 loop, while

`loop` repeats forever, until a `break`. `break` can hand a value out of the loop:

```
fn main() {
    let mut n = 1;
    let first_big = loop {
        n *= 3;
        if n > 100 {
            break n;
        }
    };
    println!("{first_big}");

    let mut countdown = 3;
    while countdown > 0 {
        println!("{countdown}…");
        countdown -= 1;
    }
    println!("go!");
}
```

You should see:

```
243
3…
2…
1…
go!
```

50.3 for and ranges

`for` walks through anything iterable: a range, an array, a list. `a..b` runs from `a` up to *but not including* `b`; `a..=b` includes `b`:

```
fn main() {
    for i in 0..3 {
        print!("{i} ");
    }
    println!();
    for i in (1..=10).step_by(3) {
        print!("{i} ");
    }
    println!();
    for i in (1..4).rev() {
        print!("{i} ");
    }
    println!();
    let cities = ["Sharjah", "Dubai", "Kyoto"];
    for (i, city) in cities.iter().enumerate() {
        println!("{i}: {city}");
    }
}
```

You should see:

```
0 1 2
1 4 7 10
3 2 1
0: Sharjah
1: Dubai
2: Kyoto
```

(`print!` is `println!` without the new line)

Sums and counts with a `for`:

```
fn main() {
    let mut total = 0;
    for n in 1..=100 {
        total += n;
    }
    let evens = (1..=100).filter(|n| n % 2 == 0).count();
    println!("{total} {evens}");
}
```

You should see:

```
5050 50
```

(the second line is a taste of chapter 60: `filter` keeps the numbers for which the little function `|n| n % 2 == 0` — a closure — is true)

50.4 continue, and labelled loops

`continue` skips to the next turn. In nested loops, a **label** (`'outer:`) says which loop a `break` or `continue` means:

```
fn main() {
    for n in 1..=10 {
        if n % 3 == 0 {
            continue;
        }
        print!("{n} ");
    }
    println!();

    'outer: for a in 1..10 {
        for b in 1..10 {
            if a * b == 42 {
                println!("found {a} × {b}");
                break 'outer;
            }
        }
    }
}
```

You should see:

```
1 2 4 5 7 8 10
found 6 × 7
```

50.5 match on numbers, a first look

`match` compares a value against patterns, top to bottom, and runs the first that fits. It must cover every possible value — `_` means "anything else". Chapter 54 shows its full power:

```
fn describe(n: i32) -> &'static str {
    match n {
        0 => "zero",
        1 | 2 | 3 => "a few",
        4..=9 => "several",
        _ if n < 0 => "negative",
        _ => "many",
    }
}

fn main() {
    for n in [0, 2, 7, -5, 100] {
        println!("{n}: {}", describe(n));
    }
}
```

You should see:

```
0: zero
2: a few
7: several
-5: negative
100: many
```

50.6 Common mistakes

1. A number as a condition — `expected bool, found integer`. Compare it: `if n != 0`.

2. Branches of different types in an `if` used as a value:

```
fn main() {
    let x = 5;
    let label = if x > 3 { "big" } else { 0 };
}
```

The compiler says:

```
error[E0308]: `if` and `else` have incompatible types
 --> src/main.rs:3:43
  |
3 |     let label = if x > 3 { "big" } else { 0 };
  |                            -----          ^ expected `&str`, found integer
  |                            |
  |                            expected because of this
```

3. Off by one with ranges. `0..10` is 0 to 9. For 1 to 10 write `1..=10`.

4. A `match` that misses cases — `non-exhaustive patterns`. Add the missing ones, or `_`.

```
fn main() {
    let n: u8 = 3;
    let s = match n {
        0 => "none",
        1 => "one",
    };
}
```

The compiler says:

```
error[E0004]: non-exhaustive patterns: `2_u8..=u8::MAX` not covered
 --> src/main.rs:3:19
  |
3 |     let s = match n {
  |                   ^ pattern `2_u8..=u8::MAX` not covered
  |
  = note: the matched value is of type `u8`
help: ensure that all possible cases are being handled by adding a match arm with a wildcard pattern or an
explicit pattern as shown
  |
5 ~         1 => "one",
6 ~         2_u8..=u8::MAX => todo!(),
  |
```

50.7 Checkpoint project — a number-guessing machine

You think of a number from 1 to 1000; the machine guesses, and you answer *higher* or *lower*. It halves the range each time — **binary search** — so it never needs more than 10 guesses ($2^{10} = 1024$). Here the "you" is played by a `secret` variable, so the program runs by itself:

```
fn main() {
    let secret = 737;
    let (mut low, mut high) = (1, 1000);
    let mut tries = 0;
    loop {
        let guess = (low + high) / 2;
        tries += 1;
        if guess < secret {
            println!("{tries:>2}: {guess:>4} → higher");
            low = guess + 1;
        } else if guess > secret {
            println!("{tries:>2}: {guess:>4} → lower");
            high = guess - 1;
        } else {
            println!("{tries:>2}: {guess:>4} → yes!");
            break;
        }
    }
}
```

You should see:

```
 1:  500 → higher
 2:  750 → lower
 3:  625 → higher
 4:  687 → higher
 5:  718 → higher
 6:  734 → higher
 7:  742 → lower
 8:  738 → lower
 9:  736 → higher
10:  737 → yes!
```

`Checkpoint.` (1) Wrap it in a `for secret in 1..=1000` loop and find the largest number of tries over all secrets (it should be 10). (2) Read the answers from the keyboard instead: `let mut line = String::new(); std::io::stdin().read_line(&mut line).unwrap();` then `match line.trim() { "h" => …, "l" => …, "y" => …, _ => … }`. Level 1 done.

Exercises

50.1 ★ Count down. Print 10 down to 1, then `lift off!`, with `(1..=10).rev()`.

50.2 ★ FizzBuzz. 1 to 15, with `Fizz`, `Buzz` and `FizzBuzz` — with `if`, then with `match (n % 3, n % 5) { (0, 0) => …, (0, _) => …, (_, 0) => …, _ => … }`.

50.3 ★ Sum of squares. The sum of the squares of 1 to 10 with a `for` loop. *Expected:* `385`.

50.4 ★★ Leap years. `fn is_leap(y: u32) -> bool`, and print the leap years from 1896 to 1912. *Expected:* `1896 1904 1908 1912` — 1900 is not.

50.5 ★★ Times table. A 5 × 5 times table, each number right-aligned in 4 characters.

50.6 ★★ Digits. `fn digit_sum(mut n: u64) -> u64` with a `while` loop and `% 10`, `/ 10`. *Expected:* `digit_sum(2026)` → `10`.

50.7 ★★★ Primes below 50. `fn is_prime(n: u32) -> bool` (trial division up to the square root, `d * d <= n`), and print the primes below 50.

50.8 ★★★ Collatz, longest. For each start from 1 to 10 000, count the Collatz steps (chapter 27's exercise); print the start with the most steps. *Expected:* `6171 takes 261 steps`.

50.9 ★★★★ A diamond. `fn diamond(n: usize)` prints a diamond of `*` of height `2n - 1`, using `" ".repeat(k)` and `"*".repeat(k)`.

有 51. Rust level 2: ownership

In this chapter

- the three rules of ownership, each with a tiny program
- moving, copying and cloning
- what happens when a value goes out of scope
- ownership through functions: giving a value away and getting it back

You need: level 1 (chapters 49–50).

This is the chapter that makes Rust *Rust*. Take it slowly: type every example, and read every compiler message. Once ownership clicks, the rest of the language is easy — and you will see why Rust programs do not crash.

有 51.1 Why ownership

Every program must give memory back when it is done with it. Python does it with a *garbage collector* that runs now and then and frees what nobody uses — convenient, but it costs time and memory. C makes you free it by hand — fast, but forget once and the program leaks; free twice and it crashes. Rust does neither. It follows three rules, checked **when you compile**, that say exactly when each value is freed:

1. Every value has one **owner** — a variable.
2. There is only one owner at a time.
3. When the owner goes out of **scope**, the value is dropped (its memory freed).

Nothing runs to check these while the program runs: they cost nothing at all.

有 51.2 Scope: where a variable lives

A variable lives from its `let` to the end of the `{ }` block it was made in. At that `}`, Rust drops its value:

```
fn main() {
    let outer = String::from("outer");
    {
        let inner = String::from("inner");
        println!("{outer} {inner}");
    }
    println!("{outer}");
}
```

You should see:

```
outer inner
outer
```

```
fn main() {
    {
        let inner = String::from("inner");
    }
    println!("{inner}");
}
```

The compiler says:

```
error[E0425]: cannot find value `inner` in this scope
 --> src/main.rs:5:16
  |
5 |     println!("{inner}");
  |                ^^^^^
  |
help: the binding `inner` is available in a different scope in the same function
 --> src/main.rs:3:13
  |
3 |         let inner = String::from("inner");
  |             ^^^^^
```

You can watch a value being dropped. `Drop` is a trait (chapter 57) — code that runs at that moment; here it just prints:

```
struct Noisy(&'static str);

impl Drop for Noisy {
    fn drop(&mut self) {
        println!("drop {}", self.0);
    }
}

fn main() {
    let _a = Noisy("a");
    {
        let _b = Noisy("b");
        println!("end of the inner block");
    }
    println!("end of main");
}
```

You should see:

```
end of the inner block
drop b
end of main
drop a
```

(values are dropped in reverse order of making: `b` at its `}`, then `a` at main's)

有 51.3 Moving

A `String` keeps its text on the **heap** — memory that can grow — and the variable holds a small record of where it is. When you assign it to another variable, Rust does not copy the text; it **moves** ownership. The old name is no longer valid:

```
fn main() {
    let a = String::from("hello");
    let b = a;
    println!("{a}");
}
```

The compiler says:

```
error[E0382]: borrow of moved value: `a`
 --> src/main.rs:4:16
  |
2 |     let a = String::from("hello");
  |         - move occurs because `a` has type `String`, which does not implement the `Copy` trait
3 |     let b = a;
  |             - value moved here
4 |     println!("{a}");
  |                ^ value borrowed here after move
  |
help: consider cloning the value if the performance cost is acceptable
  |
3 |     let b = a.clone();
  |              ++++++++
```

Why? If both `a` and `b` owned the same text, both would free it at the end — a crash in C. Rust prevents it by having only one owner: after `let b = a;`, `b` owns it, and `a` is empty.

```
let a = String::from("hello");        let b = a;

  a ──────► [h e l l o]                    a   (moved: unusable)
                                         b ──────► [h e l l o]
```

Using `b` is fine:

```
fn main() {
    let a = String::from("hello");
    let b = a;
    println!("{b}");
}
```

You should see:

```
hello
```

有 51.4 Copying: small simple values

Numbers, `bool`, `char` — and tuples and arrays of them — live entirely on the **stack**: they are small and of fixed size, so copying is as cheap as moving. They are **Copy** types: `=` makes a copy and both names stay valid:

```
fn main() {
    let x = 5;
    let y = x;
    let p = (1.5, true);
    let q = p;
    println!("{x} {y} {:?} {:?}", p, q);
}
```

You should see:

```
5 5 (1.5, true) (1.5, true)
```

Moves	Copies
`String`, `Vec`, `HashMap`, `Box` — anything that owns heap memory	`i32`, `f64`, `bool`, `char`, `usize` …
your structs, unless you derive `Copy`	tuples and arrays of Copy types
	`&T` — a borrow (chapter 52)

有 51.5 Cloning: a real copy, on purpose

When you do want two separate copies of heap data, ask for it with `.clone()`. It is explicit because it costs time and memory — you see every place it happens:

```
fn main() {
    let a = String::from("hello");
    let mut b = a.clone();
    b.push_str(" world");
    println!("{a} | {b}");
}
```

You should see:

```
hello | hello world
```

有 51.6 Ownership and functions

Passing a value to a function **moves** it into the function, just like `let b = a;`. When the function ends, the value is dropped:

```
fn shout(s: String) {
    println!("{}!", s.to_uppercase());
}

fn main() {
    let word = String::from("hello");
    shout(word);
    shout(word);
}
```

The compiler says:

```
error[E0382]: use of moved value: `word`
 --> src/main.rs:8:11
  |
6 |     let word = String::from("hello");
  |         ---- move occurs because `word` has type `String`, which does not implement the `Copy` trait
7 |     shout(word);
  |           ---- value moved here
8 |     shout(word);
  |           ^^^^ value used here after move
  |
note: consider changing this parameter type in function `shout` to borrow instead if owning the value isn't
necessary
 --> src/main.rs:1:13
  |
1 | fn shout(s: String) {
  |    -----    ^^^^^^ this parameter takes ownership of the value
  |    |
  |    in this function
help: consider cloning the value if the performance cost is acceptable
  |
7 |     shout(word.clone());
  |               ++++++++
```

A function can give ownership back by returning the value:

```
fn add_bang(mut s: String) -> String {
    s.push('!');
    s
}

fn main() {
    let word = String::from("hello");
    let word = add_bang(word);
    let word = add_bang(word);
    println!("{word}");
}
```

You should see:

```
hello!!
```

Passing in and handing back works, but is clumsy. Usually a function only needs to *look at* a value, or change it, without owning it. That is **borrowing** — the next chapter.

Copy types are copied into functions, so the caller keeps its own:

```
fn double(n: i32) -> i32 {
    n * 2
}

fn main() {
    let n = 21;
    println!("{} {n}", double(n));
}
```

You should see:

```
42 21
```

有 51.7 Moves in loops and collections

A `for` loop over a `Vec` (chapter 55's list) takes ownership of it, item by item:

```
fn main() {
    let names = vec![String::from("Sara"), String::from("Omar")];
    for n in names {
        println!("{n}");
    }
    println!("{}", names.len());
}
```

The compiler says:

```
error[E0382]: borrow of moved value: `names`
 --> src/main.rs:6:20
  |
2 |     let names = vec![String::from("Sara"), String::from("Omar")];
  |         ----- move occurs because `names` has type `Vec<String>`, which does not implement the `Copy`
trait
3 |     for n in names {
  |              ----- `names` moved due to this implicit call to `.into_iter()`
...
6 |     println!("{}", names.len());
  |                    ^^^^^ value borrowed here after move
  |
note: `into_iter` takes ownership of the receiver `self`, which moves `names`
```

Loop over `&names` — borrowing it — and you keep it:

```
fn main() {
    let names = vec![String::from("Sara"), String::from("Omar")];
    for n in &names {
        println!("{n}");
    }
    println!("{}", names.len());
}
```

You should see:

```
Sara
Omar
2
```

Taking a `String` *out* of a `Vec` by index would leave a hole, so Rust refuses; clone it, or borrow it:

```
fn main() {
    let names = vec![String::from("Sara")];
    let first = names[0];
}
```

The compiler says:

```
error[E0507]: cannot move out of index of `Vec<String>`
 --> src/main.rs:3:17
  |
3 |     let first = names[0];
  |                 ^^^^^^^^ move occurs because value has type `String`, which does not implement the
`Copy` trait
  |
help: consider borrowing here
  |
3 |     let first = &names[0];
  |                 +
help: consider cloning the value if the performance cost is acceptable
  |
3 |     let first = names[0].clone();
  |                         ++++++++
```

有 51.8 A partial move

Moving one field out of a struct (chapter 53) or tuple moves only that part; the rest stays usable:

```
fn main() {
    let pair = (String::from("left"), String::from("right"));
    let l = pair.0;
    println!("{l} {}", pair.1);
}
```

You should see:

```
left right
```

有 51.9 Common mistakes

1. `borrow of moved value` (E0382). You moved a value, then used the old name. Choose: borrow instead (`&x`, next chapter) — usually right; or `.clone()` if you truly need two.

2. Cloning everything to make the errors go away. It works, but costs memory and hides the design. Ask first: does this function need to *own* the value, or only look at it?

3. Expecting `let b = a;` to copy a String. It moves. Only Copy types are copied.

Exercises

51.1 ★ Which compile?. For each, say whether it compiles, then check: (a) `let a = 5; let b = a; println!("{a}");` (b) `let a = String::from("x"); let b = a; println!("{a}");` (c) the same with `a.clone()`.

51.2 ★ Fix with clone. Make this compile by changing one line: `let s = String::from("hi"); let t = s; println!("{s} {t}");`

51.3 ★ Give it back. `fn exclaim(s: String) -> String` adds `"!!"`; call it twice on the same text, keeping ownership.

51.4 ★★ The drop order. Make three `Noisy` values in `main`, one inside a block, and predict the order of the drop lines before you run it.

51.5 ★★ Drop early. `drop(value)` frees a value before the end of its scope. Show it with `Noisy`: the drop line comes before `end`.

51.6 ★★ Count the clones. Write `fn longest(a: String, b: String) -> String` returning the longer. Call it so that you can still print both originals afterwards — how many `clone`s does that take? (Two. Chapter 52 gets it down to none.)

51.7 ★★★ Move into a struct. `struct User { name: String }`. Make a `String`, move it into a `User`, then print the user's name — and show that the original variable can no longer be used (comment it out after reading the error).

51.8 ★★★ Swap ownership. With `std::mem::swap(&mut a, &mut b)`, swap two `String`s without cloning; then with `std::mem::take(&mut s)` take a string out of a variable, leaving an empty one.

51.9 ★★★★ Ownership on paper. For this program, write down at every line who owns each `String`, and where each is dropped; then add `Noisy` prints to check: `let a = make("a"); let b = make("b"); let c = pick(a, b); let d = make("d"); drop(c);` with `fn pick(x, y) -> returns x` (so `y` is dropped inside `pick`).

借 52. Rust level 2: borrowing and slices

In this chapter

- references: look at a value without owning it (`&`)
- change a value without owning it (`&mut`)
- the borrowing rule: many readers or one writer
- slices: borrow part of a string or an array

You need: chapter 51.

借 52.1 References: borrowing to look

A **reference** `&value` lets a function use a value while the caller keeps owning it. This is called **borrowing**: like lending a book — the friend reads it and gives it back; it was always yours.

```
fn length(s: &String) -> usize {
    s.len()
}

fn main() {
    let word = String::from("fedokawaii");
    let n = length(&word);
    println!("{word} has {n} letters");
}
```

You should see:

```
fedokawaii has 10 letters
```

Write	Means
`&word`	borrow `word`: make a reference to it
`s: &String`	`s` is a reference to a String, not a String
when `length` ends	the reference goes; `word` was never moved, nothing is dropped

Chapter 51's clumsy "pass it in and hand it back" becomes simply:

```
fn shout(s: &str) {
    println!("{}!", s.to_uppercase());
}

fn main() {
    let word = String::from("hello");
    shout(&word);
    shout(&word);
    println!("{word} is still mine");
}
```

You should see:

```
HELLO!
HELLO!
hello is still mine
```

(a `&String` is accepted wherever a `&str` is asked, so functions that only read text take `&str`: they then work with literals too — `shout("hi")`)

借 52.2 Mutable references: borrowing to change

A plain reference is read-only. To change the borrowed value, borrow it with `&mut` — and the owner must be `mut` too:

```
fn add_bang(s: &mut String) {
    s.push('!');
}

fn main() {
    let mut word = String::from("hello");
    add_bang(&mut word);
    add_bang(&mut word);
    println!("{word}");
}
```

You should see:

```
hello!!
```

```
fn add_bang(s: &String) {
    s.push('!');
}

fn main() {
    let word = String::from("hello");
    add_bang(&word);
}
```

The compiler says:

```
error[E0596]: cannot borrow `*s` as mutable, as it is behind a `&` reference
 --> src/main.rs:2:5
  |
2 |     s.push('!');
  |     ^ `s` is a `&` reference, so it cannot be borrowed as mutable
  |
help: consider changing this to be a mutable reference
  |
1 | fn add_bang(s: &mut String) {
  |                +++
```

For numbers, `*` reaches through the reference to the value (*dereferencing*):

```
fn bump(n: &mut i32) {
    *n += 1;
}

fn main() {
    let mut count = 0;
    bump(&mut count);
    bump(&mut count);
    println!("{count}");
}
```

You should see:

```
2
```

借 52.3 The rule: many readers, or one writer

At any moment, a value can have **either** any number of `&` references, **or** exactly one `&mut` — never both. Like a document: many people may read it at once, or one person may edit it, but nobody reads while someone is halfway through an edit.

```
fn main() {
    let s = String::from("hi");
    let r1 = &s;
    let r2 = &s;
    println!("{r1} {r2}");
}
```

You should see:

```
hi hi
```

```
fn main() {
    let mut s = String::from("hi");
    let r1 = &mut s;
    let r2 = &mut s;
    println!("{r1} {r2}");
}
```

The compiler says:

```
error[E0499]: cannot borrow `s` as mutable more than once at a time
 --> src/main.rs:4:14
  |
3 |     let r1 = &mut s;
  |              ------ first mutable borrow occurs here
4 |     let r2 = &mut s;
  |              ^^^^^^ second mutable borrow occurs here
5 |     println!("{r1} {r2}");
  |                -- first borrow later used here
```

```
fn main() {
    let mut s = String::from("hi");
    let reader = &s;
    s.push('!');
    println!("{reader}");
}
```

The compiler says:

```
error[E0502]: cannot borrow `s` as mutable because it is also borrowed as immutable
 --> src/main.rs:4:5
  |
3 |     let reader = &s;
  |                  -- immutable borrow occurs here
4 |     s.push('!');
  |     ^^^^^^^^^^ mutable borrow occurs here
5 |     println!("{reader}");
  |                ------ immutable borrow later used here
```

This one rule is how Rust prevents a whole class of bugs — a list changed while someone loops over it, two threads writing the same value (chapter 61) — at compile time.

借 52.4 A borrow ends at its last use

A reference counts as "alive" only up to the last line that uses it — not to the end of the block. So this compiles: the readers are done before the writer starts.

```
fn main() {
    let mut s = String::from("hi");
    let r1 = &s;
    let r2 = &s;
    println!("{r1} {r2}");
    let w = &mut s;
    w.push('!');
    println!("{s}");
}
```

You should see:

```
hi hi
hi!
```

The classic trap — growing a list while holding a reference into it:

```
fn main() {
    let mut v = vec![1, 2, 3];
    let first = &v[0];
    v.push(4);
    println!("{first}");
}
```

The compiler says:

```
error[E0502]: cannot borrow `v` as mutable because it is also borrowed as immutable
 --> src/main.rs:4:5
  |
3 |     let first = &v[0];
  |                  - immutable borrow occurs here
4 |     v.push(4);
  |     ^^^^^^^^^ mutable borrow occurs here
5 |     println!("{first}");
  |                ----- immutable borrow later used here
```

(why it matters: `push` *may move the whole list to a bigger place in memory, and* `first` *would point at the old, freed place. In C that is a crash; in Rust, a compile error)*

借 52.5 No dangling references

A reference can never outlive the value it points at. Returning a reference to a local variable — which is dropped as the function ends — is refused:

```
fn make() -> &String {
    let s = String::from("hi");
    &s
}

fn main() {
    let r = make();
}
```

The compiler says:

```
error[E0106]: missing lifetime specifier
 --> src/main.rs:1:14
  |
1 | fn make() -> &String {
  |              ^ expected named lifetime parameter
  |
  = help: this function's return type contains a borrowed value, but there is no value for it to be borrowed
from
help: consider using the `'static` lifetime, but this is uncommon unless you're returning a borrowed value
from a `const` or a `static`
  |
1 | fn make() -> &'static String {
  |               +++++++
help: instead, you are more likely to want to return an owned value
  |
1 - fn make() -> &String {
1 + fn make() -> String {
  |
```

The fix is to return the `String` itself — moving ownership out:

```
fn make() -> String {
    String::from("hi")
}

fn main() {
    let s = make();
    println!("{s}");
}
```

You should see:

```
hi
```

借 52.6 Slices: borrowing a part

A **slice** borrows a stretch of a string or array: `&s[start..end]`. The string slice type is `&str` — which is what text literals have been all along:

```
fn main() {
    let s = String::from("hello world");
    let hello: &str = &s[0..5];
    let world = &s[6..];
    println!("[{hello}] [{world}]");

    let nums = [10, 20, 30, 40, 50];
    let middle: &[i32] = &nums[1..4];
    println!("{:?} {}", middle, middle.len());
}
```

You should see:

```
[hello] [world]
[20, 30, 40] 3
```

A function that takes `&[i32]` works with arrays, `Vec`s and parts of either:

```
fn sum(xs: &[i32]) -> i32 {
    let mut total = 0;
    for x in xs {
        total += x;
    }
    total
}

fn main() {
    let arr = [1, 2, 3, 4];
    let v = vec![10, 20, 30];
    println!("{} {} {}", sum(&arr), sum(&v), sum(&arr[..2]));
}
```

You should see:

```
10 60 3
```

The first word of a sentence, returned as a slice of it — no copying:

```
fn first_word(s: &str) -> &str {
    match s.find(' ') {
        Some(i) => &s[..i],
        None => s,
    }
}

fn main() {
    let text = String::from("borrow checker");
    println!("{}", first_word(&text));
    println!("{}", first_word("single"));
}
```

You should see:

```
borrow
single
```

Careful. String slices count **bytes**, not letters. Non-English letters take 2–4 bytes; slicing through the middle of one panics. For letters, use `.chars()`.

```
fn main() {
    let s = "京都";
    println!("{} bytes, {} letters", s.len(), s.chars().count());
    println!("{}", &s[0..1]);
}
```

You should see:

```
6 bytes, 2 letters

thread 'main' (1) panicked at src/main.rs:4:22:
end byte index 1 is not a char boundary; it is inside '京' (bytes 0..3 of string)
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace
```

借 52.7 Common mistakes

1. `cannot borrow as mutable more than once` (E0499) or `… as mutable because it is also borrowed as immutable` (E0502). Finish with one borrow before starting the other: often just reorder lines, or put one use in its own `{ }` block.

2. Taking `&String` or `&Vec<T>` parameters. Take `&str` and `&[T]`: they accept more.

3. Returning a reference to something made inside the function. Return the owned value.

Exercises

52.1 ★ Borrow to measure. `fn vowels(s: &str) -> usize`, called twice on the same `String`, which is printed after.

52.2 ★ Change through &mut. `fn double_all(v: &mut [i32])` doubles every element in place (`for x in v.iter_mut() { *x *= 2 }`).

52.3 ★ Slice it. From `"2026-10-21"`, take the year, month and day as `&str` slices.

52.4 ★★ Fix the order. Make this compile by moving one line: `let mut v = vec![1]; let r = &v; v.push(2); println!("{:?}", r);` — so that it prints `[1]`.

52.5 ★★ Largest, by reference. `fn largest(xs: &[i32]) -> &i32` returning a reference to the largest element.

52.6 ★★ Longest without clones. Rewrite exercise 51.6's `longest` with `&str` parameters — no clones at all. (You will need `fn longest<'a>(a: &'a str, b: &'a str) -> &'a str`; chapter 58 explains the `'a`.)

52.7 ★★★ Split in two. `fn split_at_comma(s: &str) -> (&str, &str)` returning the parts before and after the first comma (trimmed), or `(s, "")` if there is none.

52.8 ★★★ Reverse in place. `fn reverse(v: &mut [i32])` with two indexes walking inwards and `v.swap(i, j)`.

52.9 ★★★★ Words, as slices. `fn words(s: &str) -> Vec<&str>` splitting on spaces by hand (walk the bytes with `s.char_indices()`, slice between spaces), without `split`. Check it against `s.split_whitespace().collect::<Vec<_>>()`.

描 53. Rust level 2: structs and methods

In this chapter

- structs: your own types, with named fields
- `impl`: methods and associated functions (`new`)
- `&self`, `&mut self`, `self`: borrowing applied to methods
- `#[derive(Debug, Clone, PartialEq)]`

You need: chapters 51 and 52.

描 53.1 A struct

A **struct** groups named values into one new type — Python's dataclass, with types on every field:

```
struct User {
    name: String,
    age: u32,
    active: bool,
}

fn main() {
    let mut u = User { name: String::from("Sara"), age: 31, active: true };
    u.age += 1;
    println!("{} is {} (active: {})", u.name, u.age, u.active);
}
```

You should see:

```
Sara is 32 (active: true)
```

(to change a field, the whole variable must be `mut`)

When a variable has the field's name, write it once — the **shorthand**; and `..other` fills the remaining fields from another value:

```
struct User {
    name: String,
    age: u32,
    active: bool,
}

fn main() {
    let name = String::from("Omar");
    let age = 28;
    let a = User { name, age, active: true };
    let b = User { name: String::from("Lina"), ..a };
    println!("{} {} {}", b.name, b.age, b.active);
}
```

You should see:

```
Lina 28 true
```

描 53.2 Printing a struct: derive Debug

`println!("{}", u)` does not work for your own types: Rust does not guess how to show them. Ask it to write the debug form for you with `#[derive(Debug)]`, then print with `{:?}` — or `{:#?}` for one field per line:

```
#[derive(Debug)]
struct Point {
    x: f64,
    y: f64,
}

fn main() {
    let p = Point { x: 1.5, y: -2.0 };
    println!("{:?}", p);
    println!("{:#?}", p);
}
```

You should see:

```
Point { x: 1.5, y: -2.0 }
Point {
    x: 1.5,
    y: -2.0,
}
```

描 53.3 Methods with impl

Functions that belong to a type go in an `impl` block. A **method** takes `self` — the value it is called on — as its first parameter, almost always borrowed:

```
#[derive(Debug)]
struct Rect {
    w: f64,
    h: f64,
}

impl Rect {
    fn area(&self) -> f64 {
        self.w * self.h
    }

    fn scale(&mut self, k: f64) {
        self.w *= k;
        self.h *= k;
    }

    fn fits_in(&self, other: &Rect) -> bool {
        self.w <= other.w && self.h <= other.h
    }
}

fn main() {
    let mut r = Rect { w: 3.0, h: 4.0 };
    println!("{}", r.area());
    r.scale(2.0);
    println!("{:?} {}", r, r.fits_in(&Rect { w: 10.0, h: 10.0 }));
}
```

You should see:

```
12
Rect { w: 6.0, h: 8.0 } true
```

First parameter	The method	Use for
`&self`	borrows the value, to read	most methods: `area`, `len`, `name`
`&mut self`	borrows it, to change	`scale`, `push`, `deposit`
`self`	takes ownership: the value is used up	turning it into something else: `into_string`

描 53.4 Associated functions: new

A function in `impl` *without* `self` belongs to the type, not a value — called with `::`. The habit is a `new` that builds one, checking what must be true:

```
#[derive(Debug)]
struct Temperature {
    celsius: f64,
}

impl Temperature {
    fn new(celsius: f64) -> Temperature {
        assert!(celsius >= -273.15, "below absolute zero");
        Temperature { celsius }
    }

    fn from_fahrenheit(f: f64) -> Self {
        Self::new((f - 32.0) * 5.0 / 9.0)
    }

    fn fahrenheit(&self) -> f64 {
        self.celsius * 9.0 / 5.0 + 32.0
    }
}

fn main() {
    let t = Temperature::new(31.0);
    let u = Temperature::from_fahrenheit(212.0);
    println!("{:.1} {:?}", t.fahrenheit(), u);
}
```

You should see:

```
87.8 Temperature { celsius: 100.0 }
```

(`Self` inside an `impl` means the type itself, here `Temperature`. `String::from` and `Vec::new` are associated functions too)

A struct can have unnamed fields — a **tuple struct**, reached as `.0`, `.1`. It is useful to give a plain number a meaning the compiler checks, so metres are never added to seconds by mistake:

```
#[derive(Debug, Clone, Copy)]
struct Metres(f64);
#[derive(Debug, Clone, Copy)]
struct Seconds(f64);

fn speed(d: Metres, t: Seconds) -> f64 {
    d.0 / t.0
}

fn main() {
    let d = Metres(100.0);
    let t = Seconds(9.58);
    println!("{:.2} m/s", speed(d, t));
}
```

You should see:

```
10.44 m/s
```

```
#[derive(Debug, Clone, Copy)]
struct Metres(f64);
#[derive(Debug, Clone, Copy)]
struct Seconds(f64);

fn speed(d: Metres, t: Seconds) -> f64 {
    d.0 / t.0
}

fn main() {
    println!("{}", speed(Seconds(9.58), Metres(100.0)));
}
```

The compiler says:

```
error[E0308]: arguments to this function are incorrect
  --> src/main.rs:11:20
   |
11 |     println!("{}", speed(Seconds(9.58), Metres(100.0)));
   |                     ^^^^^ -------------  ------------- expected `Seconds`, found `Metres`
   |                           |
   |                           expected `Metres`, found `Seconds`
   |
note: function defined here
  --> src/main.rs:6:4
   |
 6 | fn speed(d: Metres, t: Seconds) -> f64 {
   |    ^^^^^
help: swap these arguments
   |
11 -     println!("{}", speed(Seconds(9.58), Metres(100.0)));
11 +     println!("{}", speed(Metres(100.0), Seconds(9.58)));
   |
```

描 53.6 Deriving more

`#[derive(...)]` writes common traits (chapter 57) for you:

Derive	Gives
`Debug`	`{:?}` printing
`Clone`	`.clone()`
`Copy`	copies instead of moves (only if every field is Copy)
`PartialEq`	`==` and `!=`, field by field
`PartialOrd`	`<`, `>` — field by field, in order
`Default`	`Type::default()`: zeros, empty strings, `false`
`Hash`, `Eq`	use as a `HashMap` key or in a `HashSet`

```
#[derive(Debug, Clone, PartialEq, Default)]
struct Settings {
    theme: String,
    font_size: u32,
    beta: bool,
}

fn main() {
    let a = Settings { theme: "dark".into(), font_size: 12, ..Default::default() };
    let mut b = a.clone();
    println!("{}", a == b);
    b.font_size = 14;
    println!("{} {:?}", a == b, Settings::default());
}
```

You should see:

```
true
false Settings { theme: "", font_size: 0, beta: false }
```

(`"dark".into()` turns the `&str` into the `String` the field wants: `.into()` converts to whatever type is needed there)

描 53.7 Common mistakes

1. Forgetting `&` on `self`. `fn area(self)` takes ownership: after `r.area()`, `r` is gone. Read-only methods take `&self`.

2. Changing through `&self`. `cannot assign to self.w, which is behind a & reference` — the method needs `&mut self`, and the variable must be `mut`.

```
struct Counter {
    n: u32,
}

impl Counter {
    fn click(&self) {
        self.n += 1;
    }
}

fn main() {
    let c = Counter { n: 0 };
    c.click();
}
```

The compiler says:

```
error[E0594]: cannot assign to `self.n`, which is behind a `&` reference
 --> src/main.rs:7:9
  |
7 |         self.n += 1;
  |         ^^^^^^^^^^^ `self` is a `&` reference, so it cannot be written to
  |
help: consider changing this to be a mutable reference
  |
6 |     fn click(&mut self) {
  |               +++
```

3. `{}` for a struct. `doesn't implement std::fmt::Display` — use `{:?}` with `#[derive(Debug)]`, or write `Display` yourself (chapter 57).

Exercises

53.1 ★ A book. `struct Book { title: String, year: u16 }` with `#[derive(Debug)]`; make two and print them with `{:?}`.

53.2 ★ Circle. `struct Circle { r: f64 }` with `area` and `circumference` methods.

53.3 ★ A counter. `Counter` with `new()`, `click(&mut self)`, `reset(&mut self)` and `value(&self)`.

53.4 ★★ Account. `Account { owner: String, balance: i64 }` (in fils) with `deposit`, and `withdraw` returning `bool` (false, and no change, if too little money).

53.5 ★★ Vector maths. `#[derive(Debug, Clone, Copy, PartialEq)] struct V2 { x: f64, y: f64 }` with `add`, `scale`, `dot` and `length`.

53.6 ★★ Newtypes. `struct Fils(i64)` and `struct Dirham(f64)`, with `fn to_dirham(f: Fils) -> Dirham`. Show that passing a `Dirham` where `Fils` is wanted does not compile.

53.7 ★★★ A stack of plates. `struct Stack { items: Vec<String> }` with `push`, `pop -> Option<String>`, `peek -> Option<&String>`, `len`, `is_empty` — each method with the right kind of `self`.

53.8 ★★★ Builder. A `Request` with `url`, `method` (default `"GET"`) and `timeout` (default 30): build it with methods that take and return `self` — `Request::new(url).method("POST").timeout(5)`.

53.9 ★★★★ A matrix. `struct Matrix { rows: usize, cols: usize, data: Vec<f64> }` (row after row in one `Vec`) with `new(rows, cols)` (zeros), `get(r, c)`, `set(r, c, v)`, `transpose()`, and `matmul(&self, other)` that panics with a clear message when the sizes do not fit. Check a 2 × 3 times 3 × 2.



54. Rust level 2: enums and match

In this chapter

- enums: a value that is one of several kinds, each with its own data
- `match`: handle every kind, checked by the compiler
- `Option`: Rust's answer to "maybe nothing" — no null, ever
- `if let`, `let else`, and methods on enums
- the checkpoint for level 2: a bank with commands

You need: chapter 53.



54.1 An enum

An **enum** (*enumeration*) is a type whose value is exactly one of a list of **variants**:

```
#[derive(Debug, Clone, Copy, PartialEq)]
enum Direction {
    North,
    East,
    South,
    West,
}

fn turn_right(d: Direction) -> Direction {
    match d {
        Direction::North => Direction::East,
        Direction::East => Direction::South,
        Direction::South => Direction::West,
        Direction::West => Direction::North,
    }
}

fn main() {
    let mut d = Direction::North;
    for _ in 0..5 {
        print!("{:?} → ", d);
        d = turn_right(d);
    }
    println!("{:?}", d);
}
```

You should see:

```
North → East → South → West → North → East
```

54.2 Variants with data

Each variant can carry its own data — different for each. This is what makes Rust's enums so useful: one type, several shapes:

```
#[derive(Debug)]
enum Shape {
    Circle { r: f64 },
    Rect { w: f64, h: f64 },
    Triangle(f64, f64, f64),
}

fn area(s: &Shape) -> f64 {
    match s {
        Shape::Circle { r } => std::f64::consts::PI * r * r,
        Shape::Rect { w, h } => w * h,
        Shape::Triangle(a, b, c) => {
            let p = (a + b + c) / 2.0;
            (p * (p - a) * (p - b) * (p - c)).sqrt()
        }
    }
}

fn main() {
    let shapes = [Shape::Circle { r: 1.0 }, Shape::Rect { w: 2.0, h: 3.0 }, Shape::Triangle(3.0, 4.0, 5.0)];
    for s in &shapes {
        println!("{:?} → {:.2}", s, area(s));
    }
}
```

You should see:

```
Circle { r: 1.0 } → 3.14
Rect { w: 2.0, h: 3.0 } → 6.00
Triangle(3.0, 4.0, 5.0) → 6.00
```

(in the `match`, `Shape::Rect { w, h }` *takes the variant apart and names its fields — a pattern)*

54.3 match must cover everything

Add a variant, and every `match` that forgot it stops compiling — the compiler finds every place your program must learn about the new case:

```
enum Light {
    Red,
    Amber,
    Green,
}

fn wait(l: Light) -> u32 {
    match l {
        Light::Red => 30,
        Light::Green => 0,
    }
}

fn main() {}
```

The compiler says:

```
error[E0004]: non-exhaustive patterns: `Light::Amber` not covered
  --> src/main.rs:8:11
   |
 8 |     match l {
   |           ^ pattern `Light::Amber` not covered
   |
note: `Light` defined here
  --> src/main.rs:1:6
   |
 1 | enum Light {
   |      ^^^^^
 2 |     Red,
 3 |     Amber,
   |     ----- not covered
   = note: the matched value is of type `Light`
help: ensure that all possible cases are being handled by adding a match arm with a wildcard pattern or an
explicit pattern as shown
   |
10 ~         Light::Green => 0,
11 ~         Light::Amber => todo!(),
   |
```



54.4 Patterns

`match` patterns can test values, ranges, several options, and add conditions (**guards**):

```
fn classify(p: (i32, i32)) -> String {
    match p {
        (0, 0) => "the origin".to_string(),
        (x, 0) | (0, x) => format!("on an axis at {x}"),
        (x, y) if x == y => format!("on the diagonal at {x}"),
        (x, y) if x > 0 && y > 0 => "top right".to_string(),
        _ => "somewhere else".to_string(),
    }
}

fn main() {
    for p in [(0, 0), (5, 0), (0, -2), (3, 3), (1, 7), (-1, 4)] {
        println!("{:?}: {}", p, classify(p));
    }
}
```

You should see:

```
(0, 0): the origin
(5, 0): on an axis at 5
(0, -2): on an axis at -2
(3, 3): on the diagonal at 3
(1, 7): top right
(-1, 4): somewhere else
```

(`format!` *is* `println!` *that gives back a* `String` *instead of printing it)*

54.5 Option: maybe a value

Rust has no `null` or `None` that can sneak in anywhere. A value that may be missing has the type `Option<T>`, an enum from the standard library:

```
enum Option<T> {
    Some(T),   // there is a value, of type T
    None,      // there is not
}
```

The compiler will not let you use an `Option<i32>` as an `i32`: you must say what happens when it is `None`. The commonest bug of other languages is gone.

```
fn find_index(names: &[&str], wanted: &str) -> Option<usize> {
    for (i, n) in names.iter().enumerate() {
        if *n == wanted {
            return Some(i);
        }
    }
    None
}

fn main() {
    let names = ["Sara", "Omar", "Lina"];
    for wanted in ["Omar", "Zaid"] {
        match find_index(&names, wanted) {
            Some(i) => println!("{wanted} is at {i}"),
            None => println!("no {wanted}"),
        }
    }
}
```

You should see:

```
Omar is at 1
no Zaid
```

```
fn main() {
    let maybe: Option<i32> = Some(5);
    let n: i32 = maybe + 1;
}
```

The compiler says:

```
error[E0369]: cannot add `{integer}` to `Option<i32>`
 --> src/main.rs:3:24
  |
3 |     let n: i32 = maybe + 1;
  |                  ----- ^ - {integer}
  |                  |
  |                  Option<i32>
  |
note: `Option<i32>` does not implement `Add<{integer}>`
```

The standard library returns `Option` wherever something may be missing:

```
fn main() {
    let v = vec![3, 9, 4];
    let empty: Vec<i32> = vec![];
    println!("{:?} {:?} {:?}", v.first(), empty.first(), v.get(10));
    println!("{:?} {:?}", "hello".find('l'), v.iter().max());
}
```

You should see:

```
Some(3) None None
Some(2) Some(9)
```



54.6 Working with Option

`match` always works; these methods are shorter for common cases:

```
fn main() {
    let some: Option<i32> = Some(4);
    let none: Option<i32> = None;
    println!("{} {}", some.unwrap_or(0), none.unwrap_or(0));
    println!("{:?} {:?}", some.map(|n| n * 10), none.map(|n| n * 10));
    println!("{} {}", some.is_some(), none.is_none());
    if let Some(n) = some {
        println!("got {n}");
    }
}
```

You should see:

```
4 0
Some(40) None
true true
got 4
```

Write	Does
`x.unwrap_or(d)`	the value, or `d` if None
`x.map(f)`	`Some(f(v))`, or None
`if let Some(v) = x { … }`	run the block only when there is a value
`let Some(v) = x else { return … };`	take the value, or leave the function
`x.unwrap()`	the value — or **panic** if None. Only when None truly cannot happen
`x.expect("why")`	the same, with your message in the panic

`let … else` keeps the happy path unindented:

```
fn initials(name: &str) -> String {
    let Some(first) = name.chars().next() else {
        return String::from("?");
    };
    first.to_uppercase().to_string()
}

fn main() {
    println!("{} {}", initials("sara"), initials(""));
}
```

You should see:

```
S ?
```

54.7 Methods on enums

`impl` works for enums as for structs:

```
#[derive(Debug)]
enum Command {
    Deposit(u64),
    Withdraw(u64),
    Balance,
    Quit,
}

impl Command {
    fn parse(line: &str) -> Option<Command> {
        let words: Vec<&str> = line.split_whitespace().collect();
        match words.as_slice() {
            ["deposit", n] => Some(Command::Deposit(n.parse().ok()?)),
            ["withdraw", n] => Some(Command::Withdraw(n.parse().ok()?)),
            ["balance"] => Some(Command::Balance),
            ["quit"] => Some(Command::Quit),
            _ => None,
        }
    }
}

fn main() {
    for line in ["deposit 50", "withdraw x", "balance", "fly away"] {
        println!("{line:<12} → {:?}", Command::parse(line));
    }
}
```

You should see:

```
deposit 50   → Some(Deposit(50))
withdraw x   → None
balance      → Some(Balance)
fly away     → None
```

(`.ok()?` — if the number does not parse, return `None` from `parse` at once. Chapter 56 explains `?`; matching on `words.as_slice()` matches the list of words against shapes)

54.8 Common mistakes

1. `non-exhaustive patterns` — a variant is missing from a `match`. Handle it; reach for `_` only when "everything else" truly is one case.

2. `unwrap()` **everywhere.** Each is a possible crash. Use `match`, `if let`, `unwrap_or`, or `?`.

3. Forgetting the enum's name. `North` alone is not found; write `Direction::North` (or `use Direction::*;` to bring the names in).

54.9 Checkpoint project — a bank with commands

Level 2 in one program: a struct (the account), an enum (the commands), `match`, `Option`, and borrowing (`&mut self`). The commands come from a list here so the program runs by itself; chapter 50's checkpoint showed how to read them from the keyboard instead.

```
#[derive(Debug)]
enum Command {
    Deposit(u64),
    Withdraw(u64),
    Balance,
    History,
}

fn parse(line: &str) -> Option<Command> {
    let mut words = line.split_whitespace();
    let cmd = words.next()?;
    let amount = words.next().and_then(|w| w.parse::<u64>().ok());
    match (cmd, amount) {
        ("deposit", Some(n)) => Some(Command::Deposit(n)),
        ("withdraw", Some(n)) => Some(Command::Withdraw(n)),
        ("balance", None) => Some(Command::Balance),
        ("history", None) => Some(Command::History),
        _ => None,
    }
}

struct Account {
    balance: u64,
    history: Vec<String>,
}

impl Account {
    fn run(&mut self, c: Command) -> String {
        match c {
            Command::Deposit(n) => {
                self.balance += n;
                self.history.push(format!("+{n}"));
                format!("deposited {n}")
            }
            Command::Withdraw(n) if n > self.balance => format!("refused: only {} left", self.balance),
            Command::Withdraw(n) => {
                self.balance -= n;
                self.history.push(format!("-{n}"));
                format!("withdrew {n}")
            }
            Command::Balance => format!("balance {}", self.balance),
            Command::History => self.history.join(" "),
        }
    }
}

fn main() {
    let mut acc = Account { balance: 0, history: Vec::new() };
    for line in ["deposit 100", "withdraw 30", "withdraw 500", "lend 5", "balance", "history"] {
        let reply = match parse(line) {
            Some(cmd) => acc.run(cmd),
            None => String::from("unknown command"),
        };
        println!("> {line:<14} {reply}");
    }
}
```

You should see:

```
> deposit 100    deposited 100
> withdraw 30    withdrew 30
> withdraw 500   refused: only 70 left
> lend 5         unknown command
> balance        balance 70
> history        +100 -30
```

`Checkpoint.` Add (1) a `Transfer(u64, String)` command between named accounts — keep them in a `Vec<(String, Account)>` for now; (2) an `Undo` that reverses the last deposit or withdrawal; (3) reading commands from the keyboard until `quit`. Level 2 done: you understand ownership, the part of Rust that is truly new.

Exercises

54.1 ★ Weekday. `enum Day { Mon, ⋯ Sun }` and `fn is_weekend(d: Day) -> bool` with one `match` arm using `|`.

54.2 ★ Safe division. `fn divide(a: f64, b: f64) -> Option<f64>`, `None` for `b == 0.0`; print two results with `match`.

54.3 ★ unwrap_or. Print the first element of an empty `Vec<i32>`, or `-1`, in one line.

54.4 ★★ Coins. `enum Coin { Fils25, Fils50, Dirham }` with a `value_in_fils` method; total a list of coins.

54.5 ★★ Traffic light. `enum Light { Red, Amber, Green }` with `next(self) -> Light` and `seconds(&self) -> u32`; step through six changes.

54.6 ★★ Parse a grade. `fn grade(score: u32) -> Option<char>`: `None` above 100; `A`/`B`/`C`/`F` with range patterns (`90..=100 => ⋯`).

54.7 ★★★ A calculator enum. `enum Expr { Num(f64), Add(Box<Expr>, Box<Expr>), Mul(Box<Expr>, Box<Expr>) }` and a recursive `fn eval(e: &Expr) -> f64`. Evaluate `(2 + 3) × 4`. (`Box` puts a value on the heap, so an enum can contain itself.)

54.8 ★★★ Find a user. `struct User { id: u32, name: String }`; `fn find(users: &[User], id: u32) -> Option<&User>`; print the name for id 2, or `nobody` — with `map` and `unwrap_or`.

54.9 ★★★★ A tiny state machine. A vending machine: `enum State { Idle, HasCoins(u32), Dispensing }` and `enum Event { Coin(u32), Choose(u32), Cancel }`, with `fn step(s: State, e: Event) -> (State, String)` that accepts coins, dispenses when enough money was put in for the chosen price (giving change), refunds on `Cancel`, and refuses events that do not fit the state. Run a sequence of events and print each transition.

集 55. Rust level 3: collections

In this chapter

- `Vec<T>`: a list that grows
- `String` in depth: bytes, characters, building text
- `HashMap` and `HashSet`: look up by key, keep unique values
- `BTreeMap` for sorted keys, `VecDeque` for queues

You need: level 2 (chapters 51–54).

集 55.1 Vec: a growing list

`Vec<T>` is Rust's list: values of one type `T`, in order, on the heap, growing as needed. `vec![…]` makes one with values already in it:

```
fn main() {
    let mut v: Vec<i32> = Vec::new();
    v.push(3);
    v.push(1);
    v.push(2);
    let w = vec![10, 20, 30];
    println!("{:?} {} {:?}", v, v.len(), w);
    println!("{} {:?} {:?}", w[0], w.get(1), w.get(99));
    v.sort();
    let last = v.pop();
    println!("{:?} {:?} {}", v, last, v.contains(&1));
}
```

You should see:

```
[3, 1, 2] 3 [10, 20, 30]
10 Some(20) None
[1, 2] Some(3) true
```

Method	Does
`push(x)` · `pop()`	add at the end · take from the end (`Option`)
`v[i]` · `v.get(i)`	the item (panics past the end) · `Option<&T>` (never panics)
`insert(i, x)` · `remove(i)`	put at / take from position `i`
`len()` · `is_empty()`	size
`sort()` · `sort_by_key(f)` · `reverse()` · `dedup()`	in place
`contains(&x)` · `iter().position(…)`	search
`extend(other)` · `retain(f)` · `truncate(n)` · `clear()`	bulk changes

Looping — three ways, by what you need:

```
fn main() {
    let mut scores = vec![72, 95, 61];
    for s in &scores {
        print!("{s} ");
    }
    println!();
    for s in &mut scores {
        *s += 5;
    }
    println!("{:?}", scores);
    for s in scores {
        print!("{s} ");
    }
    println!("(scores is gone now)");
}
```

You should see:

```
72 95 61
[77, 100, 66]
77 100 66 (scores is gone now)
```

Loop over	Each item is	The Vec afterwards
`&v`	`&T`: read	still yours
`&mut v`	`&mut T`: change with `*`	still yours, changed
`v`	`T`: owned	moved: gone

集 55.2 Vec of structs, sorting

```
#[derive(Debug)]
struct Player {
    name: String,
    score: u32,
}

fn main() {
    let mut players = vec![
        Player { name: "Sara".into(), score: 88 },
        Player { name: "Omar".into(), score: 95 },
        Player { name: "Lina".into(), score: 79 },
    ];
    players.sort_by_key(|p| std::cmp::Reverse(p.score));
    for (rank, p) in players.iter().enumerate() {
        println!("{}. {:<5} {}", rank + 1, p.name, p.score);
    }
    players.retain(|p| p.score >= 80);
    println!("{} kept", players.len());
}
```

You should see:

```
1. Omar  95
2. Sara  88
3. Lina  79
2 kept
```

Floats cannot use `sort` (a `NaN` has no order); use `sort_by` with `total_cmp`:

```
fn main() {
    let mut xs: Vec<f64> = vec![2.5, -1.0, 9.75, 0.0];
    xs.sort_by(|a, b| a.total_cmp(b));
    println!("{:?}", xs);
}
```

You should see:

```
[-1.0, 0.0, 2.5, 9.75]
```

集 55.3 String in depth

A `String` is a `Vec` of bytes that are always valid UTF-8 text. Build it with `push`, `push_str`, `+`, or — usually clearest — `format!`:

```
fn main() {
    let mut s = String::from("tea");
    s.push(',');
    s.push_str(" rice");
    let t = s.clone() + " and milk";
    let u = format!("{t} — {} items", 3);
    println!("{s}\n{t}\n{u}");
    println!("{}", u.replace("tea", "coffee").to_uppercase());
}
```

You should see:

```
tea, rice
tea, rice and milk
tea, rice and milk — 3 items
COFFEE, RICE AND MILK — 3 ITEMS
```

`len()` counts **bytes**. Count and walk letters with `.chars()`:

```
fn main() {
    let s = "Kyoto 京都";
    println!("{} bytes, {} chars", s.len(), s.chars().count());
    for c in s.chars().rev() {
        print!("{c}");
    }
    println!();
    let first_three: String = s.chars().take(3).collect();
    println!("{first_three}");
}
```

You should see:

```
12 bytes, 8 chars
都京 otoyK
Kyo
```

Splitting, trimming, testing:

```
fn main() {
    let line = "  name = Sara  ";
    let parts: Vec<&str> = line.split('=').map(|p| p.trim()).collect();
    println!("{:?}", parts);
    println!("{:?}", "a,b,,c".split(',').collect::<Vec<_>>());
    println!("{:?}", "one two  three".split_whitespace().collect::<Vec<_>>());
    println!("{} {}", "fedora".starts_with("fed"), "fedora".contains("dor"));
    println!("{:?}", "key: value".split_once(": "));
    let n: i32 = " 42 ".trim().parse().unwrap();
    println!("{}", n + 1);
}
```

You should see:

```
["name", "Sara"]
["a", "b", "", "c"]
["one", "two", "three"]
true true
Some(("key", "value"))
43
```

`&str` or `String`?	
a function's parameter that only reads text	`&str`
a struct's field that holds text	`String` (the struct owns it)
a return value of text you made	`String`
a return value that is part of the input	`&str` (a slice of it)

集 55.4 HashMap: look up by key

`HashMap<K, V>` maps keys to values (Python's `dict`). It lives in `std::collections`:

```
use std::collections::HashMap;

fn main() {
    let mut phone: HashMap<String, String> = HashMap::new();
    phone.insert("Sara".into(), "050 111".into());
    phone.insert("Omar".into(), "055 222".into());
    println!("{:?}", phone.get("Sara"));
    println!("{:?}", phone.get("Zaid"));
    println!("{} {}", phone.len(), phone.contains_key("Omar"));
    phone.remove("Omar");
    println!("{}", phone.len());
}
```

You should see:

```
Some("050 111")
None
2 true
1
```

The **entry** API is how you count and group — one lookup, inserting a starting value if the key is new:

```
use std::collections::HashMap;

fn main() {
    let text = "the cat and the dog and the bird";
    let mut counts: HashMap<&str, u32> = HashMap::new();
    for w in text.split_whitespace() {
        *counts.entry(w).or_insert(0) += 1;
    }
    let mut pairs: Vec<(&str, u32)> = counts.into_iter().collect();
    pairs.sort_by(|a, b| b.1.cmp(&a.1).then(a.0.cmp(b.0)));
    println!("{:?}", pairs);

    let mut by_letter: HashMap<char, Vec<&str>> = HashMap::new();
    for w in ["apple", "avocado", "banana"] {
        by_letter.entry(w.chars().next().unwrap()).or_default().push(w);
    }
    println!("{:?}", by_letter[&'a']);
}
```

You should see:

```
[("the", 3), ("and", 2), ("bird", 1), ("cat", 1), ("dog", 1)]
["apple", "avocado"]
```

(a `HashMap` keeps no order — printing it directly shows a different order each run. Collect into a `Vec` and sort, or use a `BTreeMap`)

集 55.5 BTreeMap: sorted keys

Same methods as `HashMap`, but the keys are kept sorted — so printing and looping give a stable order:

```
use std::collections::BTreeMap;

fn main() {
    let mut stock = BTreeMap::new();
    for (item, n) in [("tea", 3), ("milk", 1), ("rice", 2), ("tea", 4)] {
        *stock.entry(item).or_insert(0) += n;
    }
    println!("{:?}", stock);
    for (item, n) in &stock {
        println!("{item:<5}{n:>3}");
    }
    println!("{:?}", stock.first_key_value());
}
```

You should see:

```
{"milk": 1, "rice": 2, "tea": 7}
milk   1
rice   2
tea    7
Some(("milk", 1))
```

集 55.6 HashSet: unique values

```
use std::collections::HashSet;

fn main() {
    let visits = ["sara", "omar", "sara", "lina", "omar"];
    let unique: HashSet<&str> = visits.iter().copied().collect();
    println!("{}", unique.len());
    let a: HashSet<i32> = [1, 2, 3, 4].into();
    let b: HashSet<i32> = [3, 4, 5].into();
    let mut both: Vec<_> = a.intersection(&b).copied().collect();
    both.sort();
    let mut either: Vec<_> = a.union(&b).copied().collect();
    either.sort();
    println!("{:?} {:?} {}", both, either, a.contains(&2));
}
```

You should see:

```
3
[3, 4] [1, 2, 3, 4, 5] true
```

集 55.7 VecDeque: a queue

A `VecDeque` adds and removes at **both** ends quickly — a queue of jobs, a sliding window:

```
use std::collections::VecDeque;

fn main() {
    let mut queue = VecDeque::new();
    queue.push_back("first");
    queue.push_back("second");
    queue.push_front("urgent");
    while let Some(job) = queue.pop_front() {
        println!("doing {job}");
    }
}
```

You should see:

```
doing urgent
doing first
doing second
```

(`while let Some(x) =` ··· loops as long as the pattern matches: here, until the queue is empty)

集 55.8 Common mistakes

1. Indexing past the end panics: `v[10]` on a 3-item Vec. Use `v.get(10)` when the index might be outside.

2. Holding a reference while changing the collection — chapter 52's rule. Finish reading first, or collect what you need into a new value.

3. Slicing a `String` by letters. `&s[0..3]` is three *bytes*. Use `.chars()`.

4. Expecting order from a `HashMap`. Use `BTreeMap`, or sort.

Exercises

55.1 ★ Sum and average. A `Vec<f64>` of five prices: print the sum, the average, and the largest (with `fold` or a loop).

55.2 ★ Reverse the words. `"one two three"` → `"three two one"` with `split_whitespace`, `rev` and `join`.

55.3 ★ Capitals. A `HashMap` of four countries and capitals; print the capital of Japan, and `unknown` for Peru.

55.4 ★★ Letter frequency. Count each letter of `"mississippi"` into a `BTreeMap<char, usize>` and print it.

55.5 ★★ Remove duplicates, keep order. `fn unique(v: &[i32]) -> Vec<i32>` with a `HashSet` of seen values.

55.6 ★★ Palindrome. `fn is_palindrome(s: &str) -> bool`, ignoring case and anything that is not a letter. *Expected:* `"A man, a plan, a canal: Panama"` → `true`.

55.7 ★★★ Anagram groups. Group `["listen", "silent", "enlist", "google", "gooegl", "cat"]` by their sorted letters in a `BTreeMap<String, Vec<&str>>`.

55.8 ★★★ Moving average. With a `VecDeque<f64>` as a window of 3, print the moving average of `[1, 2, 3, 4, 5, 6]`. *Expected:* `2 3 4 5`.

55.9 ★★★★ An inventory with history. `struct Inventory { stock: BTreeMap<String, u32>, log: Vec<String> }` with `add(&mut self, item: &str, n: u32)`, `take(&mut self, item: &str, n: u32) -> bool` (refusing to go below zero, logging the refusal too), `low(&self, limit: u32) -> Vec<&str>`, and a report printing the stock and the log.

果 56. Rust level 3: errors — panic, Result and ?

In this chapter

- two kinds of failure: bugs (panic) and expected problems (`Result`)
- `Result<T, E>`: `Ok` or `Err`, handled with `match`
- the `?` operator: pass an error up in one character
- your own error type; `Box<dyn Error>` for quick programs
- the checkpoint for level 3: a word counter that never panics

You need: chapter 55.

果 56.1 Two kinds of failure

Kind	Example	In Rust
a **bug**: something that should be impossible	an index past the end, a broken promise	`panic!` — stop at once, with a message
an **expected problem**	a missing file, bad input, no network	`Result<T, E>` — a value saying what went wrong, for the caller to handle

Python uses exceptions for both. Rust keeps them apart, and the type of every function shows whether it can fail.

```
fn main() {
    let v = vec![1, 2, 3];
    let i = 7;
    println!("{}", v[i]);
}
```

You should see:

```
thread 'main' (1) panicked at src/main.rs:4:21:
index out of bounds: the len is 3 but the index is 7
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace
```

果 56.2 Result

`Result` is an enum, like `Option`:

```
enum Result<T, E> {
    Ok(T),    // it worked: here is the value
    Err(E),   // it failed: here is why
}
```

`"42".parse::<i32>()` returns a `Result`: the number, or a reason it is not one:

```
fn main() {
    for text in ["42", "4x2", ""] {
        match text.parse::<i32>() {
            Ok(n) => println!("{text:?} → {n}"),
            Err(e) => println!("{text:?} → error: {e}"),
        }
    }
}
```

You should see:

```
"42" → 42
"4x2" → error: invalid digit found in string
"" → error: cannot parse integer from empty string
```

Reading a file returns a `Result` too — the file may not exist:

```
use std::fs;

fn main() {
    fs::write("hello.txt", "first line\nsecond line\n").unwrap();
    match fs::read_to_string("hello.txt") {
        Ok(text) => println!("{} lines", text.lines().count()),
        Err(e) => println!("could not read: {e}"),
    }
    match fs::read_to_string("missing.txt") {
        Ok(text) => println!("{} lines", text.lines().count()),
        Err(e) => println!("could not read: {e}"),
    }
}
```

You should see:

```
2 lines
could not read: No such file or directory (os error 2)
```



56.3 Shortcuts: unwrap, expect, unwrap_or

Write	Does on Err	Use when
`match` / `if let`	whatever you write	always fine
`r.unwrap_or(d)` · `r.unwrap_or_default()`	gives `d` · zero, empty	a default makes sense
`r.expect("why")`	panics with your message	failure would be a bug: a file your program just wrote
`r.unwrap()`	panics	quick experiments, tests
`r?`	returns the error to the caller	inside a function that returns `Result` — the usual choice

```
fn main() {
    let a: i32 = "12".parse().unwrap_or(0);
    let b: i32 = "twelve".parse().unwrap_or(0);
    let c: i32 = "".parse().unwrap_or_default();
    println!("{a} {b} {c}");
}
```

You should see:

```
12 0 0
```

```
fn main() {
    let port: u16 = "70000".parse().expect("the port must be a number from 0 to 65535");
    println!("{port}");
}
```

You should see:

```
thread 'main' (1) panicked at src/main.rs:2:37:
the port must be a number from 0 to 65535: ParseIntError { kind: PosOverflow }
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace
```

果 56.4 The ? operator

Most functions cannot fix an error themselves; they pass it to their caller. `?` after a `Result` does exactly that: if it is `Ok`, take the value and go on; if it is `Err`, **return** the error from this function at once.

```
use std::fs;
use std::io;

fn count_lines(path: &str) -> Result<usize, io::Error> {
    let text = fs::read_to_string(path)?;
    Ok(text.lines().count())
}

fn main() {
    fs::write("notes.txt", "a\nb\nc\n").unwrap();
    println!("{:?}", count_lines("notes.txt"));
    match count_lines("nope.txt") {
        Ok(n) => println!("{n} lines"),
        Err(e) => println!("error: {e}"),
    }
}
```

You should see:

```
Ok(3)
error: No such file or directory (os error 2)
```

Without `?`, the same function is a `match` per call:

```
fn count_lines(path: &str) -> Result<usize, io::Error> {
    let text = match fs::read_to_string(path) {
        Ok(t) => t,
        Err(e) => return Err(e),
    };
    Ok(text.lines().count())
}
```

`?` works on `Option` too, in a function that returns `Option` (chapter 54's `.ok()?`).

果 56.5 Errors of different kinds: Box

A function that can fail in two ways — reading a file (`io::Error`) and parsing a number (`ParseIntError`) — needs one error type for both. The quick answer is `Box<dyn std::error::Error>`: *any* error. `?` converts each into it for you:

```
use std::error::Error;
use std::fs;

fn sum_file(path: &str) -> Result<i64, Box<dyn Error>> {
    let text = fs::read_to_string(path)?;
    let mut total = 0;
    for line in text.lines() {
        total += line.trim().parse::<i64>()?;
    }
    Ok(total)
}

fn main() {
    fs::write("good.txt", "10\n20\n30\n").unwrap();
    fs::write("bad.txt", "10\nX\n").unwrap();
    for f in ["good.txt", "bad.txt", "none.txt"] {
        match sum_file(f) {
            Ok(t) => println!("{f}: {t}"),
            Err(e) => println!("{f}: error: {e}"),
        }
    }
}
```

You should see:

```
good.txt: 60
bad.txt: error: invalid digit found in string
none.txt: error: No such file or directory (os error 2)
```

`main` itself may return a `Result`: an `Err` from `main` prints the error and exits with code 1.

```
use std::error::Error;

fn main() -> Result<(), Box<dyn Error>> {
    let n: i32 = "42".parse()?;
    println!("{n}");
    let m: i32 = "forty".parse()?;
    println!("{m}");
    Ok(())
}
```

You should see:

```
42
Error: ParseIntError { kind: InvalidDigit }
```

(the program printed 42, then stopped at the second `?` with `Error:` … and exit code 1. Only `Ok(())` — "fine, nothing to return" — lets it finish)

果 56.6 Your own error type

For a library, callers want to know *which* error happened, to handle each differently. An enum says it exactly:

```
use std::fmt;

#[derive(Debug, PartialEq)]
enum BankError {
    Insufficient { needed: u64, available: u64 },
    AccountFrozen,
}

impl fmt::Display for BankError {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        match self {
            BankError::Insufficient { needed, available } => {
                write!(f, "needed {needed}, only {available} available")
            }
            BankError::AccountFrozen => write!(f, "the account is frozen"),
        }
    }
}

impl std::error::Error for BankError {}

fn withdraw(balance: u64, amount: u64, frozen: bool) -> Result<u64, BankError> {
    if frozen {
        return Err(BankError::AccountFrozen);
    }
    if amount > balance {
        return Err(BankError::Insufficient { needed: amount, available: balance });
    }
    Ok(balance - amount)
}

fn main() {
    println!("{:?}", withdraw(100, 30, false));
    for (amount, frozen) in [(500, false), (10, true)] {
        if let Err(e) = withdraw(100, amount, frozen) {
            println!("refused: {e}");
        }
    }
}
```

You should see:

```
Ok(70)
refused: needed 500, only 100 available
refused: the account is frozen
```

(`Display` says how the error reads to a person; `impl Error` lets it go into `Box<dyn Error>`. The `thiserror` crate writes this code for you; `anyhow` is the crate for quick programs — chapter 62 uses it)

果 56.7 Converting between errors: map_err

`map_err` turns one error into another, for example adding which line was wrong:

```
fn parse_line(n: usize, line: &str) -> Result<f64, String> {
    line.trim().parse::<f64>().map_err(|e| format!("line {n}: {line:?}: {e}"))
}

fn main() {
    let lines = ["3.5", "2", "abc"];
    let parsed: Result<Vec<f64>, String> = lines.iter().enumerate().map(|(i, l)| parse_line(i + 1,
l)).collect();
    println!("{:?}", parsed);
    let good: Result<Vec<f64>, String> = ["1", "2"].iter().enumerate().map(|(i, l)| parse_line(i + 1,
l)).collect();
    println!("{:?}", good);
}
```

You should see:

```
Err("line 3: \"abc\": invalid float literal")
Ok([1.0, 2.0])
```

(collecting `Result`s into `Result<Vec<_>, _>` stops at the first error — or gives all the values)

果 56.8 Common mistakes

1. `?` in a function that returns nothing — `the ? operator can only be used in a function that returns Result or Option`. Change the return type (`main` too: `-> Result<(), Box<dyn Error>>`).

```
fn main() {
    let n: i32 = "5".parse()?;
}
```

The compiler says:

```
error[E0277]: the `?` operator can only be used in a function that returns `Result` or `Option` (or another
type that implements `FromResidual`)
 --> src/main.rs:2:29
  |
1 | fn main() {
  | --------- this function should return `Result` or `Option` to accept `?`
2 |     let n: i32 = "5".parse()?;
  |                             ^ cannot use the `?` operator in a function that returns `()`
  |
help: consider adding return type
  |
1 ~ fn main() -> Result<(), Box<dyn std::error::Error>> {
2 |     let n: i32 = "5".parse()?;
3 +     Ok(())
  |
```

2. `unwrap` on user input or files. That is a panic waiting for a typo. Return the error, or `match` it.

3. Forgetting `Ok(...)` around the final value of a function returning `Result`: `expected Result<usize, ⋯>, found usize`.

果 56.9 Checkpoint project — a word counter that never panics

Level 3 in one program: read files, count words in a `HashMap`, sort them, and handle every problem — a missing file, an empty file — with `Result` and `?`, never `unwrap` on outside data:

```
use std::collections::HashMap;
use std::error::Error;
use std::fmt;
use std::fs;

#[derive(Debug)]
enum CountError {
    Empty(String),
}

impl fmt::Display for CountError {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        match self {
            CountError::Empty(p) => write!(f, "{p} has no words"),
        }
    }
}

impl Error for CountError {}

fn top_words(path: &str, n: usize) -> Result<Vec<(String, usize)>, Box<dyn Error>> {
    let text = fs::read_to_string(path).map_err(|e| format!("{path}: {e}"))?;
    let mut counts: HashMap<String, usize> = HashMap::new();
    for word in text.split(|c: char| !c.is_alphanumeric() && c != '\'') {
        if !word.is_empty() {
            *counts.entry(word.to_lowercase()).or_insert(0) += 1;
        }
    }
    if counts.is_empty() {
        return Err(Box::new(CountError::Empty(path.to_string())));
    }
    let mut pairs: Vec<(String, usize)> = counts.into_iter().collect();
    pairs.sort_by(|a, b| b.1.cmp(&a.1).then_with(|| a.0.cmp(&b.0)));
    pairs.truncate(n);
    Ok(pairs)
}

fn main() {
    fs::write("poem.txt", "The cat sat. The cat ate.\nThe dog did not.").unwrap();
    fs::write("empty.txt", "  \n").unwrap();
    for path in ["poem.txt", "empty.txt", "missing.txt"] {
        match top_words(path, 3) {
            Ok(words) => {
                println!("{path}:");
                for (w, c) in words {
                    println!("  {c:>3} {w}");
                }
            }
            Err(e) => println!("{path}: error: {e}"),
        }
    }
}
```

You should see:

```
poem.txt:
    3 the
    2 cat
    1 ate
empty.txt: error: empty.txt has no words
missing.txt: error: missing.txt: No such file or directory (os error 2)
```

`Checkpoint.` Take the file names from the command line (`std::env::args().skip(1)`), add a `--top N` option by hand, exit with code 1 if any file failed (`std::process::exit(1)`), and give `CountError` a second variant for a file over 100 MB (check `fs::metadata(path)?.len()` first). Level 3 done.

`Exercises`

56.1 ★ Parse or zero. `fn to_num(s: &str) -> i32` returning the number, or 0 — with `unwrap_or`.

56.2 ★ Match a Result. Parse `"3.14"` and `"pi"` as `f64` and print `ok VALUE` or `error MESSAGE` for each.

56.3 ★ First with ?. `fn double_str(s: &str) -> Result<i32, std::num::ParseIntError>` using `?`.

56.4 ★★ Sum of a list of strings. `fn sum_all(v: &[&str]) -> Result<i32, String>` — collect into `Result<Vec<i32>, _>`, with the bad value in the message.

56.5 ★★ Config line. `fn parse_kv(line: &str) -> Option<(&str, &str)>` for `key = value` lines (with `split_once` and `?`); `None` for anything else.

56.6 ★★ main returns Result. Write a `main() -> Result<(), Box<dyn Error>>` that writes a file, reads it back and prints its length — all with `?`.

56.7 ★★★ A validated age. `struct Age(u8)` with `fn new(n: i32) -> Result<Age, AgeError>` where `enum AgeError { Negative, TooOld }` with a `Display`. Try -3, 40 and 200.

56.8 ★★★ CSV sums, with line numbers. Read a small CSV `name,amount` (write it first), sum the amounts, and on a bad line return an error saying which line and why.

56.9 ★★★★ A key-value store on disk. `struct Store { path: String, data: BTreeMap<String, String> }` with `open(path) -> Result<Store, ···>` (a missing file means empty), `get`, `set`, `remove`, and `save(&self) -> Result<(), ···>` writing `key=value` lines; a bad line in the file is an error naming the line. Round-trip it.

性 57. Rust level 4: traits and generics

In this chapter

- traits: a behaviour many types can promise
- implementing standard traits: `Display`, `Default`, `From`, `Add`, `Ord`
- generics: one function or struct for many types, with trait bounds
- `impl Trait` and `dyn Trait`: choosing at compile time or at run time

You need: level 3 (chapters 55–56).

性 57.1 A trait

A **trait** names a behaviour — a set of methods — that different types can promise to have. Each type **implements** it in its own way:

```
trait Speak {
    fn speak(&self) -> String;
}

struct Cat;
struct Dog {
    name: String,
}

impl Speak for Cat {
    fn speak(&self) -> String {
        "meow".to_string()
    }
}

impl Speak for Dog {
    fn speak(&self) -> String {
        format!("{} says woof", self.name)
    }
}

fn main() {
    println!("{}", Cat.speak());
    println!("{}", Dog { name: "Rex".into() }.speak());
}
```

You should see:

```
meow
Rex says woof
```

A trait may give a **default** method, which every implementer gets for free (and may replace):

```
trait Describe {
    fn name(&self) -> String;
    fn describe(&self) -> String {
        format!("this is {}", self.name())
    }
}

struct Book(&'static str);
struct Song(&'static str);

impl Describe for Book {
    fn name(&self) -> String {
        format!("the book {}", self.0)
    }
}

impl Describe for Song {
    fn name(&self) -> String {
        self.0.to_string()
    }
    fn describe(&self) -> String {
        format!("♪ {} ♪", self.0)
    }
}

fn main() {
    println!("{}", Book("Dune").describe());
    println!("{}", Song("Sakura").describe());
}
```

You should see:

```
this is the book Dune
♪ Sakura ♪
```

性 57.2 Traits as parameters

A function can accept *any* type that implements a trait:

```
trait Area {
    fn area(&self) -> f64;
}

struct Square(f64);
struct Circle(f64);

impl Area for Square {
    fn area(&self) -> f64 {
        self.0 * self.0
    }
}

impl Area for Circle {
    fn area(&self) -> f64 {
        std::f64::consts::PI * self.0 * self.0
    }
}

fn report(shape: &impl Area) {
    println!("area {:.2}", shape.area());
}

fn main() {
    report(&Square(2.0));
    report(&Circle(1.0));
}
```

You should see:

```
area 4.00
area 3.14
```

(`&impl Area` reads: "a reference to some type that implements `Area`")

性 57.3 Standard traits you implement

Much of Rust's convenience comes from a handful of standard traits. Implement them, and your type works with `println!`, `+`, `sort`, `.into()`:

Display — how `{}` shows your type:

```
use std::fmt;

struct Money {
    fils: i64,
}

impl fmt::Display for Money {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        write!(f, "{}.{:02} AED", self.fils / 100, self.fils % 100)
    }
}

fn main() {
    let m = Money { fils: 1250 };
    println!("{m}");
    let s = m.to_string();
    println!("{} characters", s.len());
}
```

You should see:

```
12.50 AED
9 characters
```

(implementing `Display` gives `.to_string()` for free)

Add — the `+` sign; **From** — converting one type into another, which also gives `.into()`:

```
use std::ops::Add;

#[derive(Debug, Clone, Copy, PartialEq)]
struct V2 {
    x: f64,
    y: f64,
}

impl Add for V2 {
    type Output = V2;
    fn add(self, o: V2) -> V2 {
        V2 { x: self.x + o.x, y: self.y + o.y }
    }
}

impl From<(f64, f64)> for V2 {
    fn from(t: (f64, f64)) -> V2 {
        V2 { x: t.0, y: t.1 }
    }
}

fn main() {
    let a = V2::from((1.0, 2.0));
    let b: V2 = (0.5, 0.5).into();
    println!("{:?}", a + b);
}
```

You should see:

```
V2 { x: 1.5, y: 2.5 }
```

PartialOrd / Ord — comparison and sorting. Deriving compares field by field in order:

```
#[derive(Debug, PartialEq, Eq, PartialOrd, Ord)]
struct Version {
    major: u32,
    minor: u32,
    patch: u32,
}

fn main() {
    let mut vs = vec![
        Version { major: 1, minor: 10, patch: 0 },
        Version { major: 1, minor: 2, patch: 7 },
        Version { major: 0, minor: 9, patch: 9 },
    ];
    vs.sort();
    for v in &vs {
        println!("{}.{}.{}", v.major, v.minor, v.patch);
    }
    println!("{}", vs[0] < vs[1]);
}
```

You should see:

```
0.9.9
1.2.7
1.10.0
true
```

性 57.4 Generics

A **generic** function works for many types. `T` is a placeholder for "some type"; a **trait bound** (`T: PartialOrd`) says what `T` must be able to do — here, be compared:

```
fn largest<T: PartialOrd + Copy>(xs: &[T]) -> T {
    let mut best = xs[0];
    for &x in xs {
        if x > best {
            best = x;
        }
    }
    best
}

fn main() {
    println!("{}", largest(&[3, 9, 4]));
    println!("{}", largest(&[1.5, -2.0]));
    println!("{}", largest(&['k', 'y', 'o']));
}
```

You should see:

```
9
1.5
y
```

Without the bound, the compiler refuses: it cannot know that any `T` has `>`:

```
fn largest<T>(xs: &[T]) -> &T {
    let mut best = &xs[0];
    for x in xs {
        if x > best {
            best = x;
        }
    }
    best
}

fn main() {}
```

The compiler says:

```
error[E0369]: binary operation `>` cannot be applied to type `&T`
 --> src/main.rs:4:14
  |
4 |         if x > best {
  |            - ^ ---- &T
  |            |
  |            &T
  |
help: consider restricting type parameter `T` with trait `PartialOrd`
  |
1 | fn largest<T: std::cmp::PartialOrd>(xs: &[T]) -> &T {
  |             ++++++++++++++++++++++
```

Long bounds read better after a `where`:

```
use std::fmt::Display;

fn show_all<T>(items: &[T]) -> String
where
    T: Display,
{
    let parts: Vec<String> = items.iter().map(|i| format!("<{i}>")).collect();
    parts.join(" ")
}

fn main() {
    println!("{}", show_all(&[1, 2, 3]));
    println!("{}", show_all(&["a", "b"]));
}
```

You should see:

```
<1> <2> <3>
<a> <b>
```

```
#[derive(Debug)]
struct Pair<T> {
    a: T,
    b: T,
}

impl<T: PartialOrd + Copy> Pair<T> {
    fn bigger(&self) -> T {
        if self.a > self.b { self.a } else { self.b }
    }
}

fn main() {
    let p = Pair { a: 3, b: 8 };
    let q = Pair { a: 2.5, b: -1.0 };
    println!("{:?} {} {}", p, p.bigger(), q.bigger());
}
```

You should see:

```
Pair { a: 3, b: 8 } 8 2.5
```

`Vec<T>`, `Option<T>`, `Result<T, E>` and `HashMap<K, V>` are all generic structs and enums like this one.

`Word — monomorphisation.` For each type you use a generic with, the compiler writes a separate copy of the code — `largest` for `i32`, another for `f64`. So generics cost nothing when the program runs: they are as fast as code written by hand for each type.

性 57.6 dyn Trait: different types in one list

A `Vec` holds one type. To keep circles and squares in one list, store them as **trait objects**, `Box<dyn Area>` — "some type that implements `Area`, decided when the program runs":

```
trait Area {
    fn area(&self) -> f64;
    fn name(&self) -> &str;
}

struct Square(f64);
struct Circle(f64);

impl Area for Square {
    fn area(&self) -> f64 { self.0 * self.0 }
    fn name(&self) -> &str { "square" }
}

impl Area for Circle {
    fn area(&self) -> f64 { 3.14159 * self.0 * self.0 }
    fn name(&self) -> &str { "circle" }
}

fn main() {
    let shapes: Vec<Box<dyn Area>> = vec![Box::new(Square(2.0)), Box::new(Circle(1.0)),
Box::new(Square(3.0))];
    let mut total = 0.0;
    for s in &shapes {
        println!("{:<7}{:>6.2}", s.name(), s.area());
        total += s.area();
    }
    println!("total {total:.2}");
}
```

You should see:

```
square   4.00
circle   3.14
square   9.00
total 16.14
```

	impl Trait / generics	dyn Trait
the type is chosen	when compiling	when running
speed	fastest: a copy per type	one small extra step per call
mixed types in one collection	no	yes
use for	almost everything	plugins, lists of different things

性 57.7 Common mistakes

1. A missing bound. `binary operation > cannot be applied to type T` — add the trait the operation needs (`T: PartialOrd`); the compiler usually suggests it.

2. A trait not in scope. A trait's methods work only where the trait is imported: `use std::fmt::Write;` before `write!` into a `String`, `use std::io::Read;` for `read_to_string` on a file.

3. Implementing a foreign trait for a foreign type (`Display` for `Vec<i32>`) is not allowed — the *orphan rule*. Wrap it in your own type: `struct Scores(Vec<i32>);`.

Exercises

57.1 ★ A Greet trait. `trait Greet { fn hello(&self) -> String; }` for `English` and `Arabic` unit structs; print both.

57.2 ★ Display a point. `struct Point { x: i32, y: i32 }` printed by `{}` as `(3, 4)`.

57.3 ★ Generic smallest. `fn smallest<T: PartialOrd + Copy>(xs: &[T]) -> T`, tried on ints and chars.

57.4 ★★ From Celsius. `struct Celsius(f64)` and `struct Fahrenheit(f64)` with `From<Celsius> for Fahrenheit`; convert 100 °C with `.into()`.

57.5 ★★ Default settings. `#[derive(Default, Debug)]` on a settings struct, but implement `Default` by hand so the font size starts at 12.

57.6 ★★ Sum anything. `fn total<T: std::iter::Sum<T> + Copy>(xs: &[T]) -> T` with `xs.iter().copied().sum()`, on `i32` and `f64`.

57.7 ★★★ Multiply money. For chapter 57's `Money`, implement `Add` and `Mul<i64>` (`use std::ops::Mul;`), so `Money { fils: 1250 } * 3 + Money { fils: 50 }` prints `38.00 AED`.

57.8 ★★★ A zoo of trait objects. `trait Animal { fn name(&self) -> String; fn sound(&self) -> String; fn legs(&self) -> u32 { 4 } }` with three animals (one a bird with 2 legs), kept in a `Vec<Box<dyn Animal>>`; print each and the total number of legs.

57.9 ★★★★ A generic stack with Display. `struct Stack<T> { items: Vec<T> }` with `push`, `pop`, `peek`, `len`, and `impl<T: Display> Display for Stack<T>` printing `[a | b | c>` (top on the right). Use it with numbers and with strings.

58. Rust level 4: lifetimes, gently

In this chapter

- what a lifetime is: how long a reference stays valid
- when you must write one (`'a`), and when the compiler works it out
- structs that hold references
- `'static`, and owning instead of borrowing
- the checkpoint for level 4: a generic statistics library

You need: chapters 52 and 57.

58.1 What a lifetime is

Every reference is valid for some stretch of the program: from when it is made until the value it points at is dropped or changed. That stretch is its **lifetime**. The compiler checks every reference's lifetime fits inside its value's — that is how chapter 52's dangling references were caught. Almost always it works lifetimes out by itself; this chapter is about the few times it asks you.

```
fn main() {
    let r;
    {
        let x = 5;
        r = &x;
    }
    println!("{r}");
}
```

The compiler says:

```
error[E0597]: `x` does not live long enough
 --> src/main.rs:5:13
  |
4 |         let x = 5;
  |             - binding `x` declared here
5 |         r = &x;
  |             ^^ borrowed value does not live long enough
6 |     }
  |     - `x` dropped here while still borrowed
7 |     println!("{r}");
  |                - borrow later used here
```

(`x` is dropped at the inner `}`; `r` would point at nothing. The compiler draws it for you: `x` does not live long enough)

58.2 When you must write one

A function that takes two references and returns one: which input does the result borrow from? The compiler cannot tell from the signature, so it asks:

```
fn longer(a: &str, b: &str) -> &str {
    if a.len() >= b.len() { a } else { b }
}

fn main() {}
```

The compiler says:

```
error[E0106]: missing lifetime specifier
 --> src/main.rs:1:32
  |
1 | fn longer(a: &str, b: &str) -> &str {
  |              ----     ----     ^ expected named lifetime parameter
  |
  = help: this function's return type contains a borrowed value, but the signature does not say whether it
is borrowed from `a` or `b`
help: consider introducing a named lifetime parameter
  |
1 | fn longer<'a>(a: &'a str, b: &'a str) -> &'a str {
  |          ++++     ++          ++          ++
```

The answer is a **lifetime parameter**, written `'a` (any name after an apostrophe). It names a lifetime, like `T` names a type:

```
fn longer<'a>(a: &'a str, b: &'a str) -> &'a str {
    if a.len() >= b.len() { a } else { b }
}

fn main() {
    let x = String::from("kyoto");
    let y = String::from("sharjah");
    println!("{}", longer(&x, &y));
}
```

You should see:

```
sharjah
```

Read `<'a>(a: &'a str, b: &'a str) -> &'a str` as: *the result lives no longer than the shorter of* `a` *and* `b`. It does not change how long anything lives; it **describes** the connection, so the compiler can check each call:

```
fn longer<'a>(a: &'a str, b: &'a str) -> &'a str {
    if a.len() >= b.len() { a } else { b }
}

fn main() {
    let x = String::from("kyoto");
    let result;
    {
        let y = String::from("sharjah");
        result = longer(&x, &y);
    }
    println!("{result}");
}
```

The compiler says:

```
error[E0597]: `y` does not live long enough
  --> src/main.rs:10:29
   |
 9 |         let y = String::from("sharjah");
   |             - binding `y` declared here
10 |         result = longer(&x, &y);
   |                             ^^ borrowed value does not live long enough
11 |     }
   |     - `y` dropped here while still borrowed
12 |     println!("{result}");
   |                ------ borrow later used here
```



58.3 When you do not need one

The compiler fills lifetimes in by three rules (**elision**). The two that matter:

1. With exactly **one** reference coming in, a reference going out borrows from it: `fn first_word(s: &str) -> &str` needs no `'a`.
2. In a method with `&self`, a reference going out borrows from `self`.

```
struct Library {
    books: Vec<String>,
}

impl Library {
    fn first(&self) -> &str {
        &self.books[0]
    }
}

fn first_word(s: &str) -> &str {
    s.split(' ').next().unwrap_or("")
}

fn main() {
    let lib = Library { books: vec!["Dune".into(), "Emma".into()] };
    println!("{} {}", lib.first(), first_word("borrow checker"));
}
```

You should see:

```
Dune borrow
```

So you write `'a` only when a function takes **two or more** references and returns one, or in a struct that holds a reference.



58.4 Structs that hold references

A struct can borrow instead of own — useful for a view into a big text without copying it. It needs a lifetime parameter: the struct cannot outlive what it borrows.

```
#[derive(Debug)]
struct Highlight<'a> {
    text: &'a str,
    line: usize,
}

fn find<'a>(doc: &'a str, word: &str) -> Vec<Highlight<'a>> {
    let mut out = Vec::new();
    for (i, line) in doc.lines().enumerate() {
        if line.contains(word) {
            out.push(Highlight { text: line, line: i + 1 });
        }
    }
    out
}

fn main() {
    let doc = String::from("ownership\nborrowing rules\nlifetimes and borrowing");
    for h in find(&doc, "borrow") {
        println!("{}: {}", h.line, h.text);
    }
}
```

You should see:

```
2: borrowing rules
3: lifetimes and borrowing
```

(`find` takes two references but returns borrows only from `doc` — so only `doc` shares `'a`; `word` needs none)

58.5 'static

`'static` is the lifetime of the whole program. Text literals are `&'static str`: they are built into the program file and never freed:

```
fn greeting(hour: u32) -> &'static str {
    if hour < 12 { "good morning" } else { "good evening" }
}

fn main() {
    let s: &'static str = "lives forever";
    println!("{s} — {}", greeting(9));
}
```

You should see:

```
lives forever — good morning
```

> Careful. When the compiler suggests adding `'static`, it is rarely what you want. It usually means you are returning a reference to something you made — return the owned value (`String`, not `&str`) instead.

58.6 When in doubt, own it

Lifetimes are a tool for speed — borrowing instead of copying. While you learn, owning is always a correct choice: a struct with `String` fields instead of `&str` needs no lifetimes at all. Make it work with owned values, then borrow where profiling shows the copies matter.

Field type	Lifetimes	Cost
`name: String`	none	one copy of the text
`name: &'a str`	the struct needs `<'a>`	no copy, but the text must outlive the struct

命 58.7 Checkpoint project — a generic statistics library

Level 4 in one: a trait with default methods, generics with bounds, `Display`, and a function that borrows from its input with an explicit lifetime.

```
use std::fmt;

trait Stats {
    fn values(&self) -> Vec<f64>;

    fn mean(&self) -> Option<f64> {
        let v = self.values();
        if v.is_empty() { None } else { Some(v.iter().sum::<f64>() / v.len() as f64) }
    }

    fn median(&self) -> Option<f64> {
        let mut v = self.values();
        if v.is_empty() {
            return None;
        }
        v.sort_by(|a, b| a.total_cmp(b));
        let m = v.len() / 2;
        Some(if v.len() % 2 == 0 { (v[m - 1] + v[m]) / 2.0 } else { v[m] })
    }
}

struct Series<T> {
    name: String,
    data: Vec<T>,
}

impl<T: Into<f64> + Copy> Stats for Series<T> {
    fn values(&self) -> Vec<f64> {
        self.data.iter().map(|&x| x.into()).collect()
    }
}

impl<T: Into<f64> + Copy> fmt::Display for Series<T> {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        match (self.mean(), self.median()) {
            (Some(a), Some(m)) => write!(f, "{}: n={} mean={a:.2} median={m:.2}", self.name,
self.data.len()),
            _ => write!(f, "{}: empty", self.name),
        }
    }
}

fn top<'a, T: PartialOrd>(items: &'a [T], n: usize) -> Vec<&'a T> {
    let mut refs: Vec<&T> = items.iter().collect();
    refs.sort_by(|a, b| b.partial_cmp(a).unwrap());
    refs.truncate(n);
    refs
}

fn main() {
    let temps = Series { name: "temps".into(), data: vec![31.5_f32, 28.0, 35.25, 33.0] };
    let steps = Series { name: "steps".into(), data: vec![8000_u32, 12000, 4000] };
    let none: Series<u8> = Series { name: "none".into(), data: vec![] };
    println!("{temps}\n{steps}\n{none}");
    println!("{:?}", top(&steps.data, 2));
}
```

You should see:

```
temps: n=4 mean=31.94 median=32.25
steps: n=3 mean=8000.00 median=8000.00
none: empty
[12000, 8000]
```

(`T: Into<f64>` accepts any number type that converts to `f64` without loss — `f32`, `u32`, `u8`, `i32` …; `Series<f64>` works too)

`Checkpoint.` Add `std_dev` and `min_max` as default methods of `Stats`; implement `Stats` for a plain `Vec<f64>` too; and add a `fn longest_name<'a>(a: &'a Series<f64>, b: &'a Series<f64>) -> &'a str`. Level 4 done: you can read almost any Rust library's signatures now.

Exercises

58.1 ★ Which need 'a?. Without compiling, say which of these need a lifetime written: (a) `fn f(s: &str) -> &str`, (b) `fn g(a: &str, b: &str) -> String`, (c) `fn h(a: &str, b: &str) -> &str`, (d) `fn m(&self, other: &str) -> &str` in an `impl`.

58.2 ★ First line. `fn first_line(text: &str) -> &str` (no `'a` needed). Try it on a two-line string.

58.3 ★★ Pick by flag. `fn pick<'a>(a: &'a str, b: &'a str, first: bool) -> &'a str`.

58.4 ★★ Only one input matters. `fn prefix_of<'a>(text: &'a str, sep: &str) -> &'a str` returning the part before `sep` — note `sep` gets no `'a`. Show it compiles when `sep` is dropped before the result is used.

58.5 ★★★ A tokenizer that borrows. `struct Tokens<'a> { rest: &'a str }` with a method `fn next_word(&mut self) -> Option<&'a str>` returning words one at a time (skip spaces, slice up to the next space).

58.6 ★★★ Own instead. Rewrite `Highlight<'a>` of 58.5 with `text: String`, so it needs no lifetime. What does each version cost?

58.7 ★★★★ Longest common prefix. `fn common_prefix<'a>(words: &[&'a str]) -> &'a str` returning a slice of the first word — the prefix all words share (careful with non-ASCII: compare `char_indices`).

組 59. Rust level 5: modules, crates and tests

In this chapter

- split a program into modules and files; `pub`, `use`, `crate::`
- a library and a binary in one package
- crates from crates.io with `cargo add`
- tests: unit tests, `assert_eq!`, `#[should_panic]`, tests in `tests/`, doc tests

You need: level 4 (chapters 57–58).

組 59.1 Modules: rooms in a program

A **module** groups related code under a name, and decides what the rest of the program may see. Everything inside is **private** unless marked `pub`:

```
mod geometry {
    pub fn area(w: f64, h: f64) -> f64 {
        w * h
    }

    pub mod circle {
        pub fn area(r: f64) -> f64 {
            super::PI_ISH * r * r
        }
    }

    const PI_ISH: f64 = 3.14159;
}

use geometry::circle;

fn main() {
    println!("{}", geometry::area(2.0, 3.0));
    println!("{:.2}", circle::area(1.0));
}
```

You should see:

```
6
3.14
```

Write	Means
`mod name { ⋯ }`	a module
`pub`	visible outside its module
`geometry::area`	a path: the module, then the item
`use geometry::circle;`	bring a name in, to write `circle::area`
`super::` · `crate::`	the parent module · the top of this crate

Private items stay private — the compiler enforces the boundary:

```
mod bank {
    pub struct Account {
        pub owner: String,
        balance: u64,
    }
}

fn main() {
    let a = bank::Account { owner: "Sara".into(), balance: 100 };
}
```

The compiler says:

```
error[E0451]: field `balance` of struct `Account` is private
 --> src/main.rs:9:51
  |
9 |     let a = bank::Account { owner: "Sara".into(), balance: 100 };
  |                                                   ^^^^^^^ private field
```

(a private field means only the module's own code can set it — so it can guarantee rules, like "the balance only changes through `deposit`*". Give the module a* `pub fn new` *that builds one)*

組 59.2 Modules in files

In a real project each module is its own file. `mod name;` (with a semicolon) in `main.rs` tells cargo to read `src/name.rs`:

```
$ cargo new -q shapes
```

`shapes/src/geometry.rs`:

```
//! Areas of simple shapes.

/// The area of a rectangle.
pub fn rect(w: f64, h: f64) -> f64 {
    w * h
}

/// The area of a circle of radius `r`.
pub fn circle(r: f64) -> f64 {
    std::f64::consts::PI * r * r
}
```

`shapes/src/main.rs`:

```
mod geometry;

use geometry::{circle, rect};

fn main() {
    println!("{} {:.3}", rect(2.0, 3.0), circle(1.0));
}
```

```
$ cd shapes && cargo run -q
6 3.142
$ cd shapes && find src | sort
src
src/geometry.rs
src/main.rs
```

(`///` is a doc comment: `cargo doc --open` turns it into documentation pages. `//!` documents the module itself)

A module with sub-modules is a folder: `src/geometry/mod.rs`, or `src/geometry.rs` beside a `src/geometry/` folder holding the sub-modules.

組 59.3 A library and a binary

A package can hold a **library** (`src/lib.rs`: code for others — and for your tests) and a **binary** (`src/main.rs`: the program). Put the logic in the library and keep `main.rs` thin; then everything is testable:

```
$ cargo new -q --lib textkit
```

`textkit/src/lib.rs`:

```
//! Small text tools.

/// Counts the words in `text`.
///
/// ```
/// assert_eq!(textkit::word_count("one two  three"), 3);
/// ```
pub fn word_count(text: &str) -> usize {
    text.split_whitespace().count()
}

/// Turns a title into a URL slug: `"Hello, World"` → `"hello-world"`.
pub fn slugify(title: &str) -> String {
    title
        .to_lowercase()
        .split(|c: char| !c.is_alphanumeric())
        .filter(|w| !w.is_empty())
        .collect::<Vec<_>>()
        .join("-")
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn counts_words() {
        assert_eq!(word_count("a b c"), 3);
    }

    #[test]
    fn empty_has_no_words() {
        assert_eq!(word_count("   "), 0);
    }

    #[test]
    fn slug_drops_punctuation() {
        assert_eq!(slugify("Rust, Python & AI!"), "rust-python-ai");
    }
}
```

textkit/src/main.rs:

```
fn main() {
    let title = "Ownership, Borrowing & Lifetimes";
    println!("{} words → {}", textkit::word_count(title), textkit::slugify(title));
}
```

```
$ cd textkit && cargo run -q
4 words → ownership-borrowing-lifetimes
```

組 59.4 Tests

`#[cfg(test)] mod tests` holds tests that are compiled only for `cargo test`. Each `#[test]` function is one test; `assert_eq!(left, right)` checks two values are equal, and shows both when they are not:

```
$ cd textkit && cargo test -q 2>&1 | grep -v '^$'
running 3 tests
...
test result: ok. 3 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
running 1 test
.
test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
all doctests ran in 0.01s; merged doctests compilation took 0.18s
```

(three unit tests, then the doc test: the example in `word_count`'s documentation is compiled and run too — so examples in docs can never go stale)

A failing test shows both sides, like pytest:

textkit/src/lib.rs:

```
pub fn word_count(text: &str) -> usize {
    text.split(' ').count()
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn empty_has_no_words() {
        assert_eq!(word_count("   "), 0);
    }
}
```

```
$ cd textkit && cargo test -q --lib 2>&1 | grep -v '^$'
running 1 test
tests::empty_has_no_words --- FAILED
failures:
---- tests::empty_has_no_words stdout ----
thread 'tests::empty_has_no_words' (1) panicked at src/lib.rs:11:9:
assertion `left == right` failed
  left: 4
 right: 0
note: run with `RUST_BACKTRACE=1` environment variable to display a backtrace
failures:
    tests::empty_has_no_words
test result: FAILED. 0 passed; 1 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
error: test failed, to rerun pass `--lib`
```

(`split(' ')` makes a piece between every two spaces: four empty pieces. `split_whitespace` is the fix — put the first version back)

`textkit/src/lib.rs`:

```
//! Small text tools.

/// Counts the words in `text`.
///
/// ```
/// assert_eq!(textkit::word_count("one two  three"), 3);
/// ```
pub fn word_count(text: &str) -> usize {
    text.split_whitespace().count()
}

/// Turns a title into a URL slug: `"Hello, World"` → `"hello-world"`.
pub fn slugify(title: &str) -> String {
    title
        .to_lowercase()
        .split(|c: char| !c.is_alphanumeric())
        .filter(|w| !w.is_empty())
        .collect::<Vec<_>>()
        .join("-")
}
```

```
fn divide(a: i32, b: i32) -> i32 {
    if b == 0 {
        panic!("division by zero");
    }
    a / b
}

fn parse_age(s: &str) -> Result<u8, String> {
    s.parse::<u8>().map_err(|e| e.to_string())
}

#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn divides() {
        assert_eq!(divide(7, 2), 3);
    }

    #[test]
    #[should_panic(expected = "division by zero")]
    fn zero_panics() {
        divide(1, 0);
    }

    #[test]
    fn bad_age_is_an_error() {
        assert!(parse_age("old").is_err());
        assert_eq!(parse_age("42"), Ok(42));
    }

    #[test]
    fn with_question_mark() -> Result<(), String> {
        let a = parse_age("30")?;
        assert!(a > 18, "expected an adult, got {a}");
        Ok(())
    }
}
```

You should see:

```
running 4 tests
test tests::bad_age_is_an_error ... ok
test tests::divides ... ok
test tests::with_question_mark ... ok
test tests::zero_panics - should panic ... ok

test result: ok. 4 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
```

Write	Checks
`assert!(cond)` · `assert!(cond, "msg {x}")`	`cond` is true
`assert_eq!(a, b)` · `assert_ne!(a, b)`	equal · different, showing both
`#[should_panic(expected = "…")]`	the test panics, with that text
a test returning `Result`	fails on `Err` — so `?` works inside
`#[ignore]`	skipped unless `cargo test -- --ignored` (slow tests)

Integration tests go in a `tests/` folder beside `src/`: each file there is compiled as a separate crate that uses your library from outside, as a user would — only its `pub` items.

textkit/tests/slugs.rs:

```
use textkit::slugify;

#[test]
fn slug_of_a_title() {
    assert_eq!(slugify("Hello, World"), "hello-world");
}
```

```
$ cd textkit && cargo test -q --test slugs 2>&1 | grep -v '^$'
running 1 test
.
test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
```

`cargo test NAME` runs only the tests whose names contain NAME; in neovim, `Space` `t` `t` runs a file's tests and `Space` `t` `w` runs them on every save (chapter 48).

組 59.6 Crates from crates.io

crates.io holds over 200 000 libraries. `cargo add` puts one in `Cargo.toml`; the next build downloads and compiles it:

```
$ cargo add rand
    Updating crates.io index
      Adding rand v0.9.2 to dependencies
$ grep -A1 dependencies Cargo.toml
[dependencies]
rand = "0.9.2"
```

Crate	For
`serde`, `serde_json`	reading and writing JSON (and TOML, YAML…) into your structs
`clap`	command-line arguments (chapter 62)
`anyhow`, `thiserror`	errors: quick, and for libraries
`rand`	random numbers
`rayon`	data parallelism in one line (chapter 61)
`tokio`, `reqwest`	async, and HTTP
`candle`, `burn`	machine learning (chapter 64)

`Cargo.lock` records the exact versions, as `uv.lock` does for Python: commit it for programs.

組 59.7 Common mistakes

1. `function is private` (E0603) — add `pub`, to the item *and* to every module on the path.

2. A module file not found — `mod geometry;` needs `src/geometry.rs` (or `src/geometry/mod.rs`), and must be written in `main.rs` or `lib.rs` (or the parent module).

3. Tests that cannot see private functions — unit tests in `mod tests` inside the same file can (`use super::*;`); integration tests in `tests/` cannot, by design.

Exercises

59.1 ★ A module. Put `fn c_to_f` and `fn f_to_c` in `mod temperature` and call both from `main`.

59.2 ★ First tests. Three `#[test]`s for chapter 50's `is_leap`: 2024, 1900 and 2000.

59.3 ★ should_panic. A test that `vec![1, 2][5]` panics, with `expected = "index out of bounds"`.

59.4 ★★ Private by design. `mod bank` with `pub struct Account { balance: u64 }` (private field), `pub fn new()`, `pub fn deposit`, `pub fn balance(&self)`. Show from `main` that the balance can only change through `deposit`.

59.5 ★★ Test the slug. Add three more tests for `slugify`: an empty title, only punctuation, and kanji (does `is_alphanumeric` keep them? Find out).

59.6 ★★★ A library crate. `cargo new --lib units` with `pub fn km_to_miles`, `pub fn miles_to_km`, a doc test on each, unit tests that one undoes the other (within `1e-9`), and a `tests/` integration test. `cargo test` should report all three kinds.

59.7 ★★★★ The expense ledger, in Rust. Port level 3's Python `ledger.py` (chapter 33) to a Rust library: `struct Row { date: String, category: String, amount: f64 }`, `fn parse(line: &str) -> Result<Row, String>`, `fn totals(rows: &[Row]) -> BTreeMap<String, f64>`, `load` and `save` with `std::fs`, each with tests; then a thin `main.rs` reading commands. Compare the two versions: length, and what each compiler or checker caught.

60. Rust level 5: iterators and closures

In this chapter

- closures: small functions that capture their surroundings
- iterators: `map`, `filter`, `sum`, `collect` — Rust's comprehensions
- laziness: nothing runs until something consumes it
- writing your own iterator

You need: chapter 59.

60.1 Closures

A **closure** is a function without a name, written `|args| body`. It can use the variables around it — it **captures** them:

```
fn main() {
    let double = |x: i32| x * 2;
    let add = |a: i32, b: i32| a + b;
    println!("{} {}", double(21), add(2, 3));

    let rate = 0.05;
    let with_vat = |price: f64| price * (1.0 + rate);
    println!("{}", with_vat(200.0));

    let shout = |s: &str| {
        let up = s.to_uppercase();
        format!("{up}!")
    };
    println!("{}", shout("hello"));
}
```

You should see:

```
42 5
210
HELLO!
```

(types on closure parameters are usually optional: the compiler works them out from the first use. Python's `lambda` is the same idea, limited to one expression)

Closures that **change** what they capture must be `mut`; `move` makes a closure take ownership of what it captures (needed for threads, chapter 61):

```
fn main() {
    let mut count = 0;
    let mut click = || count += 1;
    click();
    click();
    println!("{count}");

    let name = String::from("Sara");
    let greet = move || format!("hello {name}");
    println!("{}", greet());
}
```

You should see:

```
2
hello Sara
```

反 60.2 Functions that take closures

A function accepts a closure with a generic bound: `Fn` (only reads what it captured), `FnMut` (changes it), or `FnOnce` (may consume it, so it can be called once):

```
fn apply_twice<F: Fn(i32) -> i32>(f: F, x: i32) -> i32 {
    f(f(x))
}

fn repeat(n: usize, mut f: impl FnMut()) {
    for _ in 0..n {
        f();
    }
}

fn main() {
    println!("{}", apply_twice(|x| x + 10, 1));
    println!("{}", apply_twice(|x| x * x, 3));
    let mut log = Vec::new();
    repeat(3, || log.push("tick"));
    println!("{:?}", log);
}
```

You should see:

```
21
81
["tick", "tick", "tick"]
```

反 60.3 Iterators

An **iterator** gives values one at a time (chapter 37's Python iterators, exactly). Every collection makes one with `.iter()`; then **adapters** transform it, and a **consumer** at the end produces the result:

```
fn main() {
    let prices = vec![3.5, 12.0, 7.25, 20.0];
    let total: f64 = prices.iter().sum();
    let cheap: Vec<f64> = prices.iter().filter(|&&p| p < 10.0).cloned().collect();
    let with_vat: Vec<f64> = prices.iter().map(|p| p * 1.05).collect();
    let count = prices.iter().filter(|p| **p > 5.0).count();
    println!("{total} {:?} {:?} {count}", cheap, with_vat);
}
```

You should see:

```
42.75 [3.5, 7.25] [3.6750000000000003, 12.600000000000001, 7.612500000000001, 21.0] 3
```

Adapter (gives another iterator)	Does
`map(f)`	each value changed by `f`
`filter(f)`	keeps values where `f` is true
`enumerate()`	`(index, value)`
`zip(other)`	pairs, side by side
`take(n)` · `skip(n)`	the first n · all but the first n
`rev()` · `chain(other)`	backwards · one after the other
`flat_map(f)` · `flatten()`	each value into many, all joined
`take_while(f)` · `skip_while(f)`	while the condition holds
`windows(n)` (on slices) · `chunks(n)`	overlapping · separate pieces

Consumer (gives a result)	Gives
`collect()`	a `Vec`, `String`, `HashMap`, … — whatever type you ask for
`sum()` · `product()` · `count()`	a number
`min()` · `max()` · `max_by_key(f)`	`Option` of a value
`find(f)` · `position(f)`	`Option`: the first match · its index
`any(f)` · `all(f)`	`bool`
`fold(start, f)`	one value, built step by step
`for x in it`	a loop is a consumer too



60.4 iter, iter_mut, into_iter

Three ways in, by what each value should be — the same choice as chapter 55's loops:

```
fn main() {
    let mut v = vec![1, 2, 3];
    let doubled: Vec<i32> = v.iter().map(|x| x * 2).collect();
    for x in v.iter_mut() {
        *x += 100;
    }
    let owned: Vec<String> = v.into_iter().map(|x| x.to_string()).collect();
    println!("{:?} {:?}", doubled, owned);
}
```

You should see:

```
[2, 4, 6] ["101", "102", "103"]
```

反 60.5 Laziness

Adapters do nothing by themselves: they describe work. Only a consumer makes values flow — and then one value at a time goes through the whole chain:

```
fn main() {
    let it = (1..=3).map(|x| {
        println!("  map {x}");
        x * 10
    });
    println!("nothing has run yet");
    let v: Vec<i32> = it.collect();
    println!("{:?}", v);

    let first_big = (1..).map(|n| n * n).find(|&sq| sq > 50);
    println!("{:?}", first_big);
}
```

You should see:

```
nothing has run yet
  map 1
  map 2
  map 3
[10, 20, 30]
Some(64)
```

(`1..` has no end — an infinite range — but `find` stops at the first match, so only 8 squares are ever made)

反 60.6 collect into anything

`collect` builds whatever type you name — a `Vec`, a `String`, a `HashMap`, even a `Result`:

```
use std::collections::{BTreeMap, HashSet};

fn main() {
    let words = ["tea", "rice", "tea", "milk"];
    let set: HashSet<&str> = words.iter().copied().collect();
    let lens: BTreeMap<&str, usize> = words.iter().map(|w| (*w, w.len())).collect();
    let initials: String = words.iter().map(|w| w.chars().next().unwrap()).collect();
    let nums: Result<Vec<i32>, _> = ["1", "2", "x"].iter().map(|s| s.parse::<i32>()).collect();
    println!("{} {:?} {initials} {}", set.len(), lens, nums.is_err());
}
```

You should see:

```
3 {"milk": 4, "rice": 4, "tea": 3} trtm true
```

反 60.7 A real chain

Word statistics in one expression each — the shape most Rust data code takes:

```
fn main() {
    let text = "the borrow checker checks the borrows of the program";
    let words: Vec<&str> = text.split_whitespace().collect();
    let longest = words.iter().max_by_key(|w| w.len()).unwrap();
    let avg = words.iter().map(|w| w.len()).sum::<usize>() as f64 / words.len() as f64;
    let caps: Vec<String> = words
        .iter()
        .filter(|w| w.len() > 3)
        .map(|w| w[..1].to_uppercase() + &w[1..])
        .collect();
    let pairs: Vec<(&str, &str)> = words.windows(2).map(|p| (p[0], p[1])).filter(|(a, _)| *a ==
"the").collect();
    println!("{longest} {avg:.2}\n{:?}\n{:?}", caps, pairs);
}
```

You should see:

```
program 4.89
["Borrow", "Checker", "Checks", "Borrows", "Program"]
[("the", "borrow"), ("the", "borrows"), ("the", "program")]
```

(and it is fast: the compiler turns such chains into the same machine code as a hand-written loop — "zero-cost abstractions")

反 60.8 Your own iterator

Implement the `Iterator` trait — one method, `next`, returning `Some(value)` or `None` when done — and every adapter above works on your type for free:

```
struct Fibonacci {
    a: u64,
    b: u64,
}

impl Iterator for Fibonacci {
    type Item = u64;
    fn next(&mut self) -> Option<u64> {
        let r = self.a;
        self.a = self.b;
        self.b += r;
        Some(r)
    }
}

fn fib() -> Fibonacci {
    Fibonacci { a: 0, b: 1 }
}

fn main() {
    println!("{:?}", fib().take(12).collect::<Vec<_>>());
    println!("{}", fib().take_while(|&x| x < 1000).filter(|x| x % 2 == 0).sum::<u64>());
}
```

You should see:

```
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89]
798
```

反 60.9 Common mistakes

1. Forgetting the consumer. `v.iter().map(|x| x * 2);` does nothing — the compiler warns *unused* `Map` *that must be used*. Add `.collect()` (or use a `for` loop for side effects).

2. `collect` **without a type.** `type annotations needed` — say what to build: `let v: Vec<_> = ⋯collect();` or `.collect::<Vec<_>>()`.

```
fn main() {
    let v = [1, 2, 3].iter().map(|x| x * 2).collect();
}
```

The compiler says:

```
error[E0283]: type annotations needed
 --> src/main.rs:2:9
  |
2 |     let v = [1, 2, 3].iter().map(|x| x * 2).collect();
  |         ^                                   ------- type must be known at this point
  |
  = note: the type must implement `FromIterator<i32>`
note: required by a bound in `collect`
```

3. Too many `&`. `filter` gets a *reference to* each item, so on `iter()` of numbers it is `&&i32`: write `|&&x| x > 2`, or `|x| *x > 2` — or use `.copied()` first.

Exercises

60.1 ★ Squares. The squares of 1 to 10, collected into a `Vec`, with `map`.

60.2 ★ Evens sum. The sum of the even numbers from 1 to 100, with `filter` and `sum`.

60.3 ★ A closure that captures. `let threshold = 18;` and a closure `is_adult` using it; count adults in `[12, 30, 17, 45]`.

60.4 ★★ Word lengths. From a sentence, a `Vec<(String, usize)>` of each word in capitals with its length.

60.5 ★★ Dot product. `fn dot(a: &[f64], b: &[f64]) -> f64` with `zip`, `map` and `sum`.

60.6 ★★ fold. The product of 1 to 10 with `fold`, then the longest word of a sentence with `fold`.

60.7 ★★★ Counter iterator. `struct Countdown(u32)` implementing `Iterator` from n down to 1; then `Countdown(5).map(|x| x * 10).collect::<Vec<_>>()`.

60.8 ★★★ Group into a map. From `["apple", "avocado", "banana", "blueberry", "cherry"]`, a `BTreeMap<char, Vec<&str>>` by first letter — with `fold`, then with a `for` loop and `entry`. Which reads better?

60.9 ★★★★ Primes, lazily. An iterator of all primes (`struct Primes { found: Vec<u64>, next: u64 }`), then: the 100th prime; the sum of primes below 1000; and the first pair of *twin primes* above 1000 (primes 2 apart) with `windows(2)` on a collected `Vec`.

61. Rust level 5: concurrency basics

In this chapter

- threads: several things at once, on several cores
- `move` closures and `join`; scoped threads
- sharing safely: `Arc` and `Mutex`; sending messages with channels
- why Rust's compiler makes data races impossible
- the checkpoint for level 5: a parallel file scanner

You need: chapter 60.

61.1 Threads

Your processor has many cores; a normal program uses one. A **thread** is a second line of work inside the same program, running at the same time on another core. `std::thread::spawn` starts one with a closure; `join` waits for it to finish and hands back its result:

```
use std::thread;

fn main() {
    let handle = thread::spawn(|| {
        let mut sum = 0u64;
        for i in 0..1_000_000 {
            sum += i;
        }
        sum
    });
    println!("the main thread goes on meanwhile");
    let result = handle.join().unwrap();
    println!("the thread computed {result}");
}
```

You should see:

```
the main thread goes on meanwhile
the thread computed 499999500000
```

Several threads, each with its own part of the work. `move` gives each closure its own copy of `i`:

```
use std::thread;

fn main() {
    let mut handles = Vec::new();
    for i in 0..4 {
        handles.push(thread::spawn(move || {
            let start = i * 250;
            (start..start + 250).map(|n: u64| n * n).sum::<u64>()
        }));
    }
    let total: u64 = handles.into_iter().map(|h| h.join().unwrap()).sum();
    println!("{total}");
    println!("{}", (0..1000u64).map(|n| n * n).sum::<u64>());
}
```

You should see:

```
332833500
332833500
```

(the order threads run in is up to the system. The results *are the same every time; only prints from inside threads may interleave differently)*



61.2 Why move

A spawned thread may outlive the function that started it, so it cannot borrow that function's variables — the compiler refuses:

```
use std::thread;

fn main() {
    let name = String::from("Sara");
    let h = thread::spawn(|| {
        println!("hello {name}");
    });
    h.join().unwrap();
}
```

The compiler says:

```
error[E0373]: closure may outlive the current function, but it borrows `name`, which is owned by the current
function
 --> src/main.rs:5:27
  |
5 |     let h = thread::spawn(|| {
  |                           ^^ may outlive borrowed value `name`
6 |         println!("hello {name}");
  |                          ---- `name` is borrowed here
  |
note: function requires argument type to outlive `'static`
 --> src/main.rs:5:13
  |
5 |       let h = thread::spawn(|| {
  |  _____________^
6 | |         println!("hello {name}");
7 | |     });
  | |______^
help: to force the closure to take ownership of `name` (and any other referenced variables), use the `move`
keyword
  |
5 |     let h = thread::spawn(move || {
  |                           ++++
```

`move` hands the closure ownership; or use **scoped threads**, which are guaranteed to finish before the scope ends — and so may borrow:

```
use std::thread;

fn main() {
    let data = vec![1, 2, 3, 4, 5, 6, 7, 8];
    let (left, right) = data.split_at(data.len() / 2);
    let (a, b) = thread::scope(|s| {
        let ha = s.spawn(|| left.iter().sum::<i32>());
        let hb = s.spawn(|| right.iter().sum::<i32>());
        (ha.join().unwrap(), hb.join().unwrap())
    });
    println!("{a} + {b} = {}", a + b);
    println!("data is still mine: {:?}", data);
}
```

You should see:

```
10 + 26 = 36
data is still mine: [1, 2, 3, 4, 5, 6, 7, 8]
```



61.3 Sharing state: Arc and Mutex

Several threads changing one value must take turns. A **Mutex** (*mutual exclusion*) lets one thread at a time `lock` the value; an **Arc** (*atomically reference-counted*) lets several threads own the same mutex:

```
use std::sync::{Arc, Mutex};
use std::thread;

fn main() {
    let counter = Arc::new(Mutex::new(0));
    let mut handles = vec![];
    for _ in 0..8 {
        let c = Arc::clone(&counter);
        handles.push(thread::spawn(move || {
            for _ in 0..1000 {
                *c.lock().unwrap() += 1;
            }
        }));
    }
    for h in handles {
        h.join().unwrap();
    }
    println!("{}", *counter.lock().unwrap());
}
```

You should see:

```
8000
```

(8 threads × 1000 = 8000, every time. The lock is released at the end of each statement, when the guard `lock()` gave back is dropped)

Type	Gives
`Arc<T>`	shared ownership across threads (read-only by itself)
`Mutex<T>`	one thread at a time may change `T`
`RwLock<T>`	many readers or one writer — chapter 52's rule, at run time
`AtomicU64` etc.	a single number changed safely without a lock

61.4 Data races are compile errors

Try to share a plain value between threads and change it — the bug that haunts C, C++ and Java programs for years — and Rust will not compile it:

```
use std::thread;

fn main() {
    let mut count = 0;
    let h = thread::spawn(|| {
        count += 1;
    });
    count += 1;
    h.join().unwrap();
}
```

The compiler says:

```
error[E0373]: closure may outlive the current function, but it borrows `count`, which is owned by the
current function
 --> src/main.rs:5:27
  |
5 |     let h = thread::spawn(|| {
  |                           ^^ may outlive borrowed value `count`
6 |         count += 1;
  |         ----- `count` is borrowed here
  |
note: function requires argument type to outlive `'static`
 --> src/main.rs:5:13
  |
5 |       let h = thread::spawn(|| {
  |  _____________^
6 | |         count += 1;
7 | |     });
  | |______^
help: to force the closure to take ownership of `count` (and any other referenced variables), use the `move`
keyword
  |
5 |     let h = thread::spawn(move || {
  |                           ++++
```

Two traits make this work: `Send` (a value may move to another thread) and `Sync` (it may be shared between threads). The compiler marks every type automatically, and refuses to send an `Rc` (non-atomic) or share a `RefCell` across threads. You rarely write them; you meet them in error messages.

61.5 Channels: send messages

Often clearer than sharing: threads **send** results to one receiver through a channel. `mpsc` means *multiple producers, single consumer*:

```
use std::sync::mpsc;
use std::thread;

fn main() {
    let (tx, rx) = mpsc::channel();
    for id in 0..3 {
        let tx = tx.clone();
        thread::spawn(move || {
            let result = (1..=10).map(|n| n * (id + 1)).sum::<u32>();
            tx.send((id, result)).unwrap();
        });
    }
    drop(tx);
    let mut results: Vec<(u32, u32)> = rx.iter().collect();
    results.sort();
    println!("{:?}", results);
}
```

You should see:

```
[(0, 55), (1, 110), (2, 165)]
```

(the loop over `rx` ends when every sender is dropped: each thread drops its clone when it finishes, and `drop(tx)` drops the original)

61.6 How much faster?

Real work, split over all the cores. `available_parallelism` says how many there are:

```
use std::thread;
use std::time::Instant;

fn is_prime(n: u64) -> bool {
    n >= 2 && (2..).take_while(|d| d * d <= n).all(|d| n % d != 0)
}

fn count(range: std::ops::Range<u64>) -> usize {
    range.filter(|&n| is_prime(n)).count()
}

fn main() {
    let n = 2_000_000u64;
    let cores = thread::available_parallelism().map(|c| c.get()).unwrap_or(4) as u64;
    let t = Instant::now();
    let one = count(0..n);
    let single = t.elapsed();

    let t = Instant::now();
    let chunk = n / cores;
    let many: usize = thread::scope(|s| {
        let hs: Vec<_> = (0..cores)
            .map(|i| s.spawn(move || count(i * chunk..if i == cores - 1 { n } else { (i + 1) * chunk })))
            .collect();
        hs.into_iter().map(|h| h.join().unwrap()).sum()
    });
    let multi = t.elapsed();
    println!("{one} primes; same with threads: {}", one == many);
    println!("faster with threads: {}", multi < single);
}
```

You should see:

```
148933 primes; same with threads: true
faster with threads: true
```

(on an 8-core machine, about 6–7 times faster. Compile with `--release` for real timings)

For data-parallel loops, the `rayon` crate does the splitting for you: change `.iter()` to `.par_iter()` and the work is spread over every core — `v.par_iter().map(f).sum()`. The async world (many network waits, chapter 38's idea) is the `tokio` crate.

61.7 Common mistakes

1. `closure may outlive the current function` — add `move`, or use `thread::scope`.

2. Forgetting to `join`. When `main` ends, the program ends — threads still running are stopped mid-work.

3. Holding a lock too long. `let guard = m.lock().unwrap();` keeps the lock until `guard` is dropped; other threads wait. Lock, change, and let it go — in a small `{ }` block if needed.

4. Two locks taken in different orders by two threads can wait for each other forever — a **deadlock**. Take locks always in the same order, or use one lock, or channels.

61.8 Checkpoint project — a parallel file scanner

Level 5 in one: modules of logic, iterators, closures, threads and a channel — searching every `.txt` file in a folder for a word, one thread per file, results sent back and sorted:

```
use std::fs;
use std::path::PathBuf;
use std::sync::mpsc;
use std::thread;

fn matching_lines(text: &str, word: &str) -> Vec<(usize, String)> {
    let w = word.to_lowercase();
    text.lines()
        .enumerate()
        .filter(|(_, l)| l.to_lowercase().contains(&w))
        .map(|(i, l)| (i + 1, l.trim().to_string()))
        .collect()
}

fn main() -> std::io::Result<()> {
    fs::create_dir_all("notes")?;
    fs::write("notes/a.txt", "buy rice\ncall Omar\n")?;
    fs::write("notes/b.txt", "Rice cooker broken\nfix bike\n")?;
    fs::write("notes/c.txt", "nothing here\n")?;

    let mut files: Vec<PathBuf> = fs::read_dir("notes")?
        .filter_map(|e| e.ok().map(|e| e.path()))
        .filter(|p| p.extension().is_some_and(|x| x == "txt"))
        .collect();
    files.sort();

    let (tx, rx) = mpsc::channel();
    for path in files {
        let tx = tx.clone();
        thread::spawn(move || {
            if let Ok(text) = fs::read_to_string(&path) {
                for (n, line) in matching_lines(&text, "rice") {
                    tx.send((path.display().to_string(), n, line)).unwrap();
                }
            }
        });
    }
    drop(tx);
    let mut hits: Vec<_> = rx.iter().collect();
    hits.sort();
    for (file, n, line) in &hits {
        println!("{file}:{n}: {line}");
    }
    println!("{} matches", hits.len());
    Ok(())
}
```

You should see:

```
notes/a.txt:1: buy rice
notes/b.txt:1: Rice cooker broken
2 matches
```

`Checkpoint.` Make it a cargo project with `matching_lines` in `src/lib.rs` (with tests), take the folder and the word from `std::env::args`, walk sub-folders too (a function that recurses into `read_dir`), and limit it to as many threads as cores (a channel of jobs that N workers take from). Time it on a big folder against `rg` (chapter 10). Level 5 done.

Exercises

61.1 ★ Two threads. Spawn two threads that each return a string; `join` both and print them.

61.2 ★ Move a Vec. Move a `Vec<i32>` into a thread that returns its sum.

61.3 ★★ Scoped halves. With `thread::scope`, find the max of each half of a slice in two threads, then the overall max.

61.4 ★★ A shared list. Four threads each push their id into an `Arc<Mutex<Vec<u32>>>`; print the sorted list.

61.5 ★★★ Workers and a channel. Three worker threads take numbers from a shared job channel (`Arc<Mutex<Receiver<u64>>>`), square them, and send results back on a second channel; feed them 1–9 and print the sum.

61.6 ★★★ An atomic counter. Redo the 8-thread counter with `std::sync::atomic::AtomicU64` (`fetch_add(1, Ordering::Relaxed)`) instead of a `Mutex`.

61.7 ★★★★ Parallel Monte Carlo. Estimate π by throwing random points in a square and counting those inside the circle — 8 threads, 10 million points in all, each thread with its own simple random generator (an *xorshift*: `x ^= x << 13; x ^= x >> 7; x ^= x << 17;` on a `u64`, seeded differently per thread). Print the estimate to 3 places.

62. A real command-line tool

In this chapter

- a word-frequency tool, first with the standard library alone, then with clap
- errors that tell the user what went wrong
- install it as a command of your own

You need: levels 1–5 (chapters 49–61).

```
$ cargo new wordfreq && cd wordfreq
$ cargo add clap --features derive
```

```
// wordfreq — the most common words in the files named, most common first
use clap::Parser;
use std::collections::HashMap;
use std::fs;
use std::path::PathBuf;

/// the most common words in some files
#[derive(Parser)]
struct Args {
    /// the files to read
    #[arg(required = true)]
    files: Vec<PathBuf>,
    /// how many words to show
    #[arg(short = 'n', long, default_value_t = 10)]
    top: usize,
    /// ignore words shorter than this
    #[arg(long, default_value_t = 1)]
    min_len: usize,
}

fn count(text: &str, min_len: usize, counts: &mut HashMap<String, usize>) {
    for word in text.split(|c: char| !c.is_alphanumeric()) {
        if word.chars().count() >= min_len.max(1) {
            *counts.entry(word.to_lowercase()).or_insert(0) += 1;
        }
    }
}

fn main() -> Result<(), Box<dyn std::error::Error>> {
    let args = Args::parse();
    let mut counts = HashMap::new();
    for f in &args.files {
        let text = fs::read_to_string(f).map_err(|e| format!("{}: {e}", f.display()))?;
        count(&text, args.min_len, &mut counts);
    }
    let mut top: Vec<(String, usize)> = counts.into_iter().collect();
    top.sort_by(|a, b| b.1.cmp(&a.1).then(a.0.cmp(&b.0)));
    for (word, n) in top.iter().take(args.top) {
        println!("{n:6}  {word}");
    }
    Ok(())
}
```

And its tests, at the end of the same file:

```
#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn counts_ignore_case_and_punctuation() {
        let mut c = HashMap::new();
        count("Tea, tea and TEA!", 1, &mut c);
        assert_eq!(c["tea"], 3);
        assert_eq!(c["and"], 1);
    }

    #[test]
    fn short_words_can_be_skipped() {
        let mut c = HashMap::new();
        count("a cup of tea", 3, &mut c);
        assert_eq!(c.len(), 2);
    }
}
```

```
$ cargo run -q -- --help
the most common words in some files

Usage: wordfreq [OPTIONS] <FILES>...

Arguments:
  <FILES>...  the files to read

Options:
  -n, --top <TOP>          how many words to show [default: 10]
      --min-len <MIN_LEN>  ignore words shorter than this [default: 1]
  -h, --help               Print help
$ cargo run -q --release -- -n 5 --min-len 5 ~/repos/fedokawaii/docs/fedokawaii-guide.md
$ cargo run -q -- /nope
Error: "/nope: No such file or directory (os error 2)"
```

(the `///` comments become the help text; the `--` separates cargo's own options from your program's)

Install it as a command of your own, in cargo's own `bin` folder (on your `PATH`):

```
$ cargo install --path .
$ wordfreq -n 3 ~/Notes/notes.md
```

Exercises

62.1 ★ Your notes' words. Run it on your notes with `--min-len 5`. Which words do you write most?

62.2 ★★ Race. Time it against a one-line Python version on the handbook (`time`), both optimised as they can be.

62.3 ★★ Stop words. Add `--stop FILE`: words listed there are skipped. A test with a string.

62.4 ★★★ Parallel. `cargo add rayon` and count each file on its own core (`files.par_iter()`), merging the maps at the end. Is it faster on many files?

63. A tiny autograd, in Rust

In this chapter

- numbers that remember how they were made
- backpropagation over that record, by hand
- a neuron, a network, XOR learnt

You need: chapters 43 and 57.

PyTorch's `loss.backward()` (chapter 44) seems like magic. It is a small idea: each number keeps a note of the operation that made it and of its inputs; walking those notes backwards, applying the chain rule at each, gives every gradient. Build it — about 120 lines — and the magic is gone for good.

```
$ cargo new tinygrad && cd tinygrad
```

```
// tinygrad — numbers that remember how they were made, and their gradients
use std::cell::RefCell;
use std::collections::HashSet;
use std::ops::{Add, Mul};
use std::rc::Rc;

enum Op {
    Leaf,
    Add(Value, Value),
    Mul(Value, Value),
    Tanh(Value),
}

struct Node {
    data: f64,
    grad: f64,
    op: Op,
}

#[derive(Clone)]
pub struct Value(Rc<RefCell<Node>>);
```

Its methods: making values, `tanh`, and `backward`, which walks the record from the end:

```
impl Value {
    pub fn new(data: f64) -> Value {
        Value(Rc::new(RefCell::new(Node { data, grad: 0.0, op: Op::Leaf })))
    }
    fn from_op(data: f64, op: Op) -> Value {
        Value(Rc::new(RefCell::new(Node { data, grad: 0.0, op })))
    }
    pub fn data(&self) -> f64 { self.0.borrow().data }
    pub fn grad(&self) -> f64 { self.0.borrow().grad }

    pub fn tanh(&self) -> Value {
        Value::from_op(self.data().tanh(), Op::Tanh(self.clone()))
    }

    /// every node this value was made from, children before parents
    fn topo(&self, seen: &mut HashSet<*const RefCell<Node>>, out: &mut Vec<Value>) {
        if !seen.insert(Rc::as_ptr(&self.0)) { return; }
        match &self.0.borrow().op {
            Op::Leaf => {}
            Op::Add(a, b) | Op::Mul(a, b) => { a.topo(seen, out); b.topo(seen, out); }
            Op::Tanh(a) => a.topo(seen, out),
        }
        out.push(self.clone());
    }

    pub fn backward(&self) {
        let mut order = Vec::new();
        self.topo(&mut HashSet::new(), &mut order);
        for v in &order { v.0.borrow_mut().grad = 0.0; }
        self.0.borrow_mut().grad = 1.0;
        for v in order.iter().rev() {
            let n = v.0.borrow();
            match &n.op {
                Op::Leaf => {}
                Op::Add(a, b) => {
                    a.0.borrow_mut().grad += n.grad;
                    b.0.borrow_mut().grad += n.grad;
                }
                Op::Mul(a, b) => {
                    let (ad, bd) = (a.data(), b.data());
                    a.0.borrow_mut().grad += bd * n.grad;
                    b.0.borrow_mut().grad += ad * n.grad;
                }
                Op::Tanh(a) => a.0.borrow_mut().grad += (1.0 - n.data * n.data) * n.grad,
            }
        }
    }

    pub fn step(&self, lr: f64) {
        let mut n = self.0.borrow_mut();
        n.data -= lr * n.grad;
    }
}
```

Then the two operators, so that `&a + &b` and `&a * &b` build nodes:

```
impl Add for &Value {
    type Output = Value;
    fn add(self, o: &Value) -> Value { Value::from_op(self.data() + o.data(), Op::Add(self.clone(),
o.clone())) }
}
impl Mul for &Value {
    type Output = Value;
    fn mul(self, o: &Value) -> Value { Value::from_op(self.data() * o.data(), Op::Mul(self.clone(),
o.clone())) }
}
```

On top of `Value`, a neuron, and the training — the same four steps as chapter 43:

```
/// one neuron: tanh(w·x + b)
struct Neuron { w: Vec<Value>, b: Value }

impl Neuron {
    fn new(inputs: usize, seed: &mut u64) -> Neuron {
        let mut r = || { *seed = seed.wrapping_mul(6364136223846793005).wrapping_add(1442695040888963407);
                         ((*seed >> 33) as f64 / (1u64 << 31) as f64) * 2.0 - 1.0 };
        Neuron { w: (0..inputs).map(|_| Value::new(r())).collect(), b: Value::new(0.0) }
    }
    fn forward(&self, x: &[Value]) -> Value {
        let mut s = self.b.clone();
        for (w, xi) in self.w.iter().zip(x) { s = &s + &(w * xi); }
        s.tanh()
    }
    fn params(&self) -> Vec<Value> { let mut p = self.w.clone(); p.push(self.b.clone()); p }
}

fn main() {
    // the gradient of one small expression, checked by hand
    let a = Value::new(2.0);
    let b = Value::new(-3.0);
    let c = &(&a * &b) + &a;            // c = a*b + a
    c.backward();
    println!("c = {}  dc/da = {}  dc/db = {}", c.data(), a.grad(), b.grad());

    // XOR with 4 hidden neurons
    let mut seed = 42;
    let hidden: Vec<Neuron> = (0..4).map(|_| Neuron::new(2, &mut seed)).collect();
    let out = Neuron::new(4, &mut seed);
    let data = [([0.0, 0.0], -1.0), ([0.0, 1.0], 1.0), ([1.0, 0.0], 1.0), ([1.0, 1.0], -1.0)];
    let mut params: Vec<Value> = hidden.iter().flat_map(|n| n.params()).collect();
    params.extend(out.params());

    for epoch in 0..=500 {
        let mut loss = Value::new(0.0);
        for (x, target) in &data {
            let x: Vec<Value> = x.iter().map(|&v| Value::new(v)).collect();
            let h: Vec<Value> = hidden.iter().map(|n| n.forward(&x)).collect();
            let diff = &out.forward(&h) + &Value::new(-target);
            loss = &loss + &(&diff * &diff);
        }
        loss.backward();
        for p in &params { p.step(0.05); }
        if epoch % 100 == 0 { println!("epoch {epoch:3}  loss {:.4}", loss.data()); }
    }
}
```

```
$ cargo run -q --release
c = -4  dc/da = -2  dc/db = 2
epoch   0  loss 4.1237
epoch 100  loss 0.1982
epoch 200  loss 0.0463
epoch 300  loss 0.0239
epoch 400  loss 0.0157
epoch 500  loss 0.0116
```

Check the first line by hand: `c = a·b + a`, so `dc/da = b + 1 = -2` and `dc/db = a = 2`. The rest is XOR — the classic problem a network without a hidden layer cannot learn — learnt.

Rust idea	Why here
`Rc`	a value used by several later values has several owners: `Rc` counts them
`RefCell`	gradients are added from many places during `backward`: `RefCell` allows change through a shared handle, checked while it runs
`enum Op`	each node is exactly one kind of operation, and `match` handles each
`impl Add for &Value`	`&a + &b` builds a node: Rust lets your types use `+` and `*`

Exercises

63.1 ★ Tests. Turn the first line's check into a `#[test]`: `c.data() == -4.0`, `a.grad() == -2.0`.

63.2 ★★ More operations. Add `Op::Pow(Value, f64)` ($d/dx\ x^n = n \cdot x^{n-1}$) and `exp`. Test each gradient by nudging, as in 43.4.

63.3 ★★ ReLU. Add `relu` and train XOR with it instead of `tanh`. Faster or slower?

63.4 ★★★ Tensors. Values are one number each, so the network is slow. Write `Tensor` — `Vec<f64>` plus a shape — with `matmul` and its backward. You will have rebuilt, in miniature, what PyTorch is.

推 64. Running models in Rust: candle and burn

In this chapter

- load the digits network PyTorch trained, and run it in Rust
- burn: a Rust framework that runs and trains on your card through Vulkan

You need: chapters 44 and 62.

Training is usually Python's job; *running* a trained model — in a command-line tool, a server, a desktop program — is where Rust shines: one small binary, no Python, fast start. The bridge is the **safetensors** file chapter 44 saved: plain tensors with names, readable from any language.

Framework	From	Runs on, here
candle	Hugging Face	the processor (its graphics backends are NVIDIA's CUDA and Apple's Metal)
burn	Tracel	the RX 9060 XT, through Vulkan (Mesa's driver, installed)

推 64.1 candle: the PyTorch model, in Rust

```
$ cargo new digits-candle && cd digits-candle
$ cargo add candle-core candle-nn anyhow
$ cp ../dl/digits.safetensors ../dl/test.safetensors .
```

```
// digits-candle — the PyTorch model's weights, run by candle: no Python at all
use candle_core::{DType, Device, D};
use candle_nn::{linear, Module, VarBuilder};

fn main() -> anyhow::Result<()> {
    let dev = Device::Cpu;
    // the weights, by the names PyTorch gave them: fc1.weight, fc1.bias, fc2.weight, fc2.bias
    let vb = unsafe { VarBuilder::from_mmaped_safetensors(&["digits.safetensors"], DType::F32, &dev)? };
    let fc1 = linear(64, 64, vb.pp("fc1"))?;
    let fc2 = linear(64, 10, vb.pp("fc2"))?;

    let test = candle_core::safetensors::load("test.safetensors", &dev)?;
    let (x, y) = (&test["x"], &test["y"]);

    let logits = fc2.forward(&fc1.forward(x)?.relu()?)?;
    let guess: Vec<u32> = logits.argmax(D::Minus1)?.to_vec1()?;
    let truth: Vec<i64> = y.to_vec1()?;
    let right = guess.iter().zip(&truth).filter(|(g, t)| **g as i64 == **t).count();
    println!("{right} of {} right ({:.1} %)", truth.len(), 100.0 * right as f64 / truth.len() as f64);
    println!("first ten: {:?}", &guess[..10]);
    Ok(())
}
```

```
$ cargo run -q --release
273 of 297 right (91.9 %)
first ten: [1, 7, 4, 6, 3, 1, 3, 9, 1, 7]
```

(for example: the same accuracy PyTorch printed at its last epoch — the same weights, the same arithmetic, another language. `unsafe` *here only says the file is mapped into memory: do not change it while the program runs)*

推 64.2 burn: tensors on the card, through Vulkan

```
$ cargo new burn-hello && cd burn-hello
$ cargo add burn --features vulkan
```

```
use burn::prelude::*;

fn main() {
    let device = Device::vulkan(Default::default());     // the card, through Vulkan
    let a = Tensor::<2>::from_data([[1.0, 2.0], [3.0, 4.0]], &device);
    let b = Tensor::<2>::ones([2, 2], &device);
    println!("{}", a.matmul(b));
}
```

```
Tensor {
  data:
[[3.0, 3.0],
 [7.0, 7.0]],
  shape:  [2, 2],
…
```

(`Tensor<2>` says how many dimensions it has: a shape mistake is a compile error, not a crash after an hour of training. The device says where it lives — here the card)

Exercises

64.1 ★ Same answer. Run 64.1 and compare with chapter 44's last test accuracy.

64.2 ★★ One picture. Print the 8 × 8 picture of the first test digit with ■ and · (values over 0.5), then the model's guess beside it.

64.3 ★★ A command. Make it a CLI with clap: `digits --index 42` shows picture 42 and the guess.

64.4 ★★★ Train in burn. Follow burn's book (`burn.dev`, the MNIST guide) and train the digits network on the card from Rust, with btop open.

橋 65. Rust inside Python: PyO3

In this chapter

- write a Python module in Rust
- build it into your uv project, call it from Python, measure the gain

You need: chapters 24 and 62.

The best of both: write the program in Python, find the slow part, rewrite only that in Rust, and call it as if it were Python. This is how `tokenizers`, `pydantic-core`, `polars` and `ruff` are made. **PyO3** is the bridge; **maturin** builds it.

```
$ cd ~/repos
$ uvx maturin new --bindings pyo3 fastwords && cd fastwords
  ✨ Done! New project created fastwords
```

(`uvx` runs a Python tool without installing it. `maturin` lays out a Rust crate that builds into a Python module: `Cargo.toml`, `src/lib.rs`, `pyproject.toml`)

Replace `src/lib.rs`:

```
use pyo3::prelude::*;

/// fastwords — word counting in Rust, called from Python
#[pymodule]
mod fastwords {
    use pyo3::prelude::*;
    use std::collections::HashMap;

    /// how many times each word appears (lower case, letters and digits only)
    #[pyfunction]
    fn count(text: &str) -> HashMap<String, usize> {
        let mut counts = HashMap::new();
        for w in text.split(|c: char| !c.is_alphanumeric()).filter(|w| !w.is_empty()) {
            *counts.entry(w.to_lowercase()).or_insert(0) += 1;
        }
        counts
    }

    /// the dot product of two lists of numbers
    #[pyfunction]
    fn dot(a: Vec<f64>, b: Vec<f64>) -> PyResult<f64> {
        if a.len() != b.len() {
            return Err(pyo3::exceptions::PyValueError::new_err("the lists differ in length"));
        }
        Ok(a.iter().zip(&b).map(|(x, y)| x * y).sum())
    }
}
```

Build it into the project's `.venv` and try it:

```
$ uv run --with maturin maturin develop --release --uv
🛠 Installed fastwords-0.1.0
$ uv run python
>>> import fastwords
>>> fastwords.dot([1, 2, 3], [4, 5, 6])
32.0
>>> fastwords.dot([1], [1, 2])
ValueError: the lists differ in length
```

(after each change to `lib.rs`, run the `maturin develop` line again. A Rust `HashMap` arrives in Python as a `dict`, a `Vec<f64>` is made from a list, and a Rust error becomes a Python exception)

And the race, in `bench.py`:

```
import collections
import re
import time

import fastwords

text = "Tea, rice and green tea. " * 400_000
t = time.perf_counter(); rust = fastwords.count(text); t_rust = time.perf_counter() - t
t = time.perf_counter()
python = collections.Counter(w.lower() for w in re.findall(r"\w+", text))
t_python = time.perf_counter() - t
assert rust["tea"] == python["tea"] == 800_000
print(f"rust {t_rust:.2f} s · python {t_python:.2f} s")
```

```
$ uv run bench.py
rust 0.57 s · python 2.82 s
```

(for example: about five times faster here, on this desktop's processor)

Exercises

65.1 ★ Build and call. Build it and call `count` on your notes.

65.2 ★★ Python tests for Rust code. `tests/test_fastwords.py` with pytest: both functions, and that `dot` raises `ValueError`. `maturin develop` then `pytest` — the same loop as ever.

65.3 ★★★ The hot loop. Take chapter 42's Game of Life, move `step` into Rust, and measure a 1 000-step run on a 200 × 200 board, before and after.

道 66. Twenty-six weeks: a plan

In this chapter

- a week-by-week plan from this book's first page to its last

You need: nothing — start on a Monday.

An hour a day, five days a week. Each week ends with something that works and is committed to GitHub, and with every ★ and ★★ exercise of its chapters done (the ★★★ and ★★★★ when you have the time — they are where proficiency comes from). Tick the weeks off in your notes.

Week	Chapters	By Friday you have
1	1–8	the desktop by heart: a week without the mouse (4.4)
2	9–14	a dotfile of your own (`local.sh`), your first repository on GitHub
3	15–22, 24	every program tried once; the Python loop of chapter 24 running
4	25–26	numbers, variables, text: the exercises of 25 and 26
5	27	level 1's checkpoint: the guessing game, with its three extensions
6	28–29	functions and lists, every function with an `assert`
7	30	level 2's checkpoint: the contacts book
8	31–32	files and errors: a program that never shows you a traceback
9	33	level 3's checkpoint: the expense tracker, in a uv project
10	34–35	classes, and pytest running on every save
11	36	level 4's checkpoint: the grade book, green in pytest and mypy
12	37–38	generators and async
13	39	level 5's checkpoint: `note`, installed as your own command
14	40–41, 42	data by hand, NumPy, and the small fun projects
15	43	level 6's checkpoint: the spiral and the digits learnt by your NumPy network
16	44–45	PyTorch on the card; a model writing in the handbook's voice
17	46–47	your fine-tuned model, and an agent for your notes
18	48–50	Rust's loop running; level 1's number-guessing machine
19	51–52	ownership and borrowing: every example typed, every error read
20	53–54	level 2's checkpoint: the bank with commands
21	55–56	level 3's checkpoint: the word counter that never panics
22	57–58	level 4's checkpoint: the generic statistics library
23	59–60	a library crate with unit, integration and doc tests
24	61	level 5's checkpoint: the parallel file scanner, timed against `rg`
25	62–63	`wordfreq` installed; tinygrad learning XOR
26	64–65	the digits model in Rust; a Python module written in Rust

Behind? Stretch a week to two. Ahead? Do the ★★★★ exercises, or write the same checkpoint in the other language: the Python expense tracker in Rust (exercise 59.7) teaches both at once.

Where to go next

You liked	Next
level 4 of Python	*Fluent Python* (Luciano Ramalho), the book that takes you from good to expert
chapter 45	Andrej Karpathy's *Neural Networks: Zero to Hero* videos, then his `nanoGPT`
chapter 46	Hugging Face's free LLM course (`huggingface.co/learn`)
chapter 47	Anthropic's documentation on tool use and agents (`docs.claude.com`)
Part V	*The Rust Programming Language* (`doc.rust-lang.org/book`), free; then *Rust for Rustaceans*
exercises	Rustlings (`rustlings.rust-lang.org`) and Exercism's Python and Rust tracks: hundreds more, checked for you
chapter 63	Karpathy's `micrograd`, the Python original of tinygrad
chapter 64	the burn book (`burn.dev`) and candle's examples
everything	build something you will use every day — and keep it on GitHub

A. Glossary

Word	Means	First in
absolute path	a path from `/` or `~`, the same from anywhere	1.2
agent	a model in a loop, calling tools you provide	21, 47
argument	a word that tells a command what to work on; a value passed to a function	1.3, 28.2
array (NumPy)	many numbers of one type, worked on at once	41.1
assert	a check that stops the program if it is false	28.6
async, await	many waits overlapping in one program	38.2
attention	each position taking what it needs from earlier ones	45
autograd	the automatic computing of gradients	44, 63
borrow	using a Rust value without taking it over: `&` or `&mut`	52.1
branch	a line of commits of its own, for an experiment	14
broadcasting	NumPy stretching a smaller array to fit a larger one	41.5
class	your own kind of value: data and methods together	34.1
closure	a small nameless function that captures its surroundings	60.1
commit	a saved snapshot of a project, with a message	14
compositor	the program that draws and places the windows: dwl	4
comprehension	a loop that builds a list, dict or set in one line	37.1
container	a program with everything it needs, kept apart	22
crate	a Rust library or program	48, 59
dataclass	a class whose `__init__`, `repr` and `==` are written for you	34.5
decorator	a function that wraps another: `@cache`, `@dataclass`	37.7
dictionary	a map from keys to values	30.1
embedding	a token turned into a learnt vector	45
enum	a Rust type whose value is one of several variants	54.1
epoch	one pass over all the training data	44
exception	Python stopping because a line cannot be done	32.1
f-string	text with values put in: `f"{x}"`	25.5
fine-tuning	teaching a trained model a little more	46
fixture	a value pytest makes for a test: `tmp_path`	35.6
function	a named piece of work, called with `()`	28.1
generator	a function that `yield`s values one at a time	37.4
generic	code for many types: `fn largest<T>`	57.4
gradient	how much the loss changes when a weight changes	43
immutable	cannot be changed in place: strings, tuples; Rust's default	26.4, 49.1
iterator	something that gives values one at a time	37.3, 60.3
lifetime	how long a Rust reference stays valid: `'a`	58.1
list	many values, in order, changeable	29.1
LoRA	fine-tuning small extra matrices instead of the whole model	46
loss	one number saying how wrong the model is	43
method	a function belonging to a value: `s.upper()`	26.3
module	a file of code you can `import` / `mod`	33.1, 59.1

Word	Means	First in
move	handing a Rust value to a new owner	51.3
Mutex, Arc	a lock for one writer at a time; shared ownership across threads	61.3
Option	Rust's maybe-a-value: `Some(x)` or `None`	54.5
overfitting	memorising the training data instead of learning	44
ownership	Rust's rule: every value has one owner	51.1
package	installable code, from PyPI or crates.io	33.4
panic	Rust stopping the program on purpose: a bug was found	49.3
path	a file's address	1.2
pipe	`|`: one command's output into the next	9
prompt	where the shell waits for your command; `>>>` for Python	2, 25.1
REPL	Python's prompt: read, evaluate, print, loop	25.1
repository	a project folder git keeps the history of	14
Result	Rust's ok-or-error: `Ok(v)` or `Err(e)`	56.2
ROCm	AMD's computing stack, PyTorch's way to the card	44
scope	where a name exists: a function's body, a `{ }` block	28.5, 51.2
set	values without repeats, with fast `in`	30.7
shell	the program that reads your commands: bash	2
slice	a part of a string, list or array: `s[1:4]`, `&s[1..4]`	26.2, 52.6
snapshot	a frozen copy of every file at a moment	8
string	text: `str` in Python; `String` and `&str` in Rust	26.1, 49.4
struct	a Rust type with named fields	53.1
tensor	an array of any number of dimensions	43
test	a small program that checks your code: pytest, `cargo test`	35.1, 59.4
thread	a second line of work running at the same time	61.1
token	a piece of text a model reads as one number	45
trait	a behaviour many Rust types can promise	57.1
transformer	the architecture of today's language models	45
tuple	a few values together, fixed	29.6
type	the kind of a value: `int`, `str`, `i32`, `String`	25.4, 49.3
type hint	`x: int` — what a name should hold, checked by mypy	36.1
variable	a name holding a value	1.4, 25.3
venv	a project's own Python and libraries	24, 33.4
workspace	one of nine desks of windows	4

解 B. Solutions: Python

The worked solution to every exercise of chapters 25–41, by its number, with what it printed. Yours may differ and still be right: what matters is the same output, and code you can read.

解 Chapter 25 — Python level 1: numbers, variables, print

25.1 ★ Seconds in a day.

```
day = 24 * 60 * 60
print(day)
print(day * 7)
```

You should see:

```
86400
604800
```

25.2 ★ Your first file.

```
print("Mohamed")
print("Sharjah")
```

You should see:

```
Mohamed
Sharjah
```

25.3 ★ Even or odd.

```
print(17 % 2, 18 % 2, 2026 % 2)
```

You should see:

```
1 0 0
```

25.4 ★★ Shopping total.

```
rice = 3 * 7.25
tea = 2 * 12.5
print(f"Total: {rice + tea:.2f}")
```

You should see:

```
Total: 46.75
```

25.5 ★★ Minutes and seconds.

```
seconds = 754
print(f"{seconds // 60} min {seconds % 60} s")
```

You should see:

```
12 min 34 s
```

25.6 ★★ Celsius to Fahrenheit.

```
c = float(input("Celsius? "))
f = c * 9 / 5 + 32
print(f"{c:.1f} °C is {f:.1f} °F")
```

You should see:

```
Celsius? 31
31.0 °C is 87.8 °F
```

25.7 ★★★ Swap without a third name.

```
a = 5
b = 9
tmp = a
a = b
b = tmp
print(a, b)
a, b = b, a
print(a, b)
```

You should see:

```
9 5
5 9
```

25.8 ★★★ A receipt.

```
name = input("Item? ")
price = float(input("Price? "))
count = int(input("How many? "))
total = price * count
print(f"{name:<12}{total:>8.2f}")
```

You should see:

```
Item? tea
Price? 8.5
How many? 3
tea            25.50
```

25.9 ★★★★ Change in coins.

```
amount = int(input("Fils? "))
c100 = amount // 100; amount = amount % 100
c50 = amount // 50; amount = amount % 50
c25 = amount // 25; amount = amount % 25
c10 = amount // 10; amount = amount % 10
c5 = amount // 5; amount = amount % 5
print(f"100×{c100} 50×{c50} 25×{c25} 10×{c10} 5×{c5} 1×{amount}")
```

You should see:

```
Fils? 385
100×3 50×1 25×1 10×1 5×0 1×0
```

解 Chapter 26 — Python level 1: text

26.1 ★ Initials.

```
first = "mohamed"
last = "eldoush"
print(f"{first[0].upper()}.{last[0].upper()}.")
```

You should see:

```
M.E.
```

26.2 ★ Shout.

```
s = input("Say something: ")
print(s.upper() + "!!!")
```

You should see:

```
Say something: hello
HELLO!!!
```

26.3 ★ The extension.

```
filename = "report.final.pdf"
print(filename.split(".")[-1])
```

You should see:

```
pdf
```

26.4 ★★ Palindrome check.

```
w = input("Word? ").lower()
print(w == w[::-1])
```

You should see:

```
Word? Level
True
```

26.5 ★★ Count the vowels.

```
s = "Fedokawaii is a desktop".lower()
print(s.count("a") + s.count("e") + s.count("i") + s.count("o") + s.count("u"))
```

You should see:

```
10
```

26.6 ★★ Date flip.

```
parts = "2026-10-21".split("-")
print("/".join([parts[2], parts[1], parts[0]]))
```

You should see:

```
21/10/2026
```

26.7 ★★★ Title case for a sentence.

```
s = input("Sentence? ").strip()
fixed = s[0].upper() + s[1:].lower()
print(f"{fixed} ({len(s.split())} words)")
```

You should see:

```
Sentence? tHE quick FOX
The quick fox (3 words)
```

26.8 ★★★ Mask a card number.

```
card = "4111222233334444"
print("*" * (len(card) - 4) + card[-4:])
```

You should see:

```
************4444
```

26.9 ★★★★ Caesar shift, one letter.

```
c = "y"
k = 3
shifted = chr((ord(c) - ord("a") + k) % 26 + ord("a"))
print(shifted)
```

You should see:

```
b
```

解 Chapter 27 — Python level 1: decisions and loops

27.1 ★ Positive, negative, zero.

```
n = float(input("Number? "))
if n > 0:
    print("positive")
elif n < 0:
    print("negative")
else:
    print("zero")
```

You should see:

```
Number? -4
negative
```

27.2 ★ Count down.

```
for n in range(10, 0, -1):
    print(n, end=" ")
print("lift off!")
```

You should see:

```
10 9 8 7 6 5 4 3 2 1 lift off!
```

27.3 ★ Sum of evens.

```
total = 0
for n in range(2, 101, 2):
    total += n
print(total)
```

You should see:

```
2550
```

27.4 ★★ FizzBuzz.

```
for n in range(1, 16):
    if n % 15 == 0:
        print("FizzBuzz")
    elif n % 3 == 0:
        print("Fizz")
    elif n % 5 == 0:
        print("Buzz")
    else:
        print(n)
```

You should see:

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

27.5 ★★ Leap year.

```
y = int(input("Year? "))
if (y % 4 == 0 and y % 100 != 0) or y % 400 == 0:
    print(y, "is a leap year")
else:
    print(y, "is not a leap year")
```

You should see:

```
Year? 1900
1900 is not a leap year
```

27.6 ★★ Vowels by loop.

```
count = 0
for ch in input("Word? ").lower():
    if ch in "aeiou":
        count += 1
print(count, "vowels")
```

You should see:

```
Word? Fedokawaii
6 vowels
```

27.7 ★★★ Password check.

```
while True:
    pw = input("Password? ")
    has_digit = False
    for ch in pw:
        if ch.isdigit():
            has_digit = True
    if len(pw) >= 8 and has_digit:
        print("saved")
        break
    print("8 characters and a digit, please")
```

You should see:

```
Password? short
8 characters and a digit, please
Password? longenough
8 characters and a digit, please
Password? longenough7
saved
```

27.8 ★★★ Is it prime?.

```
n = int(input("Number? "))
prime = n >= 2
d = 2
while d * d <= n:
    if n % d == 0:
        prime = False
        print(f"{n} = {d} × {n // d}")
        break
    d += 1
print(n, "is prime" if prime else "is not prime")
```

You should see:

```
Number? 91
91 = 7 × 13
91 is not prime
```

27.9 ★★★ A triangle of stars.

```
h = int(input("Height? "))
for row in range(1, h + 1):
    print("*" * row)
```

You should see:

```
Height? 4
*
**
***
****
```

27.10 ★★★★ The Collatz walk.

```
n = int(input("Start? "))
steps = 0
print(n, end=" ")
while n != 1:
    if n % 2 == 0:
        n = n // 2
    else:
        n = 3 * n + 1
    steps += 1
    print(n, end=" ")
print()
print(steps, "steps")
```

You should see:

```
Start? 6
6 3 10 5 16 8 4 2 1
8 steps
```

解 Chapter 28 — Python level 2: functions

28.1 ★ Square.

```
def square(n):
    return n * n

assert square(0) == 0
assert square(3) == 9
assert square(-4) == 16
print("ok")
```

You should see:

```
ok
```

28.2 ★ Greeting.

```
def greeting(name, hour):
    if hour < 12:
        return f"Good morning, {name}"
    return f"Good evening, {name}"

print(greeting("Sara", 9))
print(greeting("Sara", 20))
```

You should see:

```
Good morning, Sara
Good evening, Sara
```

28.3 ★ Is even.

```
def is_even(n):
    return n % 2 == 0

print(is_even(4), is_even(7))
```

You should see:

```
True False
```

28.4 ★★ Pay with overtime.

```
def pay(hours, rate):
    if hours <= 40:
        return hours * rate
    return 40 * rate + (hours - 40) * rate * 1.5

assert pay(30, 10) == 300
assert pay(40, 10) == 400
assert pay(45, 10) == 475
print(pay(30, 10), pay(40, 10), pay(45, 10))
```

You should see:

```
300 400 475.0
```

28.5 ★★ Clamp.

```
def clamp(x, low=0, high=100):
    if x < low:
        return low
    if x > high:
        return high
    return x

print(clamp(150), clamp(-5), clamp(42), clamp(7, high=5))
```

You should see:

```
100 0 42 5
```

28.6 ★★ Reverse words.

```
def reverse_words(text):
    return " ".join(text.split()[::-1])

assert reverse_words("one two three") == "three two one"
print(reverse_words("one two three"))
```

You should see:

```
three two one
```

28.7 ★★★ Power without .**

```
def power_loop(base, exp):
    result = 1
    for _ in range(exp):
        result *= base
    return result

def power_rec(base, exp):
    if exp == 0:
        return 1
    return base * power_rec(base, exp - 1)

for b, e in [(2, 10), (3, 0), (5, 3)]:
    assert power_loop(b, e) == b ** e == power_rec(b, e)
print(power_loop(2, 10), power_rec(5, 3))
```

You should see:

```
1024 125
```

28.8 ★★★ Grade.

```
def grade(score):
    assert 0 <= score <= 100, "score must be 0-100"
    if score >= 90:
        return "A"
    if score >= 75:
        return "B"
    if score >= 60:
        return "C"
    return "F"

print([grade(s) for s in (95, 80, 61, 12)])
```

You should see:

```
['A', 'B', 'C', 'F']
```

28.9 ★★★★ Fibonacci, two ways.

```
import time

def fib_rec(n):
    if n < 2:
        return n
    return fib_rec(n - 1) + fib_rec(n - 2)

def fib_loop(n):
    a, b = 0, 1
    for _ in range(n):
        a, b = b, a + b
    return a

print(fib_loop(30), fib_rec(30))
t = time.perf_counter(); fib_rec(25); slow = time.perf_counter() - t
t = time.perf_counter(); fib_loop(25); fast = time.perf_counter() - t
print("the loop is faster:", fast < slow)
```

You should see:

```
832040 832040
the loop is faster: True
```

解 Chapter 29 — Python level 2: lists and tuples

29.1 ★ The middle.

```
nums = [4, 8, 15, 16, 23]
print(nums[len(nums) // 2])
```

You should see:

```
15
```

29.2 ★ Average.

```
def average(nums):
    return sum(nums) / len(nums)

print(average([72, 95, 61]))
```

You should see:

```
76.0
```

29.3 ★ Shopping list.

```
shop = []
shop.append("tea")
shop.append("rice")
shop.append("milk")
shop.remove("tea")
print(sorted(shop))
```

You should see:

```
['milk', 'rice']
```

29.4 ★★ Only the long words.

```
words = ["a", "kawaii", "dwl", "fedora", "is"]
long_words = []
for w in words:
    if len(w) > 3:
        long_words.append(w)
print(long_words)
```

You should see:

```
['kawaii', 'fedora']
```

29.5 ★★ Second largest.

```
def second_largest(nums):
    first = second = None
    for n in nums:
        if first is None or n > first:
            first, second = n, first
        elif n != first and (second is None or n > second):
            second = n
    return second

print(second_largest([3, 9, 4, 9, 7]))
```

You should see:

```
7
```

29.6 ★★ Report card.

```
names = ["Sara", "Omar", "Lina"]
scores = [88, 52, 79]
for name, score in zip(names, scores):
    print(f"{name:<5}{score:>4}  {'pass' if score >= 60 else 'fail'}")
```

You should see:

```
Sara   88  pass
Omar   52  fail
Lina   79  pass
```

29.7 ★★★ Remove duplicates, keep order.

```
def unique(items):
    seen = []
    for x in items:
        if x not in seen:
            seen.append(x)
    return seen

print(unique([3, 1, 3, 2, 1]))
```

You should see:

```
[3, 1, 2]
```

29.8 ★★★ Rotate.

```
def rotate(items, k):
    k = k % len(items)
    return items[-k:] + items[:-k]

print(rotate([1, 2, 3, 4, 5], 2))
print(rotate([1, 2, 3, 4, 5], 7))
```

You should see:

```
[4, 5, 1, 2, 3]
[4, 5, 1, 2, 3]
```

29.9 ★★★★ Tic-tac-toe winner.

```
def winner(board):
    lines = [row for row in board]
    lines += [[board[r][c] for r in range(3)] for c in range(3)]
    lines.append([board[i][i] for i in range(3)])
    lines.append([board[i][2 - i] for i in range(3)])
    for line in lines:
        if line[0] != " " and line.count(line[0]) == 3:
            return line[0]
    return None

board = [["X", "O", " "],
         ["O", "X", " "],
         ["O", " ", "X"]]
print(winner(board))
print(winner([[" "] * 3 for _ in range(3)]))
```

You should see:

```
X
None
```

解 Chapter 30 — Python level 2: dictionaries and sets

30.1 ★ Capitals.

```
capitals = {"UAE": "Abu Dhabi", "Japan": "Tokyo", "Egypt": "Cairo", "France": "Paris"}
for country, capital in capitals.items():
    print(f"{country} → {capital}")
```

You should see:

```
UAE → Abu Dhabi
Japan → Tokyo
Egypt → Cairo
France → Paris
```

30.2 ★ Letter count.

```
counts = {}
for ch in "banana":
    counts[ch] = counts.get(ch, 0) + 1
print(counts)
```

You should see:

```
{'b': 1, 'a': 3, 'n': 2}
```

30.3 ★ Unique visitors.

```
visits = ["sara", "omar", "sara", "lina", "omar", "sara"]
print(len(set(visits)))
```

You should see:

```
3
```

30.4 ★★ Price list total.

```
prices = {"rice": 7.25, "tea": 12.5}
basket = {"rice": 3, "tea": 2}
total = 0
for item, n in basket.items():
    total += prices[item] * n
print(total)
```

You should see:

```
46.75
```

30.5 ★★ Invert a dictionary.

```
d = {"a": 1, "b": 2}
inv = {}
for k, v in d.items():
    inv[v] = k
print(inv)
```

You should see:

```
{1: 'a', 2: 'b'}
```

30.6 ★★ Common friends.

```
sara = {"omar", "lina", "noor", "ali"}
omar = {"lina", "ali", "zaid"}
print(sorted(sara & omar))
print(sorted(sara - omar))
```

You should see:

```
['ali', 'lina']
['noor', 'omar']
```

30.7 ★★★ Word lengths.

```
text = "the cat left the house quietly"
groups = {}
for w in text.split():
    groups.setdefault(len(w), []).append(w)
print(groups)
```

You should see:

```
{3: ['the', 'cat', 'the'], 4: ['left'], 5: ['house'], 7: ['quietly']}
```

30.8 ★★★ Anagrams.

```
words = ["listen", "silent", "enlist", "google", "gooegl", "cat"]
fam = {}
for w in words:
    fam.setdefault("".join(sorted(w)), []).append(w)
print(list(fam.values()))
```

You should see:

```
[['listen', 'silent', 'enlist'], ['google', 'gooegl'], ['cat']]
```

30.9 ★★★★ The most common word.

```
from collections import Counter

text = """The cat sat on the mat. The cat ate a fish, and the dog
watched the cat and the fish."""
stop = {"the", "a", "and", "of"}
words = [w for w in text.lower().replace(".", "").replace(",", "").split() if w not in stop]
counts = {}
for w in words:
    counts[w] = counts.get(w, 0) + 1
print(sorted(counts.items(), key=lambda kv: kv[1], reverse=True)[:3])
print(Counter(words).most_common(3))
```

You should see:

```
[('cat', 3), ('fish', 2), ('sat', 1)]
[('cat', 3), ('fish', 2), ('sat', 1)]
```

解 Chapter 31 — Python level 3: files

31.1 ★ Write and read.

```
from pathlib import Path

p = Path("me.txt")
p.write_text("Mohamed\nSharjah\nPython\n", encoding="utf-8")
print(len(p.read_text(encoding="utf-8")))
```

You should see:

```
23
```

31.2 ★ Number the lines.

```
from pathlib import Path

Path("poem.txt").write_text("old pond\na frog jumps in\nthe sound of water\n", encoding="utf-8")
with open("poem.txt", encoding="utf-8") as f:
    for i, line in enumerate(f, 1):
        print(f"{i:>3}  {line.rstrip()}")
```

You should see:

```
  1  old pond
  2  a frog jumps in
  3  the sound of water
```

31.3 ★ A diary.

```
from pathlib import Path

for entry in ["2026-10-21 learnt files", "2026-10-22 learnt CSV"]:
    with open("diary.txt", "a", encoding="utf-8") as f:
        f.write(entry + "\n")
print(Path("diary.txt").read_text(encoding="utf-8"), end="")
```

You should see:

```
2026-10-21 learnt files
2026-10-22 learnt CSV
```

31.4 ★★ Word count.

```
from pathlib import Path

def wc(path):
    text = Path(path).read_text(encoding="utf-8")
    return len(text.splitlines()), len(text.split()), len(text)

Path("t.txt").write_text("one two\nthree\n", encoding="utf-8")
print(wc("t.txt"))
```

You should see:

```
(2, 3, 14)
```

31.5 ★★ Average from a CSV.

```
import csv
from pathlib import Path

Path("marks.csv").write_text("name,mark\nSara,88\nOmar,92\nLina,79\n", encoding="utf-8")
with open("marks.csv", encoding="utf-8", newline="") as f:
    marks = [int(r["mark"]) for r in csv.DictReader(f)]
print(round(sum(marks) / len(marks), 1))
```

You should see:

```
86.3
```

31.6 ★★ Settings with defaults.

```
import json
from pathlib import Path

p = Path("my-settings.json")
settings = json.loads(p.read_text(encoding="utf-8")) if p.exists() else {"theme": "dark"}
settings["font_size"] = 14
p.write_text(json.dumps(settings, indent=2), encoding="utf-8")
print(p.read_text(encoding="utf-8"))
```

You should see:

```
{
  "theme": "dark",
  "font_size": 14
}
```

31.7 ★★★ Biggest files.

```
from pathlib import Path

d = Path("sizes")
d.mkdir(exist_ok=True)
for name, n in [("a.txt", 10), ("b.txt", 300), ("c.txt", 50), ("d.txt", 7)]:
    (d / name).write_text("x" * n)
files = [p for p in d.rglob("*") if p.is_file()]
for p in sorted(files, key=lambda p: p.stat().st_size, reverse=True)[:3]:
    print(p.name, p.stat().st_size)
```

You should see:

```
b.txt 300
c.txt 50
a.txt 10
```

31.8 ★★★ CSV to JSON.

```
import csv, json
from pathlib import Path

Path("people.csv").write_text("name,city,age\nSara,Sharjah,31\nOmar,Dubai,28\n", encoding="utf-8")
with open("people.csv", encoding="utf-8", newline="") as f:
    people = list(csv.DictReader(f))
for p in people:
    p["age"] = int(p["age"])
print(json.dumps(people, indent=1))
```

You should see:

```
[
 {
  "name": "Sara",
  "city": "Sharjah",
  "age": 31
 },
 {
  "name": "Omar",
  "city": "Dubai",
  "age": 28
 }
]
```

31.9 ★★★★ Find in files.

```
from pathlib import Path

d = Path("notes")
d.mkdir(exist_ok=True)
(d / "a.txt").write_text("buy rice\ncall Omar\n", encoding="utf-8")
(d / "b.txt").write_text("Rice cooker broken\n", encoding="utf-8")

def grep(word, folder):
    for p in sorted(Path(folder).rglob("*.txt")):
        with open(p, encoding="utf-8") as f:
            for n, line in enumerate(f, 1):
                if word.lower() in line.lower():
                    print(f"{p}:{n}: {line.rstrip()}")

grep("rice", "notes")
```

You should see:

```
notes/a.txt:1: buy rice
notes/b.txt:1: Rice cooker broken
```

解 Chapter 32 — Python level 3: errors and exceptions

32.1 ★ Safe int.

```
def to_int(text, default=0):
    try:
        return int(text)
    except ValueError:
        return default

print(to_int("42"), to_int("x"), to_int("", default=-1))
```

You should see:

```
42 0 -1
```

32.2 ★ Read or say so.

```
from pathlib import Path

try:
    print(Path("notes-missing.txt").read_text(encoding="utf-8"))
except FileNotFoundError:
    print("no notes yet")
```

You should see:

```
no notes yet
```

32.3 ★ Which error?.

```
tests = ['[1, 2][9]', '{"a": 1}["b"]', '"a" * "b"', 'int("1.5")', 'None.upper()']
for t in tests:
    try:
        eval(t)
    except Exception as e:
        print(f"{t:<15} {type(e).__name__}")
```

You should see:

```
[1, 2][9]       IndexError
{"a": 1}["b"]   KeyError
"a" * "b"       TypeError
int("1.5")      ValueError
None.upper()    AttributeError
```

32.4 ★★ Age check.

```
def check_age(age):
    if age < 0 or age > 150:
        raise ValueError(f"age must be 0-150, got {age}")
    return age

for a in (30, -2, 200):
    try:
        print("ok", check_age(a))
    except ValueError as e:
        print("refused:", e)
```

You should see:

```
ok 30
refused: age must be 0-150, got -2
refused: age must be 0-150, got 200
```

32.5 ★★ Average of good lines.

```
lines = ["3", "x", "5", "", "10"]
good, bad = [], 0
for s in lines:
    try:
        good.append(float(s))
    except ValueError:
        bad += 1
print(f"average {sum(good) / len(good)}, {bad} bad")
```

You should see:

```
average 6.0, 2 bad
```

32.6 ★★★ Retry.

```
def ask_int(prompt, low, high):
    while True:
        text = input(prompt)
        try:
            n = int(text)
        except ValueError:
            print("  a whole number, please")
            continue
        if low <= n <= high:
            return n
        print(f"  between {low} and {high}, please")

print("you chose", ask_int("1-10? ", 1, 10))
```

You should see:

```
1-10? ten
  a whole number, please
1-10? 99
  between 1 and 10, please
1-10? 7
you chose 7
```

32.7 ★★★ Your own exception.

```
class InsufficientFunds(Exception):
    pass

def withdraw(balance, amount):
    if amount > balance:
        raise InsufficientFunds(f"short by {amount - balance}")
    return balance - amount

print(withdraw(100, 30))
try:
    withdraw(100, 130)
except InsufficientFunds as e:
    print("refused:", e)
```

You should see:

```
70
refused: short by 30
```

32.8 ★★★★ A safe calculator.

```
def calc(line):
    parts = line.split()
    if len(parts) != 3:
        raise ValueError("write: NUMBER SIGN NUMBER")
    a, op, b = parts
    a, b = float(a), float(b)
    ops = {"+": a + b, "-": a - b, "*": a * b}
    if op == "/":
        return a / b
    if op not in ops:
        raise KeyError(op)
    return ops[op]

for line in ["12 / 4", "3 * x", "5 / 0", "2 ^ 3", "7"]:
    try:
        print(line, "=", calc(line))
    except ZeroDivisionError:
        print(line, ": cannot divide by zero")
    except KeyError as e:
        print(line, ": unknown sign", e)
    except ValueError as e:
        print(line, ":", e)
```

You should see:

```
12 / 4 = 3.0
3 * x : could not convert string to float: 'x'
5 / 0 : cannot divide by zero
2 ^ 3 : unknown sign '^'
7 : write: NUMBER SIGN NUMBER
```

解 Chapter 33 — Python level 3: modules, venv and uv

33.1 ★ Dice.

```
import random

random.seed(3)
for _ in range(10):
    a, b = random.randint(1, 6), random.randint(1, 6)
    print(a, b, a + b)
```

You should see:

```
2 5 7
5 2 7
3 5 8
4 6 10
5 1 6
5 1 6
4 3 7
5 2 7
2 6 8
4 5 9
```

33.2 ★ Days to go.

```
from datetime import date

today = date(2026, 10, 21)   # date.today() in your version
print((date(today.year, 12, 31) - today).days)
```

You should see:

```
71
```

33.3 ★ Statistics.

```
import statistics as st

data = [2, 4, 4, 4, 5, 5, 7, 9]
print(st.mean(data), st.median(data), round(st.stdev(data), 3))
```

You should see:

```
5 4.5 2.138
```

33.4 ★★ Your module.

```
from pathlib import Path
import subprocess, sys

Path("textutil.py").write_text('''def vowels(s):
    return sum(ch in "aeiou" for ch in s.lower())

def words(s):
    return len(s.split())

if __name__ == "__main__":
    print("demo:", vowels("Fedora"), words("a b c"))
''')
Path("main2.py").write_text('from textutil import vowels, words\nprint(vowels("Sharjah"), words("one
two"))\n')
print(subprocess.run([sys.executable, "textutil.py"], capture_output=True, text=True).stdout, end="")
print(subprocess.run([sys.executable, "main2.py"], capture_output=True, text=True).stdout, end="")
```

You should see:

```
demo: 3 3
2 2
```

33.5 ★★ Counter and defaultdict.

```
from collections import Counter, defaultdict

print(Counter("hello world").most_common(3))
groups = defaultdict(list)
for w in "tea toast rice rye milk".split():
    groups[w[0]].append(w)
print(dict(groups))
```

You should see:

```
[('l', 3), ('o', 2), ('h', 1)]
{'t': ['tea', 'toast'], 'r': ['rice', 'rye'], 'm': ['milk']}
```

33.6 ★★ A project of your own.

```
$ uv init quotes && cd quotes
$ uv add --dev pytest
$ printf 'Simple is better than complex.\nReadability counts.\n' > quotes.txt
$ cat > main.py <<'EOF'
import random
from pathlib import Path
print(random.choice(Path("quotes.txt").read_text().splitlines()))
EOF
$ uv run main.py
Readability counts.
$ grep -A2 dependency-groups pyproject.toml
[dependency-groups]
dev = [
    "pytest>=9.1.1",
$ git add -A && git commit -m "quotes: a random line"
```

(not run here line by line; the same commands are run in 33.5 and chapter 35)

33.7 ★★★ A web request.

```
import json
from urllib.request import urlopen

with urlopen("https://api.github.com/repos/python/cpython", timeout=10) as r:
    repo = json.load(r)
print(repo["stargazers_count"], "stars —", repo["description"])
```

(the numbers change daily, so no output is printed here; you should see a number of stars, then "The Python programming language")

33.8 ★★★ SQLite.

```
import sqlite3

db = sqlite3.connect(":memory:")   # "money.db" keeps it on disk
db.execute("CREATE TABLE expenses (date TEXT, category TEXT, amount REAL)")
db.executemany("INSERT INTO expenses VALUES (?, ?, ?)",
               [("2026-10-21", "food", 32.5), ("2026-10-21", "taxi", 18), ("2026-10-22", "food", 12.25)])
for cat, total in db.execute("SELECT category, SUM(amount) FROM expenses GROUP BY category ORDER BY
category"):
    print(cat, total)
```

You should see:

```
food 44.75
taxi 18.0
```

33.9 ★★★★ Back up a folder.

```
import shutil
from datetime import date
from pathlib import Path

def backup(src, dest, limit=10 * 1024 * 1024, today=None):
    src, dest = Path(src), Path(dest)
    target = dest / f"{src.name}-{(today or date.today()).isoformat()}"
    copied = skipped = 0
    for p in src.rglob("*"):
        if not p.is_file():
            continue
        if p.stat().st_size > limit:
            skipped += 1
            continue
        out = target / p.relative_to(src)
        out.parent.mkdir(parents=True, exist_ok=True)
        shutil.copy2(p, out)
        copied += 1
    return target, copied, skipped

Path("docs/sub").mkdir(parents=True, exist_ok=True)
Path("docs/a.txt").write_text("a"); Path("docs/sub/b.txt").write_text("b")
Path("docs/big.bin").write_bytes(b"0" * 2000)
print(backup("docs", "backups", limit=1000, today=date(2026, 10, 21)))
```

You should see:

```
(PosixPath('backups/docs-2026-10-21'), 2, 1)
```

解 Chapter 34 — Python level 4: classes and objects

34.1 ★ A rectangle.

```
class Rectangle:
    def __init__(self, width, height):
        self.width = width
        self.height = height

    def area(self):
        return self.width * self.height

    def perimeter(self):
        return 2 * (self.width + self.height)

r = Rectangle(3, 4)
print(r.area(), r.perimeter())
```

You should see:

```
12 14
```

34.2 ★ A nice print.

```
class Rectangle:
    def __init__(self, width, height):
        self.width, self.height = width, height

    def __str__(self):
        return f"{self.width}×{self.height} rectangle"

print(Rectangle(3, 4))
```

You should see:

```
3×4 rectangle
```

34.3 ★ As a dataclass.

```
from dataclasses import dataclass

@dataclass
class Rectangle:
    width: float
    height: float

    def area(self):
        return self.width * self.height

print(Rectangle(3, 4), Rectangle(3, 4) == Rectangle(3, 4), Rectangle(3, 4).area())
```

You should see:

```
Rectangle(width=3, height=4) True 12
```

34.4 ★★ A to-do list.

```
class TodoList:
    def __init__(self):
        self.items = []

    def add(self, text):
        self.items.append([text, False])

    def done(self, n):
        self.items[n - 1][1] = True

    def __str__(self):
        return "\n".join(f"{i}. [{'x' if d else ' '}] {t}" for i, (t, d) in enumerate(self.items, 1))

t = TodoList()
t.add("shop"); t.add("cook"); t.done(1)
print(t)
```

You should see:

```
1. [x] shop
2. [ ] cook
```

34.5 ★★ Temperature.

```
class Temperature:
    def __init__(self, celsius):
        self.celsius = celsius

    def fahrenheit(self):
        return self.celsius * 9 / 5 + 32

    def __lt__(self, other):
        return self.celsius < other.celsius

    def __repr__(self):
        return f"{self.celsius}°C"

print(sorted([Temperature(31), Temperature(-2), Temperature(18)]))
print(Temperature(100).fahrenheit())
```

You should see:

```
[-2°C, 18°C, 31°C]
212.0
```

34.6 ★★ Shapes.

```
import math

class Shape:
    def area(self):
        return 0

class Circle(Shape):
    def __init__(self, r):
        self.r = r
    def area(self):
        return math.pi * self.r ** 2

class Square(Shape):
    def __init__(self, s):
        self.s = s
    def area(self):
        return self.s ** 2

print(round(sum(s.area() for s in [Circle(1), Square(2), Shape()]), 3))
```

You should see:

```
7.142
```

34.7 ★★★ A stack.

```
class Stack:
    def __init__(self):
        self._items = []
    def push(self, x):
        self._items.append(x)
    def pop(self):
        if not self._items:
            raise IndexError("empty stack")
        return self._items.pop()
    def peek(self):
        return self._items[-1] if self._items else None
    def __len__(self):
        return len(self._items)

def balanced(text):
    pairs = {")": "(", "]": "[", "}": "{"}
    s = Stack()
    for ch in text:
        if ch in "([{":
            s.push(ch)
        elif ch in pairs:
            if s.peek() != pairs[ch]:
                return False
            s.pop()
    return len(s) == 0

print(balanced("(a[b]{c})"), balanced("(a[b)]"), balanced("(("))
```

You should see:

```
True False False
```

34.8 ★★★ Money that refuses to mix.

```
class Money:
    def __init__(self, amount, currency="AED"):
        self.amount, self.currency = amount, currency
    def __add__(self, other):
        if other.currency != self.currency:
            raise ValueError(f"cannot add {self.currency} and {other.currency}")
        return Money(self.amount + other.amount, self.currency)
    def __mul__(self, n):
        return Money(self.amount * n, self.currency)
    def __str__(self):
        return f"{self.amount:.2f} {self.currency}"

print(Money(10) * 3, Money(1) + Money(2))
try:
    Money(1) + Money(1, "USD")
except ValueError as e:
    print(e)
```

You should see:

```
30.00 AED 3.00 AED
cannot add AED and USD
```

34.9 ★★★★ An inventory.

```
from dataclasses import dataclass

@dataclass(frozen=True)
class Product:
    name: str
    price: float

class Inventory:
    def __init__(self):
        self.stock = {}
    def add(self, p, n=1):
        self.stock[p] = self.stock.get(p, 0) + n
    def remove(self, p, n=1):
        if self.stock.get(p, 0) < n:
            raise ValueError(f"only {self.stock.get(p, 0)} {p.name} left")
        self.stock[p] -= n
    def value(self):
        return sum(p.price * n for p, n in self.stock.items())
    def low(self, limit):
        return sorted((p.name for p, n in self.stock.items() if n < limit))

tea, rice = Product("tea", 12.5), Product("rice", 7.25)
inv = Inventory()
inv.add(tea, 4); inv.add(rice, 10); inv.remove(tea, 3)
print(inv.value(), inv.low(5))
try:
    inv.remove(tea, 2)
except ValueError as e:
    print(e)
```

You should see:

```
85.0 ['tea']
only 1 tea left
```

解 Chapter 35 — Python level 4: testing with pytest

35.1 ★ Your first test file.

```
def is_even(n):
    return n % 2 == 0

def test_is_even():
    assert is_even(2)
    assert not is_even(7)
    assert is_even(0)

test_is_even()
print("passed")
```

You should see:

```
passed
```

35.2 ★ Parametrize it.

```
import pytest

@pytest.mark.parametrize("n, expected", [(0, True), (1, False), (2, True), (-3, False), (10, True)])
def test_is_even(n, expected):
    assert is_even(n) == expected
```

(pytest reports `5 passed`*)*

35.3 ★ Expect an error.

```
import pytest

def test_not_a_number():
    with pytest.raises(ValueError):
        int("x")
```

35.4 ★★ Test the Account.

```
import pytest

class Account:
    def __init__(self, owner, balance=0):
        self.owner, self.balance = owner, balance
    def deposit(self, amount):
        self.balance += amount
    def withdraw(self, amount):
        if amount > self.balance:
            raise ValueError(f"only {self.balance} available")
        self.balance -= amount

def test_deposit():
    a = Account("x", 10); a.deposit(5); assert a.balance == 15

def test_withdraw():
    a = Account("x", 10); a.withdraw(4); assert a.balance == 6

def test_overdraw():
    with pytest.raises(ValueError, match="available"):
        Account("x", 10).withdraw(11)

for t in (test_deposit, test_withdraw, test_overdraw):
    t()
print("3 passed")
```

You should see:

```
3 passed
```

35.5 ★★ Files with tmp_path.

```
from wc import wc

def test_empty(tmp_path):
    f = tmp_path / "e.txt"; f.write_text("")
    assert wc(f) == (0, 0, 0)

def test_two_lines(tmp_path):
    f = tmp_path / "t.txt"; f.write_text("one two\nthree\n")
    assert wc(f) == (2, 3, 14)
```

35.6 ★★ Find the bug with a test.

```
def biggest_buggy(nums):
    m = 0
    for n in nums:
        m = max(m, n)
    return m

def biggest(nums):
    m = nums[0]
    for n in nums[1:]:
        m = max(m, n)
    return m

print(biggest_buggy([-5, -2, -9]), "← wrong: all negative")
assert biggest([-5, -2, -9]) == -2
assert biggest([3, 9, 4]) == 9
print("fixed")
```

You should see:

```
0 ← wrong: all negative
fixed
```

35.7 ★★★ Test first: Roman numerals.

```
def roman(n):
    table = [(1000, "M"), (900, "CM"), (500, "D"), (400, "CD"), (100, "C"), (90, "XC"),
             (50, "L"), (40, "XL"), (10, "X"), (9, "IX"), (5, "V"), (4, "IV"), (1, "I")]
    out = ""
    for value, sym in table:
        while n >= value:
            out += sym
            n -= value
    return out

for n, r in [(1, "I"), (4, "IV"), (9, "IX"), (14, "XIV"), (40, "XL"), (1994, "MCMXCIV"), (2026, "MMXXVI")]:
    assert roman(n) == r, (n, roman(n))
print("all 7 pass")
```

You should see:

```
all 7 pass
```

35.8 ★★★ A fixture of your own.

```
import pytest
from shop import Basket, Item

@pytest.fixture
def basket():
    b = Basket()
    b.items += [Item("rice", 7.25, 3), Item("tea", 12.5)]
    return b

def test_total(basket):
    assert basket.total() == pytest.approx(34.25)

def test_two_items(basket):
    assert len(basket.items) == 2
```

(each test receives a new basket: changing it in one test cannot affect another)

35.9 ★★★★ Tests for the expense tracker.

```
import pytest
from ledger import load, parse, save, totals

def test_parse_good():
    assert parse("2026-10-21 food 32.5") == {"date": "2026-10-21", "category": "food", "amount": 32.5}

@pytest.mark.parametrize("line", ["", "lunch", "2026-10-21 food", "2026-10-21 food x"])
def test_parse_bad(line):
    with pytest.raises(ValueError):
        parse(line)

def test_totals():
    rows = [parse("d food 1"), parse("d food 2"), parse("d taxi 5")]
    assert totals(rows) == {"food": 3, "taxi": 5}

def test_round_trip(tmp_path):
    rows = [parse("2026-10-21 food 32.5")]
    save(tmp_path / "e.csv", rows)
    assert load(tmp_path / "e.csv") == rows

def test_missing_file(tmp_path):
    assert load(tmp_path / "none.csv") == []
```

(8 tests; `--cov` should report 100 % of `ledger.py`)

解 Chapter 36 — Python level 4: type hints

36.1 ★ Hint it.

```
def pay(hours: float, rate: float) -> float:
    if hours <= 40:
        return hours * rate
    return 40 * rate + (hours - 40) * rate * 1.5

def greeting(name: str, hour: int) -> str:
    return f"Good {'morning' if hour < 12 else 'evening'}, {name}"

print(pay(45, 10), greeting("Sara", 9))
```

You should see:

```
475.0 Good morning, Sara
```

36.2 ★ Read the hints.

```
def f(xs: list[str]) -> dict[str, int]:
    return {x: len(x) for x in xs}

print(f(["tea", "rice"]))
```

You should see:

```
{'tea': 3, 'rice': 4}
```

36.3 ★ Find the bug by checking.

mypy says `Incompatible return value type (got "float", expected "int")`: `/` always gives a float. Use `n // 2`, or change the return type to `float`.

36.4 ★★ Optional.

```
def price_of(item: str, prices: dict[str, float]) -> float | None:
    return prices.get(item)

def total(items: list[str], prices: dict[str, float]) -> float:
    s = 0.0
    for it in items:
        p = price_of(it, prices)
        if p is not None:
            s += p
    return s

print(total(["tea", "cake", "rice"], {"tea": 12.5, "rice": 7.25}))
```

You should see:

```
19.75
```

36.5 ★★ Typed dataclass.

```
from dataclasses import dataclass, field

@dataclass
class Book:
    title: str
    year: int
    tags: list[str] = field(default_factory=list)

def by_tag(books: list[Book], tag: str) -> list[str]:
    return [b.title for b in books if tag in b.tags]

shelf = [Book("Dune", 1965, ["sf"]), Book("Emma", 1815, ["classic"]), Book("Solaris", 1961, ["sf"])]
print(by_tag(shelf, "sf"))
```

You should see:

```
['Dune', 'Solaris']
```

36.6 ★★★ A generic helper.

```
def chunks[T](items: list[T], n: int) -> list[list[T]]:
    return [items[i:i + n] for i in range(0, len(items), n)]

print(chunks([1, 2, 3, 4, 5], 2))
print(chunks(list("abcdef"), 3))
```

You should see:

```
[[1, 2], [3, 4], [5]]
[['a', 'b', 'c'], ['d', 'e', 'f']]
```

36.7 ★★★ Literal states.

```
from typing import Literal

State = Literal["red", "green", "amber"]

def next_state(s: State) -> State:
    return {"red": "green", "green": "amber", "amber": "red"}[s]

s: State = "red"
for _ in range(4):
    print(s, end=" → ")
    s = next_state(s)
print(s)
```

You should see:

```
red → green → amber → red → green
```

36.8 ★★★★ Type the expense tracker.

```
from typing import TypedDict

class Row(TypedDict):
    date: str
    category: str
    amount: float

def load(path: str | Path) -> list[Row]: ...
def save(path: str | Path, rows: list[Row]) -> None: ...
def parse(line: str) -> Row: ...
def totals(rows: list[Row]) -> dict[str, float]: ...
```

(inside `load`*, build each* `Row` *explicitly —* `Row(date=r["date"], category=r["category"], amount=float(r["amount"]))` *— since* `csv.DictReader` *gives* `dict[str, str]`*; then* `mypy --strict` *reports* `Success`*)*

解 Chapter 37 — Python level 5: comprehensions, iterators and generators

37.1 ★ Squares of evens.

```
print([n * n for n in range(1, 21) if n % 2 == 0])
```

You should see:

```
[4, 16, 36, 64, 100, 144, 196, 256, 324, 400]
```

37.2 ★ Lengths.

```
s = "simple is better than complex"
print({w: len(w) for w in s.split()})
```

You should see:

```
{'simple': 6, 'is': 2, 'better': 6, 'than': 4, 'complex': 7}
```

37.3 ★ Flatten.

```
nested = [[1, 2], [3], [4, 5, 6]]
print([x for row in nested for x in row])
```

You should see:

```
[1, 2, 3, 4, 5, 6]
```

37.4 ★★ Countdown generator.

```
def countdown(n):
    while n > 0:
        yield n
        n -= 1

for x in countdown(3):
    print(x, end=" ")
print(list(countdown(5)))
```

You should see:

```
3 2 1 [5, 4, 3, 2, 1]
```

37.5 ★★ Matrix transpose.

```
m = [[1, 2, 3], [4, 5, 6]]
print([[row[i] for row in m] for i in range(len(m[0]))])
print([list(col) for col in zip(*m)])
```

You should see:

```
[[1, 4], [2, 5], [3, 6]]
[[1, 4], [2, 5], [3, 6]]
```

37.6 ★★ Running average.

```
def running_avg(nums):
    total = 0
    for i, n in enumerate(nums, 1):
        total += n
        yield total / i

print(*running_avg([2, 4, 6]))
```

You should see:

```
2.0 3.0 4.0
```

37.7 ★★★ Primes forever.

```
from itertools import count, islice

def primes():
    found = []
    for n in count(2):
        if all(n % p for p in found if p * p <= n):
            found.append(n)
            yield n

print(list(islice(primes(), 15)))
```

You should see:

```
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
```

37.8 ★★★ A decorator that counts.

```
from functools import wraps

def counted(f):
    @wraps(f)
    def wrapper(*args, **kwargs):
        wrapper.calls += 1
        return f(*args, **kwargs)
    wrapper.calls = 0
    return wrapper

@counted
def hello():
    return "hi"

for _ in range(3):
    hello()
print(hello.calls)
```

You should see:

```
3
```

37.9 ★★★ Words in a big file.

```
from pathlib import Path

Path("big.txt").write_text("".join(f"line {i} {'rice' if i % 7 == 0 else 'tea'}\n" for i in range(1000)))
with open("big.txt", encoding="utf-8") as f:
    print(sum(1 for line in f if "rice" in line))
```

You should see:

```
143
```

37.10 ★★★★ Memoise by hand.

```
import time
from functools import wraps

def memo(f):
    seen = {}
    @wraps(f)
    def wrapper(*args):
        if args not in seen:
            seen[args] = f(*args)
        return seen[args]
    return wrapper

@memo
def fib(n):
    return n if n < 2 else fib(n - 1) + fib(n - 2)

t = time.perf_counter()
print(fib(35), time.perf_counter() - t < 0.05)
```

You should see:

```
9227465 True
```

解 Chapter 38 — Python level 5: async basics

38.1 ★ Hello, later.

```
import asyncio

async def greet(name, delay):
    await asyncio.sleep(delay)
    print("hello", name)

async def main():
    await asyncio.gather(greet("a", 0.3), greet("b", 0.1), greet("c", 0.2))

asyncio.run(main())
```

You should see:

```
hello b
hello c
hello a
```

38.2 ★ Results in order.

```
import asyncio

async def value(v, delay):
    await asyncio.sleep(delay)
    return v

async def main():
    print(await asyncio.gather(value("first", 0.2), value("second", 0.1)))

asyncio.run(main())
```

You should see:

```
['first', 'second']
```

38.3 ★★ Time it.

```
import asyncio, time

async def nap():
    await asyncio.sleep(0.5)

async def main():
    t = time.perf_counter()
    for _ in range(10):
        await nap()
    print(f"one by one: {time.perf_counter() - t:.1f} s")
    t = time.perf_counter()
    await asyncio.gather(*(nap() for _ in range(10)))
    print(f"together:   {time.perf_counter() - t:.1f} s")

asyncio.run(main())
```

You should see:

```
one by one: 5.0 s
together:   0.5 s
```

38.4 ★★ Timeout per job.

```
import asyncio

async def job(d):
    await asyncio.sleep(d)
    return d

async def with_timeout(d):
    try:
        async with asyncio.timeout(0.35):
            return f"{await job(d)} ok"
    except TimeoutError:
        return f"{d} timed out"

async def main():
    print(await asyncio.gather(*(with_timeout(d / 10) for d in range(1, 6))))

asyncio.run(main())
```

You should see:

```
['0.1 ok', '0.2 ok', '0.3 ok', '0.4 timed out', '0.5 timed out']
```

38.5 ★★★ A producer and a consumer.

```
import asyncio

async def producer(q):
    for i in range(5):
        await asyncio.sleep(0.1)
        await q.put(i)
    await q.put(None)

async def consumer(q):
    while (item := await q.get()) is not None:
        print("square", item * item)

async def main():
    q = asyncio.Queue()
    await asyncio.gather(producer(q), consumer(q))

asyncio.run(main())
```

You should see:

```
square 0
square 1
square 4
square 9
square 16
```

38.6 ★★★ A failure in a group.

```
import asyncio

async def ok(i):
    try:
        await asyncio.sleep(1)
        print("finished", i)
    except asyncio.CancelledError:
        print("cancelled", i)
        raise

async def bad():
    await asyncio.sleep(0.1)
    raise ValueError("boom")

async def main():
    try:
        async with asyncio.TaskGroup() as tg:
            tg.create_task(ok(1)); tg.create_task(ok(2)); tg.create_task(bad())
    except* ValueError as eg:
        print("caught:", eg.exceptions)

asyncio.run(main())
```

You should see:

```
cancelled 2
cancelled 1
caught: (ValueError('boom'),)
```

38.7 ★★★★ A polite crawler, simulated.

```
import asyncio, random, time

random.seed(4)

async def fetch(i, limit):
    async with limit:
        try:
            async with asyncio.timeout(0.15):
                await asyncio.sleep(random.uniform(0.05, 0.2))
                return True
        except TimeoutError:
            return False

async def main():
    limit = asyncio.Semaphore(4)
    t = time.perf_counter()
    results = await asyncio.gather(*(fetch(i, limit) for i in range(20)))
    print(f"{sum(results)} ok, {results.count(False)} timed out, under 2 s: {time.perf_counter() - t < 2}")

asyncio.run(main())
```

You should see:

```
13 ok, 7 timed out, under 2 s: True
```

解 Chapter 39 — Python level 5: command-line tools and packaging

39.1 ★ Add an option.

```
import argparse

p = argparse.ArgumentParser()
p.add_argument("name")
p.add_argument("--lang", choices=["en", "ar"], default="en")
for argv in (["Sara"], ["Sara", "--lang", "ar"]):
    args = p.parse_args(argv)
    print({"en": "hello", "ar": "marhaba"}[args.lang], args.name)
```

You should see:

```
hello Sara
marhaba Sara
```

39.2 ★ Exit codes.

```
import sys

def main(argv):
    try:
        float(argv[0])
        return 0
    except (ValueError, IndexError):
        return 1

print(main(["3.5"]), main(["abc"]))
```

You should see:

```
0 1
```

39.3 ★★ Sum from a pipe.

```
import io, sys

def sumcol(stream):
    return sum(float(line) for line in stream if line.strip())

print(sumcol(io.StringIO("".join(f"{i}\n" for i in range(1, 101)) + "\n")))
```

You should see:

```
5050.0
```

39.4 ★★ Many files.

```
import argparse, io

def count(files, words=False):
    total = 0
    for name, f in files:
        n = sum(len(l.split()) if words else 1 for l in f)
        total += n
        print(f"{n:>6} {name}")
    if len(files) > 1:
        print(f"{total:>6} total")

count([("a", io.StringIO("x y\nz\n")), ("b", io.StringIO("w\n"))], words=True)
```

You should see:

```
     3 a
     1 b
     4 total
```

39.5 ★★★ Subcommands.

```
import argparse, math

p = argparse.ArgumentParser(prog="calc")
sub = p.add_subparsers(dest="cmd", required=True)
for name in ("add", "mul"):
    s = sub.add_parser(name); s.add_argument("a", type=float); s.add_argument("b", type=float)
sub.add_parser("sqrt").add_argument("x", type=float)

for argv in (["add", "2", "3"], ["mul", "4", "2.5"], ["sqrt", "2"]):
    a = p.parse_args(argv)
    print({"add": lambda: a.a + a.b, "mul": lambda: a.a * a.b, "sqrt": lambda: math.sqrt(a.x)}[a.cmd]())
```

You should see:

```
5.0
10.0
1.4142135623730951
```

39.6 ★★★ Test the CLI.

```
import note

def test_add_then_grep(tmp_path, monkeypatch, capsys):
    monkeypatch.setattr(note, "STORE", tmp_path / "n.jsonl")
    assert note.main(["add", "buy", "rice"]) == 0
    assert note.main(["add", "call", "Omar"]) == 0
    assert note.main(["list", "--grep", "RICE"]) == 0
    assert capsys.readouterr().out == "2026-10-21  buy rice\n"

def test_nothing_found(tmp_path, monkeypatch):
    monkeypatch.setattr(note, "STORE", tmp_path / "n.jsonl")
    assert note.main(["list"]) == 1
```

39.7 ★★★★ wordfreq, finished.

```
import json, re
from collections import Counter

def top_words(text, n, ignore=frozenset(), min_length=1):
    words = [w for w in re.findall(r"[a-z']+", text.lower()) if w not in ignore and len(w) >= min_length]
    return Counter(words).most_common(n)

text = "The cat and the hat. The cat sat on the mat with the hat."
print(top_words(text, 3))
print(json.dumps(top_words(text, 2, ignore={"the"}, min_length=3)))
```

You should see:

```
[('the', 5), ('cat', 2), ('hat', 2)]
[["cat", 2], ["hat", 2]]
```

解 Chapter 40 — Python level 6: data with the standard library

40.1 ★ The biggest sale.

```
from load import load

rows, _ = load("sales.csv")
print(max(rows, key=lambda r: r["amount"]))
```

You should see:

```
{'date': datetime.date(2026, 10, 5), 'city': 'Abu Dhabi', 'product': 'tea', 'amount': 37.5}
```

40.2 ★ Sharjah only.

```
from load import load

rows, _ = load("sales.csv")
print(sum(r["amount"] for r in rows if r["city"] == "Sharjah"))
```

You should see:

```
36.0
```

40.3 ★★ Average per product.

```
from collections import defaultdict
from statistics import mean
from load import load

rows, _ = load("sales.csv")
g = defaultdict(list)
for r in rows:
    g[r["product"]].append(r["amount"])
print({p: round(mean(v), 2) for p, v in sorted(g.items())})
```

You should see:

```
{'milk': 7.5, 'rice': 14.5, 'tea': 21.88}
```

40.4 ★★ Share of total.

```
from collections import Counter
from load import load

rows, _ = load("sales.csv")
c = Counter()
for r in rows:
    c[r["city"]] += r["amount"]
total = sum(c.values())
shares = {k: round(v / total * 100, 1) for k, v in c.items()}
print(shares, round(sum(shares.values())))
```

You should see:

```
{'Sharjah': 24.7, 'Dubai': 45.5, 'Abu Dhabi': 29.8} 100
```

40.5 ★★★ Outliers.

```
import statistics as st
from load import load

rows, _ = load("sales.csv")
amounts = [r["amount"] for r in rows]
q1, _, q3 = st.quantiles(amounts, n=4)
limit = q3 + 1.5 * (q3 - q1)
print(round(limit, 2), [a for a in amounts if a > limit])
```

You should see:

```
46.25 []
```

40.6 ★★★ Day-on-day change.

```
from collections import defaultdict
from itertools import pairwise
from load import load

rows, _ = load("sales.csv")
d = defaultdict(float)
for r in rows:
    d[r["date"]] += r["amount"]
days = sorted(d.items())
for (d0, t0), (d1, t1) in pairwise(days):
    print(f"{d1:%m-%d} {t1:>6.2f} {t1 - t0:+.2f}")
```

You should see:

```
10-02  39.50 +19.75
10-03   6.00 -33.50
10-04  21.50 +15.50
10-05  59.25 +37.75
```

40.7 ★★★★ Your own bank statement.

Use this chapter's pieces in order: `load` (with your bank's column names and date format — `datetime.strptime(text, "%d/%m/%Y")` reads `21/10/2026`), a `defaultdict(float)` keyed by `(month, category)` with `month = d.strftime("%Y-%m")`, the bar chart of 40.6, and `json.dump` with `default=str`. Test `load` on a three-line CSV written into `tmp_path`, one row of it bad.

解 Chapter 41 — Python level 6: NumPy

41.1 ★ First array.

```
import numpy as np

a = np.arange(1, 11)
print(a.sum(), a.mean(), a[a % 2 == 0] ** 2)
```

You should see:

```
55 5.5 [  4  16  36  64 100]
```

41.2 ★ Celsius, all at once.

```
import numpy as np

print(np.array([20, 25, 31, 40]) * 9 / 5 + 32)
```

You should see:

```
[ 68.   77.   87.8 104. ]
```

41.3 ★ Shapes.

```
import numpy as np

m = np.arange(20).reshape(4, 5)
print(m[:, -1], m[1], m.T.shape)
```

You should see:

```
[ 4  9 14 19] [5 6 7 8 9] (5, 4)
```

41.4 ★★ Normalise.

```
import numpy as np

rng = np.random.default_rng(0)
x = rng.normal(5, 2, size=(5, 3))
z = (x - x.mean(axis=0)) / x.std(axis=0)
print(np.round(z.mean(axis=0), 6) + 0.0, np.round(z.std(axis=0), 6))
```

You should see:

```
[0. 0. 0.] [1. 1. 1.]
```

41.5 ★★ Distances.

```
import numpy as np

p = np.array([[3, 4], [1, 1], [-2, 0], [6, 8], [0, -1]])
d = np.sqrt((p ** 2).sum(axis=1))
print(np.round(d, 3), p[d.argmin()])
```

You should see:

```
[ 5.     1.414  2.    10.     1.   ] [ 0 -1]
```

41.6 ★★ Dice statistics.

```
import numpy as np

rng = np.random.default_rng(1)
totals = rng.integers(1, 7, size=(100_000, 2)).sum(axis=1)
counts = np.bincount(totals)[2:]
print(counts, "most common:", counts.argmax() + 2)
```

You should see:

```
[ 2747  5634  8577 10999 13718 16619 13974 11198  8165  5622  2747] most common: 7
```

41.7 ★★★ Matrix product by hand.

```
import numpy as np

def matmul(A, B):
    n, k = A.shape
    k2, m = B.shape
    assert k == k2
    C = np.zeros((n, m))
    for i in range(n):
        for j in range(m):
            for t in range(k):
                C[i, j] += A[i, t] * B[t, j]
    return C

rng = np.random.default_rng(2)
A, B = rng.normal(size=(3, 4)), rng.normal(size=(4, 2))
print(np.allclose(matmul(A, B), A @ B))
```

You should see:

```
True
```

41.8 ★★★ Least squares.

```
import numpy as np

rng = np.random.default_rng(3)
x = np.linspace(0, 10, 50)
y = 3 * x + 2 + rng.normal(0, 1, 50)
print(np.round(np.polyfit(x, y, 1), 2))
X = np.column_stack([x, np.ones_like(x)])
print(np.round(np.linalg.solve(X.T @ X, X.T @ y), 2))
```

You should see:

```
[3.03 1.85]
[3.03 1.85]
```

41.9 ★★★★ Game of Life, vectorised.

```
import numpy as np

def step(b):
    n = sum(np.roll(np.roll(b, dy, 0), dx, 1)
            for dy in (-1, 0, 1) for dx in (-1, 0, 1) if (dy, dx) != (0, 0))
    return ((n == 3) | ((b == 1) & (n == 2))).astype(int)

b = np.zeros((5, 5), dtype=int)
b[2, 1:4] = 1
print(b[1:4, 1:4]); b = step(b); print(b[1:4, 1:4]); b = step(b); print(b[1:4, 1:4])
```

You should see:

```
[[0 0 0]
 [1 1 1]
 [0 0 0]]
[[0 1 0]
 [0 1 0]
 [0 1 0]]
[[0 0 0]
 [1 1 1]
 [0 0 0]]
```

C. Solutions: Rust

The worked solution to every exercise of chapters 49–61, by its number, with what it printed.

Chapter 49 — Rust level 1: variables, types and functions

49.1 ★ Print it.

```
fn main() {
    let name = "Mohamed";
    let city = "Sharjah";
    let age = 34;
    println!("{name} lives in {city} and is {age}.");
}
```

You should see:

```
Mohamed lives in Sharjah and is 34.
```

49.2 ★ Seconds.

```
fn main() {
    let week: u32 = 7 * 24 * 60 * 60;
    let century: u64 = 100 * 365 * 24 * 60 * 60;
    println!("{week} {century} {}", u32::MAX);
}
```

You should see:

```
604800 3153600000 4294967295
```

49.3 ★ Make it change.

```
fn main() {
    let mut n = 1;
    n += 2;
    println!("{n}");
}
```

You should see:

```
3
```

49.4 ★★ Celsius.

```
fn to_fahrenheit(c: f64) -> f64 {
    c * 9.0 / 5.0 + 32.0
}

fn main() {
    for c in [0.0, 37.0, 100.0] {
        println!("{c:.1} °C = {:.1} °F", to_fahrenheit(c));
    }
}
```

You should see:

```
0.0 °C = 32.0 °F
37.0 °C = 98.6 °F
100.0 °C = 212.0 °F
```

49.5 ★★ Average of three.

```
fn average(a: i32, b: i32, c: i32) -> f64 {
    (a + b + c) as f64 / 3.0
}

fn main() {
    println!("{:.3}", average(7, 8, 8));
}
```

You should see:

```
7.667
```

49.6 ★★ Swap.

```
fn swap(p: (i32, i32)) -> (i32, i32) {
    (p.1, p.0)
}

fn main() {
    println!("{:?}", swap((1, 2)));
}
```

You should see:

```
(2, 1)
```

49.7 ★★★ Seconds to time.

```
fn hms(total: u32) -> (u32, u32, u32) {
    (total / 3600, total % 3600 / 60, total % 60)
}

fn main() {
    let (h, m, s) = hms(3725);
    println!("{h:02}:{m:02}:{s:02}");
}
```

You should see:

```
01:02:05
```

49.8 ★★★ Find the overflow.

```
fn main() {
    let n = (i32::MAX as f64).sqrt() as i32;
    println!("{n} {:?} {:?}", n.checked_mul(n), (n + 1).checked_mul(n + 1));
}
```

You should see:

```
46340 Some(2147395600) None
```

49.9 ★★★★ Body mass index, with a verdict.

```
fn bmi(kg: f64, m: f64) -> f64 {
    kg / (m * m)
}

fn verdict(b: f64) -> &'static str {
    if b < 18.5 {
        "under"
    } else if b < 25.0 {
        "normal"
    } else if b < 30.0 {
        "over"
    } else {
        "obese"
    }
}

fn main() {
    for (name, kg, m) in [("Sara", 55.0, 1.68), ("Omar", 92.0, 1.75), ("Lina", 48.0, 1.70)] {
        let b = bmi(kg, m);
        println!("{name:<8}{b:>6.1}  {}", verdict(b));
    }
}
```

You should see:

```
Sara      19.5  normal
Omar      30.0  obese
Lina      16.6  under
```

答 Chapter 50 — Rust level 1: control flow

50.1 ★ Count down.

```
fn main() {
    for n in (1..=10).rev() {
        print!("{n} ");
    }
    println!("lift off!");
}
```

You should see:

```
10 9 8 7 6 5 4 3 2 1 lift off!
```

50.2 ★ FizzBuzz.

```
fn main() {
    for n in 1..=15 {
        match (n % 3, n % 5) {
            (0, 0) => println!("FizzBuzz"),
            (0, _) => println!("Fizz"),
            (_, 0) => println!("Buzz"),
            _ => println!("{n}"),
        }
    }
}
```

You should see:

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

50.3 ★ Sum of squares.

```
fn main() {
    let mut s = 0;
    for n in 1..=10 {
        s += n * n;
    }
    println!("{s}");
}
```

You should see:

```
385
```

50.4 ★★ Leap years.

```
fn is_leap(y: u32) -> bool {
    (y % 4 == 0 && y % 100 != 0) || y % 400 == 0
}

fn main() {
    for y in 1896..=1912 {
        if is_leap(y) {
            print!("{y} ");
        }
    }
    println!();
}
```

You should see:

```
1896 1904 1908 1912
```

50.5 ★★ Times table.

```
fn main() {
    for r in 1..=5 {
        for c in 1..=5 {
            print!("{:>4}", r * c);
        }
        println!();
    }
}
```

You should see:

```
   1   2   3   4   5
   2   4   6   8  10
   3   6   9  12  15
   4   8  12  16  20
   5  10  15  20  25
```

50.6 ★★ Digits.

```
fn digit_sum(mut n: u64) -> u64 {
    let mut s = 0;
    while n > 0 {
        s += n % 10;
        n /= 10;
    }
    s
}

fn main() {
    println!("{}", digit_sum(2026));
}
```

You should see:

```
10
```

50.7 ★★★ Primes below 50.

```
fn is_prime(n: u32) -> bool {
    if n < 2 {
        return false;
    }
    let mut d = 2;
    while d * d <= n {
        if n % d == 0 {
            return false;
        }
        d += 1;
    }
    true
}

fn main() {
    for n in 1..50 {
        if is_prime(n) {
            print!("{n} ");
        }
    }
    println!();
}
```

You should see:

```
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47
```

50.8 ★★★ Collatz, longest.

```
fn steps(mut n: u64) -> u32 {
    let mut s = 0;
    while n != 1 {
        n = if n % 2 == 0 { n / 2 } else { 3 * n + 1 };
        s += 1;
    }
    s
}

fn main() {
    let mut best = (1, 0);
    for n in 1..=10_000 {
        let s = steps(n);
        if s > best.1 {
            best = (n, s);
        }
    }
    println!("{} takes {} steps", best.0, best.1);
}
```

You should see:

```
6171 takes 261 steps
```

50.9 ★★★★ A diamond.

```
fn diamond(n: usize) {
    for i in (1..=n).chain((1..n).rev()) {
        println!("{}{}", " ".repeat(n - i), "*".repeat(2 * i - 1));
    }
}

fn main() {
    diamond(4);
}
```

You should see:

```
   *
  ***
 *****
*******
 *****
  ***
   *
```

答 Chapter 51 — Rust level 2: ownership

51.1 ★ Which compile?.

```
fn main() {
    let a = 5;
    let _b = a;
    println!("(a) compiles: {a}");
    // (b) does not: a was moved into b (E0382)
    let s = String::from("x");
    let _t = s.clone();
    println!("(c) compiles: {s}");
}
```

You should see:

```
(a) compiles: 5
(c) compiles: x
```

51.2 ★ Fix with clone.

```
fn main() {
    let s = String::from("hi");
    let t = s.clone();
    println!("{s} {t}");
}
```

You should see:

```
hi hi
```

51.3 ★ Give it back.

```
fn exclaim(mut s: String) -> String {
    s.push_str("!!");
    s
}

fn main() {
    let s = exclaim(String::from("wow"));
    let s = exclaim(s);
    println!("{s}");
}
```

You should see:

```
wow!!!!
```

51.4 ★★ The drop order.

```
struct Noisy(&'static str);

impl Drop for Noisy {
    fn drop(&mut self) {
        println!("drop {}", self.0);
    }
}

fn main() {
    let _x = Noisy("x");
    {
        let _y = Noisy("y");
    }
    let _z = Noisy("z");
    println!("end");
}
```

You should see:

```
drop y
end
drop z
drop x
```

51.5 ★★ Drop early.

```
struct Noisy(&'static str);

impl Drop for Noisy {
    fn drop(&mut self) {
        println!("drop {}", self.0);
    }
}

fn main() {
    let a = Noisy("a");
    drop(a);
    println!("end");
}
```

You should see:

```
drop a
end
```

51.6 ★★ Count the clones.

```
fn longest(a: String, b: String) -> String {
    if a.len() >= b.len() { a } else { b }
}

fn main() {
    let x = String::from("tea");
    let y = String::from("coffee");
    let l = longest(x.clone(), y.clone());
    println!("{l} ({x}, {y})");
}
```

You should see:

```
coffee (tea, coffee)
```

51.7 ★★★ Move into a struct.

```
struct User {
    name: String,
}

fn main() {
    let name = String::from("Sara");
    let u = User { name };
    // println!("{name}");  // error[E0382]: borrow of moved value: `name`
    println!("{}", u.name);
}
```

You should see:

```
Sara
```

51.8 ★★★ Swap ownership.

```
fn main() {
    let mut a = String::from("left");
    let mut b = String::from("right");
    std::mem::swap(&mut a, &mut b);
    let taken = std::mem::take(&mut a);
    println!("[{a}] [{b}] [{taken}]");
}
```

You should see:

```
[] [left] [right]
```

51.9 ★★★★ Ownership on paper.

```
struct Noisy(&'static str);

impl Drop for Noisy {
    fn drop(&mut self) {
        println!("drop {}", self.0);
    }
}

fn make(s: &'static str) -> Noisy {
    Noisy(s)
}

fn pick(x: Noisy, _y: Noisy) -> Noisy {
    x
}

fn main() {
    let a = make("a");
    let b = make("b");
    let c = pick(a, b);
    println!("picked {}", c.0);
    let _d = make("d");
    drop(c);
    println!("end of main");
}
```

You should see:

```
drop b
picked a
drop a
end of main
drop d
```

答 Chapter 52 — Rust level 2: borrowing and slices

52.1 ★ Borrow to measure.

```
fn vowels(s: &str) -> usize {
    s.chars().filter(|c| "aeiouAEIOU".contains(*c)).count()
}

fn main() {
    let s = String::from("Fedokawaii");
    println!("{} {} {s}", vowels(&s), vowels(&s));
}
```

You should see:

```
6 6 Fedokawaii
```

52.2 ★ Change through &mut.

```
fn double_all(v: &mut [i32]) {
    for x in v.iter_mut() {
        *x *= 2;
    }
}

fn main() {
    let mut v = vec![1, 2, 3];
    double_all(&mut v);
    println!("{:?}", v);
}
```

You should see:

```
[2, 4, 6]
```

52.3 ★ Slice it.

```
fn main() {
    let d = "2026-10-21";
    println!("{} {} {}", &d[..4], &d[5..7], &d[8..]);
}
```

You should see:

```
2026 10 21
```

52.4 ★★ Fix the order.

```
fn main() {
    let mut v = vec![1];
    let r = &v;
    println!("{:?}", r);
    v.push(2);
}
```

You should see:

```
[1]
```

52.5 ★★ Largest, by reference.

```
fn largest(xs: &[i32]) -> &i32 {
    let mut best = &xs[0];
    for x in xs {
        if x > best {
            best = x;
        }
    }
    best
}

fn main() {
    let v = vec![3, 9, 2];
    println!("{}", largest(&v));
}
```

You should see:

```
9
```

52.6 ★★ Longest without clones.

```
fn longest<'a>(a: &'a str, b: &'a str) -> &'a str {
    if a.len() >= b.len() { a } else { b }
}

fn main() {
    let x = String::from("tea");
    let y = String::from("coffee");
    println!("{} ({x}, {y})", longest(&x, &y));
}
```

You should see:

```
coffee (tea, coffee)
```

52.7 ★★★ Split in two.

```
fn split_at_comma(s: &str) -> (&str, &str) {
    match s.find(',') {
        Some(i) => (s[..i].trim(), s[i + 1..].trim()),
        None => (s, ""),
    }
}

fn main() {
    println!("{:?} {:?}", split_at_comma("Sara, Sharjah"), split_at_comma("alone"));
}
```

You should see:

```
("Sara", "Sharjah") ("alone", "")
```

52.8 ★★★ Reverse in place.

```
fn reverse(v: &mut [i32]) {
    if v.is_empty() {
        return;
    }
    let (mut i, mut j) = (0, v.len() - 1);
    while i < j {
        v.swap(i, j);
        i += 1;
        j -= 1;
    }
}

fn main() {
    let mut v = [1, 2, 3, 4, 5];
    reverse(&mut v);
    println!("{:?}", v);
}
```

You should see:

```
[5, 4, 3, 2, 1]
```

52.9 ★★★★ Words, as slices.

```
fn words(s: &str) -> Vec<&str> {
    let mut out = Vec::new();
    let mut start = None;
    for (i, c) in s.char_indices() {
        match (c == ' ', start) {
            (false, None) => start = Some(i),
            (true, Some(st)) => {
                out.push(&s[st..i]);
                start = None;
            }
            _ => {}
        }
    }
    if let Some(st) = start {
        out.push(&s[st..]);
    }
    out
}

fn main() {
    let s = "  the borrow  checker is   kind ";
    let mine = words(s);
    println!("{:?} {}", mine, mine == s.split_whitespace().collect::<Vec<_>>());
}
```

You should see:

```
["the", "borrow", "checker", "is", "kind"] true
```

答 Chapter 53 — Rust level 2: structs and methods

53.1 ★ A book.

```
#[derive(Debug)]
struct Book {
    title: String,
    year: u16,
}

fn main() {
    let a = Book { title: "Dune".into(), year: 1965 };
    let b = Book { title: "Emma".into(), year: 1815 };
    println!("{:?}\n{:?}", a, b);
}
```

You should see:

```
Book { title: "Dune", year: 1965 }
Book { title: "Emma", year: 1815 }
```

53.2 ★ Circle.

```
struct Circle {
    r: f64,
}

impl Circle {
    fn area(&self) -> f64 {
        std::f64::consts::PI * self.r * self.r
    }
    fn circumference(&self) -> f64 {
        2.0 * std::f64::consts::PI * self.r
    }
}

fn main() {
    let c = Circle { r: 2.0 };
    println!("{:.2} {:.2}", c.area(), c.circumference());
}
```

You should see:

```
12.57 12.57
```

53.3 ★ A counter.

```
struct Counter {
    n: u32,
}

impl Counter {
    fn new() -> Self {
        Counter { n: 0 }
    }
    fn click(&mut self) {
        self.n += 1;
    }
    fn reset(&mut self) {
        self.n = 0;
    }
    fn value(&self) -> u32 {
        self.n
    }
}

fn main() {
    let mut c = Counter::new();
    c.click();
    c.click();
    print!("{} ", c.value());
    c.reset();
    println!("{}", c.value());
}
```

You should see:

```
2 0
```

53.4 ★★ Account.

```
struct Account {
    owner: String,
    balance: i64,
}

impl Account {
    fn deposit(&mut self, amount: i64) {
        self.balance += amount;
    }
    fn withdraw(&mut self, amount: i64) -> bool {
        if amount > self.balance {
            return false;
        }
        self.balance -= amount;
        true
    }
}

fn main() {
    let mut a = Account { owner: "Sara".into(), balance: 1000 };
    a.deposit(500);
    let too_much = a.withdraw(2000);
    let fine = a.withdraw(300);
    println!("{} {too_much} {fine} {}", a.owner, a.balance);
}
```

You should see:

```
Sara false true 1200
```

53.5 ★★ Vector maths.

```
#[derive(Debug, Clone, Copy, PartialEq)]
struct V2 {
    x: f64,
    y: f64,
}

impl V2 {
    fn add(self, o: V2) -> V2 {
        V2 { x: self.x + o.x, y: self.y + o.y }
    }
    fn scale(self, k: f64) -> V2 {
        V2 { x: self.x * k, y: self.y * k }
    }
    fn dot(self, o: V2) -> f64 {
        self.x * o.x + self.y * o.y
    }
    fn length(self) -> f64 {
        self.dot(self).sqrt()
    }
}

fn main() {
    let a = V2 { x: 3.0, y: 4.0 };
    println!("{:?} {:?} {} {}", a.add(a), a.scale(0.5), a.dot(a), a.length());
}
```

You should see:

```
V2 { x: 6.0, y: 8.0 } V2 { x: 1.5, y: 2.0 } 25 5
```

53.6 ★★ Newtypes.

```
#[derive(Debug)]
struct Fils(i64);
#[derive(Debug)]
struct Dirham(f64);

fn to_dirham(f: Fils) -> Dirham {
    Dirham(f.0 as f64 / 100.0)
}

fn main() {
    println!("{:?}", to_dirham(Fils(1250)));
    // to_dirham(Dirham(12.5));  // error[E0308]: mismatched types: expected `Fils`, found `Dirham`
}
```

You should see:

```
Dirham(12.5)
```

53.7 ★★★ A stack of plates.

```
struct Stack {
    items: Vec<String>,
}

impl Stack {
    fn new() -> Self {
        Stack { items: Vec::new() }
    }
    fn push(&mut self, s: &str) {
        self.items.push(s.to_string());
    }
    fn pop(&mut self) -> Option<String> {
        self.items.pop()
    }
    fn peek(&self) -> Option<&String> {
        self.items.last()
    }
    fn len(&self) -> usize {
        self.items.len()
    }
    fn is_empty(&self) -> bool {
        self.items.is_empty()
    }
}

fn main() {
    let mut s = Stack::new();
    s.push("a");
    s.push("b");
    println!("{:?}", s.peek());
    println!("{:?} {} {}", s.pop(), s.len(), s.is_empty());
}
```

You should see:

```
Some("b")
Some("b") 1 false
```

53.8 ★★★ Builder.

```
#[derive(Debug)]
struct Request {
    url: String,
    method: String,
    timeout: u32,
}

impl Request {
    fn new(url: &str) -> Self {
        Request { url: url.into(), method: "GET".into(), timeout: 30 }
    }
    fn method(mut self, m: &str) -> Self {
        self.method = m.into();
        self
    }
    fn timeout(mut self, t: u32) -> Self {
        self.timeout = t;
        self
    }
}

fn main() {
    println!("{:?}", Request::new("https://example.com").method("POST").timeout(5));
}
```

You should see:

```
Request { url: "https://example.com", method: "POST", timeout: 5 }
```

53.9 ★★★★ A matrix.

```
#[derive(Debug, Clone, PartialEq)]
struct Matrix {
    rows: usize,
    cols: usize,
    data: Vec<f64>,
}

impl Matrix {
    fn new(rows: usize, cols: usize) -> Self {
        Matrix { rows, cols, data: vec![0.0; rows * cols] }
    }
    fn from(rows: usize, cols: usize, data: &[f64]) -> Self {
        assert_eq!(data.len(), rows * cols);
        Matrix { rows, cols, data: data.to_vec() }
    }
    fn get(&self, r: usize, c: usize) -> f64 {
        self.data[r * self.cols + c]
    }
    fn set(&mut self, r: usize, c: usize, v: f64) {
        self.data[r * self.cols + c] = v;
    }
    fn transpose(&self) -> Matrix {
        let mut t = Matrix::new(self.cols, self.rows);
        for r in 0..self.rows {
            for c in 0..self.cols {
                t.set(c, r, self.get(r, c));
            }
        }
        t
    }
    fn matmul(&self, o: &Matrix) -> Matrix {
        assert!(self.cols == o.rows, "cannot multiply {}x{} by {}x{}", self.rows, self.cols, o.rows,
o.cols);
        let mut out = Matrix::new(self.rows, o.cols);
        for i in 0..self.rows {
            for j in 0..o.cols {
                let mut s = 0.0;
                for k in 0..self.cols {
                    s += self.get(i, k) * o.get(k, j);
                }
                out.set(i, j, s);
            }
        }
        out
    }
}

fn main() {
    let a = Matrix::from(2, 3, &[1.0, 2.0, 3.0, 4.0, 5.0, 6.0]);
    let p = a.matmul(&a.transpose());
    println!("{:?}", p.data);
}
```

You should see:

```
[14.0, 32.0, 32.0, 77.0]
```

54.1 ★ Weekday.

```
#[derive(Debug, Clone, Copy)]
enum Day {
    Mon,
    Tue,
    Wed,
    Thu,
    Fri,
    Sat,
    Sun,
}

fn is_weekend(d: Day) -> bool {
    match d {
        Day::Sat | Day::Sun => true,
        _ => false,
    }
}

fn main() {
    for d in [Day::Mon, Day::Tue, Day::Wed, Day::Thu, Day::Fri, Day::Sat, Day::Sun] {
        print!("{:?}:{} ", d, is_weekend(d));
    }
    println!();
}
```

You should see:

```
Mon:false Tue:false Wed:false Thu:false Fri:false Sat:true Sun:true
```

54.2 ★ Safe division.

```
fn divide(a: f64, b: f64) -> Option<f64> {
    if b == 0.0 { None } else { Some(a / b) }
}

fn main() {
    for (a, b) in [(1.0, 4.0), (1.0, 0.0)] {
        match divide(a, b) {
            Some(q) => println!("{a}/{b} = {q}"),
            None => println!("{a}/{b}: cannot divide by zero"),
        }
    }
}
```

You should see:

```
1/4 = 0.25
1/0: cannot divide by zero
```

54.3 ★ unwrap_or.

```
fn main() {
    let v: Vec<i32> = Vec::new();
    println!("{}", v.first().copied().unwrap_or(-1));
}
```

You should see:

```
-1
```

54.4 ★★ Coins.

```
enum Coin {
    Fils25,
    Fils50,
    Dirham,
}

impl Coin {
    fn value_in_fils(&self) -> u32 {
        match self {
            Coin::Fils25 => 25,
            Coin::Fils50 => 50,
            Coin::Dirham => 100,
        }
    }
}

fn main() {
    let purse = [Coin::Dirham, Coin::Fils50, Coin::Fils25, Coin::Fils25];
    let total: u32 = purse.iter().map(|c| c.value_in_fils()).sum();
    println!("{total} fils");
}
```

You should see:

```
200 fils
```

54.5 ★★ Traffic light.

```
#[derive(Debug, Clone, Copy)]
enum Light {
    Red,
    Amber,
    Green,
}

impl Light {
    fn next(self) -> Light {
        match self {
            Light::Red => Light::Green,
            Light::Green => Light::Amber,
            Light::Amber => Light::Red,
        }
    }
    fn seconds(&self) -> u32 {
        match self {
            Light::Red => 30,
            Light::Green => 25,
            Light::Amber => 4,
        }
    }
}

fn main() {
    let mut l = Light::Red;
    for _ in 0..6 {
        print!("{:?}({}) ", l, l.seconds());
        l = l.next();
    }
    println!();
}
```

You should see:

```
Red(30) Green(25) Amber(4) Red(30) Green(25) Amber(4)
```

54.6 ★★ Parse a grade.

```
fn grade(score: u32) -> Option<char> {
    match score {
        90..=100 => Some('A'),
        75..=89 => Some('B'),
        60..=74 => Some('C'),
        0..=59 => Some('F'),
        _ => None,
    }
}

fn main() {
    println!("{:?} {:?} {:?}", grade(95), grade(61), grade(120));
}
```

You should see:

```
Some('A') Some('C') None
```

54.7 ★★★ A calculator enum.

```
enum Expr {
    Num(f64),
    Add(Box<Expr>, Box<Expr>),
    Mul(Box<Expr>, Box<Expr>),
}

fn eval(e: &Expr) -> f64 {
    match e {
        Expr::Num(n) => *n,
        Expr::Add(a, b) => eval(a) + eval(b),
        Expr::Mul(a, b) => eval(a) * eval(b),
    }
}

fn main() {
    use Expr::*;
    let e = Mul(Box::new(Add(Box::new(Num(2.0)), Box::new(Num(3.0)))), Box::new(Num(4.0)));
    println!("{}", eval(&e));
}
```

You should see:

```
20
```

54.8 ★★★ Find a user.

```
struct User {
    id: u32,
    name: String,
}

fn find(users: &[User], id: u32) -> Option<&User> {
    users.iter().find(|u| u.id == id)
}

fn main() {
    let users = vec![User { id: 1, name: "Sara".into() }, User { id: 2, name: "Omar".into() }];
    for id in [2, 9] {
        println!("{}", find(&users, id).map(|u| u.name.as_str()).unwrap_or("nobody"));
    }
}
```

You should see:

```
Omar
nobody
```

54.9 ★★★★ A tiny state machine.

```
#[derive(Debug, Clone, Copy)]
enum State {
    Idle,
    HasCoins(u32),
}

#[derive(Debug, Clone, Copy)]
enum Event {
    Coin(u32),
    Choose(u32),
    Cancel,
}

fn step(s: State, e: Event) -> (State, String) {
    match (s, e) {
        (State::Idle, Event::Coin(n)) => (State::HasCoins(n), format!("credit {n}")),
        (State::HasCoins(c), Event::Coin(n)) => (State::HasCoins(c + n), format!("credit {}", c + n)),
        (State::HasCoins(c), Event::Choose(p)) if c >= p => (State::Idle, format!("dispensed, change {}", c
- p)),
        (State::HasCoins(c), Event::Choose(p)) => (State::HasCoins(c), format!("need {} more", p - c)),
        (State::HasCoins(c), Event::Cancel) => (State::Idle, format!("refund {c}")),
        (State::Idle, _) => (State::Idle, "insert coins first".into()),
    }
}

fn main() {
    let mut s = State::Idle;
    for e in [Event::Choose(150), Event::Coin(100), Event::Choose(150), Event::Coin(100),
Event::Choose(150), Event::Coin(50), Event::Cancel] {
        let (next, msg) = step(s, e);
        println!("{:<12} {:<10} → {msg}", format!("{:?}", s), format!("{:?}", e));
        s = next;
    }
}
```

You should see:

```
Idle         Choose(150) → insert coins first
Idle         Coin(100)   → credit 100
HasCoins(100) Choose(150) → need 50 more
HasCoins(100) Coin(100)  → credit 200
HasCoins(200) Choose(150) → dispensed, change 50
Idle         Coin(50)    → credit 50
HasCoins(50) Cancel      → refund 50
```

答 Chapter 55 — Rust level 3: collections

55.1 ★ Sum and average.

```
fn main() {
    let v = vec![3.5, 12.0, 7.25, 9.0, 1.75];
    let sum: f64 = v.iter().sum();
    let max = v.iter().cloned().fold(f64::MIN, f64::max);
    println!("{sum} {} {max}", sum / v.len() as f64);
}
```

You should see:

```
33.5 6.7 12
```

55.2 ★ Reverse the words.

```
fn main() {
    let s = "one two three";
    let r: Vec<&str> = s.split_whitespace().rev().collect();
    println!("{}", r.join(" "));
}
```

You should see:

```
three two one
```

55.3 ★ Capitals.

```
use std::collections::HashMap;

fn main() {
    let caps = HashMap::from([("UAE", "Abu Dhabi"), ("Japan", "Tokyo"), ("Egypt", "Cairo"), ("France",
"Paris")]);
    for c in ["Japan", "Peru"] {
        println!("{c}: {}", caps.get(c).unwrap_or(&"unknown"));
    }
}
```

You should see:

```
Japan: Tokyo
Peru: unknown
```

55.4 ★★ Letter frequency.

```
use std::collections::BTreeMap;

fn main() {
    let mut m = BTreeMap::new();
    for c in "mississippi".chars() {
        *m.entry(c).or_insert(0usize) += 1;
    }
    println!("{:?}", m);
}
```

You should see:

```
{'i': 4, 'm': 1, 'p': 2, 's': 4}
```

55.5 ★★ Remove duplicates, keep order.

```
use std::collections::HashSet;

fn unique(v: &[i32]) -> Vec<i32> {
    let mut seen = HashSet::new();
    let mut out = Vec::new();
    for &x in v {
        if seen.insert(x) {
            out.push(x);
        }
    }
    out
}

fn main() {
    println!("{:?}", unique(&[3, 1, 3, 2, 1]));
}
```

You should see:

```
[3, 1, 2]
```

55.6 ★★ Palindrome.

```
fn is_palindrome(s: &str) -> bool {
    let letters: Vec<char> = s.chars().filter(|c| c.is_alphanumeric()).flat_map(|c|
c.to_lowercase()).collect();
    letters.iter().eq(letters.iter().rev())
}

fn main() {
    println!("{} {}", is_palindrome("A man, a plan, a canal: Panama"), is_palindrome("Rust"));
}
```

You should see:

```
true false
```

55.7 ★★★ Anagram groups.

```
use std::collections::BTreeMap;

fn main() {
    let words = ["listen", "silent", "enlist", "google", "gooegl", "cat"];
    let mut groups: BTreeMap<String, Vec<&str>> = BTreeMap::new();
    for w in words {
        let mut key: Vec<char> = w.chars().collect();
        key.sort();
        groups.entry(key.into_iter().collect()).or_default().push(w);
    }
    for g in groups.values() {
        println!("{:?}", g);
    }
}
```

You should see:

```
["cat"]
["google", "gooegl"]
["listen", "silent", "enlist"]
```

55.8 ★★★ Moving average.

```
use std::collections::VecDeque;

fn main() {
    let mut w = VecDeque::new();
    for x in [1.0, 2.0, 3.0, 4.0, 5.0, 6.0] {
        w.push_back(x);
        if w.len() > 3 {
            w.pop_front();
        }
        if w.len() == 3 {
            print!("{} ", w.iter().sum::<f64>() / 3.0);
        }
    }
    println!();
}
```

You should see:

```
2 3 4 5
```

55.9 ★★★★ An inventory with history.

```
use std::collections::BTreeMap;

struct Inventory {
    stock: BTreeMap<String, u32>,
    log: Vec<String>,
}

impl Inventory {
    fn new() -> Self {
        Inventory { stock: BTreeMap::new(), log: Vec::new() }
    }
    fn add(&mut self, item: &str, n: u32) {
        *self.stock.entry(item.to_string()).or_insert(0) += n;
        self.log.push(format!("+{n} {item}"));
    }
    fn take(&mut self, item: &str, n: u32) -> bool {
        match self.stock.get_mut(item) {
            Some(have) if *have >= n => {
                *have -= n;
                self.log.push(format!("-{n} {item}"));
                true
            }
            _ => {
                self.log.push(format!("refused -{n} {item}"));
                false
            }
        }
    }
    fn low(&self, limit: u32) -> Vec<&str> {
        self.stock.iter().filter(|(_, n)| **n < limit).map(|(k, _)| k.as_str()).collect()
    }
}

fn main() {
    let mut inv = Inventory::new();
    inv.add("tea", 5);
    inv.add("rice", 2);
    inv.take("tea", 4);
    inv.take("rice", 3);
    println!("{:?}\n{:?}\n{:?}", inv.stock, inv.low(2), inv.log);
}
```

You should see:

```
{"rice": 2, "tea": 1}
["tea"]
["+5 tea", "+2 rice", "-4 tea", "refused -3 rice"]
```

答 Chapter 56 — Rust level 3: errors — panic, Result and ?

56.1 ★ Parse or zero.

```
fn to_num(s: &str) -> i32 {
    s.trim().parse().unwrap_or(0)
}

fn main() {
    println!("{} {} {}", to_num("7"), to_num(" 12 "), to_num("x"));
}
```

You should see:

```
7 12 0
```

56.2 ★ Match a Result.

```
fn main() {
    for s in ["3.14", "pi"] {
        match s.parse::<f64>() {
            Ok(v) => println!("ok {v}"),
            Err(e) => println!("error {e}"),
        }
    }
}
```

You should see:

```
ok 3.14
error invalid float literal
```

56.3 ★ First with ?.

```
use std::num::ParseIntError;

fn double_str(s: &str) -> Result<i32, ParseIntError> {
    let n: i32 = s.parse()?;
    Ok(n * 2)
}

fn main() {
    println!("{:?} {}", double_str("21"), double_str("x").is_err());
}
```

You should see:

```
Ok(42) true
```

56.4 ★★ Sum of a list of strings.

```
fn sum_all(v: &[&str]) -> Result<i32, String> {
    let nums: Result<Vec<i32>, String> = v.iter().map(|s| s.parse::<i32>().map_err(|_| format!("not a
number: {s}"))).collect();
    Ok(nums?.iter().sum())
}

fn main() {
    println!("{:?} {:?}", sum_all(&["1", "2", "3"]), sum_all(&["1", "two"]));
}
```

You should see:

```
Ok(6) Err("not a number: two")
```

56.5 ★★ Config line.

```
fn parse_kv(line: &str) -> Option<(&str, &str)> {
    let (k, v) = line.split_once('=')?;
    let (k, v) = (k.trim(), v.trim());
    if k.is_empty() { None } else { Some((k, v)) }
}

fn main() {
    for l in ["theme = dark", "font=12", "nonsense", "= x"] {
        println!("{l:?} → {:?}", parse_kv(l));
    }
}
```

You should see:

```
"theme = dark" → Some(("theme", "dark"))
"font=12" → Some(("font", "12"))
"nonsense" → None
"= x" → None
```

56.6 ★★ main returns Result.

```
use std::error::Error;
use std::fs;

fn main() -> Result<(), Box<dyn Error>> {
    fs::write("out.txt", "hello rust")?;
    let s = fs::read_to_string("out.txt")?;
    println!("{}", s.len());
    Ok(())
}
```

You should see:

```
10
```

56.7 ★★★ A validated age.

```
use std::fmt;

#[derive(Debug)]
struct Age(u8);

#[derive(Debug)]
enum AgeError {
    Negative,
    TooOld,
}

impl fmt::Display for AgeError {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        match self {
            AgeError::Negative => write!(f, "an age cannot be negative"),
            AgeError::TooOld => write!(f, "an age must be under 150"),
        }
    }
}

impl Age {
    fn new(n: i32) -> Result<Age, AgeError> {
        match n {
            ..0 => Err(AgeError::Negative),
            0..150 => Ok(Age(n as u8)),
            _ => Err(AgeError::TooOld),
        }
    }
}

fn main() {
    for n in [-3, 40, 200] {
        match Age::new(n) {
            Ok(a) => println!("{n}: {:?}", a),
            Err(e) => println!("{n}: {e}"),
        }
    }
}
```

You should see:

```
-3: an age cannot be negative
40: Age(40)
200: an age must be under 150
```

56.8 ★★★ CSV sums, with line numbers.

```
use std::fs;

fn total(path: &str) -> Result<f64, String> {
    let text = fs::read_to_string(path).map_err(|e| e.to_string())?;
    let mut sum = 0.0;
    for (i, line) in text.lines().enumerate().skip(1) {
        let (_, amount) = line.split_once(',').ok_or(format!("line {}: no comma", i + 1))?;
        sum += amount.trim().parse::<f64>().map_err(|e| format!("line {}: {e}", i + 1))?;
    }
    Ok(sum)
}

fn main() {
    fs::write("a.csv", "name,amount\ntea,12.5\nrice,7.25\n").unwrap();
    fs::write("b.csv", "name,amount\ntea,12.5\nrice,lots\n").unwrap();
    println!("{:?}\n{:?}", total("a.csv"), total("b.csv"));
}
```

You should see:

```
Ok(19.75)
Err("line 3: invalid float literal")
```

56.9 ★★★★ A key-value store on disk.

```
use std::collections::BTreeMap;
use std::error::Error;
use std::fs;
use std::io::ErrorKind;

struct Store {
    path: String,
    data: BTreeMap<String, String>,
}

impl Store {
    fn open(path: &str) -> Result<Store, Box<dyn Error>> {
        let text = match fs::read_to_string(path) {
            Ok(t) => t,
            Err(e) if e.kind() == ErrorKind::NotFound => String::new(),
            Err(e) => return Err(e.into()),
        };
        let mut data = BTreeMap::new();
        for (i, line) in text.lines().enumerate() {
            let (k, v) = line.split_once('=').ok_or(format!("{path}:{}: no '='", i + 1))?;
            data.insert(k.to_string(), v.to_string());
        }
        Ok(Store { path: path.to_string(), data })
    }
    fn get(&self, k: &str) -> Option<&String> {
        self.data.get(k)
    }
    fn set(&mut self, k: &str, v: &str) {
        self.data.insert(k.into(), v.into());
    }
    fn remove(&mut self, k: &str) -> Option<String> {
        self.data.remove(k)
    }
    fn save(&self) -> Result<(), Box<dyn Error>> {
        let text: String = self.data.iter().map(|(k, v)| format!("{k}={v}\n")).collect();
        fs::write(&self.path, text)?;
        Ok(())
    }
}

fn main() -> Result<(), Box<dyn Error>> {
    let _ = fs::remove_file("kv.txt");
    let mut s = Store::open("kv.txt")?;
    s.set("theme", "dark");
    s.set("font", "12");
    s.set("tmp", "x");
    s.remove("tmp");
    s.save()?;
    let again = Store::open("kv.txt")?;
    println!("{:?} {:?}", again.get("theme"), again.data);
    fs::write("broken.txt", "ok=1\nbroken\n")?;
    if let Err(e) = Store::open("broken.txt") {
        println!("error: {e}");
    }
    Ok(())
}
```

You should see:

```
Some("dark") {"font": "12", "theme": "dark"}
error: broken.txt:2: no '='
```

57.1 ★ A Greet trait.

```
trait Greet {
    fn hello(&self) -> String;
}

struct English;
struct Arabic;

impl Greet for English {
    fn hello(&self) -> String {
        "hello".into()
    }
}

impl Greet for Arabic {
    fn hello(&self) -> String {
        "marhaba".into()
    }
}

fn main() {
    println!("{} {}", English.hello(), Arabic.hello());
}
```

You should see:

```
hello marhaba
```

57.2 ★ Display a point.

```
use std::fmt;

struct Point {
    x: i32,
    y: i32,
}

impl fmt::Display for Point {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        write!(f, "({}, {})", self.x, self.y)
    }
}

fn main() {
    println!("{}", Point { x: 3, y: 4 });
}
```

You should see:

```
(3, 4)
```

57.3 ★ Generic smallest.

```
fn smallest<T: PartialOrd + Copy>(xs: &[T]) -> T {
    let mut m = xs[0];
    for &x in xs {
        if x < m {
            m = x;
        }
    }
    m
}

fn main() {
    println!("{} {}", smallest(&[4, 2, 8]), smallest(&['q', 'b', 'z']));
}
```

You should see:

```
2 b
```

57.4 ★★ From Celsius.

```
struct Celsius(f64);
struct Fahrenheit(f64);

impl From<Celsius> for Fahrenheit {
    fn from(c: Celsius) -> Self {
        Fahrenheit(c.0 * 9.0 / 5.0 + 32.0)
    }
}

fn main() {
    let f: Fahrenheit = Celsius(100.0).into();
    println!("{}", f.0);
}
```

You should see:

```
212
```

57.5 ★★ Default settings.

```
#[derive(Debug)]
struct Settings {
    theme: String,
    font_size: u32,
}

impl Default for Settings {
    fn default() -> Self {
        Settings { theme: "dark".into(), font_size: 12 }
    }
}

fn main() {
    let s = Settings { theme: "light".into(), ..Default::default() };
    println!("{:?} {:?}", Settings::default(), s);
}
```

You should see:

```
Settings { theme: "dark", font_size: 12 } Settings { theme: "light", font_size: 12 }
```

57.6 ★★ Sum anything.

```
fn total<T: std::iter::Sum<T> + Copy>(xs: &[T]) -> T {
    xs.iter().copied().sum()
}

fn main() {
    println!("{} {}", total(&[1, 2, 3]), total(&[0.5, 0.25]));
}
```

You should see:

```
6 0.75
```

57.7 ★★★ Multiply money.

```
use std::fmt;
use std::ops::{Add, Mul};

#[derive(Clone, Copy)]
struct Money {
    fils: i64,
}

impl fmt::Display for Money {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        write!(f, "{}.{:02} AED", self.fils / 100, self.fils % 100)
    }
}

impl Add for Money {
    type Output = Money;
    fn add(self, o: Money) -> Money {
        Money { fils: self.fils + o.fils }
    }
}

impl Mul<i64> for Money {
    type Output = Money;
    fn mul(self, n: i64) -> Money {
        Money { fils: self.fils * n }
    }
}

fn main() {
    println!("{}", Money { fils: 1250 } * 3 + Money { fils: 50 });
}
```

You should see:

```
38.00 AED
```

57.8 ★★★ A zoo of trait objects.

```
trait Animal {
    fn name(&self) -> String;
    fn sound(&self) -> String;
    fn legs(&self) -> u32 {
        4
    }
}

struct Cat;
struct Cow;
struct Crow;

impl Animal for Cat {
    fn name(&self) -> String { "cat".into() }
    fn sound(&self) -> String { "meow".into() }
}
impl Animal for Cow {
    fn name(&self) -> String { "cow".into() }
    fn sound(&self) -> String { "moo".into() }
}
impl Animal for Crow {
    fn name(&self) -> String { "crow".into() }
    fn sound(&self) -> String { "caw".into() }
    fn legs(&self) -> u32 { 2 }
}

fn main() {
    let zoo: Vec<Box<dyn Animal>> = vec![Box::new(Cat), Box::new(Cow), Box::new(Crow)];
    for a in &zoo {
        println!("{} says {}", a.name(), a.sound());
    }
    println!("{} legs", zoo.iter().map(|a| a.legs()).sum::<u32>());
}
```

You should see:

```
cat says meow
cow says moo
crow says caw
10 legs
```

57.9 ★★★★ A generic stack with Display.

```
use std::fmt;

struct Stack<T> {
    items: Vec<T>,
}

impl<T> Stack<T> {
    fn new() -> Self {
        Stack { items: Vec::new() }
    }
    fn push(&mut self, x: T) {
        self.items.push(x);
    }
    fn pop(&mut self) -> Option<T> {
        self.items.pop()
    }
    fn peek(&self) -> Option<&T> {
        self.items.last()
    }
    fn len(&self) -> usize {
        self.items.len()
    }
}

impl<T: fmt::Display> fmt::Display for Stack<T> {
    fn fmt(&self, f: &mut fmt::Formatter) -> fmt::Result {
        let parts: Vec<String> = self.items.iter().map(|i| i.to_string()).collect();
        write!(f, "[{}>", parts.join(" | "))
    }
}

fn main() {
    let mut a = Stack::new();
    a.push(1);
    a.push(2);
    a.push(3);
    a.pop();
    let mut b = Stack::new();
    b.push("tea");
    b.push("rice");
    println!("{a} {b} {:?} {}", b.peek(), b.len());
}
```

You should see:

```
[1 | 2> [tea | rice> Some("rice") 2
```

答 Chapter 58 — Rust level 4: lifetimes, gently

58.1 ★ Which need 'a?.

Only (c). (a) has one input reference; (b) returns an owned `String`; (d) returns a borrow of `self` by the method rule. In (c), write `fn h<'a>(a: &'a str, b: &'a str) -> &'a str` — or tie the result to only the one it really comes from.

58.2 ★ First line.

```
fn first_line(text: &str) -> &str {
    text.lines().next().unwrap_or("")
}

fn main() {
    println!("{}", first_line("one\ntwo"));
}
```

You should see:

```
one
```

58.3 ★★ Pick by flag.

```
fn pick<'a>(a: &'a str, b: &'a str, first: bool) -> &'a str {
    if first { a } else { b }
}

fn main() {
    println!("{} {}", pick("x", "y", true), pick("x", "y", false));
}
```

You should see:

```
x y
```

58.4 ★★ Only one input matters.

```
fn prefix_of<'a>(text: &'a str, sep: &str) -> &'a str {
    text.split(sep).next().unwrap_or(text)
}

fn main() {
    let text = String::from("key=value");
    let p;
    {
        let sep = String::from("=");
        p = prefix_of(&text, &sep);
    }
    println!("{p}");
}
```

You should see:

```
key
```

58.5 ★★★ A tokenizer that borrows.

```
struct Tokens<'a> {
    rest: &'a str,
}

impl<'a> Tokens<'a> {
    fn next_word(&mut self) -> Option<&'a str> {
        let s = self.rest.trim_start();
        if s.is_empty() {
            return None;
        }
        let end = s.find(' ').unwrap_or(s.len());
        self.rest = &s[end..];
        Some(&s[..end])
    }
}

fn main() {
    let text = String::from("  rust  owns   memory ");
    let mut t = Tokens { rest: &text };
    while let Some(w) = t.next_word() {
        print!("[{w}]");
    }
    println!();
}
```

You should see:

```
[rust][owns][memory]
```

58.6 ★★★ Own instead.

```
#[derive(Debug)]
struct Highlight {
    text: String,
    line: usize,
}

fn find(doc: &str, word: &str) -> Vec<Highlight> {
    doc.lines().enumerate().filter(|(_, l)| l.contains(word)).map(|(i, l)| Highlight { text: l.to_string(),
line: i + 1 }).collect()
}

fn main() {
    let hits = {
        let doc = String::from("ownership\nborrowing rules");
        find(&doc, "borrow")
    };
    println!("{:?}", hits);
}
```

You should see:

```
[Highlight { text: "borrowing rules", line: 2 }]
```

58.7 ★★★★ Longest common prefix.

```
fn common_prefix<'a>(words: &[&'a str]) -> &'a str {
    let Some(first) = words.first() else { return "" };
    let mut end = first.len();
    for w in &words[1..] {
        let mut e = 0;
        for ((i, a), b) in first.char_indices().zip(w.chars()) {
            if a != b {
                break;
            }
            e = i + a.len_utf8();
        }
        end = end.min(e);
    }
    &first[..end]
}

fn main() {
    println!("{:?} {:?}", common_prefix(&["fedora", "fedokawaii", "federal"]), common_prefix(&["京都", "京
橋"]));
}
```

You should see:

```
"fed" "京"
```

答 Chapter 59 — Rust level 5: modules, crates and tests

59.1 ★ A module.

```
mod temperature {
    pub fn c_to_f(c: f64) -> f64 {
        c * 9.0 / 5.0 + 32.0
    }
    pub fn f_to_c(f: f64) -> f64 {
        (f - 32.0) * 5.0 / 9.0
    }
}

fn main() {
    println!("{} {}", temperature::c_to_f(100.0), temperature::f_to_c(32.0));
}
```

You should see:

```
212 0
```

59.2 ★ First tests.

```
fn is_leap(y: u32) -> bool {
    (y % 4 == 0 && y % 100 != 0) || y % 400 == 0
}

#[cfg(test)]
mod tests {
    use super::*;
    #[test]
    fn y2024() { assert!(is_leap(2024)); }
    #[test]
    fn y1900() { assert!(!is_leap(1900)); }
    #[test]
    fn y2000() { assert!(is_leap(2000)); }
}
```

You should see:

```
running 3 tests
test tests::y1900 ... ok
test tests::y2000 ... ok
test tests::y2024 ... ok

test result: ok. 3 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
```

59.3 ★ should_panic.

```
#[cfg(test)]
mod tests {
    #[test]
    #[should_panic(expected = "index out of bounds")]
    fn out_of_bounds() {
        let v = vec![1, 2];
        let _ = v[5];
    }
}
```

You should see:

```
running 1 test
test tests::out_of_bounds - should panic ... ok

test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
```

59.4 ★★ Private by design.

```
mod bank {
    pub struct Account {
        balance: u64,
    }
    impl Account {
        pub fn new() -> Self {
            Account { balance: 0 }
        }
        pub fn deposit(&mut self, n: u64) {
            self.balance += n;
        }
        pub fn balance(&self) -> u64 {
            self.balance
        }
    }
}

fn main() {
    let mut a = bank::Account::new();
    a.deposit(40);
    // a.balance = 1_000_000;  // error[E0616]: field `balance` of struct `Account` is private
    println!("{}", a.balance());
}
```

You should see:

```
40
```

59.5 ★★ Test the slug.

```
fn slugify(title: &str) -> String {
    title.to_lowercase().split(|c: char| !c.is_alphanumeric()).filter(|w| !w.is_empty()).collect::<Vec<_>>
().join("-")
}

#[cfg(test)]
mod tests {
    use super::*;
    #[test]
    fn empty() { assert_eq!(slugify(""), ""); }
    #[test]
    fn only_punctuation() { assert_eq!(slugify("!?,"), ""); }
    #[test]
    fn kanji_are_kept() { assert_eq!(slugify("京都 Trip"), "京都-trip"); }
}
```

You should see:

```
running 3 tests
test tests::empty ... ok
test tests::kanji_are_kept ... ok
test tests::only_punctuation ... ok

test result: ok. 3 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
```

59.6 ★★★ A library crate.

```
// src/lib.rs
/// ```
/// assert!((units::km_to_miles(1.609344) - 1.0).abs() < 1e-9);
/// ```
pub fn km_to_miles(km: f64) -> f64 { km / 1.609344 }

/// ```
/// assert_eq!(units::miles_to_km(1.0), 1.609344);
/// ```
pub fn miles_to_km(mi: f64) -> f64 { mi * 1.609344 }

#[cfg(test)]
mod tests {
    use super::*;
    #[test]
    fn round_trip() {
        for x in [0.0, 1.0, 42.195, 1e6] {
            assert!((miles_to_km(km_to_miles(x)) - x).abs() < 1e-9);
        }
    }
}

// tests/marathon.rs
#[test]
fn marathon() {
    assert!((units::km_to_miles(42.195) - 26.2188).abs() < 1e-3);
}
```

(cargo test prints three sections: `unittests src/lib.rs` *1 passed,* `tests/marathon.rs` *1 passed,* `Doc-tests units` *2 passed)*

59.7 ★★★★ The expense ledger, in Rust.

```
use std::collections::BTreeMap;

#[derive(Debug, PartialEq)]
pub struct Row {
    pub date: String,
    pub category: String,
    pub amount: f64,
}

pub fn parse(line: &str) -> Result<Row, String> {
    let parts: Vec<&str> = line.split_whitespace().collect();
    let [date, category, amount] = parts.as_slice() else {
        return Err("write: DATE CATEGORY AMOUNT".into());
    };
    let amount = amount.parse::<f64>().map_err(|e| format!("amount: {e}"))?;
    Ok(Row { date: date.to_string(), category: category.to_string(), amount })
}

pub fn totals(rows: &[Row]) -> BTreeMap<String, f64> {
    let mut t = BTreeMap::new();
    for r in rows {
        *t.entry(r.category.clone()).or_insert(0.0) += r.amount;
    }
    t
}

#[cfg(test)]
mod tests {
    use super::*;
    #[test]
    fn parses() {
        assert_eq!(parse("2026-10-21 food 32.5").unwrap().amount, 32.5);
    }
    #[test]
    fn refuses() {
        assert!(parse("lunch").is_err());
        assert!(parse("d food x").is_err());
    }
    #[test]
    fn totals_by_category() {
        let rows = vec![parse("d food 1").unwrap(), parse("d food 2").unwrap(), parse("d taxi 5").unwrap()];
        let t = totals(&rows);
        assert_eq!(t["food"], 3.0);
        assert_eq!(t["taxi"], 5.0);
    }
}
```

You should see:

```
running 3 tests
test tests::parses ... ok
test tests::refuses ... ok
test tests::totals_by_category ... ok

test result: ok. 3 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.01s
```

60.1 ★ Squares.

```
fn main() {
    let v: Vec<u32> = (1..=10).map(|n| n * n).collect();
    println!("{:?}", v);
}
```

You should see:

```
[1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

60.2 ★ Evens sum.

```
fn main() {
    println!("{}", (1..=100).filter(|n| n % 2 == 0).sum::<u32>());
}
```

You should see:

```
2550
```

60.3 ★ A closure that captures.

```
fn main() {
    let threshold = 18;
    let is_adult = |age: &u32| *age >= threshold;
    println!("{}", [12, 30, 17, 45].iter().filter(|a| is_adult(a)).count());
}
```

You should see:

```
2
```

60.4 ★★ Word lengths.

```
fn main() {
    let v: Vec<(String, usize)> = "rust is fast".split_whitespace().map(|w| (w.to_uppercase(),
w.len())).collect();
    println!("{:?}", v);
}
```

You should see:

```
[("RUST", 4), ("IS", 2), ("FAST", 4)]
```

60.5 ★★ Dot product.

```
fn dot(a: &[f64], b: &[f64]) -> f64 {
    a.iter().zip(b).map(|(x, y)| x * y).sum()
}

fn main() {
    println!("{}", dot(&[1.0, 2.0, 3.0], &[4.0, 5.0, 6.0]));
}
```

You should see:

```
32
```

60.6 ★★ fold.

```
fn main() {
    let p = (1..=10u64).fold(1, |acc, n| acc * n);
    let longest = "a borrowed reference lives".split_whitespace().fold("", |best, w| if w.len() > best.len()
{ w } else { best });
    println!("{p} {longest}");
}
```

You should see:

```
3628800 reference
```

60.7 ★★★ Counter iterator.

```
struct Countdown(u32);

impl Iterator for Countdown {
    type Item = u32;
    fn next(&mut self) -> Option<u32> {
        if self.0 == 0 {
            None
        } else {
            self.0 -= 1;
            Some(self.0 + 1)
        }
    }
}

fn main() {
    println!("{:?}", Countdown(5).map(|x| x * 10).collect::<Vec<_>>());
}
```

You should see:

```
[50, 40, 30, 20, 10]
```

60.8 ★★★ Group into a map.

```
use std::collections::BTreeMap;

fn main() {
    let words = ["apple", "avocado", "banana", "blueberry", "cherry"];
    let g = words.iter().fold(BTreeMap::new(), |mut m: BTreeMap<char, Vec<&str>>, w| {
        m.entry(w.chars().next().unwrap()).or_default().push(w);
        m
    });
    println!("{:?}", g);
}
```

You should see:

```
{'a': ["apple", "avocado"], 'b': ["banana", "blueberry"], 'c': ["cherry"]}
```

60.9 ★★★★ Primes, lazily.

```
struct Primes {
    found: Vec<u64>,
    next: u64,
}

impl Iterator for Primes {
    type Item = u64;
    fn next(&mut self) -> Option<u64> {
        loop {
            let n = self.next;
            self.next += 1;
            if self.found.iter().take_while(|&&p| p * p <= n).all(|p| n % p != 0) {
                self.found.push(n);
                return Some(n);
            }
        }
    }
}

fn primes() -> Primes {
    Primes { found: Vec::new(), next: 2 }
}

fn main() {
    println!("{:?}", primes().nth(99));
    println!("{}", primes().take_while(|&p| p < 1000).sum::<u64>());
    let v: Vec<u64> = primes().skip_while(|&p| p < 1000).take(20).collect();
    println!("{:?}", v.windows(2).find(|w| w[1] - w[0] == 2));
}
```

You should see:

```
Some(541)
76127
Some([1019, 1021])
```

答 Chapter 61 — Rust level 5: concurrency basics

61.1 ★ Two threads.

```
use std::thread;

fn main() {
    let a = thread::spawn(|| String::from("from a"));
    let b = thread::spawn(|| String::from("from b"));
    println!("{} {}", a.join().unwrap(), b.join().unwrap());
}
```

You should see:

```
from a from b
```

61.2 ★ Move a Vec.

```
use std::thread;

fn main() {
    let v = vec![1, 2, 3, 4];
    let h = thread::spawn(move || v.iter().sum::<i32>());
    println!("{}", h.join().unwrap());
}
```

You should see:

```
10
```

61.3 ★★ Scoped halves.

```
use std::thread;

fn main() {
    let data = [3, 17, 8, 42, 5, 23, 9, 1];
    let (l, r) = data.split_at(4);
    let m = thread::scope(|s| {
        let a = s.spawn(|| *l.iter().max().unwrap());
        let b = s.spawn(|| *r.iter().max().unwrap());
        a.join().unwrap().max(b.join().unwrap())
    });
    println!("{m}");
}
```

You should see:

```
42
```

61.4 ★★ A shared list.

```
use std::sync::{Arc, Mutex};
use std::thread;

fn main() {
    let list = Arc::new(Mutex::new(Vec::new()));
    let hs: Vec<_> = (0..4)
        .map(|i| {
            let l = Arc::clone(&list);
            thread::spawn(move || l.lock().unwrap().push(i))
        })
        .collect();
    for h in hs {
        h.join().unwrap();
    }
    let mut v = list.lock().unwrap().clone();
    v.sort();
    println!("{:?}", v);
}
```

You should see:

```
[0, 1, 2, 3]
```

61.5 ★★★ Workers and a channel.

```
use std::sync::{mpsc, Arc, Mutex};
use std::thread;

fn main() {
    let (job_tx, job_rx) = mpsc::channel::<u64>();
    let (res_tx, res_rx) = mpsc::channel::<u64>();
    let job_rx = Arc::new(Mutex::new(job_rx));
    let mut workers = Vec::new();
    for _ in 0..3 {
        let rx = Arc::clone(&job_rx);
        let tx = res_tx.clone();
        workers.push(thread::spawn(move || {
            loop {
                let job = rx.lock().unwrap().recv();
                match job {
                    Ok(n) => tx.send(n * n).unwrap(),
                    Err(_) => break,
                }
            }
        }));
    }
    drop(res_tx);
    for n in 1..=9 {
        job_tx.send(n).unwrap();
    }
    drop(job_tx);
    for w in workers {
        w.join().unwrap();
    }
    println!("{}", res_rx.iter().sum::<u64>());
}
```

You should see:

```
285
```

61.6 ★★★ An atomic counter.

```
use std::sync::atomic::{AtomicU64, Ordering};
use std::sync::Arc;
use std::thread;

fn main() {
    let c = Arc::new(AtomicU64::new(0));
    let hs: Vec<_> = (0..8)
        .map(|_| {
            let c = Arc::clone(&c);
            thread::spawn(move || {
                for _ in 0..1000 {
                    c.fetch_add(1, Ordering::Relaxed);
                }
            })
        })
        .collect();
    for h in hs {
        h.join().unwrap();
    }
    println!("{}", c.load(Ordering::Relaxed));
}
```

You should see:

```
8000
```

61.7 ★★★★ Parallel Monte Carlo.

```
use std::thread;

fn xorshift(x: &mut u64) -> f64 {
    *x ^= *x << 13;
    *x ^= *x >> 7;
    *x ^= *x << 17;
    (*x >> 11) as f64 / (1u64 << 53) as f64
}

fn main() {
    let per = 10_000_000 / 8;
    let inside: u64 = thread::scope(|s| {
        let hs: Vec<_> = (0..8u64)
            .map(|t| {
                s.spawn(move || {
                    let mut seed = 0x9E3779B97F4A7C15 ^ (t + 1).wrapping_mul(0xBF58476D1CE4E5B9);
                    let mut hits = 0;
                    for _ in 0..per {
                        let (x, y) = (xorshift(&mut seed), xorshift(&mut seed));
                        if x * x + y * y <= 1.0 {
                            hits += 1;
                        }
                    }
                    hits
                })
            })
            .collect();
        hs.into_iter().map(|h| h.join().unwrap()).sum()
    });
    println!("{:.3}", 4.0 * inside as f64 / (per * 8) as f64);
}
```

You should see:

```
3.141
```